GRAVITATIONAL LENSING OF GRAVITATIONAL WAVES:

EFFECTS OF DIFFERENT LENS MODELS

by

Saif Ali

APPROVED BY SUPERVISORY COMMITTEE:

Lindsay J. King, Chair

Michael Kesden, Co-Chair

Mustapha Ishak-Boushak

Mohammad Akbar

Kaloyan Penev

Copyright © 2023

Saif Ali

All rights reserved

To my family,

who have always been my pillars of strength

and supported me in every step of my life.

This thesis is dedicated to you.

GRAVITATIONAL LENSING OF GRAVITATIONAL WAVES: EFFECTS OF DIFFERENT LENS MODELS

by

SAIF ALI, BS-MS, MS

DISSERTATION

Presented to the Faculty of

The University of Texas at Dallas

in Partial Fulfillment

of the Requirements

for the Degree of

DOCTOR OF PHILOSOPHY IN

PHYSICS

THE UNIVERSITY OF TEXAS AT DALLAS

May 2023

ACKNOWLEDGMENTS

First of all, I would like to express my gratitude to Dr. Lindsay J. King for her kindness and endless support throughout my PhD journey. I am grateful to Dr. Michael Kesden for providing innovative ideas and deep insights into the research work. I would also like to thank my committee members, Dr. Mohammad Akbar, Dr. Mustapha Ishak-Boushaki, and Dr. Kaloyan Penev, for providing useful comments on the proposal defense. I acknowledge that this work is supported by National Science Foundation, Division of Physics under Award Number 2011977.

Last but not least, I am thankful to my family for providing the unconditional love throughout my life.

April 2023

GRAVITATIONAL LENSING OF GRAVITATIONAL WAVES:

EFFECTS OF DIFFERENT LENS MODELS

Saif Ali, PhD
The University of Texas at Dallas, 2023

Supervising Professors: Lindsay J. King, Chair
Michael Kesden, Co-Chair

Strong gravitational lensing of gravitational waves (GWs) occurs when the GWs from a compact binary system travel near a massive object. The lensed waveform is given by the product of the lensing amplification factor F and the unlensed waveform. For many axisymmetric lens models such as the point mass and singular isothermal sphere that we consider, F can be calculated in terms of two lens parameters, the lens mass M_L and source position y. In the geometrical-optics approximation (GO), lensing in these models produces at most two discrete images which can be parameterized by two image parameters, the flux ratio I and time delay Δt_d between images. In the macrolensing regime for which Δt_d is large compared to the time T they spend within the sensitivity band of GW detectors, it is natural to parameterize lensing searches in terms of these image parameters. The functional dependence of the lensed signal on these image parameters is far simpler, facilitating data analysis for events with modest signal-to-noise ratios, and constraints on I and Δt_d can be inverted to constrain M_L and y for any lens model. Previously unexplored, we use image parameters to determine the detectability of gravitational lensing of GW in the microlensing regime ($\Delta t_d \ll T$) and find that for GW events with signal-to-noise ratios ρ and total

mass M, lensing should in principle be identifiable for flux ratios $I \gtrsim 2\rho^{-2}$ and time delays $\Delta t_d \gtrsim M^{-1}$.

We also study GW lensing using non-axisymmetric lens models, such as the singular isothermal ellipsoid (SIE) lens. The use of SIE lens model is motivated by observational constraints on the shapes of galaxies and their dark matter halos, as well as the existence of EM strong lensing configurations with more than 2 images. An SIE can produce four images when the source and lens are well aligned in projection, the cross-section for which depends on the lens ellipticity. In case of the SIE lens, the GO becomes invalid when the source position is near the non-analytic regions, such as the cusps and folds, of the source plane. Therefore, for the first time, the quasi-geometrical optics approximation (QGO) is employed in order to quantify the breakdown of the GO. Using QGO we calculate the lower bound on the lens mass, below which the GO is invalid. We also perform match-filtering analysis between GW sources (lensed by an SIE, producing four images) and (i) unlensed templates and (ii) two types of two-image lensed templates, where the images have the same or different Morse index. Analogous to axisymmetric lens models, we use image parameters to derive analytical expressions to explain the mismatch behaviors for different choices of templates. These investigations will have an impact on searches for GW lensing events in data from current and future detectors.

Orbital precession is a dramatic effect induced by spin-orbit and spin-spin couplings in a compact binary source; this precession then imprints characteristic modulations in the waveform. Search efforts are underway to detect GWs with precession during the inspiral phase of the binary system. In this study, we quantify the precessional effects by introducing five new phenomenological parameters. These parameters are expected to provide a comprehensive understanding of the amplitude and frequency modulation, especially from the geometrical perspective. We also analyse the mismatch between precessing source and non-precessing

templates to determine the minimum signal-to-noise ratio needed to detect precession for upcoming observing runs of LIGO and third generation detectors.

TABLE OF CONTENTS

ACKNOWLEDGMENTS v
ABSTRACT vi
LIST OF FIGURES xii
LIST OF TABLES xviii
CHAPTER 1 INTRODUCTION 1
1.1 Sources and detectors 3
1.2 Gravitational lensing 5
1.3 Gravitational lensing of gravitational waves 8
1.4 Outline of the dissertation 11
CHAPTER 2 FORMALISM OF GRAVITATIONAL LENSING OF GRAVITATIONAL WAVES 12
2.1 Gravitational lensing 12
2.1.1 Lens equation 16
2.2 Gravitational waves 17
2.2.1 Gravitational waves in Newtonian approximation 18
2.2.2 Gravitational waves in post-Newtonian approximation 20
2.2.3 Simulating gravitational waveforms 26
2.3 Gravitational lensing of gravitational waves 28
2.3.1 Wave optics in GW lensing 29
CHAPTER 3 DETECTABILITY OF STRONGLY LENSED GRAVITATIONAL WAVES USING MODEL-INDEPENDENT IMAGE PARAMETERS 33
3.1 Axisymmetric lens models 33
3.1.1 Point mass (PM) 33
3.1.2 Singular isothermal sphere (SIS) 34
3.2 Geometrical-optics approximation 34
3.3 Model-independent image parameters 39
3.4 Match-filtering analysis 46
3.5 Mismatch approximation in geometrical-optics limit 52

3.6 Investigating striations in the mismatch . . . 53
3.7 Summary and discussion . . . 56
CHAPTER 4 INVESTIGATING GRAVITATIONAL-WAVE SOURCES LENSED BY ELLIPTICAL LENSES . . . 60
4.1 Elliptical lensing potentials . . . 60
4.2 Quasi-geometrical optics approximation . . . 63
4.3 Validity of quasi-geometrical approximation . . . 66
4.4 Image properties of singular isothermal ellipsoid (SIE) model . . . 71
4.4.1 Relation between lens and image parameters . . . 74
4.5 Quasi-geometrical optics approximation (QGO) for non-axisymmetric lens . 79
4.6 Criteria for geometrical-optics approximation . . . 82
4.6.1 Two-image regime . . . 82
4.6.2 Four-image regime . . . 83
4.7 Waveform mismatch: unlensed and lensed templates . . . 86
4.8 Summary and discussion . . . 93
CHAPTER 5 QUANTIFYING PRECESSIONAL EFFECTS ON BINARY BLACK-HOLE SYSTEMS BY INTRODUCING A GENERIC SPINNING WAVEFORM TEMPLATE . . . 96
5.1 Introduction . . . 96
5.2 Basic formalisms and notations . . . 98
5.2.1 Gravitational waveform in the quadrupole-moment approximation . . 98
5.2.2 Taxonomy of BBH precession and nutation . . . 102
5.3 A new model of regularly precessing waveforms . . . 104
5.3.1 Defining the new model using precession parameters . . . 104
5.4 Mismatch analysis . . . 115
5.5 Discussion . . . 122
CHAPTER 6 CONCLUSIONS . . . 125
APPENDIX A THREE REFERENCE FRAMES . . . 131
A.1 Detector frame . . . 131
A.2 Sky frame . . . 131

A.3 Source frame 131
A.4 GW strain 133
APPENDIX B DERIVATION OF THE PRECESSION PARAMETERS 135
REFERENCES 137
BIOGRAPHICAL SKETCH 151
CURRICULUM VITAE

LIST OF FIGURES

1.1 Spectrogram (time-frequency representation) of the GW150914 strain data (Abbott et al., 2016). 2

1.2 Schematic of masses of announced gravitational wave detection and black holes and neutron stars previously constrained through electromagnetic observations. Credit: LIGO-Virgo/Aaron Geller/Northwestern University. 3

1.3 Sensitivity curves for different gravitational wave detectors with corresponding astrophysical sources (Sabín et al., 2015). 6

1.4 Schematic of a light bundles emitted by a distant source getting gravitationally lensed by foreground galaxy cluster. Credit: NASA/ESA. 7

1.5 Amplification factor $F(w)$ for an SIS lens model is shown as a function of dimensionless frequency $w = 8\pi G M_{Lz} f$ where M_{Lz} is the lens mass and f is the gravitational wave frequency. GW lensing is categorized into (A) perturbative, (B) 'intermediate' and (C) geometrical-optics regime (Tambalo et al., 2022). . . 10

2.1 Strong gravitational lensing around GAL-CLUS-022058s, as observed by *Hubble Space Telescope that resulted in an Einstein ring configuration (nicknamed "Molten Ring") (credit: ESA/Hubble & NASA, S. Jha).* 13

2.2 The optical image of the Bullet cluster as observed by the Hubble Space Telescope. The projected mass contours are overlaid and dominated by the dark matter (Clowe et al., 2006). 14

2.3 Schematic of a typical gravitational lensing system comprising a GW emitting BBH source on the source plane, a lens on the lens plane, and an observer. The source and lens are at angular diameter distances D_S and D_L respectively, while D_{LS} is the angular diameter distance between lens and source planes. The vectors $\boldsymbol{\eta}$ and $\boldsymbol{\xi}$ are the location of the source on the source plane and impact parameter on the lens plane, respectively. The deflection angle $\hat{\boldsymbol{\alpha}}$ is defined on the lens plane (Ali et al., 2022). 15

2.4 The top panel shows the numerical relativity waveform for a GW emitting system with parameters consistent with those recovered from GW150914 event. Shaded area shows the 90% credible region for the waveform constructed using BBH template waveforms. The inset in the top panel shows the complete evolution of BBH system. The bottom panel shows the effective separation of black holes in the BBH system in units of Schwarzschild radii $R_s = 2GM/c^2$ and the effective relative velocity given by the PN parameter $v/c = (GM\pi f/c^3)^{1/3}$, where f is the GW frequency and M is the total mass (Abbott et al., 2016). 21

2.5 Schematic diagram showing different analytical approximation and numerical techniques utilized to explain the BBH depending on the mass ratio m_2/m_1 and orbital separation $r_{12} \sim Gm/v^2$ where $m = m_1 + m_2$ is the total mass and v is the orbital velocity. (Blanchet et al., 2009). 23

2.6 A schematic of L-shaped detector and compact binary system (Apostolatos et al., 1994). 27

3.1 The magnitude of the amplification factor $F(f)$ as a function of frequency f for an SIS with lens mass $M_L = 10^3 M_\odot$ and source positions $y = 0.1$, 0.5, 1.0, and 3.0 shown in red, blue, green, and black. The solid lines are calculated using the exact result of Eq. 3.3, while the dashed lines are calculated using the geometrical-optics approximation of Eq. 3.5. 36

3.2 Waveform magnitudes as functions of frequency f for our default source parameters of $A = 0.21$, $\mathcal{M} = 20M_\odot$, $\eta = 0.25$, $t_c = \phi_c = 0$, and $D = 1\,\mathrm{Gpc}$. The orange line shows the unlensed waveform $\tilde{h}(f)$, while the blue curve shows the lensed waveform $\tilde{h}^L(f) = F(f)\tilde{h}(f)$ for the amplification factor $F(f)$ of a SIS with lens parameters $y = 0.5$ and $M_L = 10^4 M_\odot$ calculated in the geometrical-optics approximation. 38

3.3 Model-dependent lens parameter (y, M_L) and model-independent 'image parameters' $(I, \Delta t_d)$. In the geometrical-optics approximation and two image case, the two sets of parameters are interchangeable for axisymmetric lens models. 39

3.4 The flux ratio I and time delay Δt_d as parametric functions of the source position y for two-image lens models of varying lens mass M_L. The solid (dashed) curves correspond to the PM (SIS) lens model, while the red (green) curves are for a lens mass $M_L = 10^3 M_\odot$ ($1.2 \times 10^3 M_\odot$). The blue cross indicates the point ($I = 0.25$, $\Delta t_d = 2.8 \times 10^{-2}$ s) where curves for the PM and SIS models intersect. 41

3.5 The image plane is shown for a compound lens model. The macrolens (red circle) produces two macroimages (grey dots). One of the macroimages, lying closer to the microlens (blue circle), experiences microlensing that results in additional splitting into microimages (black dots) (Yeung et al., 2021). 42

3.6 The mass ratio $M_L^{\mathrm{PM}}/M_L^{\mathrm{SIS}}$ of point-mass (PM) to singular-isothermal-sphere (SIS) models producing the same flux ratio I for equal time delay Δt_d. The vertical blue dashed line corresponds to $I = 0.25$ like the vertical blue dot-dashed line in Fig. 3.4. 44

3.7 Power spectral density (PSD) of the advanced LIGO (aLIGO). 47

3.8 Contour plots of the mismatch ϵ between lensed and unlensed GWs for PM lenses (top panels) and SIS lenses (bottom panels). The left panels show ϵ as a function of the model-dependent source distance y and lens mass M_L, while the right panels show ϵ as a function of the model-independent flux ratio I and time delay Δt_d. The unlensed waveforms are calculated using the default source parameters $\mathcal{M} = 20M_\odot$, $\eta = 0.25$, and $t_c = \phi_c = 0$. 48

3.9 The difference $\Delta\epsilon = |\epsilon_{\mathrm{PM}} - \epsilon_{\mathrm{SIS}}|$ between the mismatches between lensed and unlensed waveforms for PM and SIS lens models shown in the top and bottom panels of Fig. 3.8 as functions of the source position y and lens mass M_L (left panel) and the flux ratio I and time delay Δt_d (right panel). 50

3.10 The mismatch ϵ between lensed and unlensed waveforms with source parameters $\mathcal{M} = 20M_\odot$, $\eta = 0.25$, and $t_c = \phi_c = 0$ in the SIS lens model as a function of lens mass M_L (top left panel), source position y (top right panel), time delay Δt_d (bottom left panel), and flux ratio I (bottom right panel). In the top (bottom) left panel, the blue, green, and red curves correspond to $y = 0.2$, 0.6, and 1.0 ($I = 0.2$, 0.4, and 0.6). In the top (bottom) right panel, the grey, cyan, and magenta curves correspond to $M_L = 2 \times 10^3 M_\odot$, $4 \times 10^3 M_\odot$, and $6 \times 10^3 M_\odot$ ($\Delta t_d = 0.1$ s, 0.15 s, and 0.2 s). The dotted curves in the bottom panels show the mismatch $\epsilon \approx 1 - (1 + I)^{-1/2}$ in the extreme geometrical-optics limit derived in Sec. 3.5. . 51

3.11 The mismatch ϵ between a lensed GW source and unlensed templates for an SIS lens as a function of time delay Δt_d for flux ratio $I = 0.2$. The BBH system has source parameters $\mathcal{M} = 20M_\odot$, $\eta = 0.25$, and $t_c = \phi_c = 0$. The vertical dashed and dotted lines indicate the locations of the crests and troughs listed in Table 3.1. 54

4.1 $\Delta^\pm$ in Eq. 4.30 is shown as a function of y. The Δ^+ and Δ^- in Eq. 4.30 refer to image 1 and 2 respectively. The thick green line denotes the caustic at $y = 1$ for the SIS lens. 68

4.2 $M_{Lz(min)}^{(j)}$ in Eq. 4.33 is shown as a function of y for the two image region $y < 1$ of the SIS lens model. The thick green line denotes the caustic at $y = 1$. 70

4.3 Relative uncertainty of amplification factor in geometrical-optics approximation F_{go} (Eq. (3.5)) with respect to the wave optics F_{wo} (Eq. (3.3)) as a function of the dimensionless frequency $w = 8\pi M_{Lz} f$, where f is the GW frequency. The amplification factors are calculated for an SIS lens with redshifted lens mass $M_{Lz} = 10^3 M_\odot$ and $y = 0.3, 0.5$. The dotted lines represent the approximate lower bound on GW frequency f_{low} above which the geometrical-optics approximation is valid. 72

4.4 Caustics (left panel) and critical curves (right panel) are shown for an SIE lens with redshifted lens mass $M_{Lz} = 1.05 \times 10^5 M_\odot$ and ellipticity $e = 0.2$. Different markers on the left panel locate the sources at different configurations where blue, red, and green represent $\phi = 0$ (major cusp), 45° (fold), and 90° (minor cusp) cases. Two source locations are indicated by red and green circles on the source plane (left). The two-image and four-image regimes are shown with solid and transparent markers respectively. The images corresponding to source locations are shown on the right panel. 73

4.5 Flux ratio $I^{(2)}$ and time delays $\Delta t_d^{(2)}$ in the two-image regime as a function of source position y. Fold $\phi = 45°$, major cusp $\phi = 0$, and minor cusp 90° image configurations are shown in red, blue, and green respectively. The thick solid and dashed horizontal lines are the locations of the tangential and radial caustics on the source plane. The SIE lens used has a fixed redshifted lens mass $M_{Lz} = 1.05 \times 10^5 M_\odot$ and ellipticity $e = 0.2$. 75

4.6 Flux ratio $I^{(j)}$ and time delays $\Delta t_d^{(j)}$ in the four-image regime as a function of source position y. Fold $\phi = 45°$, major cusp $\phi = 0$, and minor cusp $90°$ image configurations are shown in red, blue, and green respectively. Dotted, dashed, and dash-dotted lines indicate the second, third, and fourth arriving image respectively; the first arriving image has zero time delay. The thick solid and dashed horizontal lines are the locations of the tangential and radial caustics on the source plane. The SIE lens used has a fixed redshifted lens mass $M_{Lz} = 1.05 \times 10^5 M_\odot$ and ellipticity $e = 0.2$. 76

4.7 Eigenvalues λ_1 (blue), λ_2 (red) are shown as a function of ellipticity e in the two-image regime of the SIE lens. The eigenvalues are calculated for the T_{ab} matrix of an SIE lens with $M_{Lz} = 1.0 \times 10^5 M_\odot$ and fixed source location $y = 0.5$, $\phi = 0$. Image 1 and 2 are shown in top and bottom panel respectively. The thick blue and red lines correspond to values of $\alpha^{(j)}$ and $\beta^{(j)}$ for an SIS lens with the same M_{Lz} and y. 81

4.8 Minimum redshifted lens mass $M_{Lz(min)}$ in the two-image regime for fixed $e = 0.2$ as a function of source distance y. Blue and green correspond to major $\phi = 0$ and minor $\phi = 90°$ cusp configurations respectively, while red corresponds to fold $\phi = 45°$. The solid and dotted lines correspond to image 1 and 2 respectively. The vertical solid lines are the location of the tangential caustics $y_t(\phi)$ and dashed lines are the location of the radial caustics $y_r(\phi)$. 84

4.9 Minimum redshifted lens mass $M_{Lz(min)}$ in the two-image regime as a function of polar angle ϕ for fixed $e = 0.2$, source distances $y = 0.3$ (blue) and $y = 0.5$ (red). The solid and dotted lines correspond to image 1 and 2 respectively. 85

4.10 Minimum redshifted lens mass $M_{Lz(min)}$ in the four-image regime for fixed $e = 0.2$ and $\phi = 0$ as a function of source distance y. The solid, dotted, dashed, and dash-dotted lines correspond to image 1, 2, 3, and 4 respectively. The vertical solid line is the location of tangential caustic $y_t(\phi)$. 87

4.11 Same as Fig. 4.10 but for $\phi = 45°$. 88

4.12 Same as Fig. 4.10 but for $\phi = 90°$. 89

4.13 Mismatch ϵ as a function of source distance y for fold $\phi = 45°$. The thin solid lines are the ϵ calculated numerically while the thick solid lines are the analytical approximation in the limit time delay $\Delta t_d \to \infty$. The cyan color is the unlensed GW template whereas magenta and yellow are two-image lensed GW templates with different and same Morse index respectively. 91

5.1 Schematic depictions of the sky and source coordinate frames. The sky frame, shown in red in the bottom panel, is defined such that the unit vector $\hat{\boldsymbol{N}}$ pointing from the detector to the sky location of the GW source is along the z axis, the unit vector $\hat{\boldsymbol{V}}$ points towards the zenith at the detector location is along the y axis, and the unit vector $\hat{\boldsymbol{H}} = \hat{\boldsymbol{V}} \times \hat{\boldsymbol{N}}$ completes the orthonomal triad. The source frame is defined such that the total angular momentum $\hat{\boldsymbol{J}}$ is along the Z_S axis, $\hat{\boldsymbol{N}} \times \hat{\boldsymbol{J}}$ is along the X_S axis, and $\hat{\boldsymbol{Y}}_S = \hat{\boldsymbol{Z}}_S \times \hat{\boldsymbol{X}}_S$ completes the orthonormal triad. 101

5.2 Precession of the spherical coordinates θ_L and Φ_L that specify the direction of orbital angular momentum $\boldsymbol{L}$ with respect to the detector frame. The BBHs have equal masses and chirp mass $\mathcal{M} = 20M_\odot$. The top row shows this precession for dimensionlesss precession frequency $\tilde{\Omega} = 2$, azimuthal integration constant $\gamma_p = 0$, and dimensionless precession amplitude $\tilde{\theta} = 1$ (green), 4 (blue), and 8 (purple). The left, middle, and right panels of the top row correspond to different directions of the total angular momentum $\boldsymbol{J}$ listed as Systems 1, 2, and 3 in Table 5.1. The bottom row show the precession of $\boldsymbol{L}$ for System 2. The left panel is a copy of the middle panel of the top row, the middle panel has $\tilde{\theta} = 4$, $\gamma_p = 0$, and $\tilde{\Omega} = 1$ (green), 2 (blue), and 3 (purple), and the bottom panel has $\tilde{\theta} = 4$, $\tilde{\Omega} = 2$, and $\gamma_p = 0$ (green), $\pi/2$ (blue), and π (purple). 106

5.3 Dimensionless precession frequency $\tilde{\Omega}$ as a function of total binary mass M. In the left panel, the dot-dashed, dashed, and dotted horizontal lines show the 5th, 50th, and 95th percentiles of the $\tilde{\Omega}$ distribution for BBHs with isotropic maximal spins and mass ratios $q = 1$ (purple), $q = 0.5$ (blue), and $q = 0.2$ (green). In the right panel, the dot-dashed, dashed, and dotted horizontal lines show the 5th, 50th, and 95th percentiles of the $\tilde{\Omega}$ distribution for equal-mass BBHs with dimensionless spins magnitudes $\chi_i = 1$ (purple), $\chi_i = 0.5$ (blue), and $\chi_i = 0.2$ (green). The solid curves in both panels show the minimum value of $\tilde{\Omega}$ for which at least one precession cycle occurs after the BBHs enter the LVK sensitivity band at a GW frequency of 20 Hz. 108

5.4 The ratio of the GW amplitude B given by Eq. (5.10) in our regularly precessing model (RP) to a non-precessing (NP) model with the same BBH masses. The left, middle, and right panels correspond to directions of the total angular momentum $\hat{\boldsymbol{J}}$ listed as Systems 1, 2, and 3 in Table 5.1. The top panels fix $\tilde{\Omega} = 2$, $\gamma_p = 0$ and vary $\tilde{\theta}$, the middle panels fix $\tilde{\theta} = 4$, $\gamma_p = 0$ and vary $\tilde{\Omega}$, and the bottom panels fix $\tilde{\theta} = 4$, $\tilde{\Omega} = 2$ and vary γ_p. 109

5.5 The ratio of the GW amplitude B given by Eq. (5.10) in our regularly precessing model (RP) to a non-precessing (NP) model with the same BBH masses of $\mathcal{M} = 10M_\odot$. The left[right] panel corresponds to directions of the total angular momentum $\hat{\boldsymbol{J}}$ and the line of sight $\hat{\boldsymbol{N}}$ that produce a face-on[edge-on] case, i.e. $\theta_S, \Phi_S, \theta_J, \Phi_J = (\pi/6, \pi/4, \pi/6, \pi/4)\big[\theta_S, \Phi_S, \theta_J, \Phi_J = (\pi/6, \pi/3, 2\pi/3, \pi/3)\big]$. Both panels fix $\tilde{\Omega} = 2$, $\gamma_p = 0$ and vary $\tilde{\theta}$. 111

5.6 The evolution of the polarization phase ϕ_p given by Eq. (5.11) in our regularly precessing model (RP) with respect to the frequency. The left, middle, and right panels correspond to directions of the total angular momentum $\hat{\boldsymbol{J}}$ listed as Systems 1, 2, and 3 in Table 5.1. The top panels fix $\tilde{\Omega} = 2$, $\gamma_p = 0$ and vary $\tilde{\theta}$, the middle panels fix $\tilde{\theta} = 4$, $\gamma_p = 0$ and vary $\tilde{\Omega}$, and the bottom panels fix $\tilde{\theta} = 4$, $\tilde{\Omega} = 2$ and vary γ_p. 113

5.7 The evolution of the phase correction term $\delta\Phi$ given by Eq. (5.12) in our regularly precessing model (RP) with respect to the frequency. The left, middle, and right panels correspond to directions of the total angular momentum $\hat{\boldsymbol{J}}$ listed as Systems 1, 2, and 3 in Table 5.1. The top panels fix $\tilde{\Omega} = 2$, $\gamma_p = 0$ and vary $\tilde{\theta}$, the middle panels fix $\tilde{\theta} = 4$, $\gamma_p = 0$ and vary $\tilde{\Omega}$, and the bottom panels fix $\tilde{\theta} = 4$, $\tilde{\Omega} = 2$ and vary γ_p. 114

5.8 The mismatch ϵ given by Eq. (5.27) between a regularly precessing(RP) GW source and a non-precessing(NP) GW template with respect to the normalized precession amplitude $\tilde{\theta}$ (left panels) and the normalized precession frequency $\tilde{\Omega}$ (right panels). All four panels correspond to the direction of the total angular momentum $\hat{\boldsymbol{J}}$ listed as System 2 in Table 5.1. The sky location $\hat{\boldsymbol{N}}$ is specified by $(\theta_S, \Phi_S) = (\pi/4, 0)$, and the precession integration constant is $\gamma_p = 0$. The values of the chirp mass are $\mathcal{M} = 10M_\odot$ (top panels) and $\mathcal{M} = 20M_\odot$ (bottom panels). The green, blue, and purple curves show $\tilde{\Omega} = 1$, 2, and 3 (left panels) and $\tilde{\theta} = 1$, 4, and 8 (right panels). 117

5.9 The evolution of the mismatch ϵ given by Eq. (5.27) between a regularly precessing(RP) GW source and a non-precessing(NP) GW template with respect to the normalized precession frequency $\tilde{\Omega}$. The direction of the total angular momentum $\hat{\boldsymbol{J}}$ is listed as System 2 in Table 5.1 and the direction of the line of sight $\hat{\boldsymbol{N}}$ is specified by $(\theta_S, \Phi_S) = (\pi/4, 0)$ and for the chirp mass we have chosen $\mathcal{M} = 10M_\odot$. The purple (green) solid line denotes a value of $\tilde{\theta} = 8$ ($\tilde{\theta} = 1$). γ_p is varied from 0 to 360 for each value of $\tilde{\Omega}$ and the maximum and minimum value of the mismatch is obtained independently. The black(red) solid line is showing the maximum(minimum) envelope for the mismatch values of each individual γ_p for $\tilde{\theta} = 8$, while the dashed lines are calculated for $\tilde{\theta} = 1$. 118

5.10 The evolution of the mismatch ϵ given by Eq. (5.27) between a regularly precessing(RP) GW source and a non-precessing(NP) GW template with respect to the normalized precession frequency $\tilde{\Omega}$. The direction of the total angular momentum $\hat{\boldsymbol{J}}$ is listed as System 2 in Table 5.1 and the direction of the line of sight $\hat{\boldsymbol{N}}$ is specified by $(\theta_S, \Phi_S) = (\pi/4, 0)$ and for the chirp mass we have chosen $\mathcal{M} = 20M_\odot$. The blue(red) solid line denote that we are using the PSD of the Cosmic Explorer (aLIGO) detector for the calculation of our mismatch. The BBHs enter the Cosmic Explorer (aLIGO) sensitivity band at a GW frequency of 10Hz (20Hz). The solid lines have a value of $\tilde{\theta} = 8$ ($\tilde{\theta} = 1$) for the left (right) panel and a fixed $\gamma_p = 0$. The dot-dashed, dashed, and dotted blue horizontal lines show the value of $1/2\rho^2$ for a different choice of luminosity distance D_L. The different choices are shown in the legend. 121

6.1 Lensed gravitational waveform (orange) constructed for Point-Mass lens with fiducial lens parameters $y = 1.2$ and $M_{Lz} = 2 \times 10^3 M_\odot$. Precessing gravitational waveform (blue) constructed with fiducial precessing parameters $\tilde{\theta} = 2.0$, $\tilde{\Omega} = 10.0$, and $\gamma_p = 0.8$. 129

LIST OF TABLES

3.1 The five columns in this table list: (1) the index n of the crests and troughs, (2) the numerically determined time delay of the crest, (3) the time delay of the crest predicted by Eq. 3.28a, (4) the numerically determined time delay of the trough, and (5) the time delay of the trough predicted by Eq. 3.28b. The lens and source parameters are the same as those in Fig. 3.11. 56

4.1 The rows show the inner product Eq. (5.23) and mismatch ϵ Eq. (5.27) between different source h_s and template h_t waveforms in the extreme geometrical-optics limit $\Delta t_d \to \infty$ are shown. Here $\mu^{(i)}$ is the magnification of ith image, $I = |\mu_i|/|\mu_1|$, and $I^{(t)} = \sum_i I^{(i)}$. The columns are different templates chosen; unlensed, two image lensed template with different Morse index, and two image lensed template with same Morse index. 92

5.1 Spherical coordinates θ_J, Φ_J of the total angular momentum $\boldsymbol{J}$ in the detector frame, inclination ι_{JN} between $\hat{\boldsymbol{J}}$ and the sky location $\hat{\boldsymbol{N}}$, and polarization angle ψ_0 for zero precession amplitude. All three BBH systems have equal-mass BBHs with chirp mass $\mathcal{M} = 10M_\odot$ and sky location $\theta_S = \pi/4, \Phi_S = 0$. 107

CHAPTER 1

INTRODUCTION

After publishing the special theory of relativity in 1905 (Einstein, 1905), Einstein spent the next ten years developing the general theory of relativity, which was published in 1915 (Einstein, 1915). In special relativity he introduced the equivalence principle stating that the laws of physics should be equal regardless of inertial frame, and that the speed of light in vacuum is constant. In general relativity he extended the equivalence principle to the principle of general covariance where the laws of physics have the same form regardless of the coordinate systems. Gravitational waves (GWs) are one of the revolutionary predictions of general relativity. These are the ripples in the fabric of space-time contract and stretch space as they propagate. Einstein's general relativity predicts GWs in a similar fashion to the electromagnetic waves (EMs) predicted by the Maxwell's theory of electromagnetism.

GWs were first proposed by Oliver Heaviside in 1893 and then later by Henri Poincaré in 1905, who suggested the existence of GWs as a counterpart of EMs travelling at the speed of light. A decade later when Einstein proposed general relativity, these waves were predicted by the theory itself. Initially Einstein refused to believe the existence of these waves and in a letter to Karl Schwarzschild on February 1916 he noted that "there are no gravitational waves, that are analogous to electromagnetic waves". His doubt included questions regarding whether the waves were physical or just an artifact of of coordinate choice. It took nearly half a decade to establish the theoretical framework of GWs, when in the 1950s physicists concluded that GWs are indeed physical and can carry energy.

The attempts to measure GWs begin in the mid 1960s when Joseph Weber built the first gravitational wave detectors now known as Weber bars and, in 1969, he claimed to have detected the first GWs. He was unable to convince the physics community about the detection despite a number of published claims (Weber, 1969; Weber, 1970; Lee et al., 1976). The first evidence of GWs came in the form of indirect evidence from the binary pulsar

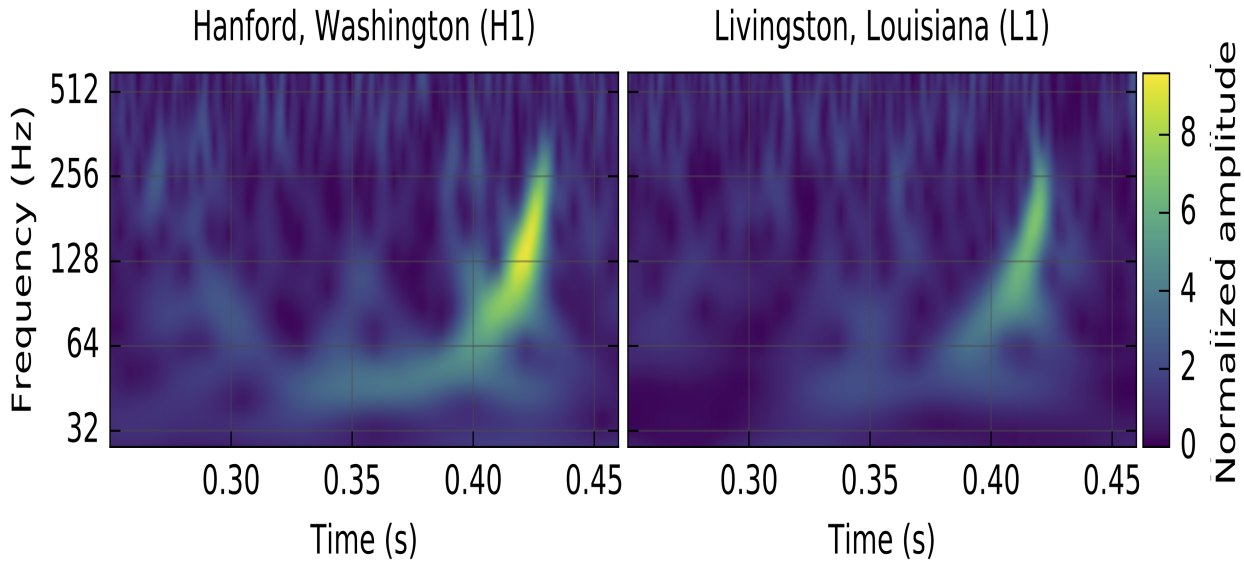


Figure 1.1. Spectrogram (time-frequency representation) of the GW150914 strain data (Abbott et al., 2016).

PSR B1913+16 observations made by Hulse and Taylor in 1974 (Hulse and Taylor, 1975). Measurements of this binary system have been consistent with the theoretical calculations, and were the first empirical test of the effects of GWs. Hulse and Taylor received the Nobel Prize in Physics for their work in 1993.

The first direct detection of GWs was made by the Laser Interferometer Gravitational-Wave Observatory (LIGO) detectors in September 14, 2015 (Abbott et al., 2016), a century after predicted by general relativity. The GW event, GW150914, was detected by the LIGO Hanford, WA, and Livingston, LA, observatories with a time difference of 10 ms due to the angle between the two detectors and the source. The GWs were produced by two black holes (BHs), with masses $29M_{\odot}$ and $36M_{\odot}$, merging at a distance of 1.3 billion light years away from the Earth. The chirp signal received from this event increased in frequency from 35 to 250 Hz over 5 orbits in a period of 0.2 seconds as seen in Fig. 1.1. From the detected gravitational waves, we can infer the various astrophysical properties of the merging objects like the masses and spins of the individual objects, distance to the merger etc. Since the

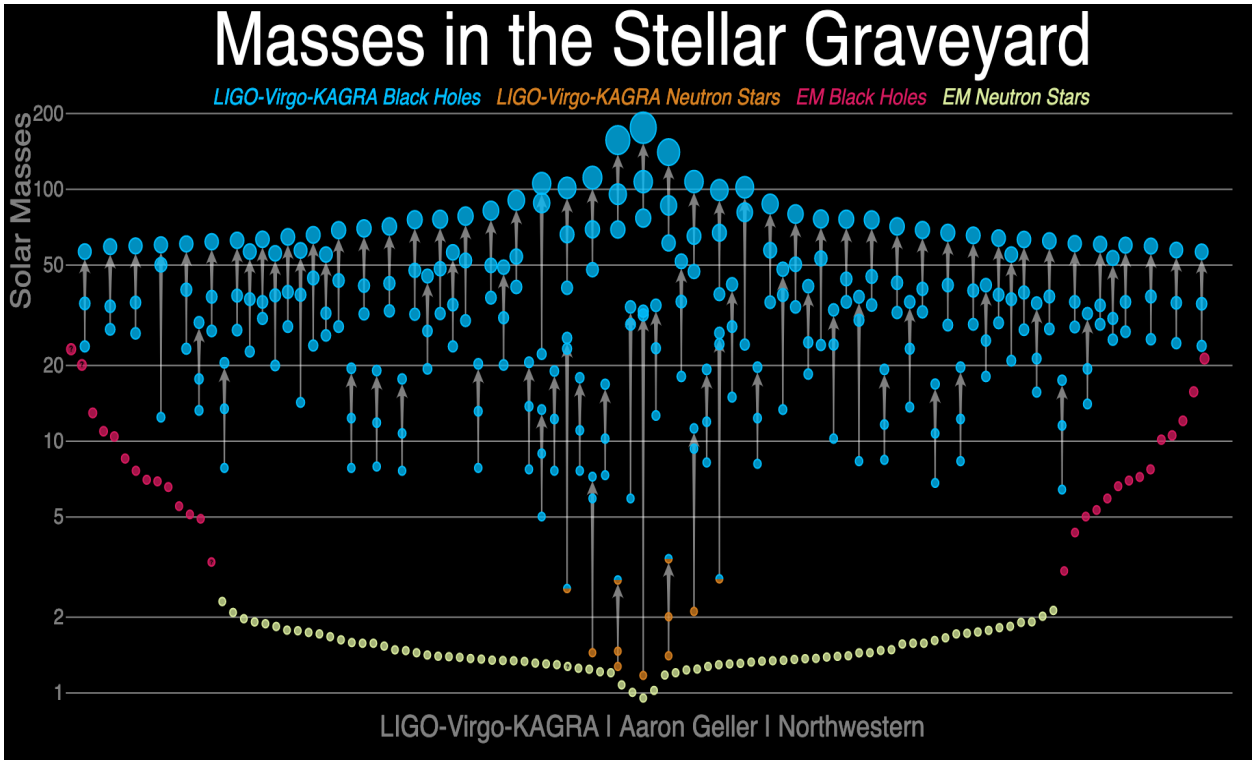


Figure 1.2. Schematic of masses of announced gravitational wave detection and black holes and neutron stars previously constrained through electromagnetic observations. Credit: LIGO-Virgo/Aaron Geller/Northwestern University.

first detection of GW150914 in 2015 (Abbott et al., 2016), Advanced LIGO (Aasi et al., 2015) and Advanced Virgo (Acernese et al., 2015) have reported about 90 events (Fig. 1.2), most of which are mergers between stellar-mass BHs, during their first three observing runs (Abbott et al., 2021). In addition to BH mergers, other events including binary neutron star (NS) mergers (Abbott et al., 2017), and NS-BH mergers (Abbott et al., 2021) have also been observed.

1.1 Sources and detectors

Generally speaking, any accelerating mass can generate GWs, including accelerating mass here on the Earth. However, only objects with very large masses can create GWs with enough energy that could lead to confident detection. Therefore, we have to look deep

into space to find GWs. GWs can be broadly categorized into four different types based on their source: Continuous, Compact Binary Inspiral, Stochastic, and Burst. Continuous GWs are produced by a single spinning NS. The GWs are generated due to deformities on the surface on NS. These deformities are basically bumps extending no more than 10 cm above the surface, that result in spherically asymmetric spinning objects. The emitted GWs are continuous in the sense that the emitted waves have the same frequency and amplitude. Compact Binary Inspiral GWs are produced by massive and compact binary objects that are orbiting each other in quasi-circular orbits. Compact binary systems can be further classified into binary NS, binary BHs, and NS-BH binary, where each system creates unique GW patterns. Stochastic GWs are the relic GWs from the Big Bang that are too faint for any current operational GW detector to observe. However, they are presumed to be passing by the Earth all the time and are mixed together to form what is called "Stochastic Signal". Burst GWs are expected to be produced by short duration unknown or unexpected sources such as a supernovae or gamma ray bursts. However, it is too difficult to model the GWs emitted by such systems (Cutler and Thorne, 2002; Riles, 2013).

Like EMs, GWs also have a spectrum. As we discussed earlier, the type of GWs produced depends on the source that generated them. Therefore, GWs can be present at a range of frequencies that depends on the properties of these sources. A specific type of detector is needed to detect GWs with a specific range of frequency. Currently, we have three categories of detectors: Ground-based detectors, space-based detectors, and pulsar timing arrays.

- The most common are the ground-based detectors. These detectors are sensitive to GWs in the frequency range $\mathcal{O}(10\text{–}10^3)$ Hz. The most notable ground-based detectors are Advanced LIGO (Aasi et al., 2015) and Advanced Virgo (Acernese et al., 2015), which work in collaboration. Kamioka Gravitational Wave Detector (KAGRA) (Somiya, 2012; Aso et al., 2013; Akutsu et al., 2021) has joined the preexisting LIGO-Virgo collaboration to form the Advanced LIGO-Virgo-Kagra (LVK) network. This

collaboration is supported by GEO600 (Grote and LIGO Scientific Collaboration, 2010) in Germany. The Einstein Telescope (Hild et al., 2011), a third generation (3G) detector, is constructed underground to minimize the impact of the seismic noise. It enables the detector to observe frequencies in a range $\mathcal{O}(10 - 10^4)$ Hz. These detectors are sensitive to Compact Binary Inspirals, Supernovae, and Continuous GWs (Fig: 1.3).

- Space-based detectors work on similar principles to ground-based detectors, but are sensitive to lower frequency GWs than their ground-based counterparts. The two main reasons behind the sensitivity to lower frequencies is, (1) space-based detectors can have much longer arms, and (2) they are unaffected by seismic noise. Space-based detectors such as the Laser Interferometer Space Antenna (LISA) (Barausse et al., 2020) is sensitive to millihertz GWs while Deci-Hertz Interferometer Gravitational Wave Observatory (DECIGO) (Kawamura et al., 2021) and Big Bang Observer (BBO) are sensitive to decihertz GWs. Space-based detectors are sensitive to massive BH binaries, galactic white-dwarf binaries, and Extreme Mass-Ratio Inspirals (EMRI) (Fig: 1.3).

- Pulsar timing arrays (PTAs) can be thought of as naturally occurring interferometers, consisting of set of pulsars which is analysed to search for correlated signatures in the pulse arrival times altered due to the passage of GWs through the space between the pulsars and observers on Earth, slightly increasing or decreasing their distances. They are sensitive to much lower frequencies than the space-based and ground-based detectors. Supermassive BH binaries, Stochastic GWs are the sources of PTAs (Fig: 1.3).

1.2 Gravitational lensing

Gravitational lensing is a fascinating phenomenon that occurs when the path of the light bundles are distorted and deflected by the curvature of spacetime caused by massive objects

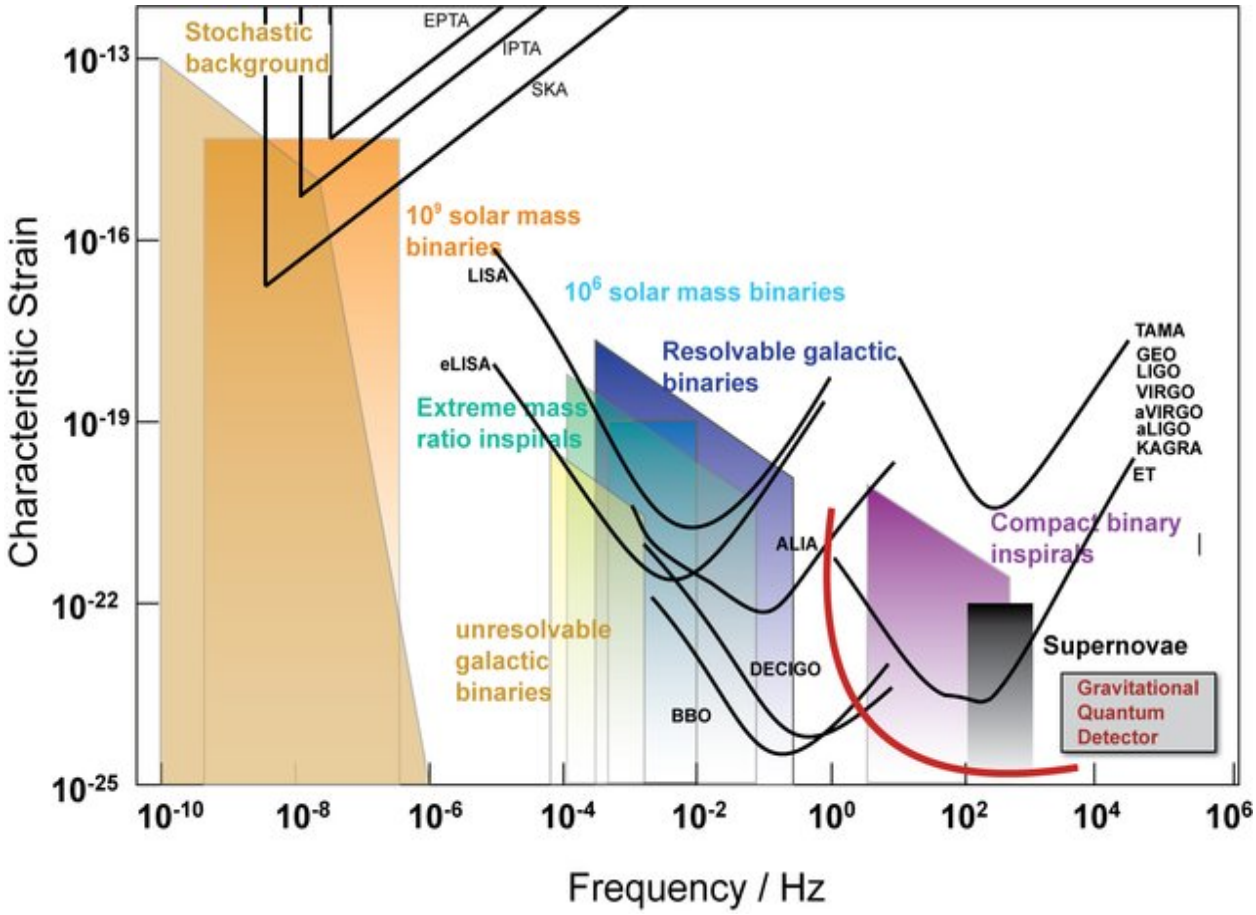


Figure 1.3. Sensitivity curves for different gravitational wave detectors with corresponding astrophysical sources (Sabín et al., 2015).

(Fig. 1.4). It was first predicted by Albert Einstein in his theory of general relativity (GR), which revolutionized our understanding of the universe. Einstein predicted that the angle by which a point mass lens (such as a star) deflects light rays is $4GM/c^2\xi$ where G, M, ξ and c are Newton's gravitational constant, lens mass, impact parameter of light ray and speed of light respectively. This prediction was spectacularly verified by Eddington and his team during a solar eclipse in 1919, when the apparent positions of stars close in projection to the sun were altered (Dyson et al., 1920).

One of the key applications of gravitational lensing is in studying the distribution of dark matter in the universe (Dalal and Kochanek, 2002). Dark matter is an elusive substance

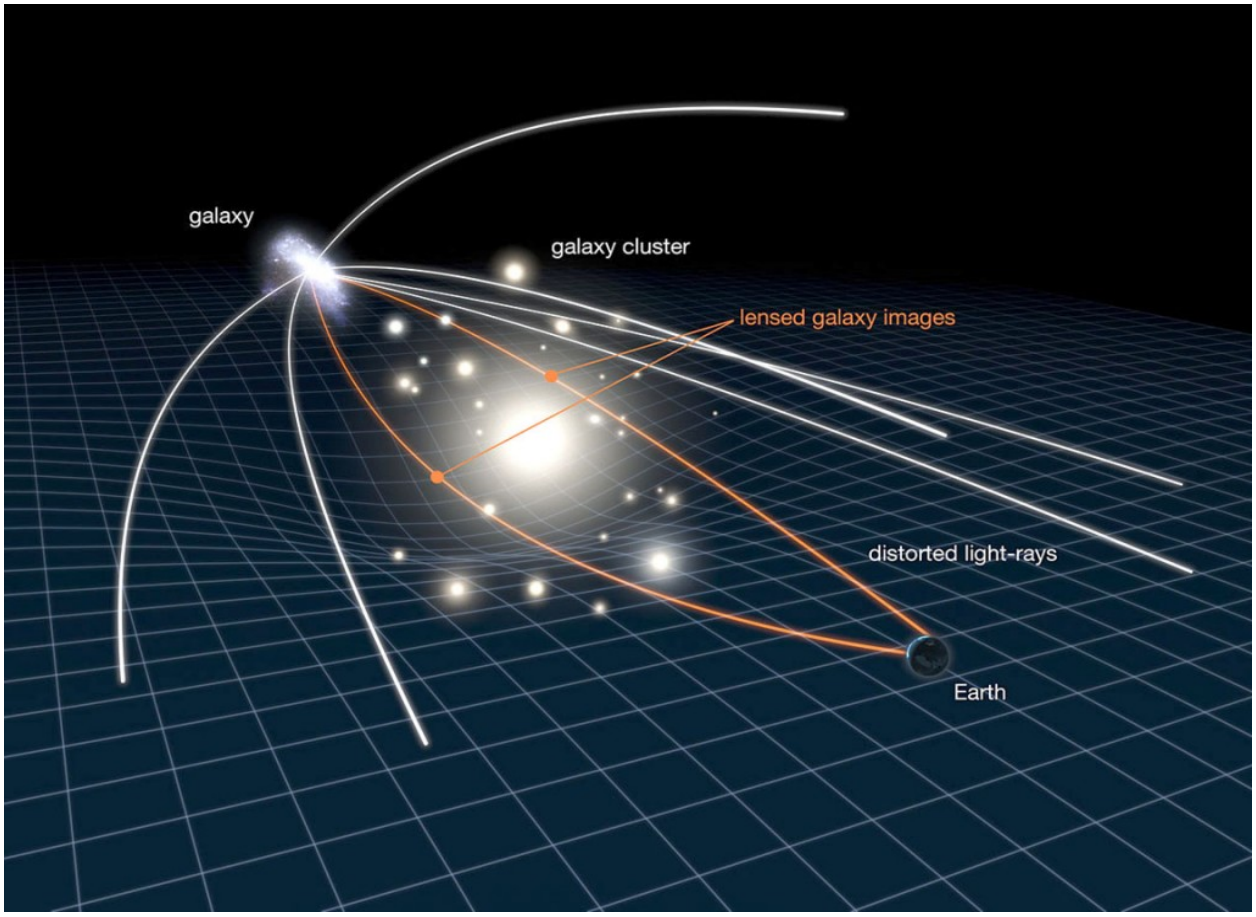


Figure 1.4. Schematic of a light bundles emitted by a distant source getting gravitationally lensed by foreground galaxy cluster. Credit: NASA/ESA.

that does not interact with light or other forms of electromagnetic radiation, making it invisible to telescopes. However, it exerts a gravitational pull on visible matter, causing it to cluster together in certain regions of space. Gravitational lensing can reveal the presence of dark matter by showing the effects of its gravitational pull on the light passing through it. Gravitational lensing can also be used to study the properties of distant galaxies and quasars due to the magnification effect (Mortlock et al., 1999; Courbin et al., 2002; Claeskens and Surdej, 2002). By analyzing the magnified and distorted images produced by a lensing object, astronomers can learn about the size, shape, and structure of these distant lens objects. Another application of gravitational lensing is in measuring the expansion rate of the

universe (Treu and Marshall, 2016; Oguri, 2019; Treu and Marshall, 2016). This is achieved by studying the way in which lensing objects magnify and distort the light from distant supernovae, which can be used as "standard candles" to determine their distance from us. By comparing the observed brightness of these supernovae to their expected brightness based on their distance, astronomers can determine the rate at which the universe is expanding. We discuss gravitational lensing in Sec. 2.1 in more detail.

1.3 Gravitational lensing of gravitational waves

GW lensing is highly complementary to EM lensing where GWs can be gravitationally lensed upon encountering massive objects, such as a galaxy or galaxy cluster, acting as a lens (Lawrence, 1971; Ohanian, 1974; Nakamura and Deguchi, 1999; Nakamura, 1998; Takahashi and Nakamura, 2003; Oguri, 2018a; Li et al., 2018). For massive lenses, GW lensing can be manifested in different ways. One of the cases could be the 'amplification' of the GWs (Will and Wiseman, 1996; Dai et al., 2017; Hannuksela et al., 2019) or in other case multiple copies (or 'images') of the GWs (Sereno et al., 2010; Takahashi, 2017; Oguri, 2019; Ezquiaga et al., 2021) are formed in the frequency band of the detector; this phenomenon is also referred to as strong lensing of GWs. For less massive lenses, interference between the images of GWs can take place that modulate the observed GWs; this is referred as microlensing (Dai et al., 2018; Jung and Shin, 2019; Liao et al., 2019; Basak et al., 2022; Gao et al., 2022).

Even though the lensing phenomenon is similar for both GWs and EMs, detecting and analyzing gravitational lensing of GWs is a challenging task, requiring precise measurements and sophisticated analysis techniques. The major differences between the GW and EM lensing arises from the perspective of wave-optics effects. In the case of EM lensing, the geometrical-optics approximation is valid for the vast majority of the observations. The approximation is satisfied by the wavelength λ of the EMs, which are much smaller than the Schwarzchild radius R_s of the lens. For GW lensing, this approximation is not always

satisfied, since detectors such as the LVK network can detect GW at frequencies f in the range $10\,\mathrm{Hz} \lesssim f \lesssim 10^4\,\mathrm{Hz}$, that is lower than even the lowest-frequency radio telescopes. These GWs have wavelengths longer than the Schwarzschild radii of lenses with masses $M_L \lesssim 10^4 M_\odot$, leading to non-negligible wave-optics effects. Moreover, the GWs emitted by the compact binaries have phase coherence resulting in the interference between lensed GWs when the signals overlap at the observer.

In the frequency domain, the modulation of GWs due to gravitational lensing is characterized by a multiplicative factor known as the amplification factor. The amplification factors is calculated using the diffraction integral (Takahashi and Nakamura, 2003; Bulashenko and Ubach, 2021; Tambalo et al., 2022), described in Sec. 2.3, performed on the lens plane. However, in the limit where the geometrical-optics approximation is valid, this factor can be expressed as a sum of terms, each of which corresponds to a discrete image with its own magnification and time delay. The amplification factor as a function of dimensionless frequency $w = 8\pi G M_{Lz} f$ is shown in Fig. 1.5, where M_{Lz} is the lens mass and f is the gravitational wave frequency (Tambalo et al., 2022). It shows different regimes of GW lensing that can be attributed to different limits of the dimensionless frequency w. We will discuss it in more detail in Chapter 4.

Although gravitational lensing has not been detected in any of the GW events observed in the first three runs of the LVK detector network (Abbott et al., 2021), recent estimates (Oguri, 2018b) suggest that $\sim 10^3$ strongly lensed GW events could be observed each year by future third-generation detectors like the proposed Cosmic Explorer (CE) (Evans et al., 2021) and Einstein Telescope (ET) (Maggiore et al., 2020). The detection of GW lensing could lead several of new astrophysical studies. The GW lensing will allow us, to measure the speed of GWs (Baker and Trodden, 2017; Collett and Bacon, 2017; Mukherjee et al., 2020), provide understanding about the elusive intermediate-mass (Lai et al., 2018) or primordial black holes (Diego, 2020; Oguri and Takahashi, 2020), and constrain source and lens distribution

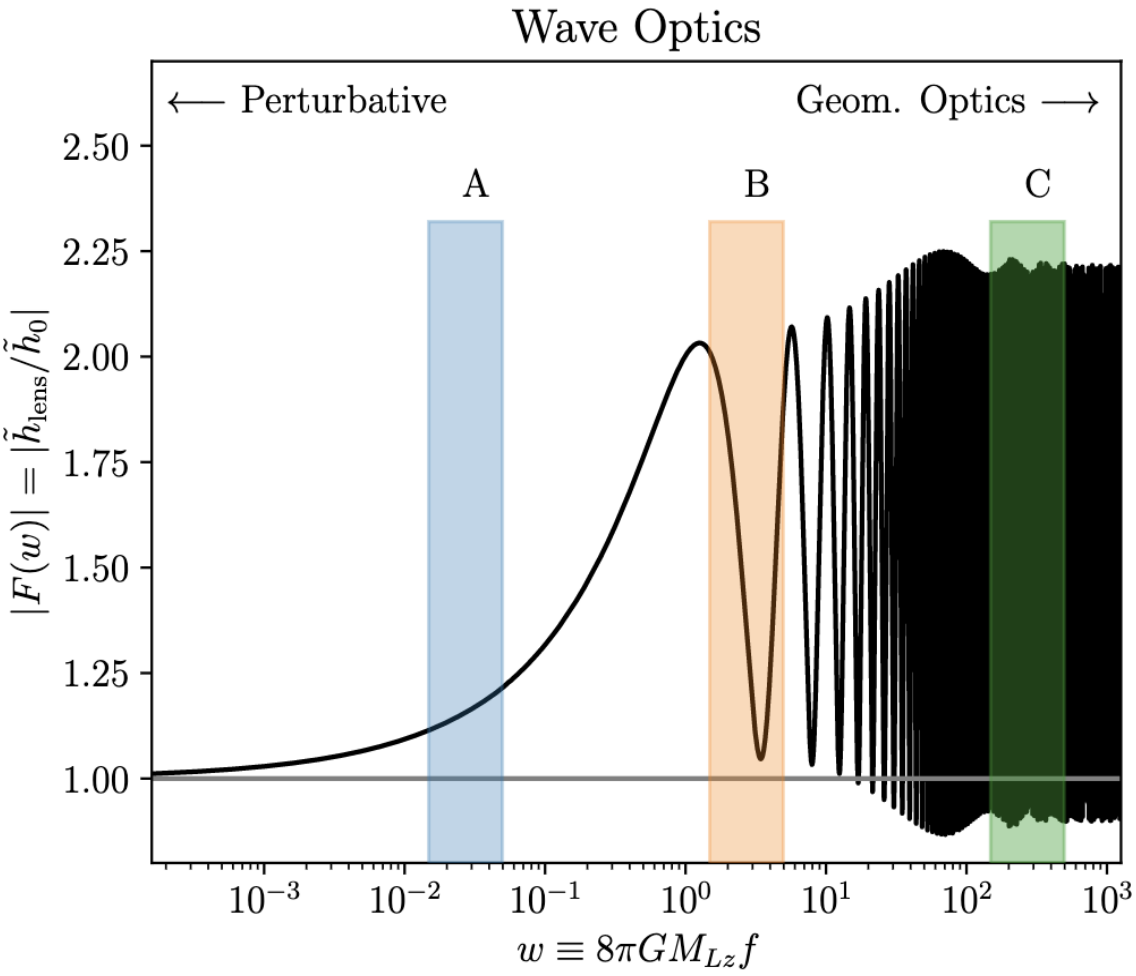


Figure 1.5. Amplification factor $F(w)$ for an SIS lens model is shown as a function of dimensionless frequency $w = 8\pi G M_{Lz} f$ where M_{Lz} is the lens mass and f is the gravitational wave frequency. GW lensing is categorized into (A) perturbative, (B) 'intermediate' and (C) geometrical-optics regime (Tambalo et al., 2022).

(Xu et al., 2022). Follow up study of lensed GW event with EM lensing surveys could enable us to locate the host galaxy to sub-arcsecond precision (Yu et al., 2020; Hannuksela et al., 2020) and lead to precision cosmography (Sereno et al., 2011; Liao et al., 2017; Cao et al., 2019; Li et al., 2019; Wempe et al., 2022).

1.4 Outline of the dissertation

This dissertation is organised as follows. In Chapter 2 we begin with discussing basic theory related to gravitational lensing and modelling of GWs using the post-Newtonian approximation. This is followed by discussion of gravitational lensing of GWs in the wave-optics and geometrical-optics limit. Chapter 3 presents a detailed analysis of the point mass (PM) and singular isothermal sphere (SIS) axisymmetric lens-mass profiles and introduce the model-independent image parameters. We discuss the results showed in (Ali et al., 2022): "Detectability of strongly lensed gravitational waves using model-independent image parameters". In Chapter 4 we consider the non-axisymmetric lens such as the Singular Isothermal Ellipsoid (SIE) lens. We explain the quasi-geometrical optics approximation and its utilization to quantify the breakdown of the geometrical-optics approximation in particular regions of the SIE lens. Finally, in Chapter 5 we follow the prescription (Gangardt et al., 2021) where a set of phenomenological parameters were introduced to allow understanding of the BH spin precession and nutation from the geometrical perspective. We introduce a 'toy' model for regularly precessing GWs, that is parameterized by two parameters, the precession amplitude $\langle\theta_{LJ}\rangle$ and precession frequency $\langle\Omega_{LJ}\rangle$, and understand the modulations imprinted by these parameters on the GWs.

CHAPTER 2

FORMALISM OF GRAVITATIONAL LENSING OF GRAVITATIONAL WAVES

In this chapter, we begin by reviewing basic theory pertaining to gravitational lensing. We then describe the gravitational waves modelling, and finally we present the formalism of the gravitational lensing of gravitational waves.

2.1 Gravitational lensing

Without the complete set of equations of relativity, Albert Einstein argued that massive objects deflect light using arguments based on the principle of equivalence. However, the general theory of relativity is needed is needed to obtain the correct value of the deflection of light by a massive object (Einstein, 1916). A massive object such as a galaxy or a cluster of galaxies can act as a lens or 'deflector' of light. The light bundles from a background source are deflected by the presence of an intervening lens before reaching the observer. This phenomenon is known as gravitational lensing. Gravitational lensing can be broadly categorized into strong lensing and Weak lensing.

- Strong Lensing: When the gravitational lensing effect is "strong" enough to produce visible distortions, multiple images, or Einstein rings Fig. 2.1, we refer to it as strong gravitational lensing (Bartelmann, 2010).

- Weak Lensing: In the case of weak lensing, the projected angular separation between the source and lens is large and as a result the distortions are much smaller. The detection of these small distortions are only possible through statistical analysis on an ensemble of extended sources. This way of analysing weak lensing effects allows us to reconstruct the dark matter distribution in the foreground lens as in Fig. 2.2.



Figure 2.1. Strong gravitational lensing around GAL-CLUS-022058s, as observed by *Hubble Space Telescope that resulted in an Einstein ring configuration (nicknamed "Molten Ring") (credit: ESA/Hubble & NASA, S. Jha).*

In gravitational lensing, we use the *thin lens approximation* which is valid because the source-lens and lens-observer distances are much greater than the physical size of the lens. Fig. 2.3 shows the typical layout of a gravitationally lensed system. The plane where the source and lens reside are called the source and lens plane respectively. The optic axis is

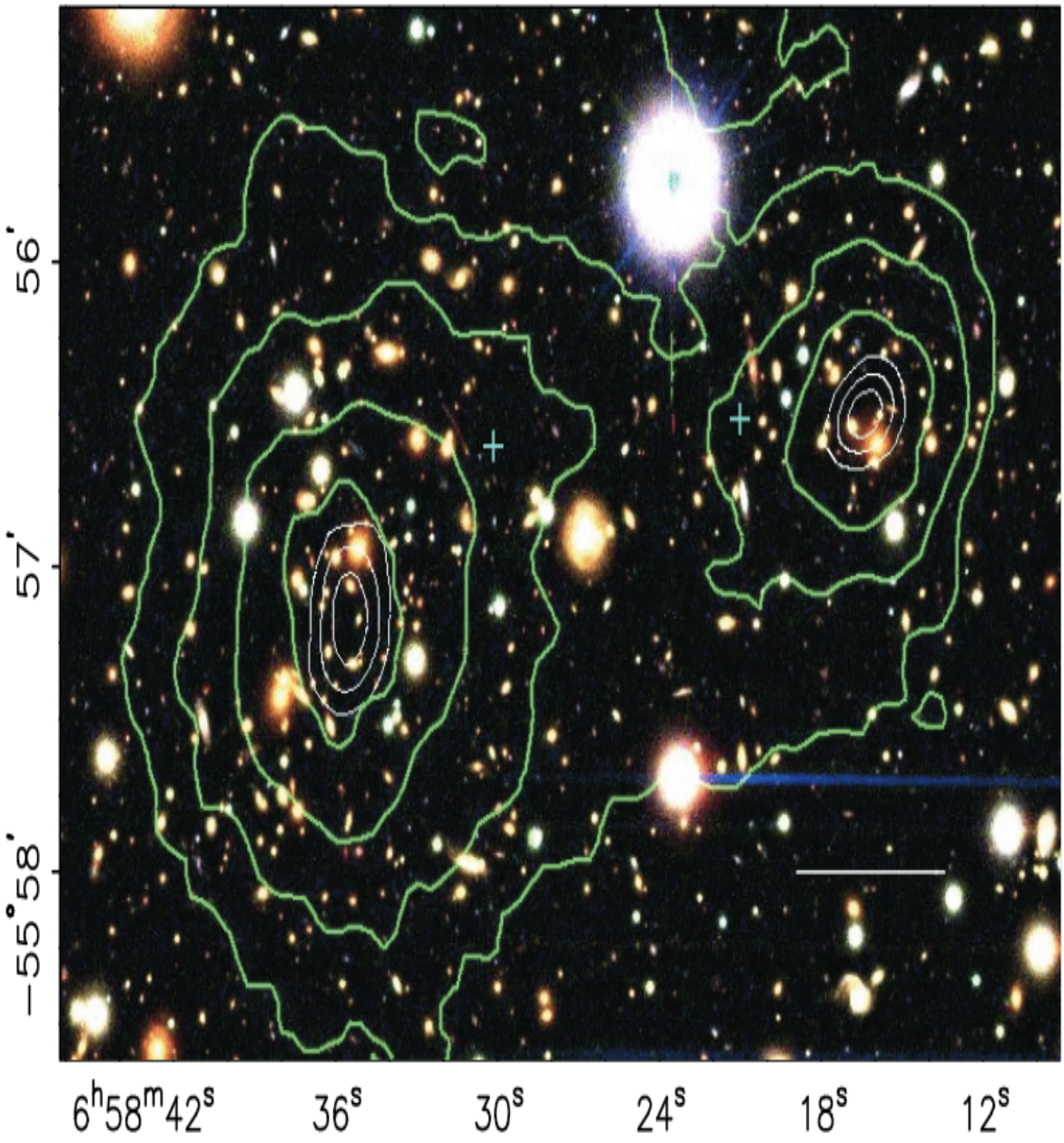


Figure 2.2. The optical image of the Bullet cluster as observed by the Hubble Space Telescope. The projected mass contours are overlaid and dominated by the dark matter (Clowe et al., 2006).

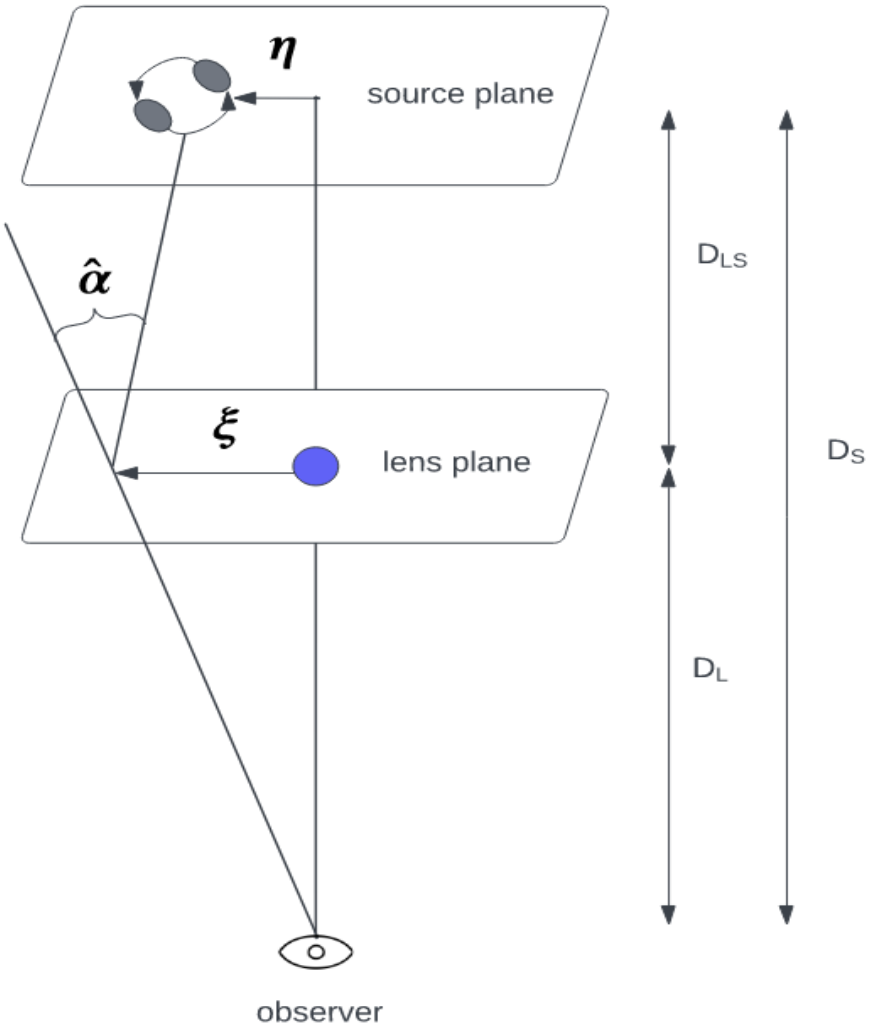


Figure 2.3. Schematic of a typical gravitational lensing system comprising a GW emitting BBH source on the source plane, a lens on the lens plane, and an observer. The source and lens are at angular diameter distances D_S and D_L respectively, while D_{LS} is the angular diameter distance between lens and source planes. The vectors $\boldsymbol{\eta}$ and $\boldsymbol{\xi}$ are the location of the source on the source plane and impact parameter on the lens plane, respectively. The deflection angle $\hat{\boldsymbol{\alpha}}$ is defined on the lens plane (Ali et al., 2022).

defined as the line passing through the lens and the observer, perpendicular to both lens and source planes.

2.1.1 Lens equation

In this figure, D_L, D_S are the comoving angular diameter distances of the lens and the source from the observer respectively. D_{LS} is the comoving angular diameter distance between the lens and the source. In Fig. 2.3 consider a source at angular position $\boldsymbol{\beta}$ on the source plane at a distance $\boldsymbol{\eta} = \boldsymbol{\beta} D_S$ from the optic axis. The light ray coming from the source has an impact parameter $\boldsymbol{\xi} = \boldsymbol{\theta} D_L$ on the lens plane and has a deflection angle $\hat{\boldsymbol{\alpha}}$. As a result of this deflection the observer receives the light from the source as if it was coming from angular position $\boldsymbol{\theta}$. From the geometric construction, we can write down the lens equation as follows:

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})\,, \tag{2.1}$$

where the reduced deflection angle $\boldsymbol{\alpha}(\boldsymbol{\theta})$ is defined as

$$\boldsymbol{\alpha}(\boldsymbol{\theta}) = \frac{D_{LS}}{D_S}\hat{\boldsymbol{\alpha}}(\boldsymbol{\theta})\,. \tag{2.2}$$

We can write down the non-dimensional form of the Eq. (2.1), by defining a length scale ξ_0 on the lens plane, which corresponds to $\eta_0 = \xi_0 D_S / D_L$ on the source plane. We can then define the dimensionless vectors

$$\boldsymbol{x} = \frac{\boldsymbol{\xi}}{\xi_0}\,, \tag{2.3}$$

$$\boldsymbol{y} = \frac{\boldsymbol{\eta}}{\eta_0}\,, \tag{2.4}$$

$$\boldsymbol{\alpha}(\boldsymbol{x}) = \frac{D_L D_{LS}}{\xi_0 D_S}\hat{\boldsymbol{\alpha}}(\xi_0 \boldsymbol{x})\,. \tag{2.5}$$

We can finally write the dimensionless form of Eq. (2.1) as

$$\boldsymbol{y} = \boldsymbol{x} - \boldsymbol{\alpha}(\boldsymbol{x})\,. \tag{2.6}$$

This equation is known as *lens equation*. The deflection angle $\hat{\boldsymbol{\alpha}}$ of a ray traveling from the source at $\boldsymbol{y}$ and intersecting the lens plane at $\boldsymbol{x}$ is determined from the gradient of the lensing potential $\Psi(\boldsymbol{x})$.

Lensing potential

The lensing potential $\Psi(\boldsymbol{x})$ is given by the projection of the three-dimensional Newtonian potential Φ on the lens plane and it is given by

$$\hat{\Psi}(\boldsymbol{\theta}) = \frac{D_{LS}}{D_L D_S} \frac{2}{c^2} \int dz\, \Phi(D_L \boldsymbol{\theta}, z)\,, \tag{2.7}$$

where the dimensionless form of the above equation can be written as

$$\Psi = \frac{D_L^2}{\xi_0^2} \hat{\Psi}. \tag{2.8}$$

As mentioned before, the gradient of Ψ gives the scaled deflection angle $\boldsymbol{\alpha}$

$$\boldsymbol{\nabla}_x \Psi(\boldsymbol{x}) = \boldsymbol{\alpha}(\boldsymbol{x})\,. \tag{2.9}$$

Moreover, the *convergence* κ is given by the Laplacian of Ψ

$$\boldsymbol{\nabla}_x^2 \Psi(\boldsymbol{x}) = 2\kappa(\boldsymbol{x})\,. \tag{2.10}$$

Convergence κ introduces isotropic distortion where the images are re-scaled by a constant value in all directions and it is given by

$$\kappa(\boldsymbol{x}) \equiv \frac{\Sigma(\boldsymbol{x})}{\Sigma_c} \tag{2.11}$$

where *critical surface mass density* Σ_c is a characteristic value calculated for a given lensing system and is defined as $\Sigma_c \equiv c^2/4\pi G D_S/(D_L D_{LS})$. The surface mass density $\Sigma(\boldsymbol{\theta})$ is

$$\Sigma(\boldsymbol{\theta}) = \frac{D_L D_{LS}}{c^2 D_S} \int_{-\infty}^{+\infty} dz\, \nabla^2 \Phi\,. \tag{2.12}$$

2.2 Gravitational waves

Gravitational waves (GWs) are the direct result of the field equations of general relativity proposed by Albert Einstein in 1915. Interestingly, it took us almost 100 years to detect

gravitational waves after Einstein's publication. The first detection of gravitational waves was made on September 14, 2015, by the LIGO and Virgo collaborations (Abbott et al., 2016). The first gravitational wave signal was named GW150914 and detected from a binary black hole (BBH) system of individual masses 36 and 29 $M_\odot$. Since then Advanced LIGO (Aasi et al., 2015) and Advanced Virgo (Acernese et al., 2015) have reported about 90 events. Most of the mergers are between stellar-mass black holes, during their first three observing runs (Abbott et al., 2021). The preexisting ground-based GW detectors, Advanced LIGO and Virgo, are accomapnied by the Kamioka Gravitational Wave Detector (KAGRA) (Somiya, 2012; Aso et al., 2013; Akutsu et al., 2021); together they form Advanced LIGO-Virgo-Kagra (LVK) network.

In this chapter, we will establish the formulation of gravitational waves, especially in the inspiral phase using the Post-Newtonian approximation. From here on, we will be using units in which $G = c = 1$.

2.2.1 Gravitational waves in Newtonian approximation

In the lowest order of Post-Newtonian (PN) approximation, the inspiral of two compact objects can be described using point particles. In this case, the Newtonian quadrupole formula gives us the gravitational waves. So, the orbital frequency w for a binary system of total mass $M = M_1 + M_2$ at orbital separation r is given by Kepler's Third law (Cutler and Flanagan, 1994)

$$\Omega = \frac{M^{1/2}}{r^{3/2}} \,. \tag{2.13}$$

For a circular orbit, the inspiral rate is given by

$$\frac{dr}{dt} = -\frac{r}{E}\frac{dE}{dt} = -\frac{64}{5}\frac{\mu M^2}{r^3} \,, \tag{2.14}$$

where $\mu \equiv (M_1 + M_2)/M$ is the reduced mass. The GW strain in time-domain $h(t)$ is given by (Cutler and Flanagan, 1994)

$$h(t) = \frac{(384/5)^{1/2}\pi^{2/3} A(\theta_S, \phi_S, \theta_L, \phi_L)\mu M}{Dr(t)} \cos\left(\int dt\, 2\pi f\right) . \tag{2.15}$$

Here, $A(\theta_S, \phi_S, \theta_L, \phi_L)$ is a function of the detector and binary orientation angles, which we will discussed later. The GW frequency is $f = \Omega/\pi$, and D is the distance to the source. We can obtain the function $r(t)$ by integrating the Eq (2.14) (Cutler and Flanagan, 1994)

$$r(t) = \left(\frac{256}{5}\mu M^2\right)^{1/4} (t_c - t)^{1/4} , \tag{2.16}$$

where t_c is known as coalescence time where $r \to 0$. Utilizing Eq. (2.13) and Eq. (2.14), we can write the time evolution of GW frequency f as

$$\frac{df}{dt} = \frac{96}{5}\pi^{8/3}\mathcal{M}^{5/3} f^{11/3} , \tag{2.17}$$

where $\mathcal{M} = \mu^{3/5} M^{2/5}$ is known as the chirp mass. We can also define the phase as

$$\phi(t) = \int dt\, 2\pi f = -2\left(\frac{1}{5}\mathcal{M}^{-1}(t_c - t)\right)^{5/8} + \phi_c , \tag{2.18}$$

where ϕ_c, a constant of integration, is known as coalescence phase such that $\phi \to \phi_c$ as $t \to t_c$. It is convenient to write the GW strain as a function of frequency, which can be obtained by using the Fourier transform of $h(t)$ as

$$\tilde{h}(f) = \int_{-\infty}^{\infty} dt e^{(2\pi i f t)} h(t) . \tag{2.19}$$

We now can rewrite Eq. (2.17) and Eq. (2.18) in frequency domain. To the lowest PN order they are written as

$$t(f) = t_c - 5(8\pi f)^{-8/3}\mathcal{M}^{-5/3} , \tag{2.20}$$

$$\phi(f) = \phi_c - 2(8\pi\mathcal{M} f)^{-5/3} . \tag{2.21}$$

Using the stationary phase approximation, the GW strain $\tilde{h}(f)$ in the frequency domain can be written as (Cutler and Flanagan, 1994)

$$\tilde{h}(f) = \frac{A}{D}\mathcal{M}^{5/6} f^{-7/6} e^{i\Psi(f)}\,,\ f \geq 0\,, \tag{2.22}$$

and the phase $\Psi(f)$ of the GW is given by

$$\Psi(f) = 2\pi f t_c - \phi_c - \frac{\pi}{4} + \frac{3}{4}(8\pi\mathcal{M}f)^{-5/3}\,. \tag{2.23}$$

2.2.2 Gravitational waves in post-Newtonian approximation

In the Sec. 2.2.1 we discussed GWs in the Newtonian regime. We now extend the analysis further to include post-Newtonian effects.

Post-Newtonian approximation

Newtonian physics is useful in describing many astrophysical systems but it fails terribly in the general relativistic regime. The post-Newtonian (PN) approximation is an approximation scheme developed such that in lowest order it explains the Newtonian regime and in the higher order perturbations it explains general relativistic phenomena. Precisely speaking, it is used to find the approximate solutions of the Einstein field equations using small parameters. The lowest order gives us Newton's law of gravitation and higher order perturbations explain the deviation from this law. We know that GWs are produced by compact binary systems consisting of two massive, dense objects such as black holes or neutron stars and as they orbit each other they go through different phases known as inspiral, merger, and ringdown (inset in top panel of Fig. 2.4).

In Fig. 2.4 the solid line in the top panel represents the waveform generated using numerical relativity for a BBH system with parameters consistent with those recovered from

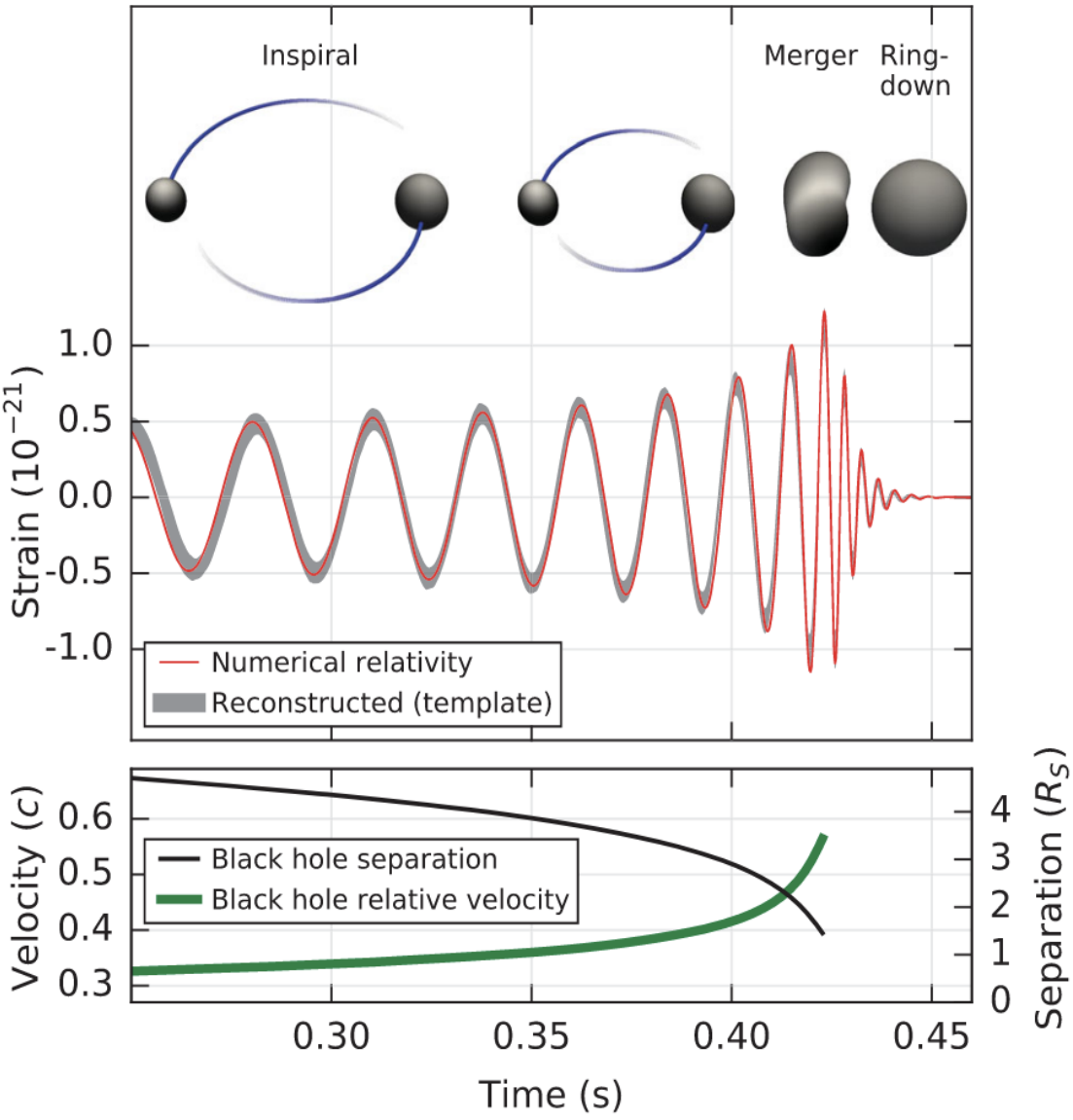


Figure 2.4. The top panel shows the numerical relativity waveform for a GW emitting system with parameters consistent with those recovered from GW150914 event. Shaded area shows the 90% credible region for the waveform constructed using BBH template waveforms. The inset in the top panel shows the complete evolution of BBH system. The bottom panel shows the effective separation of black holes in the BBH system in units of Schwarzschild radii $R_s = 2GM/c^2$ and the effective relative velocity given by the PN parameter $v/c = (GM\pi f/c^3)^{1/3}$, where f is the GW frequency and M is the total mass (Abbott et al., 2016).

GW150914 event. It is confirmed to 99.99% by an independent calculation based on (Campanelli et al., 2006). The shaded area shows 90% credible region for the waveform reconstructed using the BBH template waveforms. The bottom panel shows the effective separation of black holes in the BBH system in units of Schwarzschild radii $R_s = 2GM/c^2$ and the effective relative velocity given by the PN parameter $v/c = (GM\pi f/c^3)^{1/3}$, where f is the GW frequency and M is the total mass. For GW150914, the total mass inferred was $M = 70M_\odot$ and to reach an orbital frequency 75 Hz, which is half of GW freqeuncy, the objects must be very close and very compact. For a BBH system with equal masses, the objects would be only ≈ 350 km apart to generate GWs with such frequency.

Fig. 2.5 is a schematic description of different analytical approximation methods and numerical techniques that are employed to explain different phases of a compact binary system's evolution. Different methods are classified according to mass ratio m_2/m_1 and orbital separation r_{12} of the BBH system. The PN approximation is valid for BBH systems where the separation is large $r_{12} \gg Gm/c^2$, where $m = m_1 + m_2$ is the total mass of BBH and individual masses are comparable $m_1 \sim m_2$. The black hole perturbation theory is valid in the other limit, where the individual masses of BHs are very similar to each other and the mass ratio is extreme $m_1 \ll m_2$. The numerical relativity technique is valid for small separation and comparable mass ratio, with a small overlap in PN regime. *In our research we mainly focus on the inspiral phase; the PN approximation is appropriate for this phase as the orbital velocity* $v^2 \sim Gm/r_{12}$ *is small due to large* r_{12}. Once the black holes start merging in the merger phase, the PN approximation becomes invalid and numerical relativity/black hole perturbation theory must be used. In the ringdown phase, when the black holes have merged together to form a single object, black hole perturbation theory is used.

Gravitational wave formulation in PN approximation

The PN approximation provides the most accurate description of the GWs emitted during the inspiral phase of the stellar-mass BBH system. In the PN approximation, we have M/r

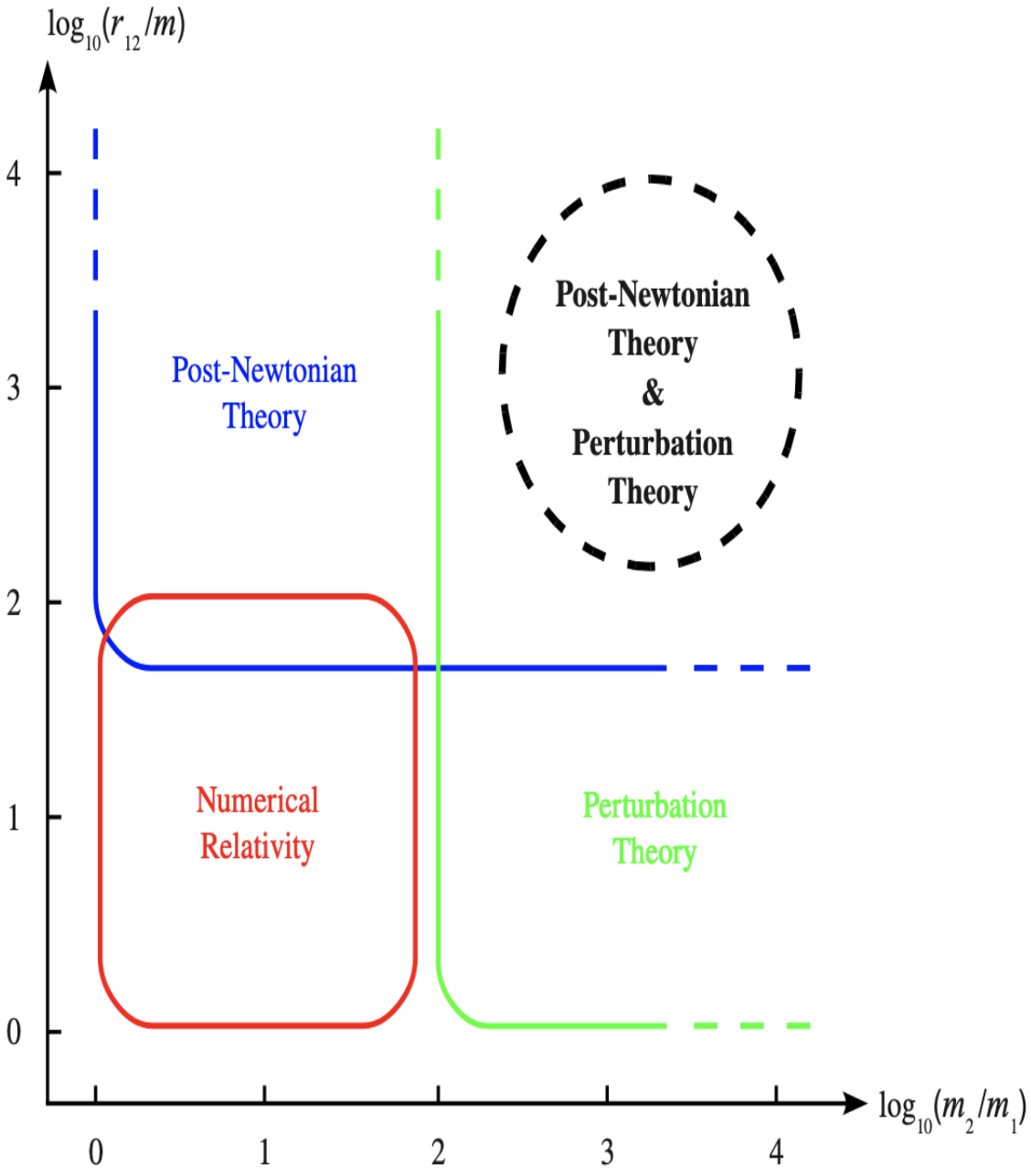


Figure 2.5. Schematic diagram showing different analytical approximation and numerical techniques utilized to explain the BBH depending on the mass ratio m_2/m_1 and orbital separation $r_{12} \sim Gm/v^2$ where $m = m_1 + m_2$ is the total mass and v is the orbital velocity. (Blanchet et al., 2009).

as the correction term with nPN order referring to the correction of order $(v/c)^{2n} \sim (M/r)^n$ to Newtonian gravity. The waveform in Eq. 2.15 is the lowest order correction M/r which gives us the waveform in the Newtonian regime. We will be using 1.5PN order of correction for our analysis given in (Cutler and Flanagan, 1994). In the 1.5PN order of correction, the orbital frequency given in Eq. 2.13 modifies into (Cutler and Flanagan, 1994)

$$\Omega(r) = \frac{M^{1/2}}{r^{3/2}} \left[1 + \left(\frac{-3}{2} + \frac{\mu}{2M}\right)\frac{M}{r} + O\left(\left(\frac{M}{r}\right)^2\right)\right] . \tag{2.24}$$

Also, the GW frequency Eq. 2.17 can be written as (Cutler and Flanagan, 1994)

$$\frac{df}{dt} = \frac{96}{5}\pi^{8/3}\mathcal{M}^{5/3}f^{11/3}\left[1 - \left(\frac{743}{336} + \frac{11\mu}{4M}(\pi Mf)^{2/3} + 4\pi(\pi Mf) + O((\pi Mf)^{4/3})\right)\right] . \tag{2.25}$$

We can also re-define the PN correction term as $x = (M/r)^{1/2} = (\pi Mf)^{2/3}$, which can be referred to as PN expansion parameter (Cutler and Flanagan, 1994). According to (Cutler and Flanagan, 1994), this is a better choice of parameter because we directly measure the frequency of the gravitational wave using the detector and it is also gauge invariant, which is not the case with the orbital radius r. In 1.5PN correction order, Eqs. 2.20 and 2.21 can be written as

$$t(f) = t_c - 5(8\pi f)^{-8/3}\mathcal{M}^{-5/3}\left[1 + \frac{4}{3}\left(\frac{743}{336} + \frac{11\mu}{4M}x - \frac{32}{5}\pi x^{3/2} + O(x^2)\right)\right] , \tag{2.26}$$

$$\phi(f) = \phi_c - 2(8\pi\mathcal{M}f)^{-5/3}\left[1 + \frac{5}{3}\left(\frac{743}{336} + \frac{11\mu}{4M}x - 10\pi x^{3/2}\right) + O(x^2)\right] , \tag{2.27}$$

where $t \to t_c$ and $\phi \to \phi_c$ as $f \to \infty$ (or $r \to 0$). The GW phase Eq. 2.23 also gets modified into

$$\Psi(f) = 2\pi f t_c - \phi_c - \frac{\pi}{4} + \frac{3}{4}(8\pi\mathcal{M}f)^{-\frac{5}{3}}\left[1 + \frac{20}{9}\left(\frac{743}{336} + \frac{11\eta}{4}\right)x - 16\pi x^{\frac{3}{2}}\right] . \tag{2.28}$$

The expression for GW strain remains same as before Eq. 2.22

$$\tilde{h}(f) = \frac{A}{D}\mathcal{M}^{5/6}f^{-7/6}e^{i\Psi(f)},\ f \geq 0. \tag{2.29}$$

but the GW phase now accounts for the PN corrections and given by Eq. 2.28.

The GW strain Eq. 2.29 is invalid at very high frequency because in real BBH systems the inspiraling black holes will merge at some orbital frequency and stop emitting GWs. For binaries consisting of black hole-black hole or black hole-neutron star, the transition from inspiral to final plunge occurs at the inner most stable circular orbit (ISCO) (Kidder et al., 1992). The ISCO is roughly defined at $r \approx 6M$ (Cutler and Flanagan, 1994). Also for neutron star-neutron star mergers, the two stars merge and form a single object at $r \approx 6M$ (Cutler and Flanagan, 1994). Using PN expansion parameter, $x = M/r = (\pi M f)^{2/3}$, we define *cut-off* frequency f_{cut} at $r = 6M$ as

$$f_{cut} = \frac{1}{6^{3/2}\pi M}\,, \tag{2.30}$$

such that compact objects in binaries merge and stop emitting GWs i.e. $\tilde{h}(f > f_{cut}) = 0$. Therefore, the waveform for GWs now takes the following form

$$\tilde{h}(f) = \begin{cases} \tilde{A} f^{-7/6} e^{i\Psi(f)}\,, & 0 < f < f_{cut}\,, \\ 0\,, & f_{cut} < f\,, \end{cases} \tag{2.31}$$

where $\tilde{A} = A\mathcal{M}^{5/6}/D$.

The waveform obtained from the PN approximation perfectly describes the inspiraling phase of the binaries. However, the approximation completely breaks down in the merger and ringdown phase. As noted above, for the purpose of gravitational lensing of GWs we restrict the waveform to the inspiral phase only, the reason for which will be explained in the next chapter. Other sophisticated waveform models exist which can be used if one wants to consider post-inspiral phases also. One can employ models such as IMRPhenom (García-Quirós et al., 2020), SEOB (Buonanno and Damour, 1999), and NRSur7dq4 (Varma et al., 2019) for more accurate and comprehensive gravitational waveforms required in GW data analysis.

2.2.3 Simulating gravitational waveforms

The gravitational waveform during the inspiral phase of binary coalescence is shown in Eq. 2.31, however the explicit form of the amplitude is not described. The GW strain $\tilde{h}(f)$ detected by a ground-based detector such as LIGO is a linear combination of two polarizations $\tilde{h}_+$ and $\tilde{h}_\times$ and given by

$$\tilde{h}(f) = F_+ \tilde{h}_+(f) + F_\times \tilde{h}_\times(f)\,, \tag{2.32}$$

where F_+ and $F_\times$ are the detector beam-pattern coefficients and given by (Apostolatos et al., 1994)

$$F_+(\theta_S, \phi_S, \psi) = \frac{1}{2}(1 + \cos\theta_S^2)\cos 2\phi_S \cos 2\psi - \cos\theta_S \sin 2\phi_S \sin 2\psi\,, \tag{2.33a}$$

$$F_\times(\theta_S, \phi_S, \psi) = \frac{1}{2}(1 + \cos\theta_S^2)\cos 2\phi_S \sin 2\psi + \cos\theta_S \sin 2\phi_S \cos 2\psi\,, \tag{2.33b}$$

where θ_S, ϕ_S specify the sky localization vector $\hat{\boldsymbol{N}}$ with respect to the $\{\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}\}$ defined with respect to the L-shaped detector in Fig. 2.6. The polarization angle ψ between the principal+ direction and the direction of constant azimuth is given by (Apostolatos et al., 1994)

$$\psi = \arctan\left(\frac{\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{Z}} - (\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}})(\hat{\boldsymbol{Z}}\cdot\hat{\boldsymbol{N}})}{\hat{\boldsymbol{N}}\cdot(\hat{\boldsymbol{L}}\times\hat{\boldsymbol{Z}})}\right)\,, \tag{2.34}$$

where $\hat{\boldsymbol{L}}$ is unit vector in the direction of the binary's orbital angular momentum.

The GW polarizations in the frequency domain can be written as (Finn and Chernoff, 1993; Takahashi and Nakamura, 2003)

$$\tilde{h}_+(f) = A f^{-7/6}[1 + (\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}})^2]e^{i\Psi(f)}\,, \tag{2.35a}$$

$$\tilde{h}_\times(f) = -2iA f^{-7/6}(\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}})e^{i\Psi(f)}\,, \tag{2.35b}$$

where amplitude A is

$$A = \sqrt{\frac{5}{96}}\frac{\pi^{-2/3}\mathcal{M}^{5/6}}{D}\,, \tag{2.36}$$

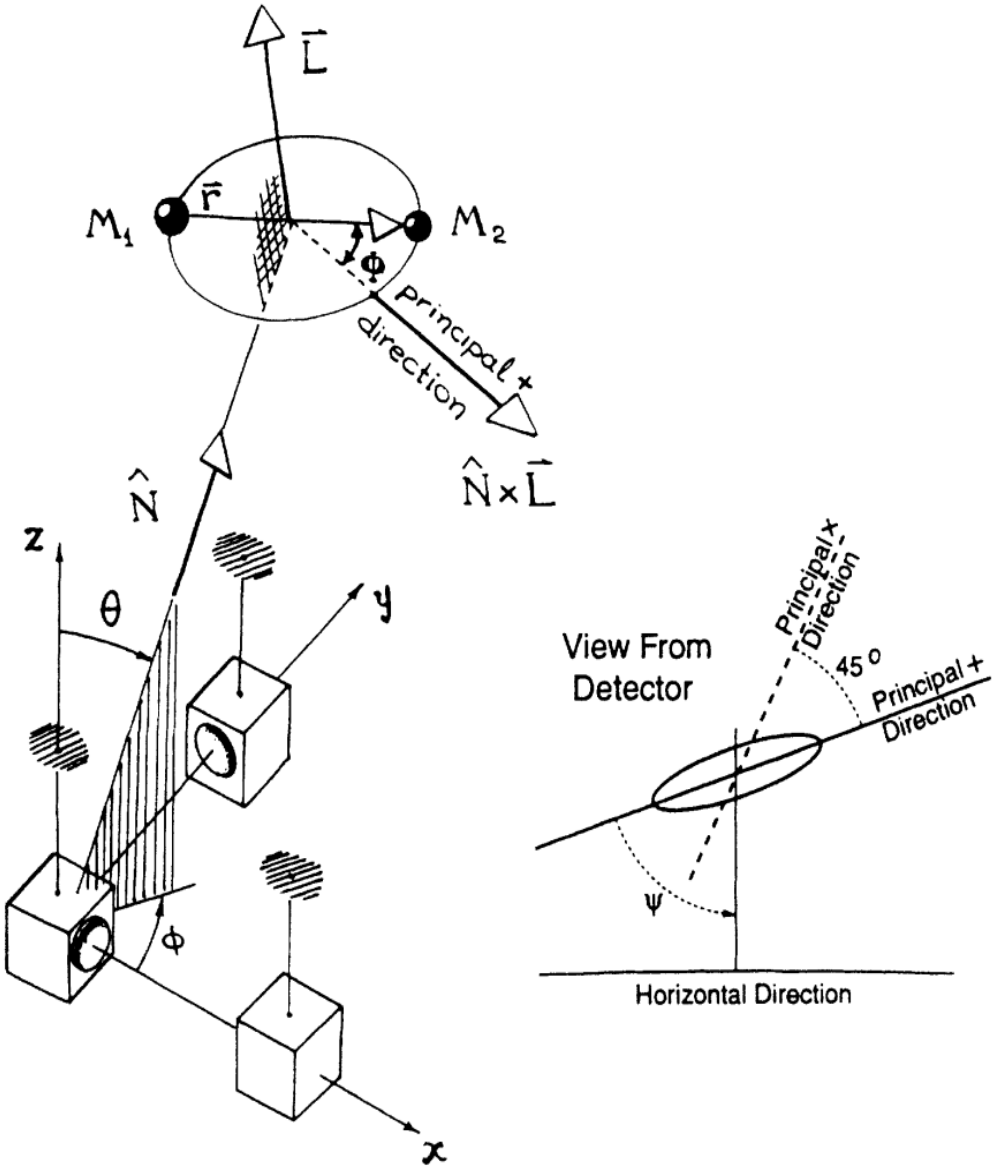


Figure 2.6. A schematic of L-shaped detector and compact binary system (Apostolatos et al., 1994).

where $\mathcal{M}$ is the binary's chirp mass, D_L is the luminosity distance to the source, and $\Psi(f)$ is the GW phase given by Eq. 2.23. Inserting Eqs. 2.35 into Eq. 2.32 yields

$$\tilde{h}(f) = \tilde{A} f^{-7/6} e^{i(\Psi(f) - \phi_p)}\,, \tag{2.37}$$

where the effective GW amplitude $\tilde{A}$ is

$$\tilde{A} = A\{4(\hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{N}})^2 F_\times^2 + [1 + (\hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{N}})^2]^2 F_+^2\}^{1/2}\,, \tag{2.38}$$

and the polarization phase ϕ_p is

$$\phi_p = \arctan\left[\frac{2(\hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{N}})F_\times}{(1 + (\hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{N}})^2)F_+}\right], \tag{2.39}$$

From Eq. 2.37 we notice that the amplitude of an inspiraling gravitational waveform has a straightforward power-law dependence $|\tilde{h}(f)| \propto f^{-7/6}$ on the frequency f of the GW. Therefore, the log-log plot of GW amplitude is a straight line with negative slope of 7/6. We require **nine** parameters to completely define the inspiraling GW strain; two detector angles θ_S, ϕ_S defining the unit vector $\hat{\boldsymbol{N}}$ specifying the binary, two binary source angles θ_L, ϕ_L defining the unit vector $\hat{\boldsymbol{L}}$ associated with the orbital angular momentum of the binary, luminosity distance D_L between the source and the observer, chirp mass $\underline{\mathcal{M}}$ and symmetric mass ratio η of the binary source, and time t_c and phase ϕ_c at which the compact objects in the binary system coalesce.

2.3 Gravitational lensing of gravitational waves

GWs traveling through the Universe, after being emitted from the coalescing binaries, encounter massive objects such as galaxies or galaxy clusters that can acts a lens. These lenses can deflect the GWs resulting in gravitational lensing of GWs. The phenomenon of gravitational lensing of GWs is highly complementary to EM waves lensing, with differences arising the way in which the theory of gravitational lensing is implemented to GW and EM waves. The two major differences in the EM waves and GW lensing are from the wave-optics point of view. The gravitational lensing of EM waves is treated exclusively in the geometrical-optics limit; the validity of this approximation lies in the fact that the wavelength λ of the

EM waves is much smaller than the Schwarzchild radius R_s of the lens. However, this approximation is not always true for GWs. The wavelength of the GWs detected by a space interferometer such as the Laser Interferometer Space Antenna (LISA) (Barausse et al., 2020) is extremely large $\lambda \sim 1$AU, compared to that of visible light $\lambda \sim 1\mu$m. The GWs wavelength are even longer than the Schwarzchild radii of lenses with masses $M_L \leq 10^4 M_\odot$, leading to non-negligible wave-optics effects. Another difference between GW and EM waves lensing is the coherent nature of GWs emitted from coalescing binaries, which results in interference between the lensed GWs if they overlap in arrival time at the detector. In summary, characteristics of GW lensing in comparison with EM lensing are: (1) the need for a full wave-optics treatment over a significant fraction of parameter space, and (2) the possibility of interference effects.

Moreover, the absorption of GWs while traveling through the Universe is negligible, unlike EM waves which are modulated by the dense regions of a galaxy, or completely obscured by EM emission from the galaxy lens. Strong lensing of GWs, thus, allows us to observe the faint image which is generically expected to be formed near the central region of the lens (Schneider et al., 1992). This has the potential to vastly increase our understanding of the centers of galaxies and their dark matter halos (Tambalo et al., 2022), which would be otherwise challenging with lensing of EM waves only. GWs can also be modeled accurately for the known source parameters such as masses, spins, and distance to the GW source. As discussed in Chapter 2, several analytical and numerical techniques exist to precisely model the waveforms. The precise modeling of EM sources is rather difficult. Therefore, gravitational lensing of GWs can be studied for general systems without making prior assumptions on those systems.

2.3.1 Wave optics in GW lensing

We briefly review wave optics in the context of gravitational lensing of gravitational waves (Schneider et al., 1992; Takahashi and Nakamura, 2003; Nakamura and Deguchi, 1999). Let

us consider a spacetime metric where GWs are propagating under the gravitational potential of the lens

$$ds^2 = -(1+2U)dt^2 + (1-2U)dr^2 = g^{(B)}_{\mu\nu}dx^\mu dx^\nu \,, \tag{2.40}$$

where $U(r) \ll 1$ is the gravitational potential of the lens object. Now, we perturb the background metric tensor $g^{(B)}_{\mu\nu}$ with a linear perturbation $h_{\mu\nu}$ such that

$$g_{\mu\nu} = g^{(B)}_{\mu\nu} + h_{\mu\nu} \,. \tag{2.41}$$

Using transverse traceless and Lorenz gauge conditions along with the assumptions that the wavelength λ is much smaller than the typical radius of curvature of the background, we have the propagation equation

$$\partial_\mu \left(\sqrt{-g^{(B)}} g^{(B)\mu\nu} \partial_v \phi \right) \,, \tag{2.42}$$

where ϕ is a scalar such that

$$h_{\mu\nu} = \phi \, e_{\mu\nu} \,, \tag{2.43}$$

and $e_{\mu\nu}$ is the polarization tensor of the GW. The Eq. 2.42 can be rewritten for the scalar in the frequency domain $\tilde{\phi}(f, \boldsymbol{r})$ as (Schneider et al., 1992; Takahashi and Nakamura, 2003)

$$\left(\nabla^2 + \omega^2\right) \tilde{\phi} = 4\omega^2 U \tilde{\phi} \,, \tag{2.44}$$

where angular frequency $\omega = 2\pi f$. We can now conveniently define the amplification factor as follows (Takahashi and Nakamura, 2003):

$$F(f) = \frac{\tilde{\phi}^L(f)}{\tilde{\phi}^{UL}(f)} \,, \tag{2.45}$$

where $\tilde{\phi}^L(f)$ and $\tilde{\phi}^{UL}(f)$ are the lensed and the unlensed gravitational wave amplitudes. The unlensed amplitude $\tilde{\phi}^{UL}(f)$ is defined when $U = 0$ in the metric (Eq. 2.40). We can also

define the amplification factor for a typical lensing system as shown in Fig. 2.3 as (Schneider et al., 1992; Takahashi and Nakamura, 2003)

$$F(f) = \frac{D_S \xi_0^2}{D_L D_{LS}} \frac{f}{i} \int d^2\boldsymbol{x} \exp[2\pi i f t_d(\boldsymbol{x}, \boldsymbol{y})], \tag{2.46}$$

where $t_d(\boldsymbol{x}, \boldsymbol{y})$ is the arrival time from the source to the observer. The amplification factor $F(f)$ is unity in the absence of a lens.

The deflection of GWs by the lens causes a time delay between the lensed GWs as observed by the observer. The time delay has two components. The first one is a geometric component which is due to the difference in the path lengths of the non-deflected (path traveled by the GWs in the absence of the lens) and the deflected GWs. The second one is due to the slowing down of the photons due to the gravitational potential of lens. This is called *Shapiro time delay* (Meneghetti, 2021). The arrival time t_d introduced at the observer due to a lens present at redshift z_L is given by (Schneider et al., 1992; Takahashi and Nakamura, 2003)

$$t_d(\boldsymbol{x}, \boldsymbol{y}) = \frac{D_S \xi_0^2}{D_L D_{LS}} (1 + z_L) \left[\frac{1}{2} |\boldsymbol{x} - \boldsymbol{y}|^2 - \Psi(\boldsymbol{x}) + \phi_m(\boldsymbol{y}) \right] , \tag{2.47}$$

where the term $\phi_m(\boldsymbol{y})$ is chosen such that the minimum value of the time delay is 0. The gravitational potential $\Psi(\boldsymbol{x})$ for a particular lens model is determined by Eq. 2.10.

In the limit $f \gg t_d^{-1}$, geometrical optics is valid and the stationary points of the time delay $t_d(\boldsymbol{x}, \boldsymbol{y})$ contribute to the integral of the amplification factor Eq. 2.46. The images are formed at discrete points that are stationary points on the time delay surface

$$\nabla_x t_d(\boldsymbol{x}, \boldsymbol{y}) = 0 \, , \tag{2.48}$$

This is known as *Fermat's principle* (Schneider, 1985). Only these points contribute to the $F(f)$ in Eq. 2.46), as a result the integral on the lens plane is transformed into the sum over these images (Schneider et al., 1992; Takahashi and Nakamura, 2003; Takahashi, 2004)

$$F(f) = \sum_j |\mu_j|^{1/2} \exp\left(2\pi i f t_d(\boldsymbol{x}_j, \boldsymbol{y}) - i\pi n_j\right) , \tag{2.49}$$

where μ_j is the magnification of the jth image calculated using lens equation Eq. 2.6

$$\mu_j = \frac{1}{\det\left(\partial \boldsymbol{y}/\partial \boldsymbol{x}_j\right)}\,, \tag{2.50}$$

and n_j is known as the *Morse index* which has topological origin. It can be 0, 1/2, or 1 depending on whether $\boldsymbol{x}_j$ is a minimum, saddle, or maximum point respectively on the time delay surface, $t_d(\boldsymbol{x}, \boldsymbol{y})$.

CHAPTER 3

DETECTABILITY OF STRONGLY LENSED GRAVITATIONAL WAVES USING MODEL-INDEPENDENT IMAGE PARAMETERS

The content of this chapter is taken from (Ali et al., 2022), which is currently under review in the Physical Review D journal. Evangelos Stoikos, Evan Meade, Michael Kesden, and Lindsay King were the collaborators on this research project. SA (lead author) undertook the bulk of the coding, analysis and paper writing, with direction and guidance from MK and LK. ES engaged in code checking and discussions and EM undertook calculations related to distributions of lensing time delays.

3.1 Axisymmetric lens models

In Chapter 2 we discussed the amplification factor $F(f)$ which describes the effect of gravitational lensing on GWs. In this section we discuss two axisymmetric lens models, the point mass (PM) and the singular isothermal sphere (SIS) lens. In the geometrical-optics limit, these lens models produce at most two images.

3.1.1 Point mass (PM)

The PM is the simplest mass distribution for a gravitational lens. The surface mass density is $\Sigma(\boldsymbol{\xi}) = M_L \delta^2(\boldsymbol{\xi})$ where M_L is the lens mass. This leads to the lensing potential is $\psi(\boldsymbol{x}) = \ln x$. The normalization constant ξ_0 as defined in Sec. 2.1.1 is chosen to be the Einstein radius given by

$$\xi_0 = \left(\frac{4 M_L D_L D_{LS}}{D_S} \right)^{1/2} . \tag{3.1}$$

The amplification factor is then given by (Peters, 1974; Takahashi and Nakamura, 2003)

$$F(f) = \exp\left\{ \frac{\pi w}{4} + i \frac{w}{2} \left[\ln\left(\frac{w}{2} \right) - 2\phi_m(y) \right] \right\} \Gamma\left(1 - \frac{i}{2} w \right) {}_1F_1\left(\frac{i}{2} w, 1; \frac{i}{2} w y^2 \right) , \tag{3.2}$$

where $w = 8\pi M_{Lz} f$, $\phi_m(y) = (x_m - y)^2/2 - \ln x_m$, $x_m = (y + (y^2+4)^{1/2})/2$, and ${}_1F_1(a, b; c)$ is the confluent hypergeometric function. The redshifted lens mass M_{Lz} is given by $M_{Lz} = M_L(1 + z_L)$.

3.1.2 Singular isothermal sphere (SIS)

The SIS density profile $\Sigma(\boldsymbol{\xi}) = \sigma_v^2/2\xi$, where σ_v is the velocity dispersion. It is the most simple profile that can effectively describe the flat rotation curves of galaxies (Gavazzi et al., 2007). The lensing potential $\psi(x) = x$ given by Eq. 2.10 and the amplification factor (Takahashi and Nakamura, 2003; Matsunaga and Yamamoto, 2006)

$$\begin{aligned} F(f) &= -iw\mathrm{e}^{iwy^2/2} \int_0^\infty dx\, x J_0(wxy) \exp\left[iw\left(\frac{1}{2}x^2 - x + \phi_m(y)\right)\right], \\ &= e^{\frac{i}{2}w(y^2+2\phi_m(y))} \sum_{n=0}^{\infty} \frac{\Gamma(1+\frac{n}{2})}{n!} (2we^{i\frac{3\pi}{2}})^{\frac{n}{2}} {}_1F_1\left(1+\frac{n}{2}, 1; -\frac{i}{2}wy^2\right), \end{aligned} \tag{3.3}$$

by Eq. (2.46), where $w = 8\pi M_{Lz} f$, $\phi_m(y) = y + 1/2$, J_0 is the Bessel function of zeroth order. The Einstein radius for SIS lens can be defined as

$$\xi_0 = \frac{4\pi\sigma_v^2 D_L D_{LS}}{D_S}, \tag{3.4}$$

and $M_{Lz} = \pi\sigma_v^2(1 + z_L)\xi_0$ is the redshifted lens mass inside the Einstein radius. From here on, we will refer redshifted lens mass M_{Lz} as lens mass and denote it as M_L.

3.2 Geometrical-optics approximation

As discussed previously in Chapter 2, in the limit where geometrical-optics approximation is valid, a discrete number of images are formed at locations $\boldsymbol{x}_j$ for a source at a position $\boldsymbol{y}$. These image locations $\boldsymbol{x}_j$ are the stationary points on the time delay surface $t_d(\boldsymbol{x}, \boldsymbol{y})$ given by Eq. 2.47. For the SIS lens, the caustics is located at $y = 1$ (Schneider et al., 1992; Takahashi and Nakamura, 2003); the caustics are the locations of the sources that produces

images with infinite magnifications given by Eq. 2.50. The number of images vary across the caustic; two images for source inside the caustic ($y < 1$) and one image for the source outside the caustic ($y > 1$). The amplification factor $F(f)$ in the geometrical-optics approximation of Eq. 2.49 is given by (Takahashi and Nakamura, 2003)

$$F(f) = \begin{cases} |\mu_+|^{1/2} - i|\mu_-|^{1/2} e^{2\pi i f \Delta t_d}, & y < 1 \\ |\mu_+|^{1/2}, & y > 1\,, \end{cases} \tag{3.5}$$

where

$$\mu_\pm = \pm 1 + \frac{1}{y}\,, \tag{3.6a}$$

$$\Delta t_d = 8 M_L y\,. \tag{3.6b}$$

A PM lens can deflect photons by an arbitrarily large angle for sufficiently small impact parameter, therefore two images are formed for all source positions y. $F(f)$ is given by (Takahashi and Nakamura, 2003)

$$F(f) = |\mu_+|^{1/2} - i|\mu_-|^{1/2} e^{2\pi i f \Delta t_d}\,, \tag{3.7}$$

where

$$\mu_\pm = \frac{1}{2} \pm \frac{y^2+2}{2y(y^2+4)^{1/2}}\,, \tag{3.8a}$$

$$\Delta t_d = 2 M_L \left\{ y(y^2+4)^{1/2} + 2 \ln \left[\frac{(y^2+4)^{1/2} + y)}{(y^2+4)^{1/2} - y)} \right] \right\}\,. \tag{3.8b}$$

Both the SIS and PM lens models can be parameterized by two lens parameters, the source position y and lens mass M_L.

In Fig. 3.1 we show the amplification factor $F(f)$ for the SIS as a function of GW frequency f. We assume that the time delay Δt_d is much less than the observing time of the GW detector as is appropriate for the lens mass shown in Fig. 3.1. In the low-frequency limit, $F(f)$ converges to unity because diffraction effects prevent long-wavelength

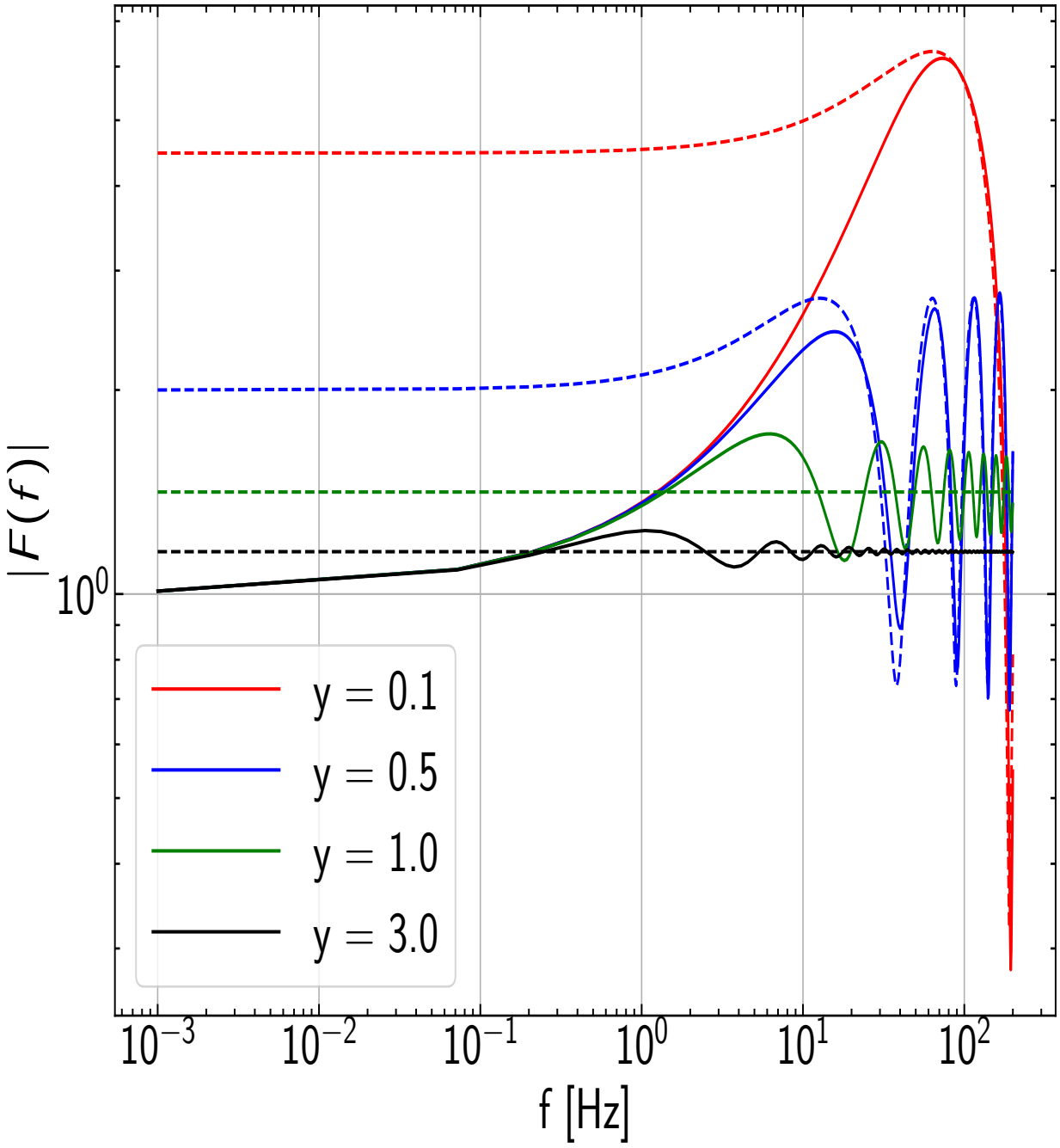


Figure 3.1. The magnitude of the amplification factor $F(f)$ as a function of frequency f for an SIS with lens mass $M_L = 10^3 M_\odot$ and source positions $y = 0.1$, 0.5, 1.0, and 3.0 shown in red, blue, green, and black. The solid lines are calculated using the exact result of Eq. 3.3, while the dashed lines are calculated using the geometrical-optics approximation of Eq. 3.5.

GWs from being affected by the presence of a lens (Bontz and Haugan, 1981; Takahashi and Nakamura, 2003). In the high-frequency limit, the geometrical-optics approximation is valid and oscillatory behavior is observed due to interference between the coherent lensed images. For two images, the amplification factor's amplitude $|F(f)|$ from Eq. 3.5 is

$$|F(f)|^2 = |\mu_+| + |\mu_-| + 2|\mu_+\mu_-|^{1/2} \sin(2\pi f \Delta t_d)\,. \tag{3.9}$$

The total magnification $|\mu|$ in the geometrical-optics approximation is given by the first and second term of Eq. 3.9 as $|\mu| = |\mu_+| + |\mu_-|$, whereas the third terms quantifies the interference between the two images and causes the oscillatory behavior as observed in Fig. 3.1. The amplitude and frequency of the oscillation can be approximately calculated from the coefficient $2|\mu_+\mu_-|^{1/2}$ and argument $2\pi f \Delta t_d$ of the sinusoid term. The lens parameter y, M_L tune the amplitude and the period of the oscillation; increasing M_L results in higher oscillation period as $\Delta t_d \propto M_L$ while increasing y leads to lower amplitude of the oscillation $2|\mu_+\mu_-|^{1/2}$ as seen in Fig. 3.1.

The lensed waveform $\tilde{h}^L(f)$ is

$$\tilde{h}^L(f) = F(f)\tilde{h}(f) \tag{3.10}$$

where $\tilde{h}(f)$ is the unlensed waveform and $F(f)$ is the amplification factor.

Fig. 3.2 shows the unlensed waveform $\tilde{h}(f)$ and lensed waveform $\tilde{h}^L(f) = F(f)\tilde{h}(f)$ as functions of frequency f. As discussed before, the unlensed waveform given by Eq. 2.37 is parameterized by the 9 parameters; we choose fiducial values of $\theta_S = \phi_S = \theta_L = \phi_L = 0$, $\mathcal{M} = 20M_\odot$, $\eta = 0.25$, $t_c = \phi_c = 0$, and $D = 1\,\mathrm{Gpc}$ for these source parameters. It is proportional to $f^{-7/6}$ according to Eq. 2.37 and appears as a straight line in this log-log plot. The lensed waveform is calculated for a SIS with lens parameters $y = 0.5$ and $M_L = 10^4 M_\odot$ and displays oscillatory behavior due to interference between the two terms in Eq. 3.5. For $\mathcal{M} = 20M_\odot$, the cut-off frequency given by Eq. 2.30 is $f_{cut} \approx 96\mathrm{Hz}$.

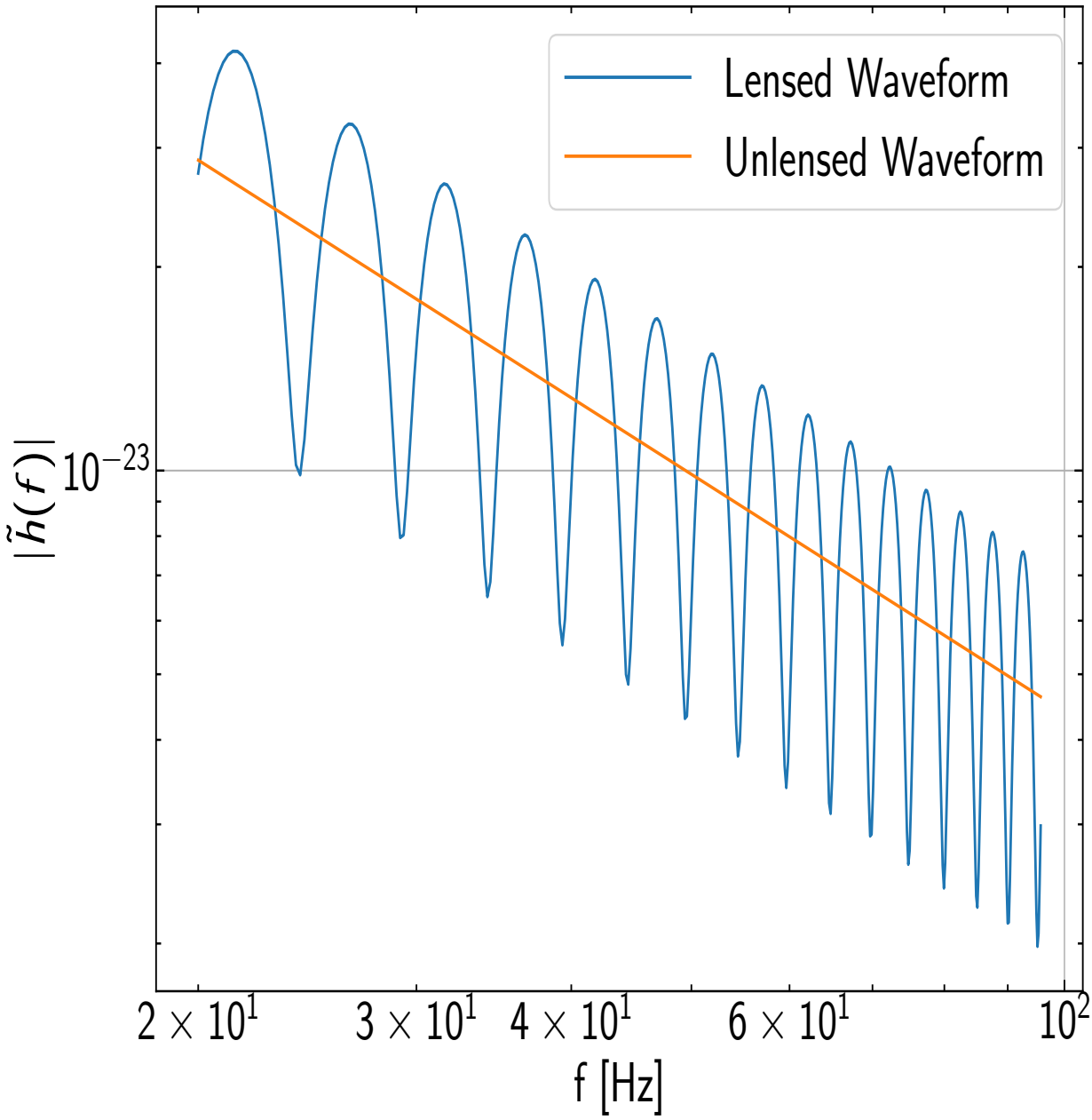


Figure 3.2. Waveform magnitudes as functions of frequency f for our default source parameters of $A = 0.21$, $\mathcal{M} = 20M_\odot$, $\eta = 0.25$, $t_c = \phi_c = 0$, and $D = 1\,\mathrm{Gpc}$. The orange line shows the unlensed waveform $\tilde{h}(f)$, while the blue curve shows the lensed waveform $\tilde{h}^L(f) = F(f)\tilde{h}(f)$ for the amplification factor $F(f)$ of a SIS with lens parameters $y = 0.5$ and $M_L = 10^4 M_\odot$ calculated in the geometrical-optics approximation.

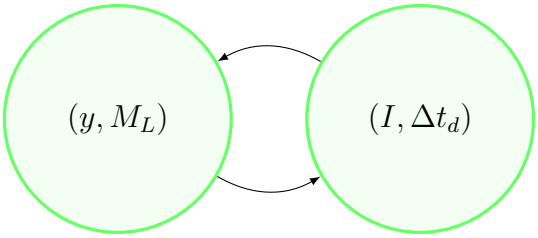


Figure 3.3. Model-dependent lens parameter (y, M_L) and model-independent 'image parameters' $(I, \Delta t_d)$. In the geometrical-optics approximation and two image case, the two sets of parameters are interchangeable for axisymmetric lens models.

3.3 Model-independent image parameters

When the amplification factor $F(f)$ is given by Eq. 3.2, i.e. the source position is such that two images are formed and the geometrical-optics approximation is valid, we can express this factor directly in terms of the flux ratio $I = |\mu_-|/|\mu_+|$ and time delay Δt_d between the images. These "image parameters" are model-independent in that they fully specify the amplification factor for any two-image lens model up to an overall normalization that is observationally degenerate with the GW amplitude $\tilde{A}$. The relationship between these image parameters and lens parameters like the source position y and lens mass M_L is model dependent; these relations for the SIS and PM lens models are given by Eqs. 3.6 and 3.8 respectively. The Eqs. 3.6 and 3.8 provide the mappings from model-dependent lens parameters (y, M_L) to model-independent 'image' parameters $(I, \Delta t_d)$. These mappings can be inverted, either analytically or numerically, to obtain the flux ratio $I(y)$ and time delay $\Delta t_d(y, M_L)$ as a function of source position y and lens mass M_L. The analytical inversion for the SIS is given by

$$I(y) = \frac{1-y}{1+y}\,, \tag{3.11}$$

$$\Delta t_d(y, M_L) = 8 M_L y\,. \tag{3.12}$$

For the SIS lens, we have from Eq. 3.11, $I = 0$ at $y = 1$ as the number of images goes from two to one at caustics $y = 1$. The second arriving image, which is at the saddle point of

the time delay surface, vanishes for $y \geq 1$; only first arriving image, located at the minimum of the time delay surface exists outside the caustics. On the other hand, as $y \to 0$, the $I \to 1$ as both the images have comparable magnifications when the source is located very close to the optics axis. For more complex lens models, such as SIE which will be discussed in the next chapter, $y \to 0$ leads to Einstein ring configuration.

In Fig. 3.4, we plot the flux ratio I and time delay Δt_d as parametric functions of the source position y for our two axisymmetric lens models and two choices of the lens mass M_L. We chose two different fidual values of lens masses as $M_L = 10^3 M_\odot$ and $1.2 \times 10^3 M_\odot$. The blue cross indicates specific, potentially observable values ($I = 0.25$ and $\Delta t_d = 2.8 \times 10^{-2}$ s) of the model-independent image parameters. According to Eqs. 3.6 and 3.8, these values can be obtained in both the SIS and PM lens models, albeit with different lens parameters, $M_L \approx 1.2 \times 10^3 M_\odot$ and $y \approx 0.6$ for the SIS lens and $M_L \approx 10^3 M_\odot$ and $y \approx 0.7$ for the PM lens. This analysis reveals that apart from any observational errors associated with measuring the image parameters I and Δt_d, there is a $\sim 20\%$ model dependence with which lens parameters like M_L and y can be reconstructed.

Before we proceed further, we would like to clarify the context in which we use 'microlensing' and 'macrolensing' events in our research. In the literature (Pagano et al., 2020; Yeung et al., 2021), a microlensing event is an additional lensing event that happens on top of a macrolensing event. If strong gravitational lensing of GWs occurs due to a sufficiently large lens such as a galaxy or galaxy cluster, multiple images will be formed with different arrival times, magnifications, and Morse indices. These images are referred to as 'macroimages' and the event is known as 'macrolensing'. If one of the macroimages is in close vicinity to a stellar-mass object such as a star or a stellar-mass black hole, the macroimage can further split into 'microimages'; this event will be called 'microlensing'. Fig. 3.5 shows a compound lens model with a macrolens (red circle) and a microlens (blue circle) producing macroimages (grey dots) and microimages (black dots).

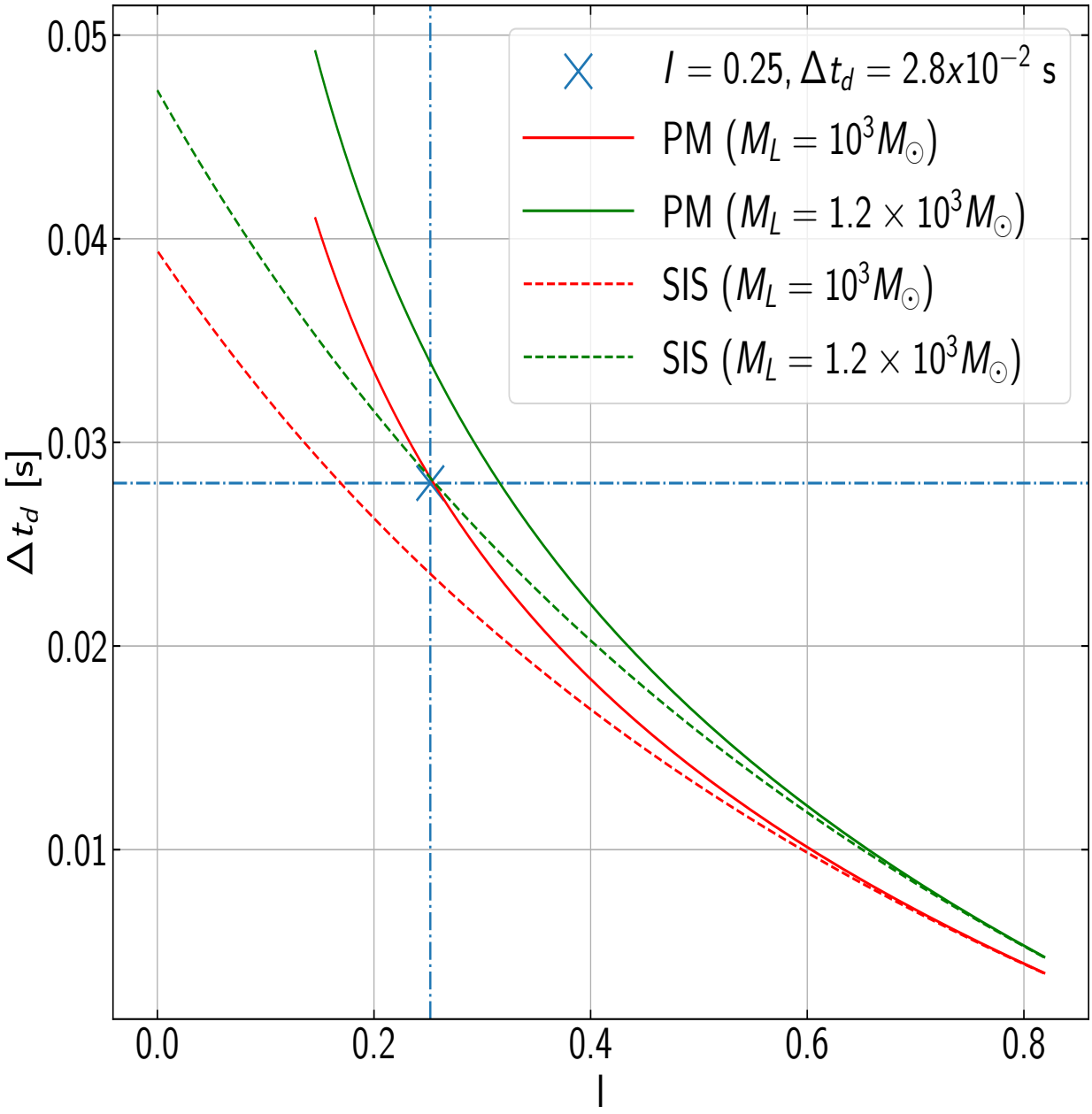


Figure 3.4. The flux ratio I and time delay Δt_d as parametric functions of the source position y for two-image lens models of varying lens mass M_L. The solid (dashed) curves correspond to the PM (SIS) lens model, while the red (green) curves are for a lens mass $M_L = 10^3 M_\odot$ ($1.2 \times 10^3 M_\odot$). The blue cross indicates the point ($I = 0.25$, $\Delta t_d = 2.8 \times 10^{-2}$ s) where curves for the PM and SIS models intersect.

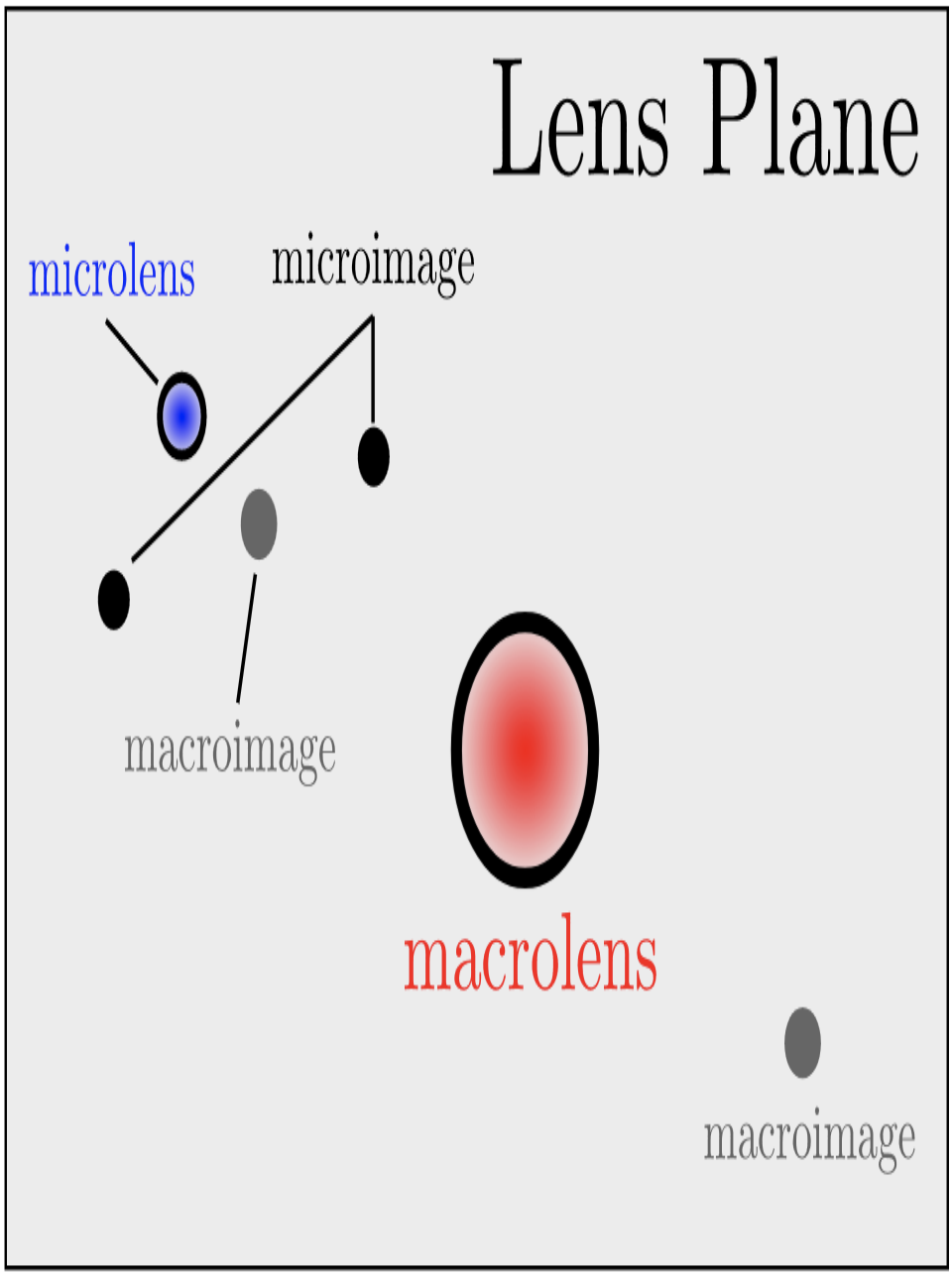


Figure 3.5. The image plane is shown for a compound lens model. The macrolens (red circle) produces two macroimages (grey dots). One of the macroimages, lying closer to the microlens (blue circle), experiences microlensing that results in additional splitting into microimages (black dots) (Yeung et al., 2021).

In our work we slightly modify the definition and classify an event as either macrolensing or microlensing event based on the whether the time delay

$$\Delta t_d \approx M_L = 0.05\,\mathrm{s}\left(\frac{M_L}{10^4\,M_\odot}\right)\,, \tag{3.13}$$

between images is longer or shorter than the time (Cutler and Flanagan, 1994)

$$\begin{aligned} T &\approx 5(8\pi f)^{-8/3}\mathcal{M}^{-5/3} \\ &= 1.45\,\mathrm{s}\left(\frac{f}{20\,\mathrm{Hz}}\right)^{-8/3}\left(\frac{\mathcal{M}}{20\,M_\odot}\right)^{-5/3} \qquad (3.14a) \\ &= 33.8\,\mathrm{d}\left(\frac{f}{10^{-4}\,\mathrm{Hz}}\right)^{-8/3}\left(\frac{\mathcal{M}}{10^6\,M_\odot}\right)^{-5/3}\,, \qquad (3.14b) \end{aligned}$$

that it takes for quasi-circular binaries of chirp mass $\mathcal{M}$ emitting GWs at frequency f to merge. As galaxies have masses $M_L > 10^{10}M_\odot$, they typically generate lensed images with time delays of hours to days when acting as gravitational lenses. This implies that galactic lenses are deeply in the macrolensing regime for LVK sources with typical parameters $f \gtrsim 20\,\mathrm{Hz}$ and $\mathcal{M} \approx 20\,M_\odot$, though not necessarily for LISA sources with typical parameters $f \gtrsim 10^{-4}\,\mathrm{Hz}$ and $\mathcal{M} \approx 10^6 M_\odot$. For LVK sources, the microlensing regime applies to lens masses $M_L \lesssim 10^5 M_\odot$ relevant to intermediate-mass black holes and dark-matter halos too small to host galaxies. In this thesis, we focus on the microlensing of LVK sources by such intermediate-mass objects. However, our treatment is also applicable to the microlensing of LISA sources by more massive galactic lenses.

To further illustrate that the same image parameters can be produced by different choices of lens parameters in different lens models, in Fig. 3.6 we show the ratio of the lens mass M_L^{PM} in the PM model to the lens mass M_L^{SIS} in the SIS model that produce the same flux ratio $I \equiv |\mu_-|/|\mu_+|$ and time delay Δt_d. To obtain this ratio, we invert Eqs. 3.6 and 3.8 to obtain the source distances y^{SIS} and y^{PM} in the SIS and PM models respectively. We then equate Eqs. 3.6b and 3.8b and insert these source distances to obtain the ratio $M_L^{\mathrm{PM}}/M_L^{\mathrm{SIS}}$.

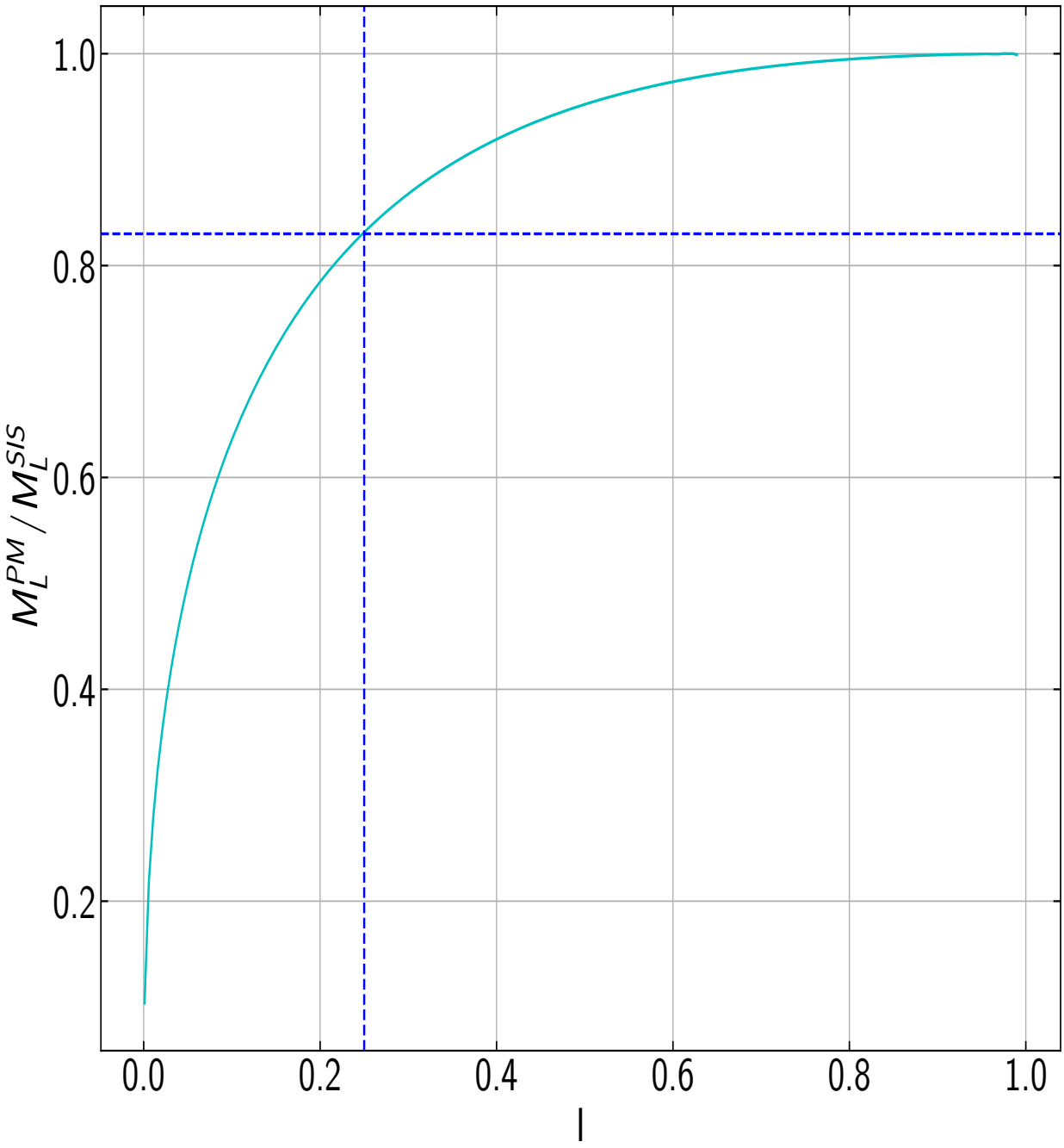


Figure 3.6. The mass ratio $M_L^{\rm PM}/M_L^{\rm SIS}$ of point-mass (PM) to singular-isothermal-sphere (SIS) models producing the same flux ratio I for equal time delay Δt_d. The vertical blue dashed line corresponds to $I = 0.25$ like the vertical blue dot-dashed line in Fig. 3.4.

The fact that this ratio is less than unity could be interpreted as indicated that the more compact PM model is a more powerful lens than the SIS model; it can produce the same image parameters at a smaller lens mass. The difference between this mass ratio and unity provides a lower bound on the systematic fractional uncertainty with which the lens mass can be measured in microlensing events due to model uncertainty, even if the image parameters were measured with arbitrary precision.

A recent study (Wright and Hendry, 2022) investigated whether a GW150914-like GW event microlensed by a point mass with $M_L^{PM} = 10^3 M_\odot$ and source position $y^{\rm PM} = 1$ could be distinguished from lensing by a SIS lens model. Parameter estimation was performed for this event in both the correct PM and incorrect SIS models. The input parameters were accurately recovered in the PM model, but in the SIS model the best-fit parameters were $M_L^{SIS} = 1468 M_\odot$ and $y^{\rm SIS} = 0.72$. Assuming that the geometrical-optics approximation held for this event, we would predict image parameters

$$I = \frac{|\mu_-^{\rm PM}|}{|\mu_+^{\rm PM}|} = \frac{3-\sqrt{5}}{3+\sqrt{5}} = 0.146 \tag{3.15a}$$

$$\Delta t_d = 2M_L \left[\sqrt{5} + 2\ln\left(\frac{\sqrt{5}+1}{\sqrt{5}-1}\right)\right] = 41.6\,{\rm ms}\,, \tag{3.15b}$$

for this event. A SIS model with lens parameters

$$y^{\rm SIS} = \frac{1-I}{1+I} = 0.745 \tag{3.16a}$$

$$M_L^{SIS} = \frac{\Delta t_d}{8 y^{\rm SIS}} = 1396 M_\odot\,, \tag{3.16b}$$

would yield the same image parameters. We see that these estimates are very close to the best-fit parameters in the SIS model, despite the fact that $w = 8\pi M_L f = 0.25$ is not that large at the bottom of the LVK sensitivity band at $f = 20\,{\rm Hz}$, suggesting that wave-optics effects may be significant. The validity of these estimates supports our contention that the use of image parameters is still valuable in the microlensing regime despite the absence of two widely separated images.

3.4 Match-filtering analysis

In this section, we perform a matched-filtering analysis to quantify the difference between lensed and unlensed GWs. The mismatch ϵ between two waveforms h_1 and h_2 is defined as (Cutler and Flanagan, 1994)

$$\epsilon(h_1, h_2) \equiv 1 - \max_{t_c, \phi_c} \frac{\langle h_1 | h_2 \rangle}{\sqrt{\langle h_1 | h_1 \rangle \langle h_2 | h_2 \rangle}} \,. \tag{3.17}$$

The noise-weighted inner product $\langle h_1 | h_2 \rangle$ between the waveforms h_1 and h_2 is defined as

$$\langle h_1 | h_2 \rangle = 4\,\mathrm{Re} \int_{f_{\rm low}}^{f_{\rm cut}} df \frac{h_1(f) h_2^*(f)}{S_n(f)} \,, \tag{3.18}$$

where $S_n(f)$ is the noise power spectral density (PSD). We use the analytical expression for the PSD of advanced LIGO (aLIGO) given in (Sathyaprakash and Schutz, 2009)

$$\frac{Sn(f)}{S_0} = \begin{cases} \infty, & f < f_s \\ (x)^{-4.14} - 5(x)^{-2} + \frac{111(1-x^2+0.5x^4)}{1+0.5xx^2}, & f_s < f, \end{cases} \tag{3.19}$$

where $x = f/f_0$, $f_s = 20$ Hz, $S_0 = 1.0 \times 10^{-49}$ Hz, and $f_0 = 215$ Hz. The noise curve is shown in Fig. 3.7. We use the PYCBC.FILTER package (Nitz et al., 2022) to evaluate the mismatch between waveforms. The two waveforms h_1 and h_2 can be distinguished when their mismatch $\epsilon \gtrsim \rho^{-2}$, where $\rho = \langle h | h \rangle^{1/2}$ is the SNR of a waveform h (Finn, 1992; Finn and Chernoff, 1993; Cutler and Flanagan, 1994).

Fig. 3.8 shows contour plots of the mismatch ϵ between lensed GW source and unlensed templates as functions of the model-dependent lens parameters y and M_L (left panels) and the model-independent image parameters I and Δt_d (right panels) for PM lenses (top panels) and SIS lenses (bottom panels). We use the same source parameters for the lensed and unlensed templates and the noise PSD appropriate for a single two-armed detector with aLIGO design sensitivity. The mismatches as functions of the lens parameters are qualitatively similar for the PM (top left panel) and SIS (bottom left panel) models. For a vanishing lens mass

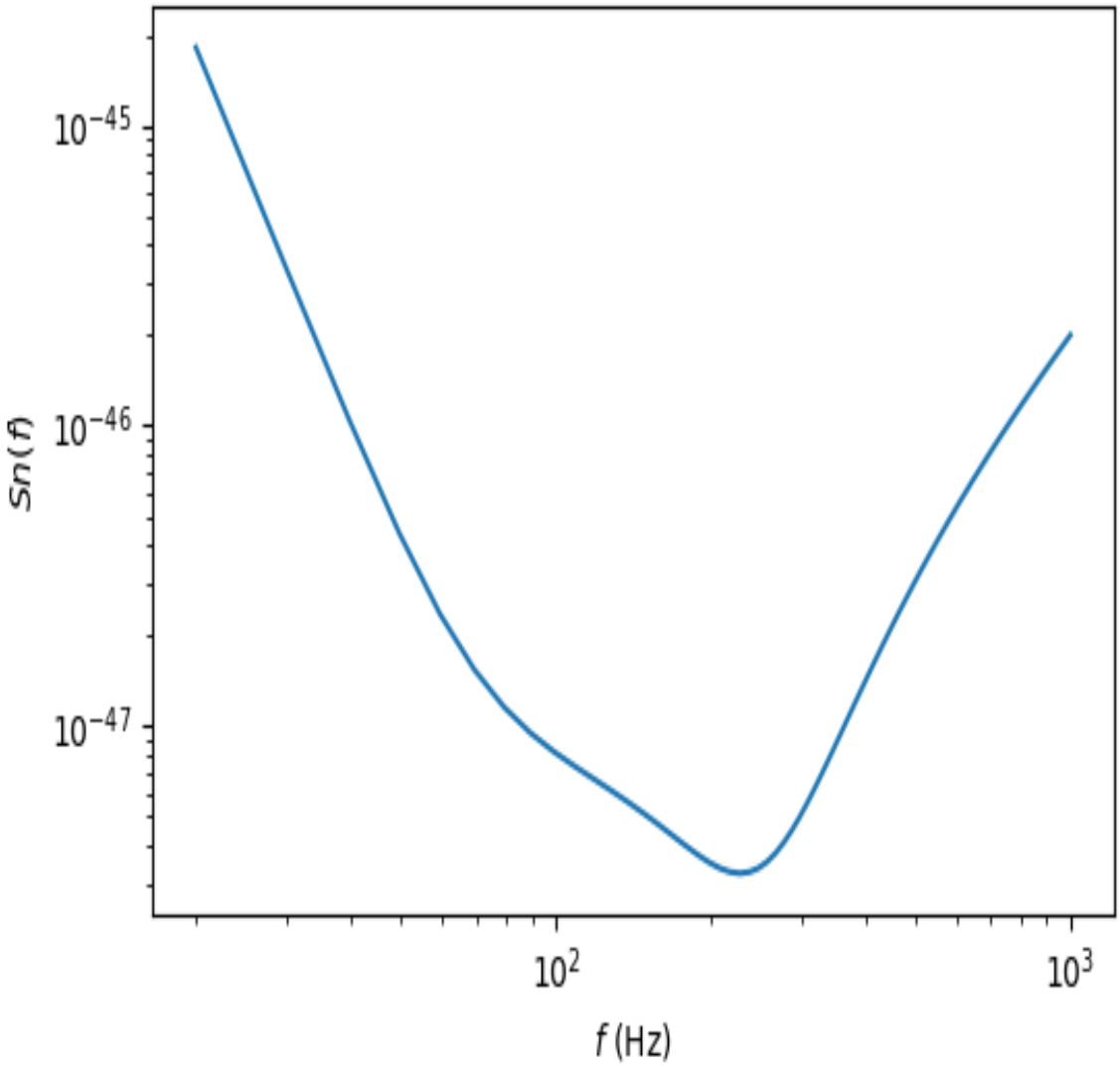


Figure 3.7. Power spectral density (PSD) of the advanced LIGO (aLIGO).

($M_L \to 0$), diffraction causes the amplification factor to approach unity ($F \to 1$) and the mismatch to vanish ($\epsilon \to 0$) according to Eq. 5.27 for $h_1 = h_2$. The biggest difference between the two models occurs in the limit $y \to 1$, where $|\mu_-| \to 0$ for the SIS model according to Eq. 3.6a but $|\mu_-| \to 0.17$ for the PM model according to Eq. 3.8a. This accounts for the much smaller mismatches for the SIS model compared to the PM model in this limit.

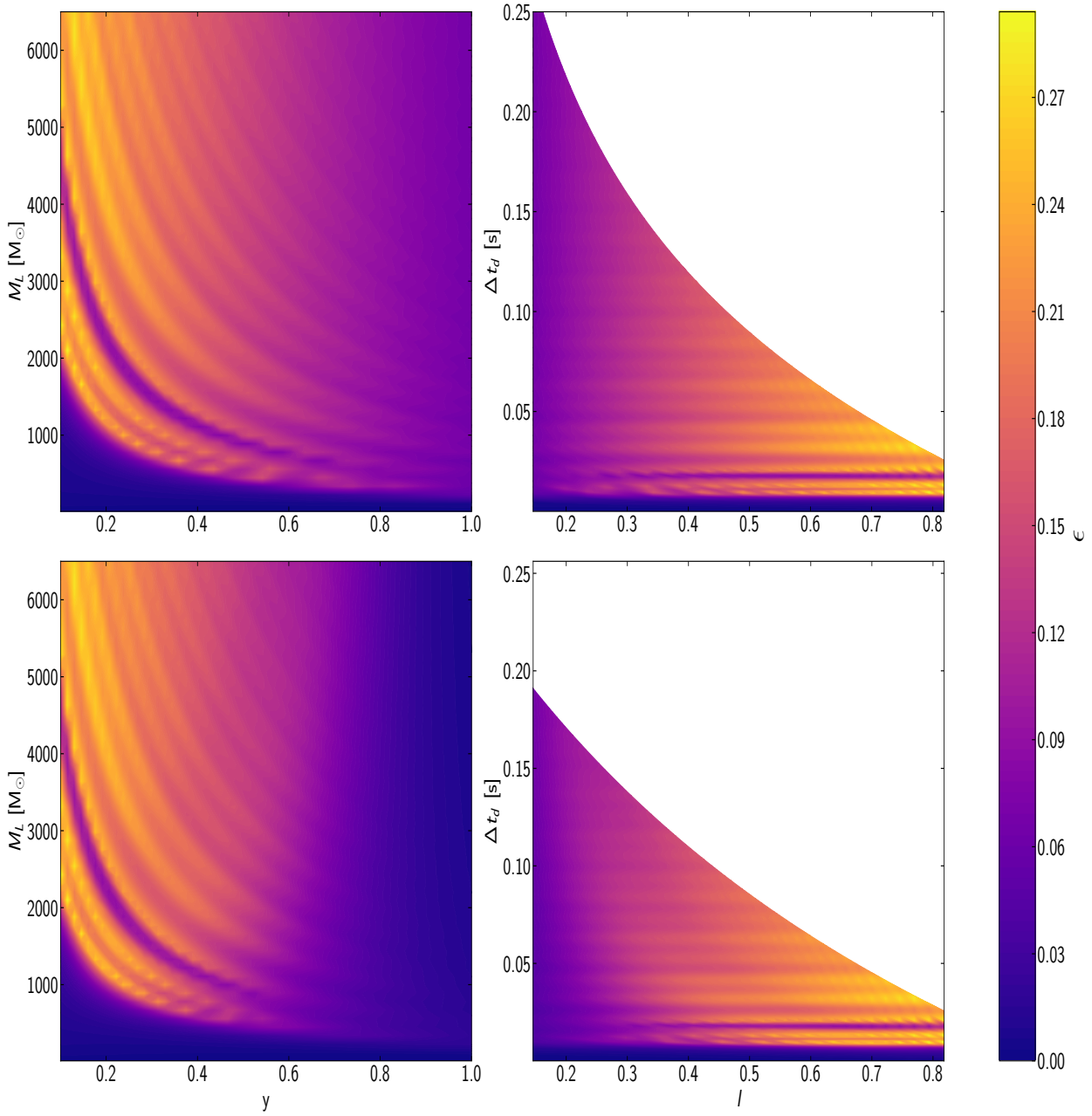


Figure 3.8. Contour plots of the mismatch ϵ between lensed and unlensed GWs for PM lenses (top panels) and SIS lenses (bottom panels). The left panels show ϵ as a function of the model-dependent source distance y and lens mass M_L, while the right panels show ϵ as a function of the model-independent flux ratio I and time delay Δt_d. The unlensed waveforms are calculated using the default source parameters $\mathcal{M} = 20M_\odot$, $\eta = 0.25$, and $t_c = \phi_c = 0$.

The PM and SIS models become qualitatively indistinguishable when the mismatch is expressed as a function of the image parameters of I and Δt_d as shown in the right panels of Fig. 3.8. Although the top and bottom panels appear the same in the regions where they overlap, the PM model extends to larger values of the time delay Δt_d than the SIS model at small flux ratios I. This occurs because the mappings Eqs. 3.6 and 3.8 between the lens parameter space in the left panels and the image parameter space in the right panels are model-dependent.

The better agreement between the mismatches in the PM and SIS lens models when expressed as functions of the image parameters I and Δt_d can be seen even more clearly in Fig 3.9. This figure shows the difference $\Delta\epsilon = |\epsilon_{\rm PM} - \epsilon_{\rm SIS}|$ between the mismatches for PM and SIS lens models shown in the top and bottom panels of Fig. 3.8. The left panel of Fig 3.9 shows large differences $\Delta\epsilon \gtrsim 0.15$ between the two lens models for source distances $y \gtrsim 0.7$ where the flux ratio I vanishes in the SIS model but not the PM model. However, when the mismatch difference is expressed as a function of the image parameters in the right panel, the amplification factors and thus the mismatches only have significant differences for $\Delta t_d \lesssim 0.02$s where the geometrical-optics approximation breaks down.

The right panels of Fig. 3.8 also reveal that the crests and troughs of the oscillations in ϵ occur on lines of constant time delay Δt_d. We examine these oscillations and the general dependence of the mismatch on our lens and image parameters for the SIS model in Fig. 3.10. The top left (right) panels of Fig. 3.10 show vertical (horizontal) slices of the bottom left panel of Fig. 3.8 along lines of constant y (M_L). The dependence of the mismatch ϵ on these lens parameters is non-monotonic and difficult to interpret, exhibiting oscillations of varying amplitude, frequency, and phase and a deep valley. The dependence of ϵ on the image parameters Δt_d and I shown in the bottom panels is far easier to interpret. The bottom left panel shows that $\epsilon \to 0$ in the diffraction limit $\Delta t_d \to 0$, while in the opposite extreme geometrical-optics limit $\Delta t_d \to \infty$, $\epsilon \to 1 - (1+I)^{-1/2}$ as derived in Sec. 3.5.. The crests

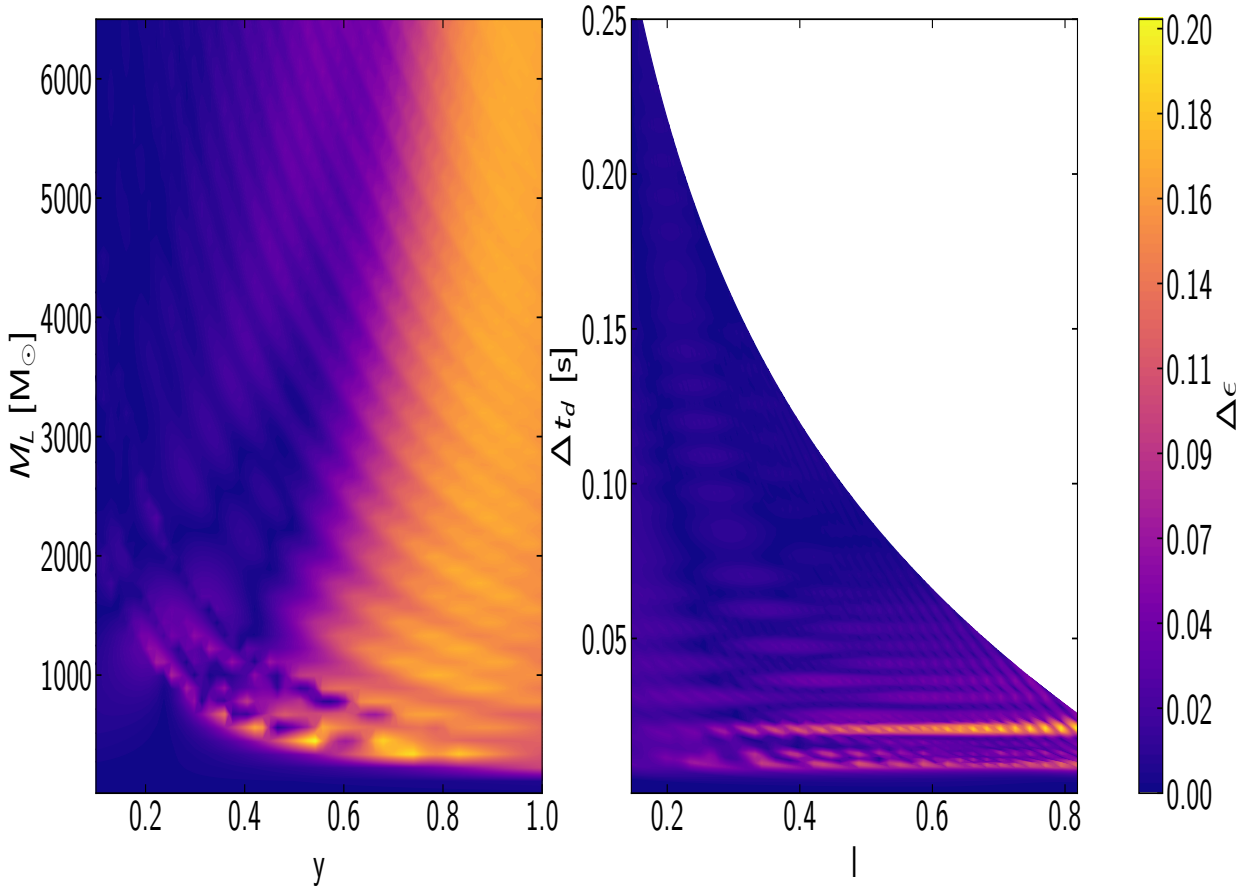


Figure 3.9. The difference $\Delta\epsilon = |\epsilon_{\rm PM} - \epsilon_{\rm SIS}|$ between the mismatches between lensed and unlensed waveforms for PM and SIS lens models shown in the top and bottom panels of Fig. 3.8 as functions of the source position y and lens mass M_L (left panel) and the flux ratio I and time delay Δt_d (right panel).

and troughs of the oscillations, as well as the deep valley at $\Delta t_d \approx 0.015\,\mathrm{s}$ are all aligned in this panel, reinforcing our contention that these oscillations are purely functions of the time delay Δt_d. They are absent in the bottom right panel, where Δt_d is held constant and the mismatch depends smoothly on the flux ratio I in reasonable agreement with the extreme geometrical-optics limit $\epsilon \to 1 - (1+I)^{-1/2}$.

We investigate the oscillations in the mismatch ϵ as a function of the time delay Δt_d in Sec. 3.6. These oscillations result from the number and location of the peaks of the amplification factor $F(f)$ shown in Fig. 3.1 within the sensitivity band of the detector changing

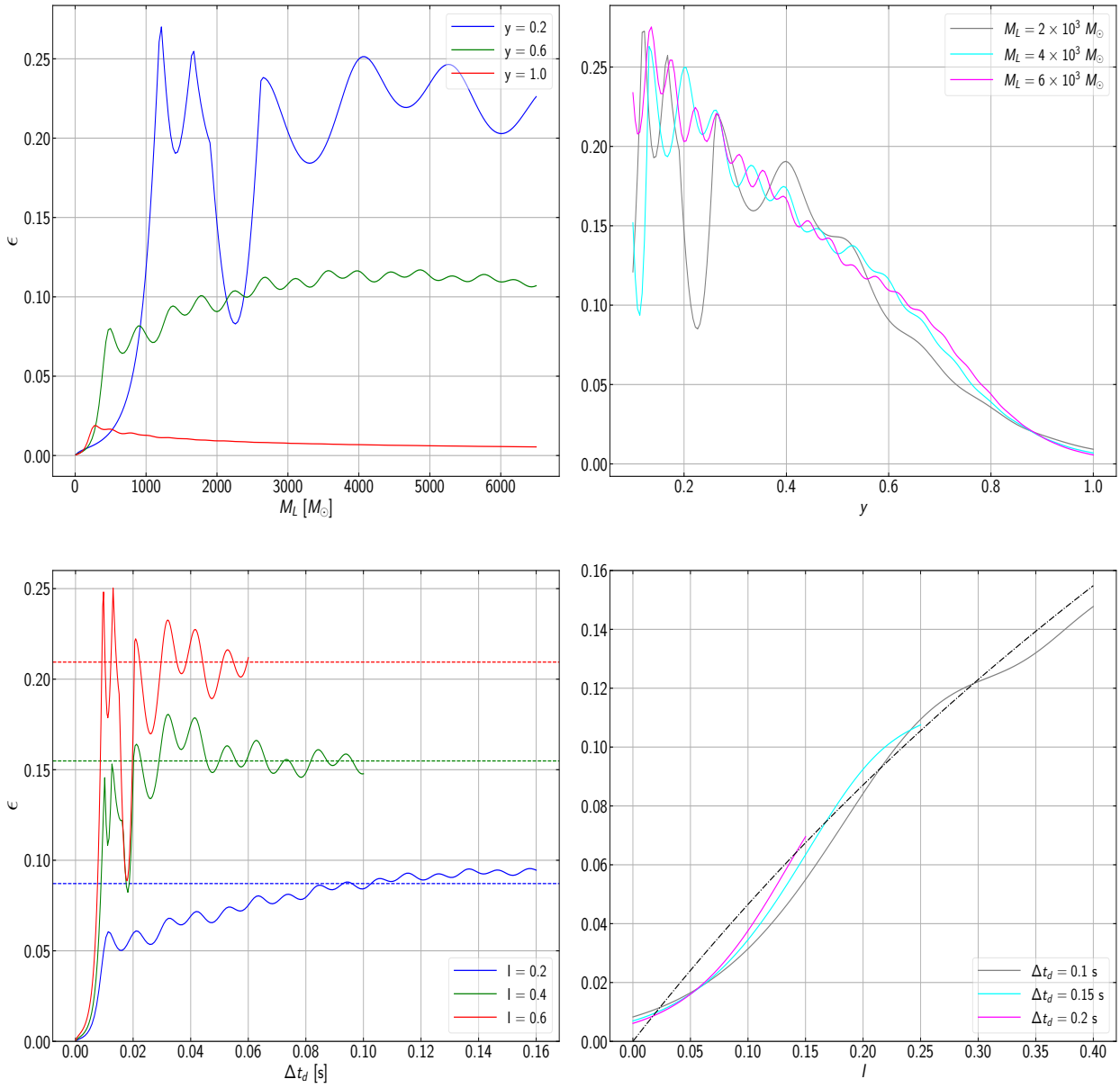


Figure 3.10. The mismatch ϵ between lensed and unlensed waveforms with source parameters $\mathcal{M} = 20M_\odot$, $\eta = 0.25$, and $t_c = \phi_c = 0$ in the SIS lens model as a function of lens mass M_L (top left panel), source position y (top right panel), time delay Δt_d (bottom left panel), and flux ratio I (bottom right panel). In the top (bottom) left panel, the blue, green, and red curves correspond to $y = 0.2$, 0.6, and 1.0 ($I = 0.2$, 0.4, and 0.6). In the top (bottom) right panel, the grey, cyan, and magenta curves correspond to $M_L = 2 \times 10^3 M_\odot$, $4 \times 10^3 M_\odot$, and $6 \times 10^3 M_\odot$ ($\Delta t_d = 0.1$ s, 0.15 s, and 0.2 s). The dotted curves in the bottom panels show the mismatch $\epsilon \approx 1 - (1 + I)^{-1/2}$ in the extreme geometrical-optics limit derived in Sec. 3.5.

as Δt_d varies. For our approximate waveform of Eq. 2.37, the sharpest feature in the detector response is a cutoff in the strain $\tilde{h}(f)$ above a frequency $f_{\rm cut} = (6^{3/2}\pi M_z)^{-1}$ where $M_z = (1+z)M$ is the the redshifted source mass for a binary at redshift z with total mass M. According to Eq. 3.27, this cutoff couples to the peaks of the amplification factor to create oscillations in the mismatch with frequency as a function of the time delay with frequency $f_{\rm cut}$ and amplitude proportional to $I^{3/2}/(1+I)$. Numerical-relativity simulations reveal that true waveforms transition smoothly to a ringdown rather than experiencing such a sharp cutoff (Mroue et al., 2013; Aasi et al., 2014; Boyle et al., 2019), but the merger should still imprint a feature leading to oscillations in the mismatch between a lensed GW source and unlensed templates. Our analysis suggests that these oscillations may be significant for large flux ratios I (small source distances y), implying that serendipity between the lens parameters and source mass may facilitate the discovery of lensing in GW events.

3.5 Mismatch approximation in geometrical-optics limit

We define the overlap $\mathcal{O}$ between a lensed waveform $\tilde{h}^L(f)$ and unlensed template $\tilde{h}(f)$ as

$$\mathcal{O} \equiv \frac{\langle \tilde{h}^L(f)|\tilde{h}(f)\rangle}{\sqrt{\langle \tilde{h}^L(f)|\tilde{h}^L(f)\rangle\langle \tilde{h}(f)|\tilde{h}(f)\rangle}}\,. \tag{3.20}$$

For a lensed waveform with the amplification factor $F(f)$ of Eq. 3.7 appropriate for a two-image lens in the geometrical-optics approximation and an unlensed waveform with the same source parameters, this overlap becomes

$$\mathcal{O} = \frac{A_1}{\sqrt{A_2 A_3}}, \tag{3.21}$$

where

$$A_1 \equiv \int df \frac{|\tilde{h}(f)|^2}{S_n(f)}(|\mu_+|^{1/2} + |\mu_-|^{1/2}\sin 2\pi f\Delta t_d)\,, \tag{3.22a}$$

$$A_2 \equiv \int df \frac{|\tilde{h}(f)|^2}{S_n(f)}(|\mu_+| + |\mu_-| + 2|\mu_+\mu_-|^{1/2}\sin 2\pi f\Delta t_d)\,, \tag{3.22b}$$

$$A_3 \equiv \int df \frac{|\tilde{h}(f)|^2}{S_n(f)}\,. \tag{3.22c}$$

In the extreme geometrical-optics limit $\Delta t_d \to \infty$, the oscillatory terms do not contribute to the integrals implying $A_1 \to |\mu_+|^{1/2} A_3$, $A_2 \to (|\mu_+| + |\mu_-|) A_3$, and

$$\mathcal{O} \to \frac{|\mu_+|^{1/2}}{\sqrt{|\mu_+| + |\mu_-|}} = \frac{1}{\sqrt{1+I}} \tag{3.23}$$

where $I = |\mu_-|/|\mu_+|$ is the flux ratio between the two images. Although the match 1 - ϵ is normally calculated by maximizing the overlap over the coalescence time t_c and phase ϕ_c as in Eq. 5.27, it is independent of these parameters in the limit $\Delta t_d \to \infty$. This implies that the mismatch takes the limiting value

$$\epsilon \to 1 - \mathcal{O} = 1 - \frac{1}{\sqrt{1+I}}. \tag{3.24}$$

3.6 Investigating striations in the mismatch

In this Section, we examine the oscillations in the mismatch ϵ as a function of the time delay Δt_d seen in the bottom left panel of Fig. 3.10. We reproduce the $I = 0.2$ curve from this figure in Fig. 3.11 above. In the geometrical-optics approximation, lensing induces an oscillatory contribution to the GW phase that is poorly matched by a linear change to the GW phase resulting from a shift in the coalesence time t_c and phase ϕ_c according to Eq. 2.28. As such, the mismatch is well approximated as $\epsilon \approx 1 - \mathcal{O}$, where the overlap $\mathcal{O}$ is given by Eqs. 3.20 - 3.22. The crests and troughs of the oscillations in the mismatch as a function of the time delay Δt_d will therefore occur where

$$\begin{aligned} \frac{d\epsilon}{d\Delta t_d} &= \mathcal{O}\left(\frac{1}{2A_2}\frac{dA_2}{d\Delta t_d} - \frac{1}{A_1}\frac{dA_1}{d\Delta t_d}\right) \\ &= \mathcal{O}\left(\frac{|\mu_+|^{1/2}}{A_2} - \frac{1}{A_1}\right)\frac{dA_1}{d\Delta t_d} \end{aligned} \tag{3.25}$$

vanishes.

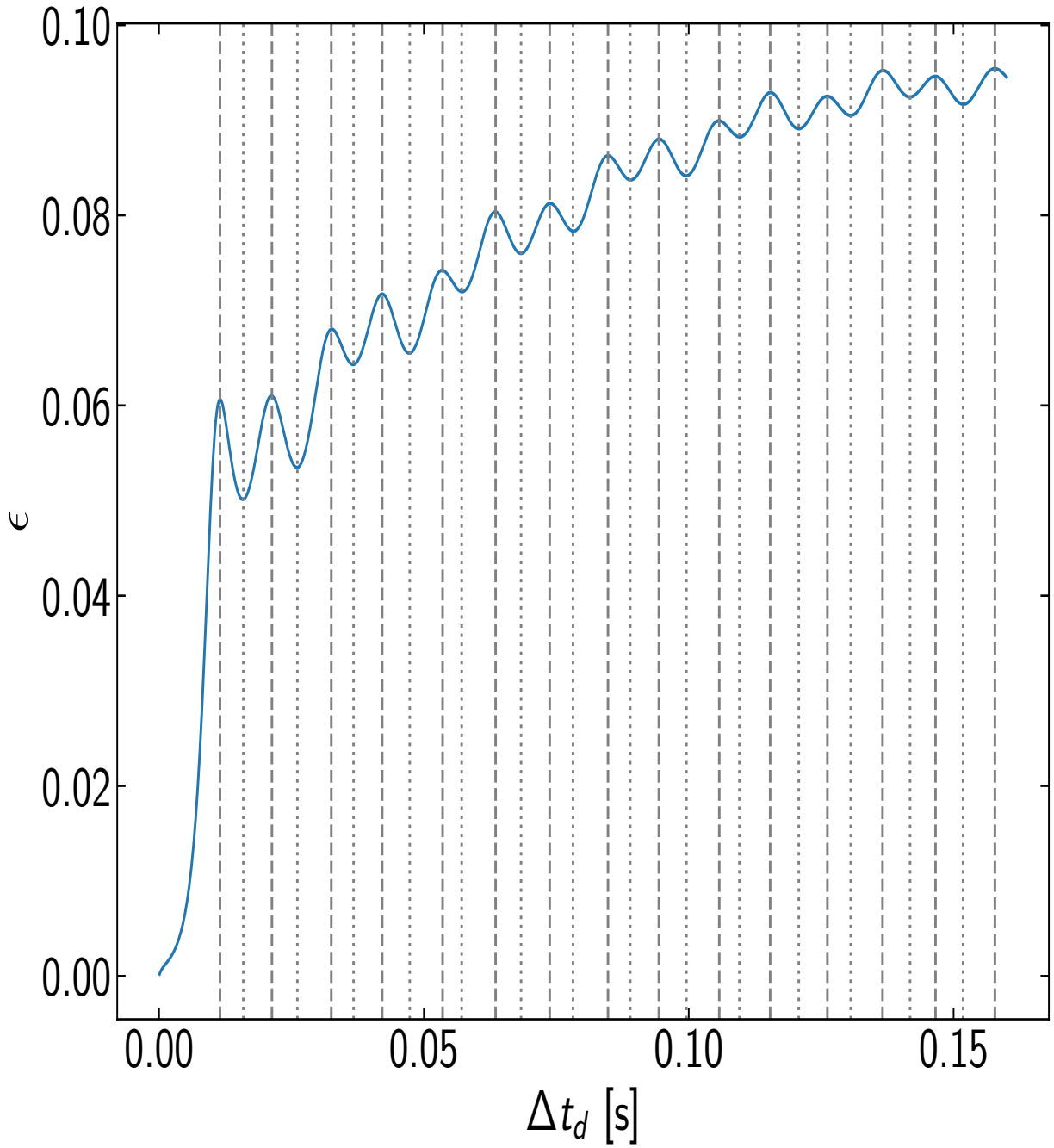


Figure 3.11. The mismatch ϵ between a lensed GW source and unlensed templates for an SIS lens as a function of time delay Δt_d for flux ratio $I = 0.2$. The BBH system has source parameters $\mathcal{M} = 20M_\odot$, $\eta = 0.25$, and $t_c = \phi_c = 0$. The vertical dashed and dotted lines indicate the locations of the crests and troughs listed in Table 3.1.

We show further that

$$\frac{dA_1}{d\Delta t_d} = |\mu_-|^{1/2} \int_{f_{\rm low}}^{f_{\rm cut}} df \frac{|\tilde{h}(f)|^2}{S_n(f)} 2\pi f \cos 2\pi f \Delta t_d \tag{3.26a}$$

$$\approx \frac{|\mu_-|^{1/2}}{2\pi \Delta t_d^2} \left\langle \frac{|\tilde{h}(f)|^2}{S_n(f)} \right\rangle (w \sin w + \cos w) \Big|_{w_{\rm low}}^{w_{\rm cut}} \tag{3.26b}$$

$$\approx |\mu_-|^{1/2} \frac{A_3}{\Delta t_d} \sin w_{\rm cut} \,, \tag{3.26c}$$

where we have defined $w \equiv 2\pi f \Delta t_d$, approximated $|\tilde{h}(f)|^2/S_n(f)$ by a constant in angular brackets equal to its average value, and assumed $w_{\rm cut} \gg w_{\rm low} \gg 1$ as will be true for small redshifted source masses in the geometrical-optics limit. Inserting these into Eq. 3.25 and using the extreme geometrical-optics approximation described in Sec. 3.5, we find

$$\frac{d\epsilon}{d\Delta t_d} = -\frac{I^{3/2}}{1+I} \frac{\mathcal{O}}{\Delta t_d} \sin w_{\rm cut} \,. \tag{3.27}$$

This result implies that the crests [troughs] in the mismatch ϵ will occur at time delays $\Delta t_{d,c}$ [$\Delta t_{d,t}$] for which $w_{\rm cut} = 2n\pi$ [$(2n+1)\pi$], i.e.

$$\Delta t_{d,c} = n f_{\rm cut}^{-1} = n \times 10.47\,{\rm ms} \left(\frac{M_z}{45.9 M_\odot} \right), \tag{3.28a}$$

$$\Delta t_{d,t} = \left(n + \frac{1}{2} \right) f_{\rm cut}^{-1} = \left(n + \frac{1}{2} \right) 10.47\,{\rm ms} \left(\frac{M_z}{45.9 M_\odot} \right). \tag{3.28b}$$

Fig. 3.11 reproduces the blue curve shown in the bottom left panel of Fig. 3.10 corresponding to the mismatch ϵ between a lensed GW source and unlensed templates in the SIS model for flux ratio $I = 0.2$ and source parameters $\mathcal{M} = 20 M_\odot$, $\eta = 0.25$, and $t_c = \phi_c = 0$. The dashed (dotted) vertical lines indicate the values of the time delay at which maxima (minima) of the mismatch occur. These numerically determined values are listed in Table 3.1, where they are compared to the values predicted by Eqs. 3.28a and 3.28b. We see that there is excellent agreement between our numerical results and analytical predictions, and that this agreement improves for larger times delays Δt_d at which the extreme geometrical-optics approximation becomes increasingly valid.

Table 3.1. The five columns in this table list: (1) the index n of the crests and troughs, (2) the numerically determined time delay of the crest, (3) the time delay of the crest predicted by Eq. 3.28a, (4) the numerically determined time delay of the trough, and (5) the time delay of the trough predicted by Eq. 3.28b. The lens and source parameters are the same as those in Fig. 3.11.

	Crests		Troughs	
n	Numerical Δt_d [ms]	Predicted $\Delta t_{d,c}$ [ms]	Numerical Δt_d [ms]	Predicted $\Delta t_{d,t}$ [ms]
1	11.51	10.47	15.91	15.71
2	21.31	20.94	26.12	26.18
3	32.52	31.41	36.72	36.65
4	42.13	41.88	47.33	47.12
5	53.53	52.36	57.13	57.59
6	63.54	62.83	68.34	68.06
7	73.75	73.30	78.15	78.53
8	84.75	83.77	88.95	89.01
9	94.36	94.24	99.56	99.48
10	105.8	104.7	109.6	109.9
11	115.4	115.2	120.8	120.4
12	126.2	125.7	130.6	130.9
13	136.6	136.1	141.8	141.4
14	146.6	146.6	151.8	151.8
15	157.8	157.1	-	-

3.7 Summary and discussion

In the geometrical-optics approximation, strong gravitational lensing produces multiple images. The lensing amplification factor $F(f)$ can be expressed as a summation over these images, each with its own magnification μ_j, time delay $t_{d,j}$, and Morse index n_j as given by Eq. 2.49. In any given lens model, these image parameters can be calculated as functions of the lens parameters such as the source position $\boldsymbol{y}$ and the lens mass M_L. Lensing of transient events can further be classified as either microlensing or macrolensing, depending on whether the time delay between images is shorter or longer than the duration of the transient. In the latter case, two distinct GW events are observed and it is natural to characterize these events using image parameters. We propose that, even in the microlensing regime, observational

searches for strong lensing in GW events, as well as theoretical studies of their feasibility, should be conducted in terms of these model-independent image parameters rather than the model-dependent lens parameters provided that the geometrical-optics approximation still holds.

In this thesis, we investigate this proposal by considering two well-known axisymmetric lens models: the singular isothermal sphere (SIS) and the point mass (PM). The SIS model has a density profile $\rho \propto r^{-2}$ appropriate for a galaxy or stellar cluster, while the PM model could describe an individual star or compact object. In the geometrical-optics approximation, the SIS produces two images for source positions $y < 1$, while the PM always produces two images. For the case of two images, the amplification factor is given by Eq. 3.7 for both models, but the mappings between the image parameters (the flux ratio $I = |\mu_-|/|\mu_+|$ and the time delay Δt_d) and the lens parameters (the source distance y and the lens mass M_L) are model dependent and given by Eqs. 3.6 and 3.8 respectively. This implies that uncertainty in the lens model will translate into uncertainty in the lens parameters, even if the image parameters could be measured with arbitrary precision. We illustrate this in Fig. 3.4, where the SIS and PM models can generate images with identical flux ratios and time delays despite have lens parameters that differ by $\sim 20\%$. This represents a conservative estimate of the systematic error in the lens parameters associated with uncertainty in the lens model, as the PM model is more compact than any real galaxy or halo. Fig. 3.6 reveals that this systematic error gets even larger as the flux ratio decreases.

Lensed GW events with finite signal-to-noise ratios ρ can be distinguished from unlensed templates if the minimum mismatch ϵ between the lensed signal and members of the template bank exceeds ρ^{-2} (Finn, 1992; Finn and Chernoff, 1993; Cutler and Flanagan, 1994). The oscillatory features induced in the waveform by lensing in the geometrical-optics limit depicted in Fig. 3.2 have little degeneracy with the source parameters specifying the unlensed waveform in Eqs. 2.31 and 2.28, so we approximate this minimum mismatch by that between

lensed and unlensed waveforms with the same source parameters. In this approximation, the mismatch can be approximated by $\epsilon \approx 1 - (1 + I)^{-1/2}$ according to Eq. 3.24 and lensing should in principle be identifiable for flux ratios $I \gtrsim 2\rho^{-2}$ and time delays $\Delta t_d \gtrsim f_{\rm cut}^{-1}$. We further showed in the right panel of Fig. 3.9 that the amplification factor of Eq. 3.7 is an excellent approximation to both the SIS and PM lens models in the geometrical-optics limit, suggesting that it should be suitable for model-independent searches for strong lensing in GW events.

The geometrical-optics approximation describes microlensing for lens masses in the range $M_{\rm min} \lesssim M_L \lesssim M_{\rm max}$, where

$$M_{\rm min} = (8\pi f)^{-1} = 400 M_\odot \left(\frac{f}{20\,{\rm Hz}}\right)^{-1} , \tag{3.29}$$

$$\begin{aligned} M_{\rm max} &= 5(8\pi f)^{-8/3}\mathcal{M}^{-5/3} \\ &= 3\times 10^5 M_\odot \left(\frac{f}{20\,{\rm Hz}}\right)^{-8/3}\left(\frac{\mathcal{M}}{20\,M_\odot}\right)^{-5/3} . \end{aligned} \tag{3.30}$$

This complements electromagnetic observations of strong lensing which are only sensitive to lens masses $M_L \gtrsim 10^6 M_\odot$ (Mao and Schneider, 1998; Dalal and Kochanek, 2002; Gilman et al., 2020). GW lensing observations thus extend our sensitivity to isolated low-mass dark-matter halos by roughly three orders of magnitude. This is particularly important because cosmological observations challenge the ΛCDM model on these small scales (Bullock and Boylan-Kolchin, 2017; Gilman et al., 2020). This has motivated investigations of alternatives to cold dark matter which suppress the matter power spectrum or halo mass function on small scales.

Although gravitational lensing has not been detected in any of the GW events observed in the first three runs of the LVK detector network (Abbott et al., 2021), recent estimates (Oguri, 2018b) suggest that $\sim 10^3$ strongly lensed GW events could be observed each year by future third-generation detectors like the proposed Cosmic Explorer (CE) (Evans et al., 2021) and Einstein Telescope (ET) (Maggiore et al., 2020). The detection of gravitational

lensing in GW events would be of tremendous scientific interest because it could test general relativity (Baker and Trodden, 2017; Collett and Bacon, 2017; Mukherjee et al., 2020), probe the distribution of dark matter (Guo and Lu, 2022; Cao et al., 2022), and improve the precision of cosmological constraints (Goyal et al., 2021) We propose that GW templates based on model-independent image parameters will be a valuable tool in this effort. In the Chapter 4 , we will investigate how effectively these GW templates can be used to identify additional images created by non-axisymmetric lenses (Ali et al., 2023).

CHAPTER 4

INVESTIGATING GRAVITATIONAL-WAVE SOURCES LENSED BY ELLIPTICAL LENSES

The content of this chapter is taken from (Ali et al., 2023), done with my collaborators, Evangelos Stoikos, Michael Kesden, and Lindsay King. SA (lead author) setup the elliptical lensing code pipeline and generated the results appeared in the Chapter. ES undertook checks on the codes and the figures generated by them. He also contributed by engaging in discussions and content writing. MK and LK are the research supervisors, who formulated the fundamental idea that led to the results presented in the Chapter. We are going to submit the work to Physical Review D in April 2023.

4.1 Elliptical lensing potentials

In the previous Chapter, we considered axisymmetric lens models such as the Singular Isothermal Sphere (SIS) and Point Mass (PM) lenses. Axisymmetric lenses are extensively used in the literature to model gravitational lenses as they are computationally simple to work with and explain lensing related effects to some extent. However these lens models do not completely explain the observed gravitational lensing scenarios. As summarized in (Schneider et al., 1992), the properties of axisymmetric and asymmetric lens models are rather different.

For realistic modeling of gravitational lenses, elliptical lens models are used that are consistent with the majority of galaxies seen in the Universe (Ohanian, 1983; Turner et al., 1984; Blandford and Narayan, 1986). Elliptical lens models have mass distributions such that contours of constant surface mass density are ellipses, with fixed parameters such as eccentricity and position angle. However, early work (Bourassa et al., 1973; Bourassa and Kantowski, 1975; Schramm, 1990) showed that the elliptical lens models can be very challenging; even

the calculation of the deflection angle is very non-trivial. Due to the complex nature of elliptical lens models, they have been mainly used for the precise numerical modelling of individual lens systems (Young et al., 1981; Young, 1981; Narasimha et al., 1982, 1984).

Several quasi-elliptical lens models have been introduced as an alternative to elliptical lens models. The quasi-elliptical lens models are constructed either with elliptical isopotential curves (Kovner, 1987a,b; Kochanek, 1991a; Kochanek and Blandford, 1987; Kochanek et al., 1989; Blandford and Kochanek, 1987) or with a combination of external shear and axisymmetric lens models (Chang and Refsdal, 1979, 1984; Kovner, 1987b; Kochanek, 1991b). These models share the general properties with elliptical lens models but are computationally much easier to handle. These quasi-elliptical models are best used for performing model fitting or statistical analysis of gravitational lensing. There are some caveats to quasi-elliptical models, especially the ones with elliptical isopotentials, where a large ellipticity could lead to a nonphysical distribution of matter (Kassiola and Kovner, 1993).

In this work, we use the Singular Isothermal Ellipsoid (SIE) (Kormann et al., 1994; Asada et al., 2003; Oguri, 2010) model, which is a natural generalization of the SIS model. The SIE model belongs to the family of lens models with homoeoidal elliptical surface mass distributions. The term 'isothermal' implies that the surface mass density is inversely proportional to the distance from the center of the lens. The lens we consider lacks any finite-sized core in the vicinity of the lens center; this results in a central singularity (infinite surface mass density at the center $\Sigma(\xi \to 0) \to \infty$) and hence the term 'singular' in SIE.

The ellipticity is introduced into the projected surface mass density through the convergence $\kappa(r)$ by replacing r with $\tilde{r}$ (Oguri, 2010)

$$r \to \tilde{r} \equiv \sqrt{\frac{\tilde{x}^2}{(1-e)} + (1-e)\tilde{y}^2}\,, \tag{4.1}$$

where e is the ellipticity, and the rotated frame is defined as

$$\tilde{x} = x\cos\theta_e + y\sin\theta_e\,, \tag{4.2}$$

$$\tilde{y} = -x\sin\theta_e + y\cos\theta_e\,. \tag{4.3}$$

In the chosen convention, the x-axis is in the West direction and the y-axis is in the North direction, and the position angle θ_e is measured East of North (Oguri, 2010).

The convergence κ of the isothermal ellipsoid model with a finite core radius r_{core} is given by

$$\kappa = \frac{4\pi\sigma^2(D_{LS}/D_S)}{2\sqrt{r_{core}^2 + q\tilde{x}^2 + \tilde{y}^2/q}}\,, \tag{4.4}$$

where the axis ratio q is defined as $q \equiv 1 - e$. The lensing potential Ψ Eq. 2.10 is then given by (Oguri, 2010)

$$\Psi = \tilde{x}\phi_{\tilde{x}} + \tilde{y}\phi_{\tilde{y}} + qb(q)s(q)\ln\left[\frac{(1+q)s(q)}{(p(q)+s(q))^2 + (1-q^2)\tilde{x}^2}\right]\,, \tag{4.5}$$

$$\phi_{\tilde{x}} = \frac{qb(q)}{\sqrt{1-q^2}}\arctan\left[\frac{\sqrt{1-q^2}\tilde{x}}{p(q)+s(q)}\right]\,, \tag{4.6}$$

$$\phi_{\tilde{y}} = \frac{qb(q)}{\sqrt{1-q^2}}\mathrm{arctanh}\left[\frac{\sqrt{1-q^2}\tilde{y}}{p(q)+q^2s(q)}\right]\,, \tag{4.7}$$

where

$$p(q) = \sqrt{q^2(s(q)^2 + \tilde{x}^2) + \tilde{y}^2}\,, \tag{4.8}$$

$$b(q) = \frac{4\pi\sigma^2 D_{LS}}{\sqrt{q}D_S}\,, \tag{4.9}$$

$$s(q) = \frac{r_{core}}{\sqrt{q}}\,. \tag{4.10}$$

In the limit of $q \to 1$ or $e \to 0$, the SIE model becomes SIS model and the Eqs. 4.6 and 4.7 reduces to

$$\phi_{\tilde{x}} = \frac{b(q)\tilde{x}}{p(q)+s(q)}\,, \tag{4.11}$$

$$\phi_{\tilde{y}} = \frac{b(q)\tilde{y}}{p(q)+s(q)}\,, \tag{4.12}$$

and the isothermal ellipsoid lensing potential in Eq. 4.5 transforms to

$$\begin{aligned}
\lim_{q\to 1}\Psi &= \frac{4\pi\sigma^2 D_{LS}}{D_S}\lim_{q\to 1}\sqrt{\frac{q}{1-q^2}}\left[\frac{\tilde{x}^2\sqrt{1-q^2}}{\sqrt{q^2\tilde{x}^2+\tilde{y}^2}}+\frac{\tilde{y}^2\sqrt{1-q^2}}{\sqrt{q^2\tilde{x}^2+\tilde{y}^2}}\right], \\
&= \frac{4\pi\sigma^2 D_{LS}}{D_S}\lim_{q\to 1}\sqrt{q}\left[\frac{\tilde{x}^2}{\sqrt{q^2\tilde{x}^2+\tilde{y}^2}}+\frac{\tilde{y}^2}{\sqrt{q^2\tilde{x}^2+\tilde{y}^2}}\right], \\
&= \frac{4\pi\sigma^2 D_{LS}}{D_S}\sqrt{\tilde{x}^2+\tilde{y}^2} = \frac{4\pi\sigma^2 D_{LS}}{D_S}\, r\,, \\
\lim_{q\to 1}\Psi &= \frac{4\pi\sigma^2 D_{LS}}{D_S}\, r
\end{aligned} \tag{4.13}$$

SIS lensing potential, where $r = \sqrt{\tilde{x}^2+\tilde{y}^2}$.

In the case of Singular Isothermal Ellipsoid (SIE) lens model, the core radius $r_{core} = 0$, therefore, Eq. 4.4 reduces to

$$\kappa = \frac{4\pi\sigma^2(D_{LS}/D_S)}{2\sqrt{q\tilde{x}^2+\tilde{y}^2/q}}\,. \tag{4.14}$$

and the lensing potential Ψ Eq. 4.5 is

$$\begin{aligned}
\Psi &= \tilde{x}\phi_{\tilde{x}} + \tilde{y}\phi_{\tilde{y}}\,, \\
\Psi &= \frac{4\pi\sigma^2 D_{LS}}{D_S}\sqrt{\frac{q}{1-q^2}}\left[\tilde{x}\,\arctan\left\{\frac{\sqrt{1-q^2}\tilde{x}}{\sqrt{q^2\tilde{x}^2+\tilde{y}^2}}\right\}+\tilde{y}\,\mathrm{arctanh}\left\{\frac{\sqrt{1-q^2}\tilde{y}}{\sqrt{q^2\tilde{x}^2+\tilde{y}^2}}\right\}\right].
\end{aligned} \tag{4.15}$$

We will discuss more about the parameterization of the SIE lens and the properties of the images formed later in the chapter.

4.2 Quasi-geometrical optics approximation

In Chapter 2 and 3 we discussed the gravitational lensing of GWs. We quantified the effect of gravitational lensing on GWs through the amplification factor $F(f)$. The amplification factor is calculated using the diffraction integral Eq. 2.46, which is valid throughout the range of GW frequency f. However, it can be computationally expensive to calculate $F(f)$ for very high f; where in any case we can employ geometrical-optics approximation. Under

this approximation, the integral is transformed into a summation Eq. 2.49, consisting of terms depending on the time-delay, magnification, and Morse index of different images.

We have already discussed that the diffraction effect is important for $f \lesssim \Delta t_d^{-1}$; in this low frequency limit the wavelength of the GW is so long that it does not get affected by the presence of the lens and the $F(f) \to 1$. For $f \gg \Delta t_d^{-1}$, the geometrical-optics approximation is valid and the $F(f)$ is given by Eq. 2.49. The computationally-intensive nature of the $F(f)$ prevents the full analysis of the GW lensing in the wave-optics regime, however, in the limit $f \geq \Delta t_d^{-1}$, we can calculate the $F(f)$ using the quasi-geometrical optics approximation. In this approximation, the geometrical optics terms are accompanied by the correction terms arising from the the effects of finite wavelength (Takahashi, 2004).

Let us define the dimensionless frequency w as

$$\begin{aligned} w &= \frac{D_S}{D_{LS} D_L} \xi_0^2 (2\pi f)(1 + z_L) \,, \\ &= 8\pi M_{Lz} f \,, \end{aligned} \tag{4.16}$$

where the redshifted lens mass is defined as $M_{Lz} = (1 + z_L) M_L$ for a lens at redshift z_L. We can rewrite the amplification factor F in terms of w as

$$F(w) = \frac{w}{2\pi i} \int d^2\boldsymbol{x} \, \exp(iwT(\boldsymbol{x}, \boldsymbol{y})) \,, \tag{4.17}$$

and dimensionless time-delay $T(\boldsymbol{x}, \boldsymbol{y})$, also known as *Fermat potential* is

$$T(\boldsymbol{x}, \boldsymbol{y}) = \frac{1}{2} |\boldsymbol{x} - \boldsymbol{y}|^2 - \Psi(\boldsymbol{x}) - \phi_m(\boldsymbol{y}) \,, \tag{4.18}$$

where $\psi(\boldsymbol{x})$ is the dimensionless gravitational potential of the lens and $\phi_m(\boldsymbol{y})$ makes the minimum arrival time equal to zero.

The leading order correction to the geometrical-optics approximation can be obtained by expanding the amplification factor Eq. 4.17 in powers of w^{-1}. The quasi-geometrical optics regime is defined as the geometrical optics regime with the leading order correction. We

will begin with the quasi-geometrical optics regime for the axisymmetric lens following the prescription provided in (Takahashi, 2004). We first expand the dimensionless time delay function $T(\boldsymbol{x},\boldsymbol{y})$ Eq.4.18 around image position $\boldsymbol{x}^{(j)}$

$$T(\boldsymbol{x},\boldsymbol{y}) = T^{(j)} + \frac{1}{2!}\sum_{a,b}\partial_a\partial_b T^{(j)}\tilde{x}_a\tilde{x}_b + \frac{1}{3!}\sum_{a,b,c}\partial_a\partial_b\partial_c T^{(j)}\tilde{x}_a\tilde{x}_b\tilde{x}_c + \frac{1}{4!}\sum_{a,b,c,e}\partial_a\partial_b\partial_c\partial_e T^{(j)}\tilde{x}_a\tilde{x}_b\tilde{x}_c\tilde{x}_e + \mathcal{O}(\tilde{x}^5)\,, \tag{4.19}$$

where $\tilde{\boldsymbol{x}} = \boldsymbol{x} - \boldsymbol{x}^{(j)}$ and $T^{(j)} = T(\boldsymbol{x}^{(j)},\boldsymbol{y})$. Here indices $a, b \in \{1,2\}$ corresponding to such that the two components of the jth image location vector $\boldsymbol{x}^{(j)}$ on the lens plane is represented by $x_a^{(j)}$. The second term in Eq. 4.19 is

$$\begin{aligned}\sum_{a,b}\partial_a\partial_b T^{(j)}\tilde{x}_a\tilde{x}_b &= \left(1 - \frac{d^2\psi^{(j)}}{dx^2}\right)\tilde{x}_1^2 + \left(1 - \frac{1}{|x^{(j)}|}\frac{d\psi^{(j)}}{dx}\right)\tilde{x}_2^2\,,\\ &= 2(\alpha^{(j)}\tilde{x}_1^2 + \beta^{(j)}\tilde{x}_2^2)\,,\end{aligned} \tag{4.20}$$

where $\alpha^{(j)} \equiv (1 - d^2\psi^{(j)}/dx^2)/2$ and $\beta^{(j)} \equiv (1 - |x^{(j)}|^{-1}d\psi^{(j)}/dx)/2$. Inserting Eq. 4.19 into 4.17, expanding Eq. 4.17 in powers of w^{-1}, and re-scaling Eq. 4.17 by $\boldsymbol{x}' = \sqrt{w}\tilde{\boldsymbol{x}}$ yields

$$F(w) = \frac{1}{2\pi i}\sum_j e^{iwT^{(j)}}\int d^2\boldsymbol{x}'\, e^{i(\alpha^{(j)}x'^2_1 + \beta^{(j)}x'^2_2)}\left(1 + \frac{iI_1(\boldsymbol{x}^{(j)}}{6\sqrt{w}},\boldsymbol{y}) + \frac{I_2(\boldsymbol{x}^{(j)},\boldsymbol{y})}{w} + \mathcal{O}(w^{-3/2})\right) \tag{4.21}$$

where

$$I_1(\boldsymbol{x}^{(j)},\boldsymbol{y}) = \sum_{a,b,c}\partial_a\partial_b\partial_c T(\boldsymbol{x}^{(j)},\boldsymbol{y})x'_a x'_b x'_c\,, \tag{4.22}$$

$$I_2(\boldsymbol{x}^{(j)},\boldsymbol{y}) = -\frac{1}{72}\left(\sum_{a,,b,c}\partial_a\partial_b\partial_c T(\boldsymbol{x}^{(j)},\boldsymbol{y})x'_a x'_b x'_c\right)^2 + \frac{i}{24}\sum_{a,,b,c,d}\partial_a\partial_b\partial_c\partial_d T(\boldsymbol{x}^{(j)},\boldsymbol{y})x'_a x'_b x'_c x'_d\,. \tag{4.23}$$

The first term in Eq. 4.21 is the amplification factor in the geometrical-optics approximation in Eq. 2.49. The second term in Eq. 4.21 integrates out to zero because the integrand in

Eq. 4.22 is an odd function of x'_a. The third term at the order $\mathcal{O}(w^{-1})$ gives the correction to the geometrical-optics approximation. After performing the integrals, the resulting amplification factor can be written as (Takahashi, 2004)

$$F(f) = \sum_j |\mu^{(j)}|^{1/2} \left(1 + i\frac{\Delta^{(j)}}{w}\right) e^{\left(iwT(\boldsymbol{x}^{(j)},\boldsymbol{y}) - i\pi n^{(j)}\right)} + \mathcal{O}(w^{-2})\,, \tag{4.24}$$

where $\Delta^{(j)}$ is a real number and given by

$$\Delta^{(j)} = \frac{1}{16}\left(\frac{1}{2\alpha^{(j)2}}\frac{d^4\psi^{(j)}}{dx^4} + \frac{5}{12\alpha^{(j)3}}\frac{d^3\psi^{(j)^2}}{dx^3} + \frac{1}{\alpha^{(j)2}|x^{(j)}|}\frac{d^3\psi^{(j)}}{dx^3} + \frac{\alpha^{(j)} - \beta^{(j)}}{\alpha^{(j)}\beta^{(j)}}\frac{1}{|x^{(j)}|^2}\right), \tag{4.25}$$

From Eq. 4.24 we notice that the geometrical-optics approximation is a valid approximation when the limit $\Delta^{(j)} \ll w$ is satisfied by all the images.

4.3 Validity of quasi-geometrical approximation

In this section we will discuss the validity of the quasi-geometrical optics approximation (QGO). We consider the SIS lens for validation purposes, as analytical equations exist for both the wave optics and geometrical optics regimes. The amplification factor for the SIS lens is given by (Matsunaga and Yamamoto, 2006)

$$F(w) = e^{\frac{i}{2}w(y^2 + 2\phi_m(y))} \sum_{n=0}^{\infty} \frac{\Gamma(1+\frac{n}{2})}{n!}(2we^{i\frac{3\pi}{2}})^{\frac{n}{2}}{}_1F_1\left(1 + \frac{n}{2}, 1; -\frac{i}{2}wy^2\right)\,, \tag{4.26}$$

where $\phi_m(y) = y + 1/2$, J_0 is the Bessel function of zeroth order, $\xi_0 = 4\pi\sigma_v^2 D_L D_{LS}/D_S$ is the Einstein radius, and $M_{Lz} = \pi\sigma_v^2(1+z_L)\xi_0$ is the redshifted lens mass inside the Einstein radius. As already discussed in Chapter 3, the amplification factor in the geometrical-optics approximation is given by (Takahashi and Nakamura, 2003)

$$F(f) = \begin{cases} |\mu_+|^{1/2} - i|\mu_-|^{1/2}e^{2\pi i f \Delta t_d}\,, & y < 1 \\ |\mu_+|^{1/2}\,, & y > 1\,, \end{cases} \tag{4.27}$$

where

$$\mu_{\pm} = \pm 1 + \frac{1}{y}\,, \tag{4.28a}$$

$$\Delta t_d = 8 M_L y\,. \tag{4.28b}$$

For $y < 1$, two images are formed at $x = y - 1$ and $x = y + 1$ and for $y > 1$, only one image exists at $x = y + 1$. We now calculate the QGO correction term given by Eq. 4.25 for the SIS lens. The lensing potential for the SIS lens is given by $\Psi(x) = x$, so

$$\frac{d^n \Psi(x)}{dx^n} = \begin{cases} 1\,, & n = 1 \\ 0\,, & n > 1\,. \end{cases} \tag{4.29}$$

Using Eq. 4.25, we can calculate the analytical expression for the QGO correction term for the SIS lens model as

$$\Delta^{(j)} = \frac{1}{8|x^{(j)}|^2(|x^{(j)}| - 1)}\,. \tag{4.30}$$

For $y < 1$, two images are formed at $x = y - 1$ and $x = y + 1$ and for $y > 1$, only one image exists at $x = y + 1$. Substituting image locations into the equation above, we obtain two terms of QGO $\Delta^{\pm}$ corresponding to the two images

$$\Delta^{\pm} = \frac{\pm 1}{8y(y \pm 1)^2}\,, \tag{4.31}$$

where Δ^{+} represents the image with zero time delay. We show the $\Delta^{\pm}$ in Fig. 4.1 with image 1 and 2 corresponding to Δ^{+} and Δ^{-} respectively. The QGO correction term $\Delta^{(j)} \to \infty$ for both the images at $y \to 0$, which can be seen from Eq. 4.30. Physically speaking, the singularity at the center of the SIS lens blows up the magnification of both of the images which causes this behavior. However, in the limit $y \to 1$, the QGO for image 2 blows up, $\Delta^{-} \to \infty$; this is because image 2 vanishes for $y > 1$ and only image 1 exists outside the caustic boundary.

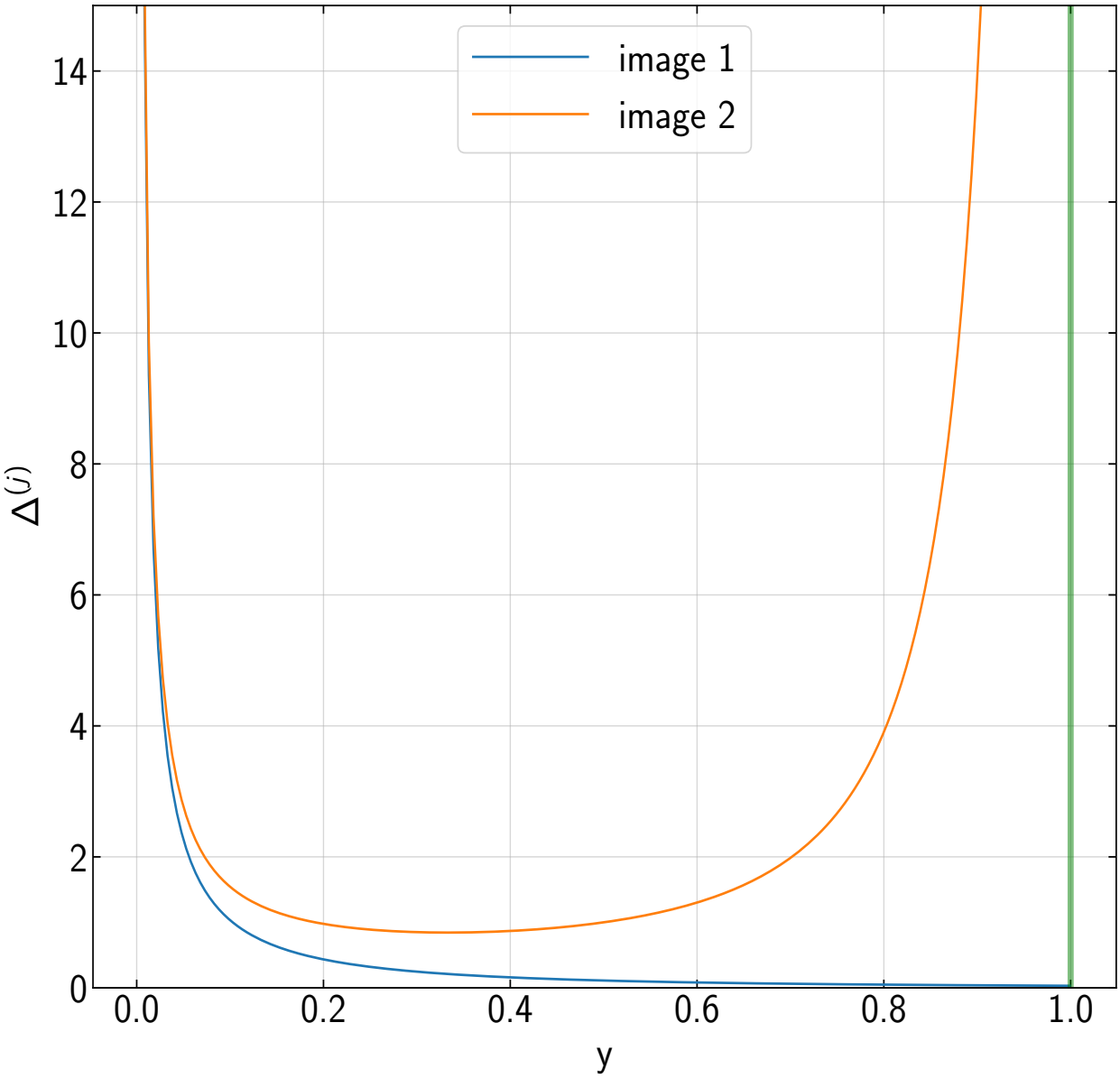


Figure 4.1. $\Delta^{\pm}$ in Eq. 4.30 is shown as a function of y. The Δ^{+} and Δ^{-} in Eq. 4.30 refer to image 1 and 2 respectively. The thick green line denotes the caustic at $y = 1$ for the SIS lens.

The QGO correction term, $\Delta^{(j)}$, by itself does not provide any insights into the physical properties of the lens. Therefore, we can slightly modify our approach to gain an understanding of the physics. We know that QGO is valid when

$$\Delta^{(j)} \ll w \implies M_{Lz}^{(j)} \gg \frac{\Delta^{(j)}}{8\pi f}\,, \tag{4.32}$$

for all of the images. The right hand side of Eq. 4.32 depends on the lens model through $\Delta^{(j)}$ and GW frequency f. For a ground-based detector network such LVK, the lower limit of the sensitivity band is set at around $f_{min} = 20$Hz. We can therefore rewrite Eq. 4.32 to obtain the minimum redshifted lens mass $M_{Lz(min)}^{(j)}$ for the images such that the geometric-optics approximation is valid for lens masses satisfying $M_{Lz}^{(j)} > M_{Lz(min)}^{(j)}$. We define $M_{Lz(min)}^{(j)}$ as

$$M_{Lz(min)}^{(j)} = \frac{|\Delta^{(j)}|}{8\pi f_{min}}\,. \tag{4.33}$$

In Fig. 4.2 we plot $M_{Lz(min)}^{(j)}$ as a function of y for the two image regime of the SIS lens model and it behaves similar to $\Delta^{(j)}$. *However, this treatment is much richer in a way that it directly provides information on the lowest lens mass required for different values of y, above which the geometrical-optics approximation is valid.* The geometrical-optics approximation is valid in the whole range of y for image 1, with the M_{Lz} as low as $10^4 M_\odot$. For image 2, $M_{Lz} = 10^4 M_\odot$ will guarantee valid geometrical-optics approximation for $y \sim 0$ to 0.9; whereas for $y \sim y_t$, from Eq. 4.31, $M_{Lz(min)}^{(2)} \to \infty$.

Of note is that the QGO can also be utilized to approximately identify the regime where diffraction effects can be neglected and where the amplification factor $F(w)$ calculated using the geometrical-optics approximation stays valid. Various studies (Bulashenko and Ubach, 2021; Tambalo et al., 2022; Bondarescu et al., 2022) have broadly categorized GW lensing into three distinct regimes based on the value of w:

- The perturbative regime is defined for $w \ll 1$, where the amplification of the GWs is small and the amplification factor can be written as $F(w) \approx 1 + w^h$, with the asymptotic properties of the lensing potential $\Psi(\boldsymbol{x})$ determining the exponent h.

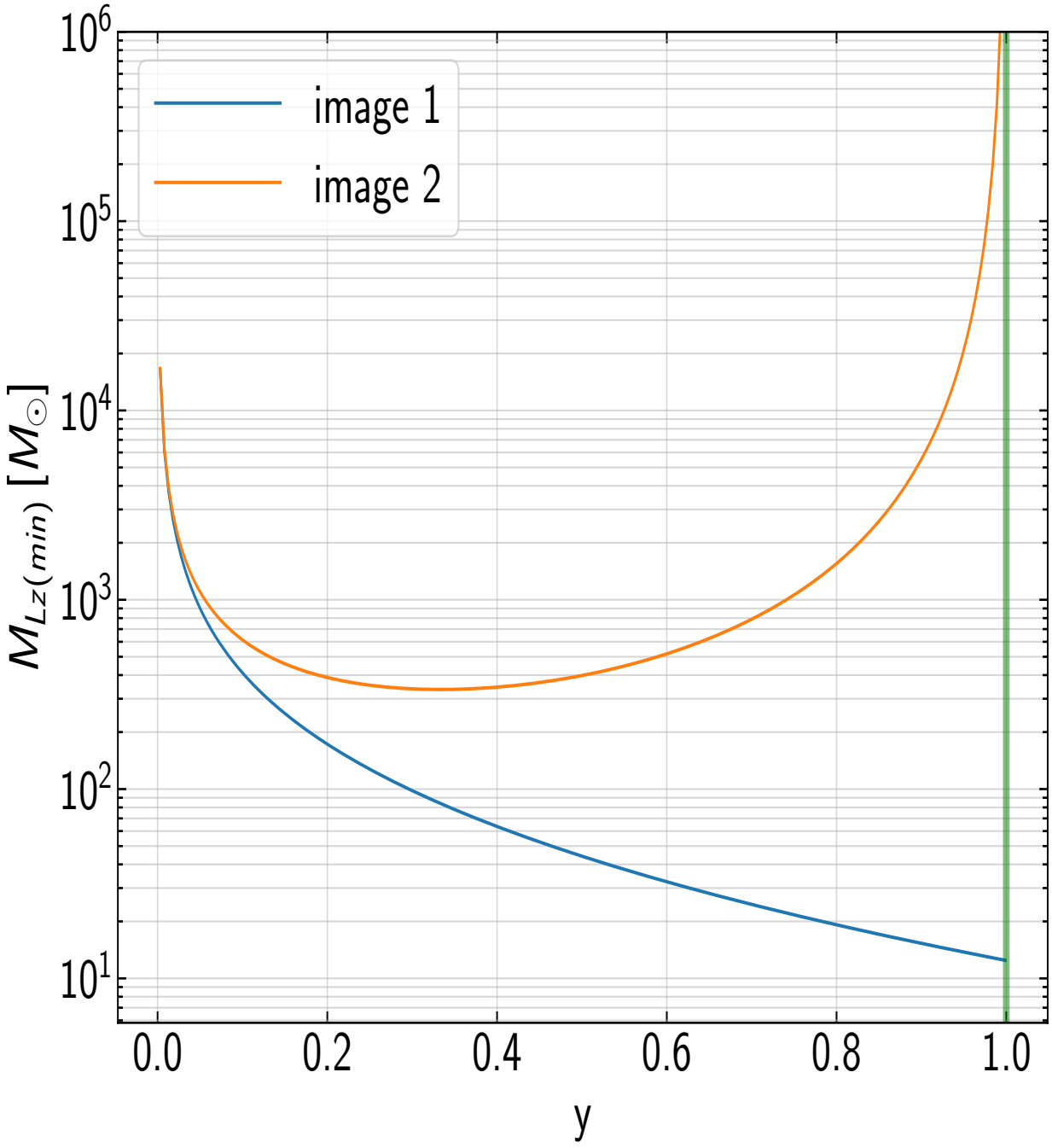


Figure 4.2. $M^{(j)}_{Lz(min)}$ in Eq. 4.33 is shown as a function of y for the two image region $y < 1$ of the SIS lens model. The thick green line denotes the caustic at $y = 1$.

- The intermediate amplification regime occurs at $w \sim 1$. This is typically where the first peak of $F(w)$ is observed.

- Finally, the geometrical-optics limit, which is valid for $w \gg 1$, and characterized by the high-amplitude oscillation of $F(w)$.

In Fig. 4.3 we plot the relative uncertainty $\left|\frac{F_{go}}{F_{wo}} - 1\right|$ between F_{wo} and F_{go} as a function of w for lens mass $M_{Lz} = 10^3 M_\odot$. The amplification factor F_{wo} is calculated using the full wave-optics equation given by Eq. 3.3 and F_{go} is in the geometrical-optics limit given by Eq. 3.5. The figure is shown for $y = 0.3$ in red and $y = 0.5$ in blue. The colored vertical dashed lines represent $\max\{\Delta^{\pm}\}$ for the corresponding y.

From Fig. 4.3 we notice that in the limit $w \gg 1$ the deviation of F_{go} from F_{wo} is small, indicating that the geometrical-optics approximation is a good approximation. However, in the small frequency limit $w \ll 1$, $F_{wo} \to 1$ and F_{go} deviates from the F_{wo}. QGO, which is defined for $w \sim 1$, represents a smooth transition from the wave-optics to the geometrical-optics regime. The vertical dashed lines are the w' calculated for respective y using QGO and it provides an approximate value of w where the transition occurs. We define w' as

$$w' = \max\{\Delta^{\pm}\}\,. \tag{4.34}$$

This shows that QGO is a good approximation to provide an estimate of the value of w where the transition from wave-optics to geometrical-optics is taking place.

4.4 Image properties of singular isothermal ellipsoid (SIE) model

Since we are working with two-dimensional lens and source planes, We define the components of the vectors in the lens plane $\boldsymbol{x} = (x_1, x_2)$ and the source plane $\boldsymbol{y} = (y_1, y_2)$. Using Eq. 2.8 and 4.15, we define the dimensionless SIE potential as

$$\psi(\boldsymbol{x}) = \sqrt{\frac{q}{1-q^2}}\left[x_1 \tan^{-1}\left(\frac{\sqrt{1-q^2}x_1}{\sqrt{q^2x_1^2 + x_2^2}}\right) + x_2 \tanh^{-1}\left(\frac{\sqrt{1-q^2}x_2}{\sqrt{q^2x_1^2 + x_2^2}}\right)\right] \tag{4.35}$$

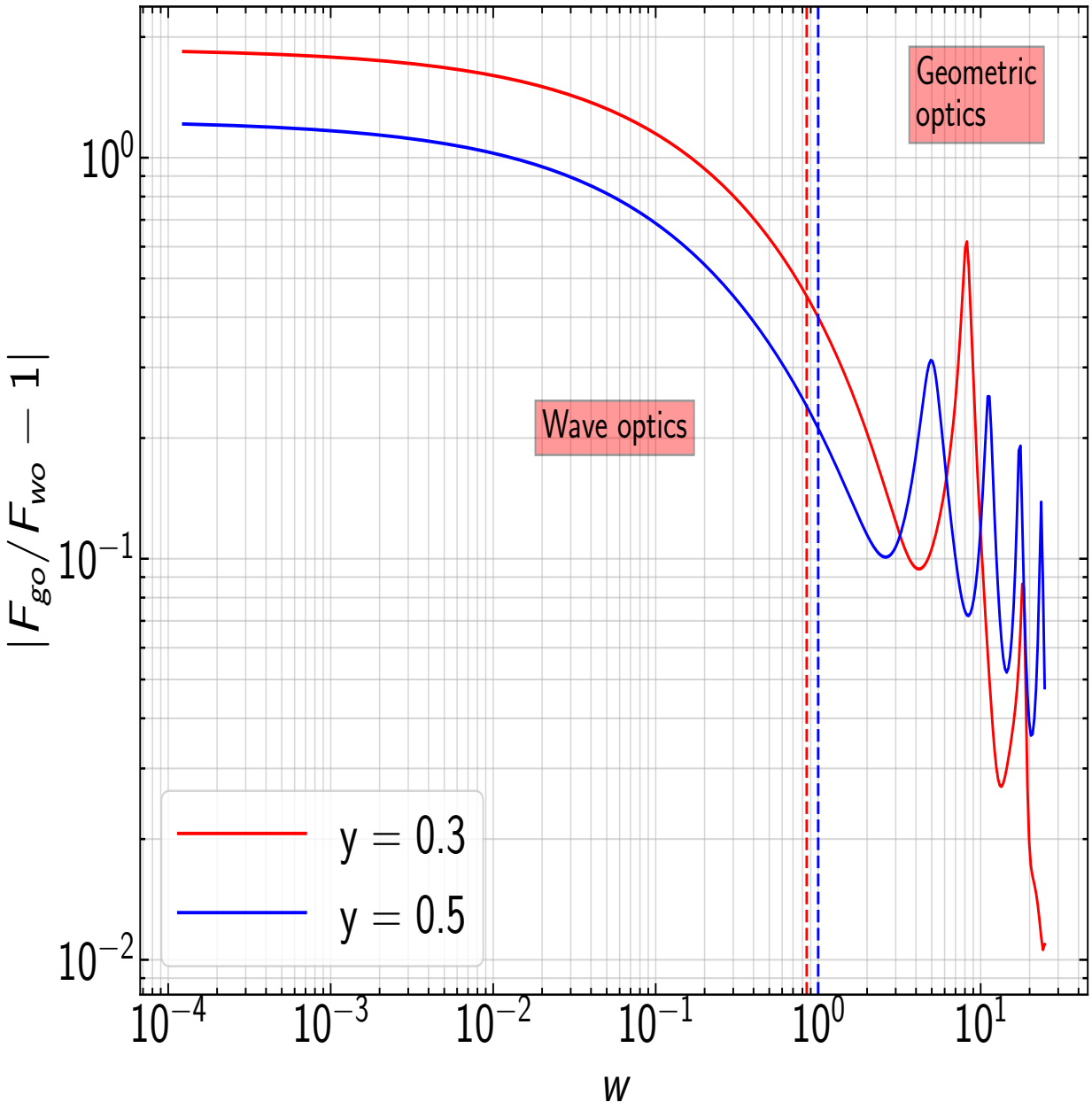


Figure 4.3. Relative uncertainty of amplification factor in geometrical-optics approximation F_{go} (Eq. (3.5)) with respect to the wave optics F_{wo} (Eq. (3.3)) as a function of the dimensionless frequency $w = 8\pi M_{Lz} f$, where f is the GW frequency. The amplification factors are calculated for an SIS lens with redshifted lens mass $M_{Lz} = 10^3 M_\odot$ and $y = 0.3, 0.5$. The dotted lines represent the approximate lower bound on GW frequency f_{low} above which the geometrical-optics approximation is valid.

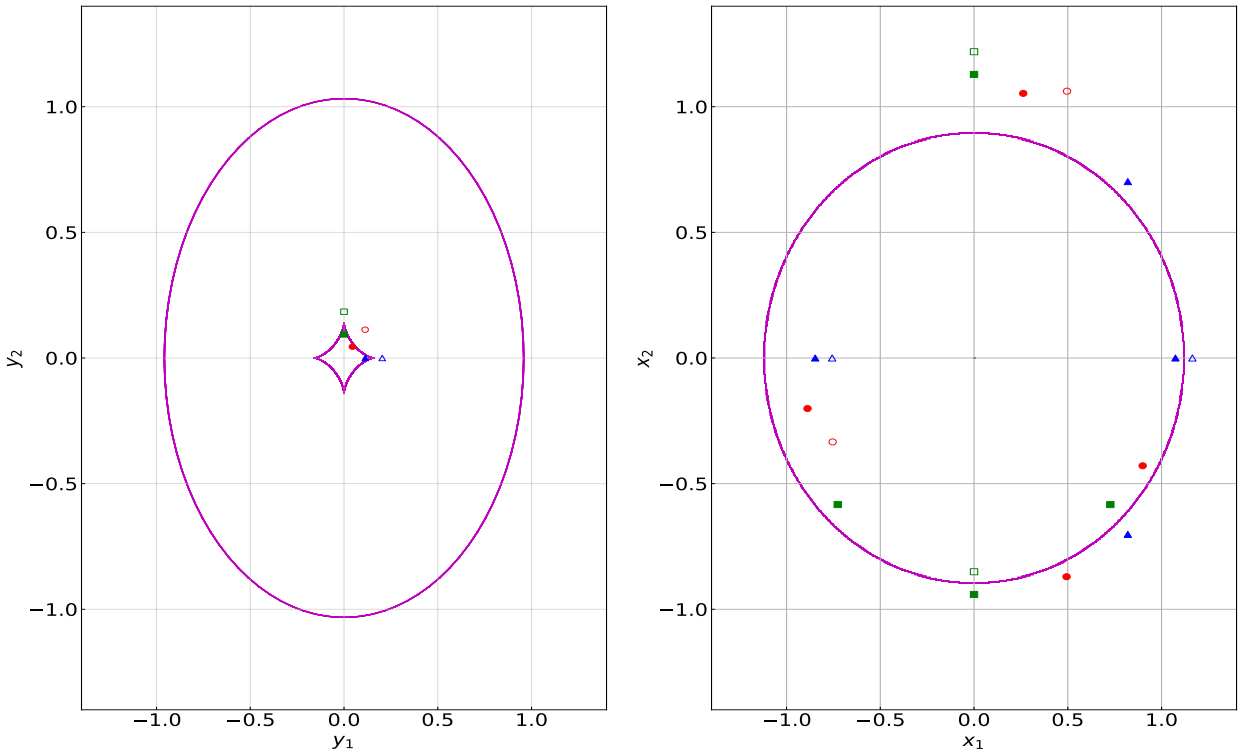


Figure 4.4. Caustics (left panel) and critical curves (right panel) are shown for an SIE lens with redshifted lens mass $M_{Lz} = 1.05 \times 10^5 M_\odot$ and ellipticity $e = 0.2$. Different markers on the left panel locate the sources at different configurations where blue, red, and green represent $\phi = 0$ (major cusp), 45° (fold), and 90° (minor cusp) cases. Two source locations are indicated by red and green circles on the source plane (left). The two-image and four-image regimes are shown with solid and transparent markers respectively. The images corresponding to source locations are shown on the right panel.

where the axis ratio for ellipticity e is defined as $q = 1 - e$. We fix the source redshift $z_S = 1$ and the lens redshift $z_L = 0.5$.

For axisymmetric lens models such as the SIS, lens parameters consist of (y, M_{Lz}), where y is the source position and M_{Lz} is the redshifted lens mass defined inside the Einstein radius ξ_0. The SIE lens breaks the axisymmetry and introduces two new parameters. The SIE lens is parameterized by four lens parameters, (M_{Lz}, e, y, ϕ). Here y, ϕ denote the polar coordinates of the source position.

Fig. (4.4) shows the caustics (left panel) and critical curves (right panel) for an SIE lens with $M_{Lz} = 1.05 \times 10^5 M_\odot$ and $e = 0.2$. The outer caustic on the left panel is smooth and

known as a 'pseudo-caustic' or radial caustic while the inner caustic is astroid shaped and has cusps; this is known as a 'true caustic' or a tangential caustic (Keeton et al., 2000). The location of the source on the source plane (left panel) determines the number of images produced on the image plane (right panel). The number of images changes from one to two, similar to the SIS lens model, as the source crosses the radial caustic. Upon crossing the tangential caustic, the number of images increases from two to four. We categorize the source plane into two distinct regimes based on the number of images produced. The region bounded by the radial and tangential caustics is the two-image regime and the region inside the tangential caustics is the four-image regime.

In Fig. (4.4), the transparent and solid markers are placed in the two-image and four-image regime respectively. The images produced by the different source locations are shown on the image plane (right panel). The source locations are further classified into different image configurations: major cusp along ($\phi = 0$) direction shown in blue, fold along ($\phi = 45°$) direction shown in red, and minor cusp along ($\phi = 90°$) direction shown in green. Without loss of generality, we restrict our analysis to the first quadrant of the source plane. We use GLAFIC (Oguri, 2010) for parametric modelling of the SIE lens profile.

4.4.1 Relation between lens and image parameters

In Chapter 3 we discussed the lens parameters, y, M_L in the case of the axisymmetric lens, which are model-dependent and the model-independent image parameters I, Δt_d. We also showed the analytical mappings between the two set of parameters. We can extend the same concepts to the SIE lens. In the case of the SIE lens, we have fixed four model-dependent lens parameters, (M_{Lz}, e, y, ϕ), whereas the model-independent image parameters can be either two for the two-image regime $(I^{(2)}, \Delta t_d^{(2)})$ or six in the four-image regime $(\{I^{(i)}\}, \{\Delta t_d^{(i)}\})$, where $i = (2, 3, 4)$.

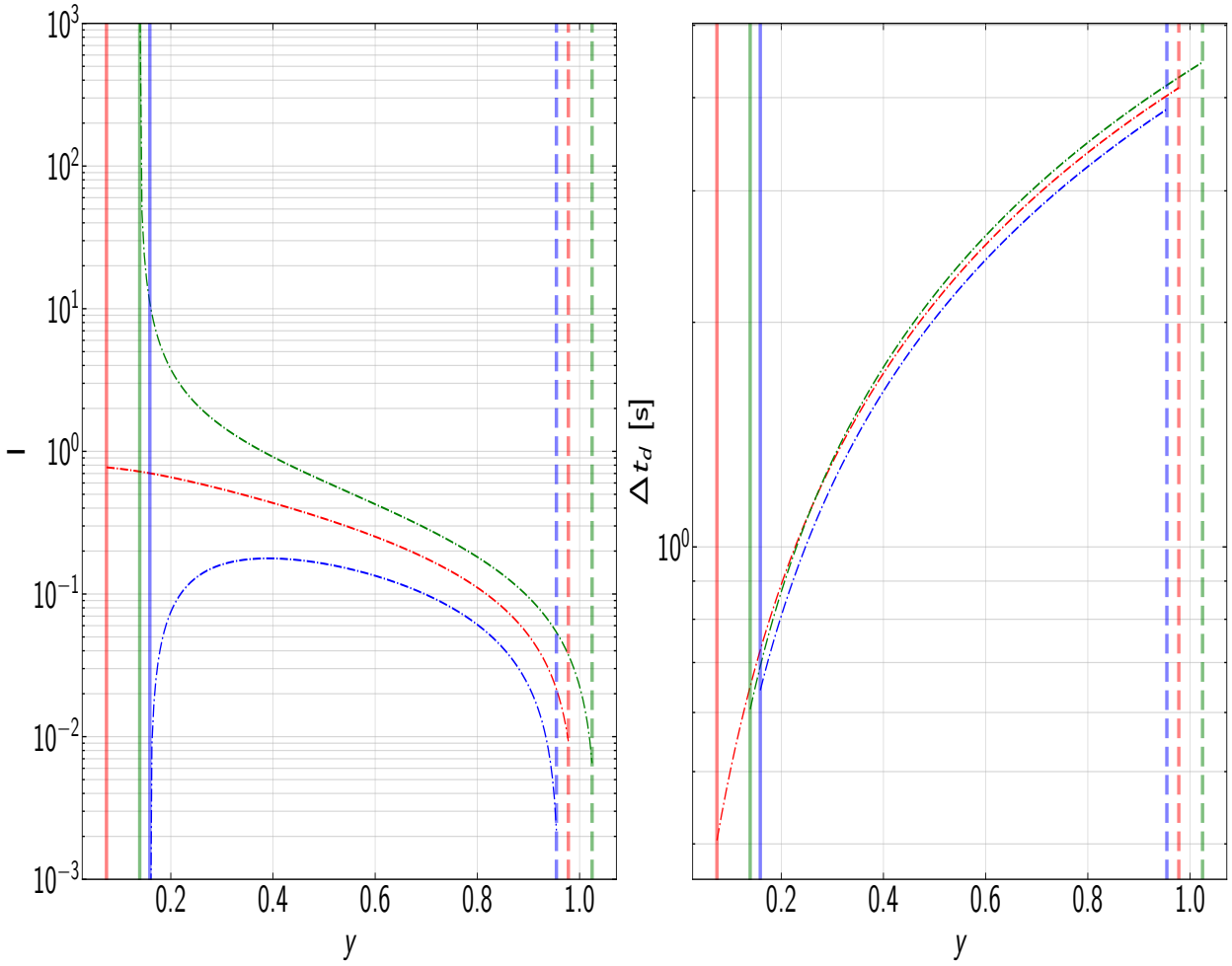


Figure 4.5. Flux ratio $I^{(2)}$ and time delays $\Delta t_d^{(2)}$ in the two-image regime as a function of source position y. Fold $\phi = 45°$, major cusp $\phi = 0$, and minor cusp 90° image configurations are shown in red, blue, and green respectively. The thick solid and dashed horizontal lines are the locations of the tangential and radial caustics on the source plane. The SIE lens used has a fixed redshifted lens mass $M_{Lz} = 1.05 \times 10^5 M_\odot$ and ellipticity $e = 0.2$.

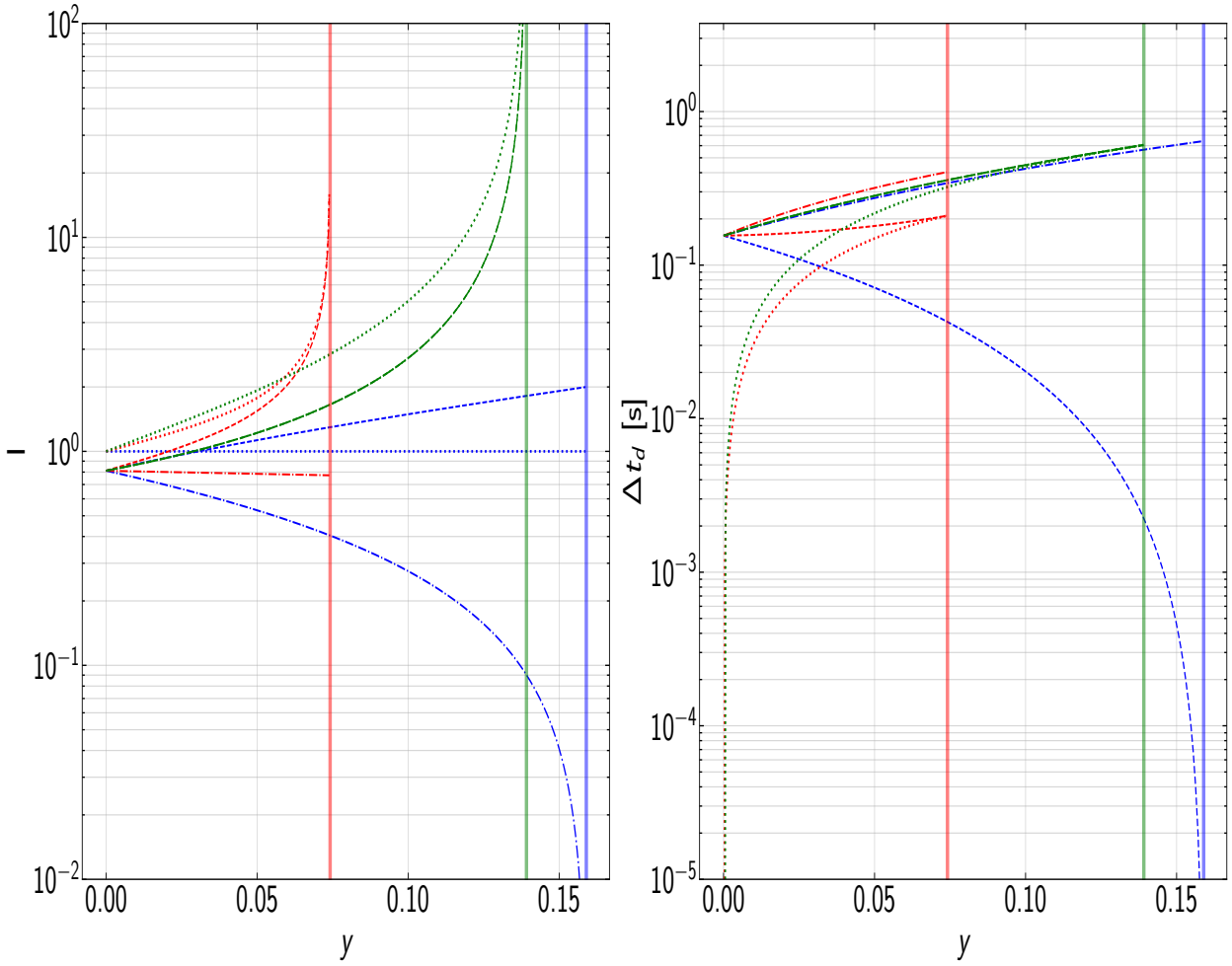


Figure 4.6. Flux ratio $I^{(j)}$ and time delays $\Delta t_d^{(j)}$ in the four-image regime as a function of source position y. Fold $\phi = 45°$, major cusp $\phi = 0$, and minor cusp 90° image configurations are shown in red, blue, and green respectively. Dotted, dashed, and dash-dotted lines indicate the second, third, and fourth arriving image respectively; the first arriving image has zero time delay. The thick solid and dashed horizontal lines are the locations of the tangential and radial caustics on the source plane. The SIE lens used has a fixed redshifted lens mass $M_{Lz} = 1.05 \times 10^5 M_\odot$ and ellipticity $e = 0.2$.

Unlike the case of the axisymmetric lens, for the SIE we sort the images according to the arrival time; we globally shift the time delay such that the first image has zero time delay. Therefore, the time delay of each image $\Delta t_d^{(j)}$ is basically its arrival time (with respect to the first image). We also define the flux ratio $I^{(j)}$ of the image with respect to the first image as $I^{(j)} = |\mu^{(j)}|/|\mu^{(1)}|$. In Fig. (4.5) and Fig. 4.6 we show the behavior of the image parameters I and Δt_d as a function of y for two-image and four-image regimes respectively. The SIE lens has a fixed $M_{Lz} = 1.05 \times 10^5 M_\odot$ and $e = 0.2$ while ϕ takes discrete values for different cusp or fold image configurations. We use solid, dotted, dashed, and dash-dotted line styles for image 1, 2, 3, and 4 respectively. The vertical solid and dashed lines represent the location of the tangential caustics $y_t(\phi)$ and radial caustics $y_r(\phi)$ respectively. These caustic locations are a function of ϕ for fixed M_L and e.

In the left panel of Fig. 4.5, we plot $I^{(2)}$ as a function of y for cusp and fold configurations in the two-image regime. From Fig. (4.4) we note that, for all image configurations, the second image inside the critical curve vanishes in the limit $y \to y_r$ and we transition to the one-image regime. Therefore, we have $I^{(2)} \to 0$ for $y \geq y_r$. In the other limit, $y \to y_t$, we observe contrasting behavior where $I^{(2)} \to 0$ for $\phi = 0$ and $I^{(2)} \to \infty$ for $\phi = 90^\circ$. As y approaches y_t, for $\phi = 0$, image 1 gets closer to the critical curve before image 2, as the major axis is along the x_1-axis. This results in $|\mu^{(1)}| \to \infty$ as $y \to y_t$. However, in the same limit for $\phi = 90^\circ$, the $|\mu^{(2)}| \to \infty$ as image 2 gets near the critical curve before image 1. For the fold case $\phi = 45^\circ$, as $y \to y_t$, both the magnifications remain finite and $I^{(2)} \approx 0.8$. To summarize, the contrasting behavior in the major and minor cusp configurations arises due to the non-axisymmetry of the SIE lens. The right panel of Fig. 4.5 shows time delay $\Delta t_d^{(2)}$ between two images as a function of y. The $\Delta t_d^{(2)}$ decreases as $y \to y_t$ with maximum value $\Delta t_d^{(2)} \approx 4$s for $\phi = 90^\circ$ and minimum value $\Delta t_d^{(2)} \approx 0.4$s for $\phi = 45^\circ$. The dependence of $\Delta t_d^{(2)}$ on y is weak and decreases as $y \to 0$ for all configurations.

In the left panel of Fig. 4.6 we plot the flux ratio $I^{(j)}$ as a function of source position y for different image configurations in the four-image regime. We will begin by discussing the behavior of $I^{(j)}$ with respect to y in the limit of $y \to y_t$:

- For $\phi = 0$, image 1 and 2 are formed off-axis, outside the critical curve, with identical magnifications and positive parity. Image 3 and 4 are formed along the major axis with highest and lowest negative parity magnifications, respectively. From Fig. 4.6, $I^{(2)} = 1$ (blue dotted) as $|\mu^{(1)}| = |\mu^{(2)}|$. The magnifications of image 1 and 3 are comparable at $y \sim y_t$, hence $I^{(3)}$ stays finite but $I^{(4)} \to 0$ due to $|\mu^{(4)}| \ll |\mu^{(1)}|$.

- For $\phi = 90°$, as $y \to y_t$, image 3 and 4 are formed off-axis inside the critical curve with identical magnifications of negative parity. Image 1 and 2 are formed along the minor axis with lowest and highest positive parity magnifications, respectively. For $\phi = 90°$ configuration, as $|\mu^{(3)}| = |\mu^{(4)}|$ for all values of y, we observe $I^{(3)} = I^{(4)}$ (overlapped green dashed and dash-dotted). $I^{(2)}$ (green dotted) also follows the same behavior as $I^{(3)}$ and $I^{(4)}$ but stays above them since $|\mu^{(2)}| > |\mu^{(3)}| = |\mu^{(4)}|$.

- For $\phi = 45°$, image 2 and 3 are formed simultaneously near the critical curve with highest, comparable magnifications. Image 2 and 3 have positive and negative parity respectively. Image 1 has positive parity magnification and image 4 has negative parity with lowest magnification - but still comparable to the image 1 magnification. For $y \sim y_t$, $|\mu^{(2)}| \approx |\mu^{(3)}| \gg |\mu^{(4)}|$, therefore $I^{(2)}$ (red dotted) and $I^{(3)}$ (red dashed) have large values whereas $I^{(4)}$ stays finite due to $|\mu^{(1)}| \sim |\mu^{(4)}|$.

In the limit $y \to 0$, images form the so-called Einstein cross configuration where $|\mu^{(1)}| = |\mu^{(2)}|$ and $|\mu^{(3)}| = |\mu^{(4)}|$. We observe Einstein cross configuration in the left panel of Fig. 4.6 where, at $y = 0$, dotted lines intersect at $I^{(2)} \to 1$ and dashed and dashed-dotted lines intersect at $I^{(3)} = I^{(4)} \approx 0.8$.

In the right panel of Fig. 4.6 we show the $\Delta t_d^{(j)}$ for cusps and fold configurations as a function of y. In the limit of $y \to y_t$:

- For $\phi = 0$, image 1, 2, and 3 are formed simultaneously while image 4 has a finite time delay. As y decreases, image 1 and 2 move away from image 3, sharing magnifications and time delays, therefore $\Delta t_d^{(1)} = \Delta t_d^{(2)} = 0$ (not shown). Meanwhile, image 3 moves inward along the major axis as $y \to 0$ and $\Delta t_d^{(3)}$ (blue dashed) gradually increases as y decreases and at $y = 0$, $\Delta t_d^{(3)} = \Delta t_d^{(4)}$.

- For $\phi = 90°$, image 2, 3, and 4 are formed simultaneously, therefore $\Delta t_d^{(2)} = \Delta t_d^{(3)} = \Delta t_d^{(4)}$. As y decreases, off-axis images, namely image 3 and 4 move away from image 2, sharing magnifications and time delays, therefore $\Delta t_d^{(3)} = \Delta t_d^{(4)}$ (overlapped green dashed and dash-dotted). $\Delta t_d^{(2)}$ (green dotted) gradually decreases as $y \to 0$ since image 2 moves downward along the minor axis with decreasing y. At $y = 0$, Einstein cross configuration is reached, therefore $\Delta t_d^{(1)} = \Delta t_d^{(2)} = 0$ and $\Delta t_d^{(3)} = \Delta t_d^{(4)}$.

- For $\phi = 45°$ image 2 and 3 are formed simultaneously in the third quadrant with identical time delay $\Delta t_d^{(2)} = \Delta t_d^{(3)}$. As y decreases, image 1 and 2 slowly align along the minor axis with similar time delays and image 3 and 4 align along the major axis with similar time delay; thereby forming an Einstein cross. Therefore, in the limit $y \to 0$, we observe $\Delta t_d^{(1)} = \Delta t_d^{(2)}$ (red dotted) $\to 0$ and $\Delta t_d^{(3)}$ (red dashed) $= \Delta t_d^{(4)}$ (red dash-dotted).

4.5 Quasi-geometrical optics approximation (QGO) for non-axisymmetric lens

In Sec. 4.2 we discussed the QGO for the axisymmetric lens, which is valid for lens models such as the SIS. However, it is not valid for non-axisymmetric lenses like the SIE. In this section, we will discuss the QGO for non-axisymmetric lenses.

Following (Takahashi, 2004) we expand $T(\boldsymbol{x},\boldsymbol{y})$ around the image position $\boldsymbol{x}^{(j)}$ as

$$T(\boldsymbol{x},\boldsymbol{y}) = T^{(j)} + \frac{1}{2!}\sum_{a,b} T_{ab}\tilde{x}_a\tilde{x}_b + \frac{1}{3!}\sum_{a,b,c}\partial_a\partial_b\partial_c T^{(j)}\tilde{x}_a\tilde{x}_b\tilde{x}_c + \frac{1}{4!}\sum_{a,b,c,e}\partial_a\partial_b\partial_c\partial_e T^{(j)}\tilde{x}_a\tilde{x}_b\tilde{x}_c\tilde{x}_e + \mathcal{O}(\tilde{x}^5)\,. \tag{4.36}$$

The second term in the expression above is a 2×2 matrix given by $T_{ab} = \partial_a\partial_b T(\boldsymbol{x}_j,\boldsymbol{y})$. The eigenvector matrix A_{ab} satisfies $\sum_{a,b} T_{ab}A_{ac}A_{bd} = \lambda_k\delta_{cd}$, where λ_k is the eigenvalue of T_{ab}. Changing the variable from $\tilde{x}_a$ to $z_a = \sum_b A_{ab}\tilde{x}_b$ and rewriting Eq. (4.36),

$$T(\boldsymbol{x},\boldsymbol{y}) = T^{(j)} + \frac{1}{2!}\sum_a \lambda_a z_a^2 + \sum_{a,b,c} M_{abc}z_a z_b z_c + \sum_{a,b,c,d} N_{abcd}z_a z_b z_c z_d + \mathcal{O}(z^5), \tag{4.37}$$

where

$$M_{abc} = \frac{1}{3!}\sum_{d,e,f}\partial_d\partial_e\partial_f t_d^{(j)}A_{ad}A_{be}A_{cf}\,, \tag{4.38}$$

$$N_{abcd} = \frac{1}{4!}\sum_{e,f,g,h}\partial_e\partial_f\partial_g\partial_h t_d^{(j)}A_{ae}A_{bf}A_{cg}A_{dh}\,. \tag{4.39}$$

The QGO correction term $\Delta^{(j)}$ in Eq. 4.24 for a non-axisymmetric lens is given by (Takahashi, 2004)

$$\Delta^{(j)} = \frac{15}{2}\left(\frac{M_{111}^2}{\lambda_{(1)}|\lambda_{(1)}|^2} + 3\frac{M_{111}M_{122}}{|\lambda_{(1)}|^2\lambda_{(2)}} + 3\frac{M_{112}M_{222}}{\lambda_{(1)}|\lambda_{(2)}|^2} + \frac{M_{222}^2}{\lambda_{(2)}|\lambda_{(2)}|^2}\right) - 3\left(\frac{N_{1111}}{|\lambda_{(1)}|^2} + 2\frac{N_{1122}}{\lambda_{(1)}\lambda_{(2)}} + \frac{N_{2222}}{|\lambda_{(2)}|^2}\right). \tag{4.40}$$

We compare the QGO for axisymmetric and non-axisymmetric lenses in the limit of ellipticity $e\to 0$. The SIE lens transforms into the SIS lens as $e\to 0$, therefore, the second term in Eqs. 4.19 and 4.36 are identical. For an SIE lens with $M_{Lz} = 1.0\times 10^5 M_\odot$, we calculate the eigenvalues λ_1, λ_2 of the matrix T_{ab} in the two-image regime for the fixed source location $y = 0.5$, $\phi = 0$. In the limit $e\to 0$, we have from Eq. 4.20

$$\alpha^{(j)} \to \frac{1}{2}\lambda_1 \tag{4.41}$$

$$\beta^{(j)} \to \frac{1}{2}\lambda_2\,. \tag{4.42}$$

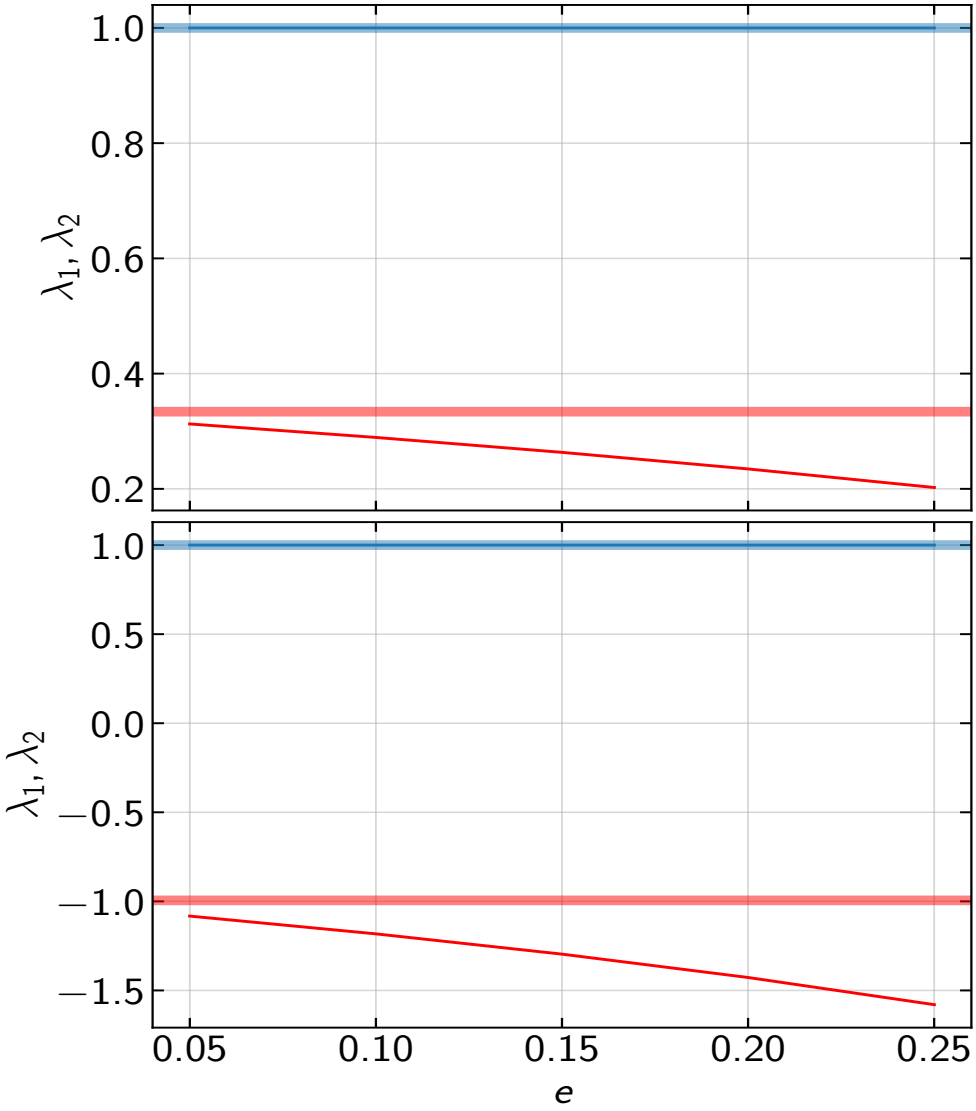


Figure 4.7. Eigenvalues λ_1 (blue), λ_2 (red) are shown as a function of ellipticity e in the two-image regime of the SIE lens. The eigenvalues are calculated for the T_{ab} matrix of an SIE lens with $M_{Lz} = 1.0 \times 10^5 M_\odot$ and fixed source location $y = 0.5$, $\phi = 0$. Image 1 and 2 are shown in top and bottom panel respectively. The thick blue and red lines correspond to values of $\alpha^{(j)}$ and $\beta^{(j)}$ for an SIS lens with the same M_{Lz} and y.

In Fig. 4.7 we show the eigenvalues λ_1 (blue), λ_2 (red) as a function of e. This analysis is performed in the two-image regime of the SIE lens with $M_{Lz} = 1.0 \times 10^5 M_\odot$ and fixed source location $y = 0.5$, $\phi = 0$. The two eigenvalues of image 1 and 2 are shown in the top and bottom panel respectively. The thick blue and red lines correspond to values of $\alpha^{(j)}$ and

$\beta^{(j)}$ for an SIS lens with the same M_{Lz} and y. We observe that, regardless of e, $\alpha^{(j)} = \lambda_1 = 1$ whereas $\lambda_2 \to \beta^{(j)}$ as $e \to 0$. This check is a crucial first step in order to make sure that the code pipelines are working properly for the calculation of $\Delta^{(j)}$ in Eq. 4.40.

4.6 Criteria for geometrical-optics approximation

In this section, we will use the Eq. 4.33 discussed in Sec. 4.2 for the SIE lens. To reiterate, the $M_{Lz(min)}$ provides an approximate value for the minimum redshifted lens mass $M^{(j)}_{Lz(min)}$ required for the geometric-optics approximation to be valid. We define $M^{(j)}_{Lz(min)}$ as

$$M^{(j)}_{Lz(min)} = \frac{|\Delta^{(j)}|}{8\pi f_{min}}\,, \tag{4.43}$$

where $f_{min} = 20$Hz is the lower limit of the sensitivity band of the current ground-based LVK detector network. In the case of the SIE lens, in order to obtain $M^{(j)}_{Lz(min)}$, we have to calculate $\Delta^{(j)}$ using Eq. 4.40 for the jth image.

4.6.1 Two-image regime

In Fig. 4.8 we show $M_{Lz(min)}$ as a function of y for major $\phi = 0$ (blue) and minor $\phi = 90°$ (green) cusp and fold $\phi = 45°$ (red) image configurations in the two-image regime of the SIE lens. Image 1, with Morse index $n^{(j)} = 0$, and image 2, with $n^{(j)} = 1/2$, are shown with solid and dotted lines respectively. $M^{(2)}_{Lz(min)} \to \infty$ as $y \to y_r$ for all image configurations; as image 2 vanishes in the one-image regime $y > y_r$. However, image 1 continues to exist for $y > y_r$, therefore $M^{(1)}_{Lz(min)}$ remains finite as $y \to y_r$. In the other limit, for $y \to y_t$, we observe contrasting behavior for major and minor cusps. For $\phi = 0$ as $|\mu^{(1)}| \to \infty$, $M^{(1)}_{Lz(min)} \to \infty$, whereas for $\phi = 90°$, $|\mu^{(2)}| \to \infty$, therefore $M^{(2)}_{Lz(min)} \to \infty$. The $M^{(2)}_{Lz(min)} = 3.6 \times 10^3 M_\odot$ and $M^{(1)}_{Lz(min)} = 8 \times 10^2 M_\odot$ stay finite for $\phi = 0$ and $\phi = 90°$, as the magnifications of the respective images stay finite. For $\phi = 45°$, since magnification of both image 1 and 2 stays finite, $M^{(1)}_{Lz(min)} = 8.2 \times 10^3$, $M^{(2)}_{Lz(min)} = 2.6 \times 10^4$ stays finite too. We conclude

that in the two-image regime, $M_{Lz} \geq 10^5 M_\odot$ should be an adequate lens mass to satisfy the geometrical-optics approximation for both images in the range of $y \sim 0.2$ to 0.9 for $\phi = 0$, 45°, and 90°.

Fig. 4.9 shows the $M_{Lz(min)}$ in the two-image regime as a function of the polar angle ϕ for fixed $e = 0.2$. The blue and red color corresponds to fixed source distances $y = 0.3$ and $y = 0.5$ respectively. The $M_{Lz}(min)$ varies approximately within three orders of magnitude for a vast range of ϕ. The dependence of $M_{Lz(min)}$ for both of the images on ϕ is uniform with no significant feature, unlike the case of y seen in Fig. 4.8.

4.6.2 Four-image regime

In Figs. 4.10, 4.11, and 4.12, we show $M_{Lz(min)}$ as a function of y in the four-image regime for $\phi = 0$, 45° and 90° respectively. The solid and dotted lines correspond to image 1 and 2 respectively, and is consistent with the line styles used for the two-image regime. The dashed and dash-dotted lines are used for image 3 and 4 respectively. In the limit $y \to y_t$ we notice continuous behavior of the images across the different image regimes.

- In Fig. 4.10, the $M^{(1,2,3)}_{Lz(min)} \to \infty$ (solid, dotted overlapped by solid, dashed) as $y \to y_t$. This is due to the fact that $|\mu^{(3,4)}| \sim 0$ at $y \sim y_t$, while $|\mu^{(1)}| \to \infty$ as $y \to y_t$. Image 4 in the four-image regime masquerades as image 2 in the two-image regime, since images are sorted according to increasing time delay. Therefore, $M^{(4)}_{Lz(min)}$ (dash-dotted) in the four-image regime and $M^{(2)}_{Lz(min)}$ (dotted in Fig. 4.8) in the two-image regime are continuous across $y = y_t$.

- In Fig. 4.11, $M^{(2,3)}_{Lz(min)} \to \infty$ (dotted, dashed) at $y \sim y_t$ because image 2 and 3 are formed simultaneously near critical curves with high magnifications. $M^{(1)}_{Lz(min)}$ (solid) is continuous across the y_t. Image 4 in the four-image regime masquerades as image 2 in the two-image regime, therefore $M^{(4)}_{Lz(min)}$ (dash-dotted) and $M^{(2)}_{Lz(min)}$ (dotted in Fig. 4.8) are continuous across $y = y_t$.

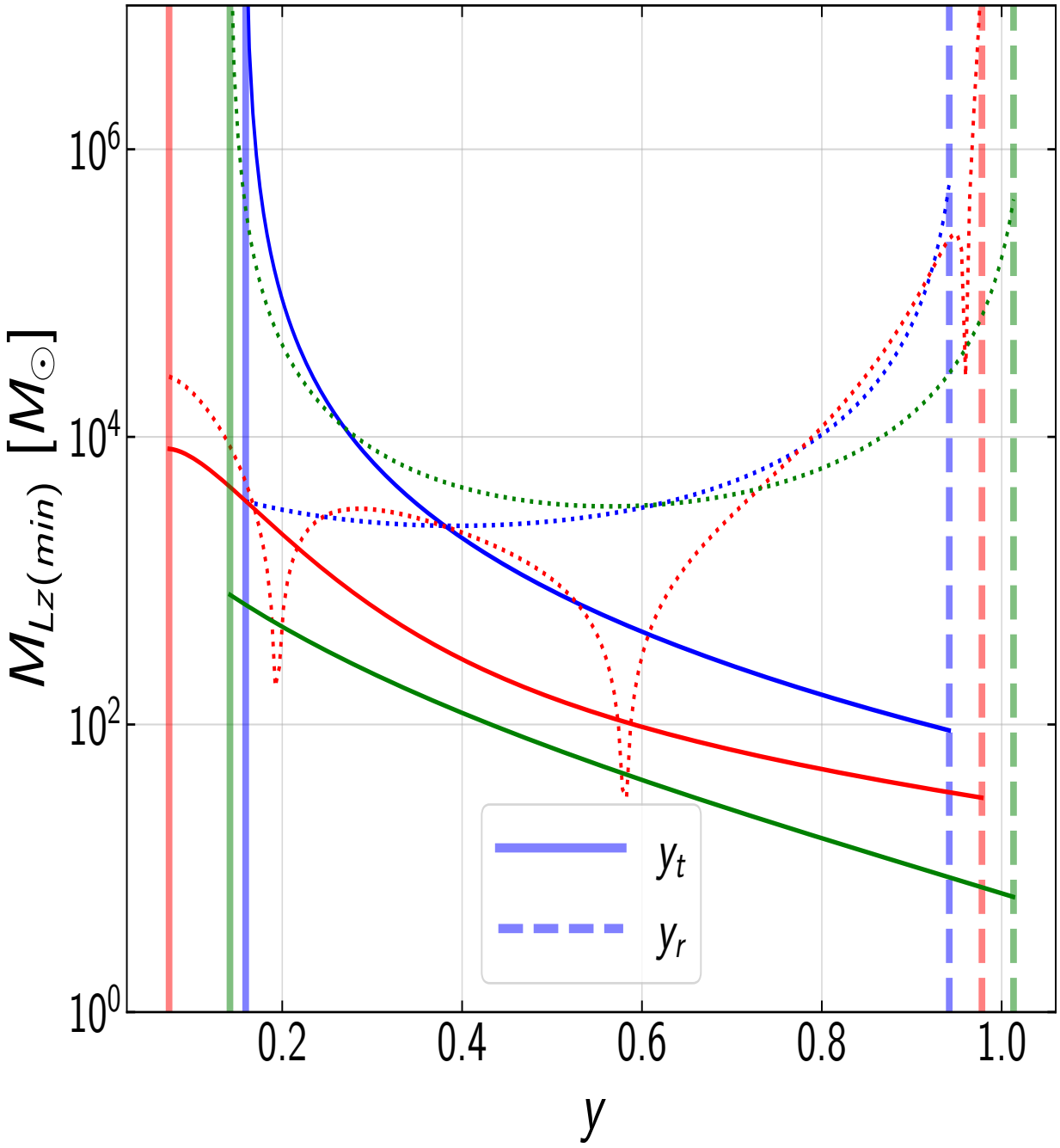


Figure 4.8. Minimum redshifted lens mass $M_{Lz(min)}$ in the two-image regime for fixed $e = 0.2$ as a function of source distance y. Blue and green correspond to major $\phi = 0$ and minor $\phi = 90°$ cusp configurations respectively, while red corresponds to fold $\phi = 45°$. The solid and dotted lines correspond to image 1 and 2 respectively. The vertical solid lines are the location of the tangential caustics $y_t(\phi)$ and dashed lines are the location of the radial caustics $y_r(\phi)$.

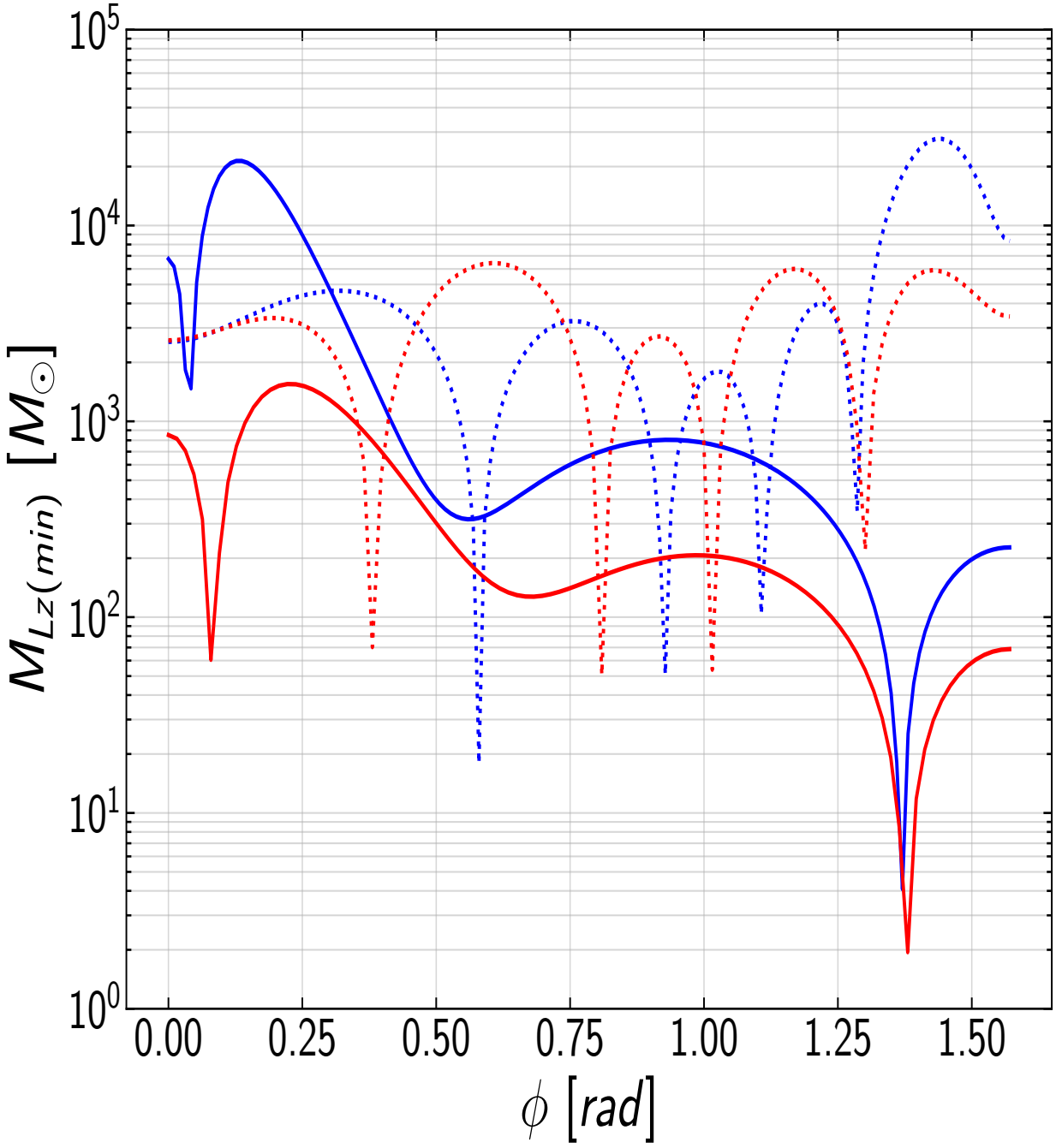


Figure 4.9. Minimum redshifted lens mass $M_{Lz(min)}$ in the two-image regime as a function of polar angle ϕ for fixed $e = 0.2$, source distances $y = 0.3$ (blue) and $y = 0.5$ (red). The solid and dotted lines correspond to image 1 and 2 respectively.

- In Fig. 4.12, we observe that $M^{(3,4)}_{Lz(min)} \to \infty$ (dashed, dash-dotted overlapped by dashed) as $y \to y_t$; this is because the $|\mu^{(3,4)}| \to 0$ in the limit $y \to y_t$ and cease to exist for $y > y_t$. $M^{(2)}_{Lz(min)} \to \infty$ (dotted) on both sides of $y = y_t$ because $|\mu^{(2)}| \to \infty$ near $y \sim y_t$. $M^{(1)}_{Lz(min)}$ (solid) is continuous across the $y = y_t$.

As discussed earlier, in the limit $y \to 0$, images are oriented in the Einstein cross configurations . In this configuration, magnifications and time delays are identical for image 1, 2 and for image 3, 4. Therefore, in Figs. 4.10, 4.11, and 4.12 we notice that, for $y \to 0$, $M^{(1)}_{Lz(min)}$ (solid) $= M^{(2)}_{Lz(min)}$ (dotted) while $M^{(3)}_{Lz(min)}$ (dashed) $= M^{(4)}_{Lz(min)}$ (dash-dotted). In the four-image regime, the $M_{Lz(min)}$ value will depend on the ϕ when all four images simultaneously satisfy the geometrical-optics approximation. If the lens mass is above $M_{Lz} \gtrsim 2.5 \times 10^5 M_\odot$, all four images will satisfy the geometrical-optics approximation in the range of $y \sim 0$ to 0.13 for $\phi = 0$ and $y \sim 0$ to 0.12 for $\phi = 90°$. For cusp cases, it is difficult to find M_{Lz} such that the four images satisfy geometrical-optics approximation simultaneously for the whole range of y. However, for $\phi = 45°$, $M_{Lz} \gtrsim 2.5 \times 10^5 M_\odot$ can lead to valid geometrical-optics approximation for $y \sim 0$ to y_t.

4.7 Waveform mismatch: unlensed and lensed templates

The current GW search pipelines employ a match-filtering method and do not account for lensed GW in the template bank. However, the detection of lensed GW is highly probable with the increasing sensitivity of the current LVK network in the upcoming O4 run along with the advent of 3G (CE and ET) and space-based detectors (LISA). Therefore, the inclusion of lensed GW templates will be indispensable for the future analysis of the lensing signature in GW events. In this section, we perform match-filtering analysis of the GW event lensed by four images. The analysis is performed with the unlensed template and two-image lensed templates.

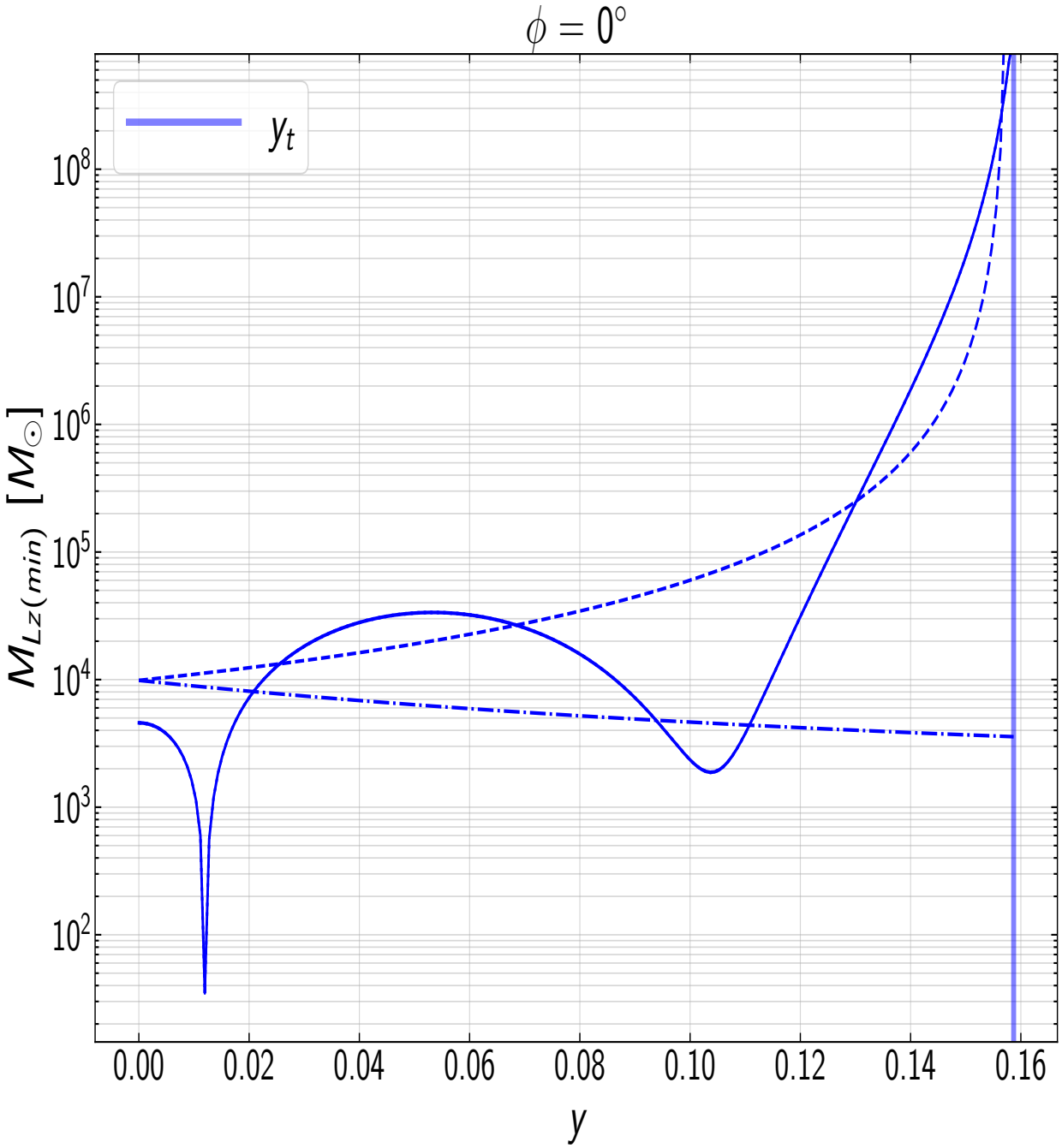


Figure 4.10. Minimum redshifted lens mass $M_{Lz(min)}$ in the four-image regime for fixed $e = 0.2$ and $\phi = 0$ as a function of source distance y. The solid, dotted, dashed, and dash-dotted lines correspond to image 1, 2, 3, and 4 respectively. The vertical solid line is the location of tangential caustic $y_t(\phi)$.

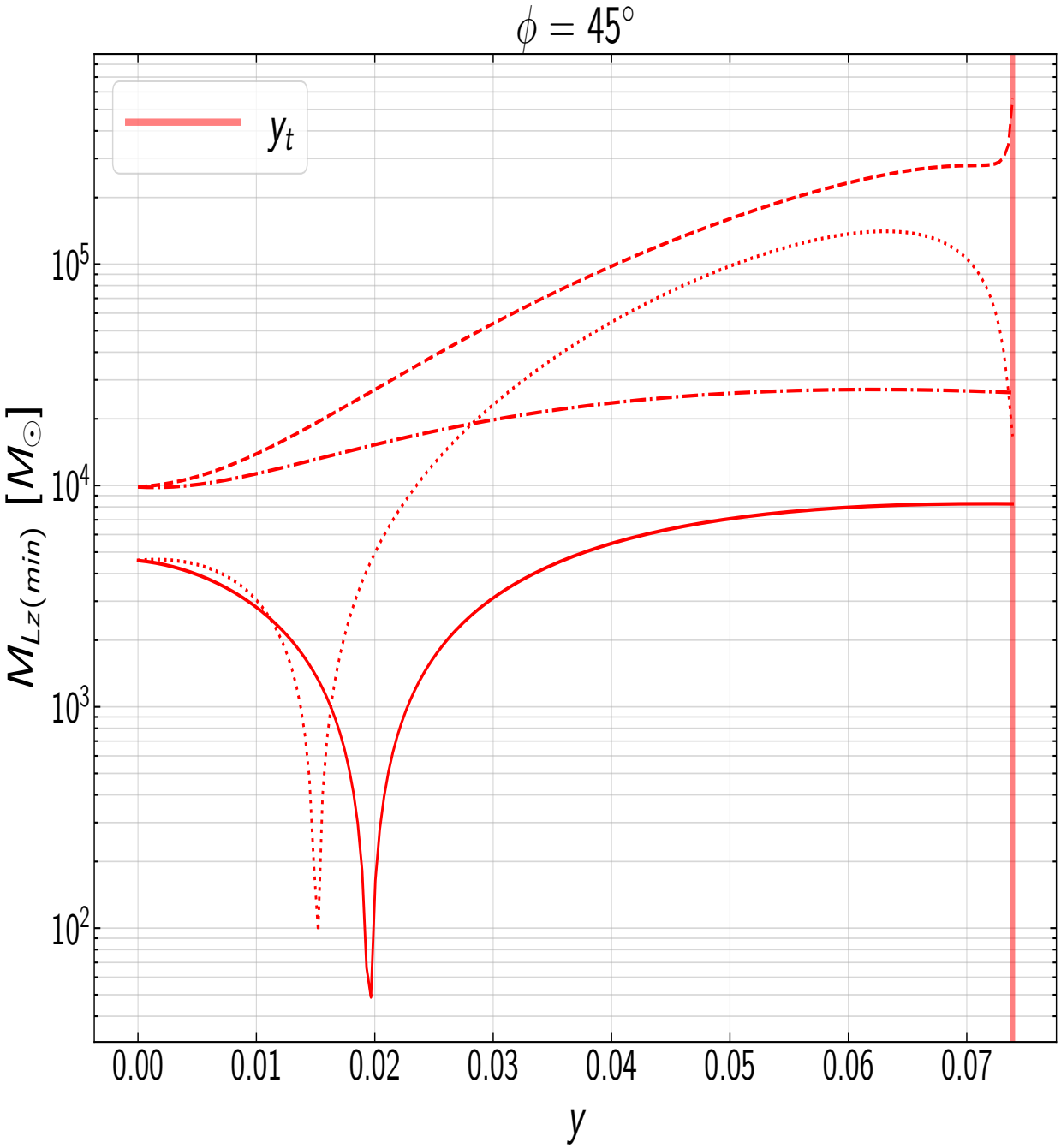


Figure 4.11. Same as Fig. 4.10 but for $\phi = 45°$.

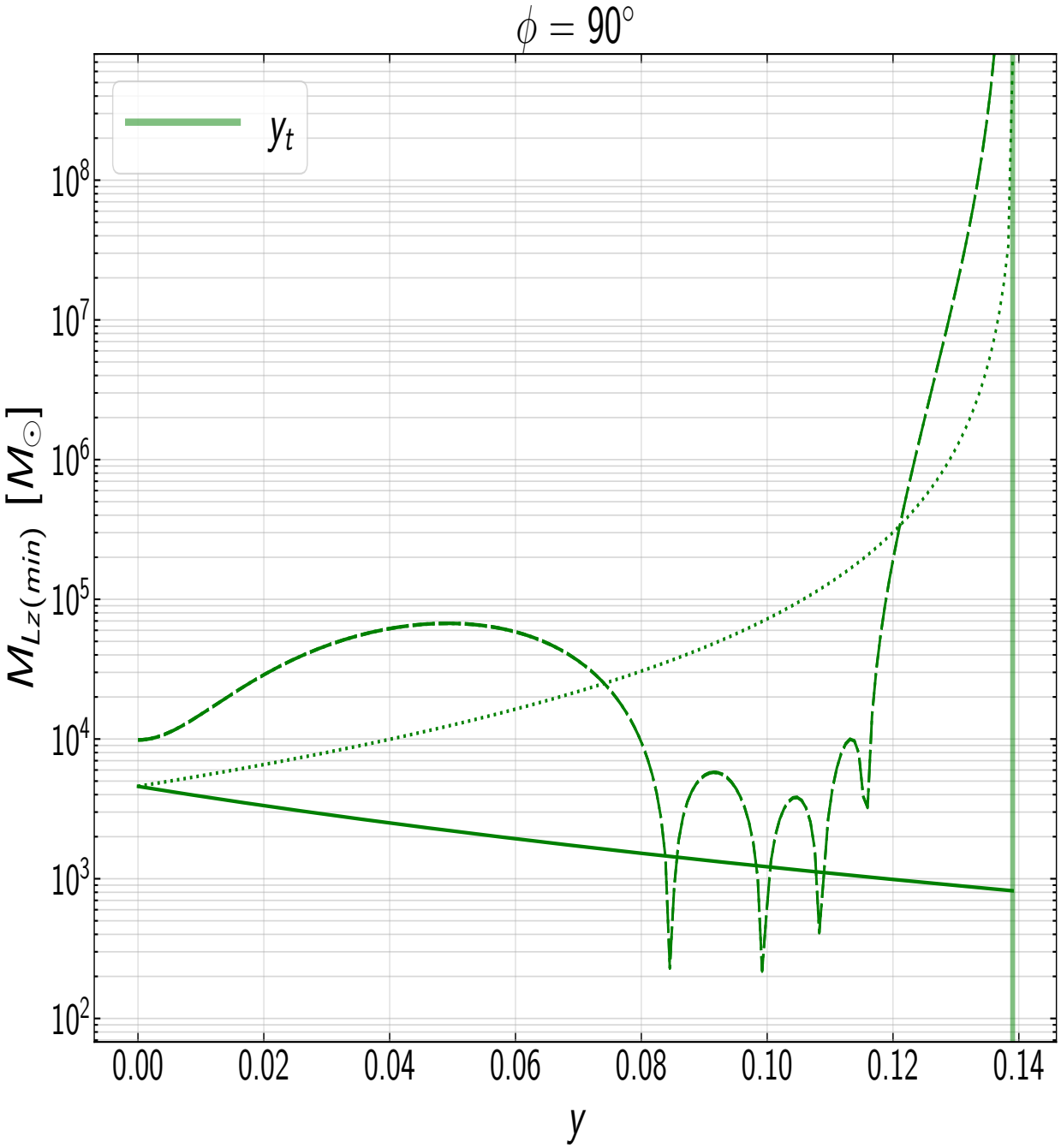


Figure 4.12. Same as Fig. 4.10 but for $\phi = 90°$.

The mismatch ϵ distinguishes two waveforms $h_1(f)$ and $h_2(f)$ and is given by (Cutler and Flanagan, 1994)

$$\epsilon(h_1, h_2) \equiv 1 - \max_{t_c, \phi_c} \mathcal{O}(h_1, h_2)\,, \tag{4.44}$$

where

$$\mathcal{O}(h_1, h_2) = \frac{\langle h_1 | h_2 \rangle}{\sqrt{\langle h_1 | h_1 \rangle \langle h_2 | h_2 \rangle}}\,. \tag{4.45}$$

The noise-weighted inner product $\langle h_1 | h_2 \rangle$ is given by

$$\langle h_1 | h_2 \rangle = 4\,\mathrm{Re} \int_{f_{\rm low}}^{f_{\rm cut}} df \frac{h_1^*(f) h_2(f)}{S_n(f)}\,, \tag{4.46}$$

The $S_n(f)$ is the noise power spectral density (PSD). In (Ali et al., 2022), mismatch analysis is performed between lensed and unlensed GW for the two-image case of SIS and PM lens using model-dependent and model-independent lens parameters. Therefore, we do not account for the two-image regime of the SIE lens and focus solely on the four-image regime. For the mismatch calculation, we employ unlensed template and two-image lensed templates for GW lensed into four images. The two-image lensed templates are further categorized into two templates: one having images with the same Morse index and one having images with different Morse index. For the template with different Morse index, each image is the brightest minima and saddle image.

In the case of the four-image regime, image 1 and 2 are minima with $n^{(j)} = 0$ and image 3 and 4 are saddles with $n^{(j)} = 1/2$. We fix the lens parameter $e = 0.2$ and $\phi = 45^\circ$ and choose $M_{Lz} = 1.5 \times 10^6 M_\odot$ such that all the images satisfy geometrical-optics approximation for $y \sim 0$ to y_t. We generate unlensed GW with BBH source of symmetric mass ratio $\eta = 0.25$ and chirp mass $\mathcal{M} = 15 M_\odot$ corresponding to total mass $M = 35 M_\odot$. The time t_{band} spent by such BBH in the sensitivity band is long enough such that $t_{band} > \max\{\Delta t_d^{(j)}\}$ where $j = (1, 2, 3, 4)$.

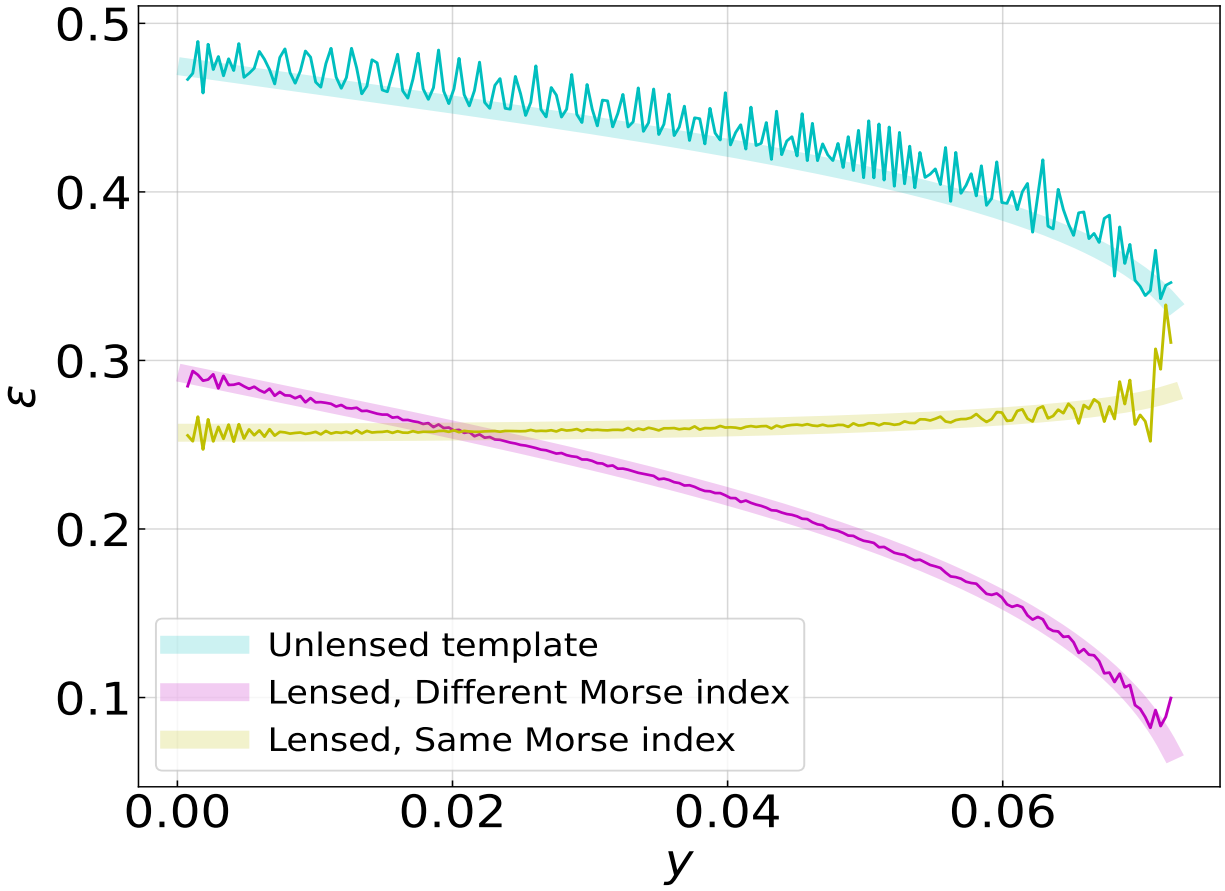


Figure 4.13. Mismatch ϵ as a function of source distance y for fold $\phi = 45°$. The thin solid lines are the ϵ calculated numerically while the thick solid lines are the analytical approximation in the limit time delay $\Delta t_d \to \infty$. The cyan color is the unlensed GW template whereas magenta and yellow are two-image lensed GW templates with different and same Morse index respectively.

We follow the prescription provided in our previous work (Ali et al., 2022) to obtain the analytic expressions for ϵ in the extreme geometrical-optics limit $\Delta t_d \to \infty$. The signal-to-noise ratio (SNR) ρ for the unlensed GW h_u is $\langle h_u|h_u\rangle = \rho^2$ and $\langle h_s|h_s\rangle = \rho^2 \sum_i |\mu^{(i)}|$. In the limit $\Delta t_d \to \infty$,

- for the unlensed template h_u,

$$\langle h_u|h_s\rangle = \rho^2 \max\{|\mu^{(i)}|^{1/2}\}\,, \tag{4.47}$$

Table 4.1. The rows show the inner product Eq. (5.23) and mismatch ϵ Eq. (5.27) between different source h_s and template h_t waveforms in the extreme geometrical-optics limit $\Delta t_d \to \infty$ are shown. Here $\mu^{(i)}$ is the magnification of ith image, $I = |\mu_i|/|\mu_1|$, and $I^{(t)} = \sum_i I^{(i)}$. The columns are different templates chosen; unlensed, two image lensed template with different Morse index, and two image lensed template with same Morse index.

	Unlensed	Lensed, different Morse index	Lensed, same Morse index
$\langle h_t \vert h_t \rangle$	ρ^2	$\rho^2(\vert\mu^{(2)}\vert + \vert\mu^{(3)}\vert)$	$\rho^2(\vert\mu^{(1)}\vert + \vert\mu^{(2)}\vert)$
$\langle h_s \vert h_t \rangle$	$\rho^2 \max\{\vert\mu^{(i)}\vert^{1/2}\}$	$\rho^2(\vert\mu^{(2)}\vert + \vert\mu^{(3)}\vert)$	$\rho^2(\vert\mu^{(1)}\vert + \vert\mu^{(2)}\vert)$
$\epsilon(h_s, h_t)$	$1 - \left(\frac{\max\{I^{(i)}\}}{I^{(t)}}\right)^{1/2}$	$1 - \left(\frac{I^{(2)}+I^{(3)}}{I^{(t)}}\right)^{1/2}$	$1 - \left(\frac{1+I^{(2)}}{I^{(t)}}\right)^{1/2}$

- for the two image lensed template with same Morse index $h_t^{'}$

$$\langle h_t^{'} | h_t^{'} \rangle = \langle h_t^{'} | h_s \rangle = \rho^2(|\mu^{(1)}| + |\mu^{(2)}|)\,, \tag{4.48}$$

- and for the two image lensed template with different Morse index $h_t^{''}$,

$$\langle h_t^{''} | h_t^{''} \rangle = \langle h_t^{''} | h_s \rangle = \rho^2(|\mu^{(2)}| + |\mu^{(4)}|)\,, \tag{4.49}$$

The overlap $\mathcal{O}$ between a four image lensed GW source and different types of templates can be calculated as

$$\mathcal{O}(h_s, h_u) = \frac{\max|\mu^{(i)}|^{1/2}}{\sqrt{\sum_i |\mu^{(i)}|}} = \left(\frac{\max\{I^{(i)}\}}{I^{(t)}}\right)^{1/2}, \tag{4.50}$$

$$\mathcal{O}(h_s, h_t^{'}) = \frac{|\mu^{(1)}| + |\mu^{(2)}|}{\sqrt{(|\mu^{(1)}| + |\mu^{(2)}|)\sum_i |\mu^{(i)}|}} = \left(\frac{1 + I^{(2)}}{I^{(t)}}\right)^{1/2}, \tag{4.51}$$

$$\mathcal{O}(h_s, h_t^{''}) = \frac{|\mu^{(2)}| + |\mu^{(3)}|}{\sqrt{(|\mu^{(2)}| + |\mu^{(3)}|)\sum_i |\mu^{(i)}|}} = \left(\frac{I^{(2)} + I^{(3)}}{I^{(t)}}\right)^{1/2}, \tag{4.52}$$

where $I^{(t)} \equiv \sum_i I^{(i)}$ is the total flux ratio. The corresponding ϵ is given by $\epsilon = 1 - \mathcal{O}$. The above results are summarized in Table 4.1.

In Fig. 4.13 we show the mismatch ϵ between four image lensed GW source and three different types of templates for the source and lens parameters previously mentioned. The mismatch with the unlensed template ϵ^U is shown in cyan, while yellow and magenta are

used for the mismatches compared to the two image lensed template with the same ϵ' and different Morse index ϵ'' respectively. The thin solid lines are the mismatches calculated numerically using the PYCBC.FILTER package (Nitz et al., 2022). The thick solid cyan, yellow, and magenta lines in Fig. 4.13 represent Eqs. 4.50, 4.51, and 4.52 respectively.

For $y \sim 0$, the images are oriented in an Einstein cross configuration where $|\mu^{(1)}|\,(= |\mu^{(2)}|) > |\mu^{(3)}|\,(= |\mu^{(4)}|)$, therefore template with same Morse index consisting of image 1 and image 2 have lower mismatch than the template with different Morse index consisting of image 2 and image 3. The ϵ'' intersects ϵ' at $y \sim 0.02$, and for $y > 0.02$, $\epsilon'' < \epsilon'$. ϵ'' follows a downward trend as $y \to y_t$ as the images in the templates always consist of the two brightest images out of the four. In contrary, the ϵ' increases with y because one of the images in the template has the lowest magnification. The ϵ^U is highest for all values of y, compared to both ϵ' and ϵ''. For $y \sim 0$, the ϵ^U is greater than ϵ'' by a factor of 1.6, and as $y \sim y_t$, $\epsilon^U/\epsilon'' \approx 4.9$. In the limit $\Delta t_d \to \infty$, the analytical equations perfectly describe the numerically calculated ϵ. For $y \sim 0$, we have $I^{(2)} = 1$ and $I^{(3)} = I^{(4)} < I^{(2)}$ and using Eqs. 4.52 and 4.51, we can clearly see that for $\mathcal{O}(h_s, h_t') > \mathcal{O}(h_s, h_t'') \implies \epsilon'' > \epsilon'$.

4.8 Summary and discussion

Gravitational lensing of GWs is quantified by using the amplification factor $F(w)$. In the wave-optics regime $w \ll 1$, the diffraction effect is significant and the $F(w) \to 1$. In the other limit $w \gg 1$, the geometrical-optics approximation is valid and the $F(w)$ is determined by the stationary points, which are image positions on the lens plane. The $F(w)$ can be expressed as the summation over these images, each with its own magnification $\mu^{(j)}$, time delay $t_d^{(j)}$, and Morse index $n^{(j)}$ as given by Eq. 2.49. In the limit $w \sim 1$, the quasi-geometrical optics approximation (QGO) can be employed (Takahashi, 2004), which includes correction, originating from the order $1/w$, to the geometrical-optics approximation. The correction to the geometrical-optics approximation can be obtained analytically by a series expansion of

Eq. 4.17 for axisymmetric and non-axisymmetric lens models. The QGO correction term can be utilized to obtain the lens mass $M_{Lz(min)}$ in Eq. 4.33 above which the geometrical-optics approximation is valid. We employ this criteria in different regimes of the source plane and investigate the behavior of the $M_{Lz(min)}$ especially near the non-analytical points such as cusps and fold of the Singular Isothermal Ellipsoid (SIE) lens.

The axisymmetric lens, Singular Isothermal Sphere (SIS), produces two images for source position $y < 1$. We begin by validating the QGO for the SIS lens in this two-image regime. In Fig. 4.3 we showed the relative uncertainty between the amplification factor calculated using the diffraction integral Eq. 3.3 and in the geometrical-optics limit Eq. 4.27. We calculated the QGO correction term for different y and found that it approximately indicates the limit $w \sim 1$ where QGO is valid. The SIE is parameterized by four lens parameters: polar coordinates of source location y, ϕ, redshifted lens mass M_{Lz} inside Einstein radius, and the ellipticity e of the lens. In Fig. 4.4 we show the two-image and four-image regime bounded by the caustics (left panel) of the SIE lens and the orientation of the images (right panel) for three different image configurations: major cusp $\phi = 0$, fold $\phi = 45°$, and minor cusp $\phi = 90°$. In Figs. 4.5 and 4.6, the behavior of the image parameters, flux ratio I and time delay Δt_d, are shown as a function of y. We order the images according to their arrival time, and in Fig. 4.5, we observe the contrasting behavior in I between the two cusp cases as $y \to y_t$, where y_t is the location of tangential caustics for different ϕ. By comparing Figs. 4.5 and 4.6 we see that I and Δt_d maintain continuous behavior across the y_t.

We investigate the $M_{Lz(min)}$ in the two-image and four-image regime of the SIE lens. In Fig. 4.8 we analyze $M_{Lz(min)}$ as a function of y in the two-image regime for major cusp, fold, and minor cusp. We notice that, for j^{th} image, $M^{(j)}_{Lz(min)} \to \infty$ either when $\mu^{(j)} \to 0$, for the limit $y \to y_r$ (radial caustics), or $\mu^{(j)} \to \infty$, for the limit $y \to y_t$. For $\phi = 0$, 45°, and 90°, both of the images satisfy the geometrical-optics approximation simultaneously if the lens mass M_{Lz} is greater than $10^5 M_\odot$ in the range of $y \sim 0.2$ to 0.9. In the four-image

regime, at $y \sim y_t$, $\mu^{(j)} \to 0$ for images that vanish outside in the two-image regime and the rest of the images show continuous behavior across y_t. At $y \sim 0$, the images are oriented in an Einstein cross configuration where images 1 (image 3) share I and Δt_d with image 2 (image 4). Therefore, for all ϕ, we have $M^{(1)}_{Lz(min)} = M^{(2)}_{Lz(min)}$ and $M^{(3)}_{Lz(min)} = M^{(4)}_{Lz(min)}$. The $M_{Lz(min)}$ is of similar order ($\sim 10^5 M_\odot$) in both two-image and four-image regimes; however the range of y varies with ϕ as seen in Figs. 4.10, 4.11, and 4.12.

In Fig. 4.13, we present a match-filtering analysis between the four image lensed source and several templates. We calculate the mismatches with an unlensed ϵ^U GW template, a two-image lensed GW template with same Morse index ϵ', and a two-image lensed GW template with different Morse index ϵ''. Currently, the match-filtering pipeline for lensed GWs searches does not include lensed templates. However, from Fig. 4.13 we can see that the two-image lensed template with different Morse index can reduce the ϵ by a factor of $1.6 - 4.9$ depending on the value of y. We also observe that for $y \sim 0$, $\epsilon' < \epsilon''$, however for most of the values of y, $\epsilon' > \epsilon''$. This highlights the importance of the Morse index in the choice of two-image lensed templates. We also derive analytical expressions for ϵ, summarized in Table. 4.1, in the extreme geometrical-optics limit where $\Delta t_d \to \infty$. In Fig. 4.13 we can see that the analytical expressions perfectly describe the numerically calculated mismatches.

CHAPTER 5

QUANTIFYING PRECESSIONAL EFFECTS ON BINARY BLACK-HOLE SYSTEMS BY INTRODUCING A GENERIC SPINNING WAVEFORM TEMPLATE

This Chapter discusses the work done in (Stoikos et al., 2023) with my collaborators, Evangelos Stoikos (lead author), Nathan Steinle, Michael Kesden, and Lindsay King. We are going to submit the manuscript to Physical Review D in April 2023. The majority of the work, that includes generating figures, deriving equations and content writing, is done by ES. As a co-author, my work includes setting up the precession pipeline, and verifying the codes and the results generated by the pipeline.

5.1 Introduction

In the most recent O3b run, the LVK collaboration used four pipelines to search for GW signals. Out of the four, three of them, GstLAL (Hanna et al., 2020; Messick et al., 2017; Cannon et al., 2021), Multi-band template analysis (Adams et al., 2016; Aubin et al., 2021) and PyCBC (Usman et al., 2016; Nitz et al., 2017) perform their searches by using already constructed CBC waveform templates, while the remaining method, cWB (Klimenko and Mitselmakher, 2004; Klimenko et al., 2016) is doing so by making minimal assumption about the behavior of the sources. The CBC search pipelines utilize match-filter analysis (Wainstein and Zubakov, 1970; Owen and Sathyaprakash, 1999) to compare the signal received on the ground-based detectors with a bank of carefully constructed templates as a means to extract the information embedded in the signal about the source's properties. These templates strongly depend on the individual masses of the BHs as well as their spin components in order to accurately model the GW signals. Even though the masses of the BHs and some aligned combinations of their spins are well constrained, the short duration of the signals

as well as the subtle effects of the spins on the templates make them difficult to measure individually.

Current observations of BBH mergers confirm that half of the individual spin magnitudes lie below $\chi_i \approx 0.26$ are typically misaligned with the orbital angular momentum $\mathbf{L}$ (The LIGO Scientific Collaboration et al., 2021). In turn, $\mathbf{L}$ also precesses and nutates around the total angular momentum $\mathbf{J}$, whose direction remains approximately constant during the inspiral part of the CBC (Apostolatos et al., 1994; Zhao et al., 2017). This spin and orbital precession plays a key role in BBH dynamics (Apostolatos et al., 1994) by imprinting modulations to the amplitude and the phase of the GW signal (Cutler and Flanagan, 1994).

These modulations enter the GW templates at the 1.5PN and 2PN order as spin-orbit and spin-spin coupling effects respectively (Thorne, 1980; Cutler and Flanagan, 1994; Apostolatos et al., 1994; Kidder, 1995). State-of-the-art waveform model families used in GW data analysis, such as IMRPhenomXPHM (Pratten et al., 2021), SEOBNRv4PHM (Ossokine et al., 2020), and NRSur7dq4 (Varma et al., 2019), model these effects accurately. However, it is fairly challenging to analytically quantify how the two spin vectors generically affect the GW template. As current detectors cannot unambiguously constrain both spins, see e.g., (Hoy et al., 2022), the two spins are typically represented by combinations of the BBH masses and spins that approximate the effects. Two main such parameters are the effective inspiral spin $\chi_{\rm eff}$ (Damour, 2001) which is conserved through the PN inspiral (Racine, 2008) of the coalescence and influences its length, and the effective precession spin $\chi_{\rm p}$ (Hannam et al., 2014; Schmidt et al., 2015; Gerosa et al., 2021) which approximates the in-plane spin component responsible for spin precession.

In our previous analysis (Gangardt et al., 2021), we developed a different set of phenomenological parameters that encompass the essential features of spin precession and nutation. These five parameters provide a geometric framework to characterize the various configurations in which the evolution of $\mathbf{L}$ about $\mathbf{J}$ impact the BBH dynamics. In our current analysis, we take one step forward and introduce the regular precessing parameters,

the precession amplitude $\langle\theta_{LJ}\rangle$ and precession frequency $\langle\Omega_{LJ}\rangle$, as functions of the GW frequency into a new 'toy' model GW template. We then show how these parameters imprint a distinct signature on the template's various amplitude and phase terms and finally we try to extract the signal-to-noise ratios needed to detect precession in the next LVK run as well as in future 3G detectors.

5.2 Basic formalisms and notations

In this section, we briefly review the inspiral waveform in the Newtonian, quadrupole-moment approximation (Apostolatos et al., 1994; Finn and Chernoff, 1993; Takahashi and Nakamura, 2003) and our taxonomy of spin precession (Gangardt et al., 2021). We will incorporate the latter into the former in the following section to construct a toy model of a precessing frequency-domain gravitational waveform.

5.2.1 Gravitational waveform in the quadrupole-moment approximation

In the frequency domain, a GW signal observed by a ground-based detector can be expressed as (Apostolatos et al., 1994; Cutler and Flanagan, 1994; Takahashi and Nakamura, 2003)

$$\tilde{h}(f) = F_+\tilde{h}_+(f) + F_\times\tilde{h}_\times(f)\,, \tag{5.1}$$

where F_+ and $F_\times$ are the detector beam-pattern coefficients and h_+ and $h_\times$ are the projections of the GW field onto the two principal polarization states. For an L-shaped GW detector, these beam-pattern coefficients can be expressed as (Forward, 1978; Apostolatos et al., 1994)

$$\begin{aligned} F_+(\theta_S,\Phi_S,\psi) &= \frac{1}{2}(1+\cos^2\theta_S^2)\cos 2\Phi_S\cos 2\psi - \cos\theta_S\sin 2\Phi_S\sin 2\psi\,, \\ &= C\cos(2\psi+\alpha)\,, \qquad (5.2a) \\ F_\times(\theta_S,\Phi_S,\psi) &= \frac{1}{2}(1+\cos^2\theta_S)\cos 2\Phi_S\sin 2\psi + \cos\theta_S\sin 2\Phi_S\cos 2\psi\,, \\ &= C\sin(2\psi+\alpha)\,, \qquad (5.2b) \end{aligned}$$

where the spherical angles θ_S, Φ_S specify the sky location $\hat{\boldsymbol{N}}$ of the source with respect to the detector frame $\{\hat{\boldsymbol{X}}_D, \hat{\boldsymbol{Y}}_D, \hat{\boldsymbol{Z}}_D\}$ defined with respect to the arms of the L-shaped detector. This frame, along with the sky and source frames we will use later, are described in greater detail in Appendix A.

The polarization angle ψ between the principal + direction and the direction of constant azimuth is given by (Apostolatos et al., 1994)

$$\psi = \tan^{-1}\left[\frac{\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{Z}}_D - (\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}})(\hat{\boldsymbol{Z}}_D\cdot\hat{\boldsymbol{N}})}{\hat{\boldsymbol{N}}\cdot(\hat{\boldsymbol{L}}\times\hat{\boldsymbol{Z}}_D)}\right], \tag{5.3}$$

where $\hat{\boldsymbol{L}}$ is the unit vector in the direction of the BBH orbital angular momentum. The beam-pattern amplitude and phase are

$$C = \left[\frac{1}{4}(1+\cos\theta_S^2)^2\cos^2 2\Phi_S + \cos^2\theta_S \sin^2 2\Phi_S\right]^{1/2}, \tag{5.4a}$$

$$\alpha = \tan^{-1}\left(\frac{2\cos\theta_S \tan 2\Phi_S}{1+\cos\theta_S^2}\right). \tag{5.4b}$$

The beam-pattern amplitude has a range $0 \leq C \leq 1$ and saturates the upper (lower) inequality for $\hat{\boldsymbol{N}} = \pm\hat{\boldsymbol{Z}}_D$ [$\hat{\boldsymbol{N}} = 2^{-1/2}(\pm\hat{\boldsymbol{X}}_D \pm \hat{\boldsymbol{Y}}_D)$].

The frequency-domain GW polarizations are (Cutler and Flanagan, 1994; Takahashi and Nakamura, 2003)

$$\tilde{h}_+(f) = Af^{-7/6}[1+(\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}})^2]e^{i\Psi}, \tag{5.5a}$$

$$\tilde{h}_\times(f) = -2iAf^{-7/6}(\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}})e^{i\Psi}, \tag{5.5b}$$

where A is

$$A = \sqrt{\frac{5}{96}}\frac{1}{D_L}\frac{\mathcal{M}^{5/6}}{\pi^{2/3}}, \tag{5.6}$$

D_L is the luminosity distance between the detector and the source and $\mathcal{M}$ is the chirp mass. For a binary with total mass $M = m_1 + m_2$ and mass ratio $q = m_2/m_1 \leq 1$, the symmetric mass ratio is $\eta = q/(1+q)^2$ and the chirp mass is $\mathcal{M} = \eta^{3/5}M$.

The spin-independent GW phase at 2PN order is (Poisson and Will, 1995)

$$\Psi(f) = 2\pi f t_c - \phi_c - \frac{\pi}{4} + \frac{3}{128}\eta^{-1}u^{-5}\left[1 + \frac{20}{9}\left(\frac{743}{336} + \frac{11}{4}\eta\right)u^2 - 16\pi u^3 + 10\Gamma u^4\right], \quad (5.7)$$

where $u \equiv (\pi M f)^{1/3} = (M/r)^{1/2}$ is our PN parameter, t_c and ϕ_c are the time and GW phase at binary coalescence, and

$$\Gamma \equiv \frac{3058673}{1016064} + \frac{5429}{1008}\eta + \frac{617}{114}\eta^2\,. \quad (5.8)$$

Inserting Eqs. (5.5) into (5.1) yields

$$\tilde{h}(f) = Be^{i[\Psi(f) - \phi_p + 2\delta\Phi]}\,, \quad (5.9)$$

where the GW amplitude B is given by

$$\begin{aligned} B &= Af^{-7/6}\{4(\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}})^2F_\times^2 + [1 + (\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}})^2]^2F_+^2\}^{1/2}\,, \quad (5.10)\\ &= ACf^{-7/6}\{4(\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}})^2\sin^2(2\psi+\alpha) + [1 + (\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}})^2]^2\cos^2(2\psi+\alpha)\}^{1/2}\,, \end{aligned}$$

the polarization phase ϕ_p is

$$\phi_p = \tan^{-1}\left\{\frac{2(\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}})F_\times}{[1 + (\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}})^2]F_+}\right\} = \tan^{-1}\left[\frac{2(\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}})\tan(2\psi+\alpha)}{1 + (\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}})^2}\right], \quad (5.11)$$

and $\delta\Phi$ is an additional contribution to the GW phase for precessing BBH systems given by integrating (Apostolatos et al., 1994)

$$\frac{d\delta\Phi}{df} = \left[\frac{\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}}}{1 - (\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}})^2}\right](\hat{\boldsymbol{L}}\times\hat{\boldsymbol{N}})\cdot\frac{d\hat{\boldsymbol{L}}}{df}\,. \quad (5.12)$$

As we are only interested in the inspiral portion of the waveform during which precession can occur, we cut off our waveforms at a GW frequency

$$f_{\rm cut} = \frac{1}{6^{3/2}\pi M} = 4.3\times10^3\,{\rm Hz}\left(\frac{M}{M_\odot}\right)^{-1} \quad (5.13)$$

equivalent to the quadrupole frequency at the innermost stable circular orbit of a BH of mass M (Bildsten and Cutler, 1992; Cutler and Flanagan, 1994).

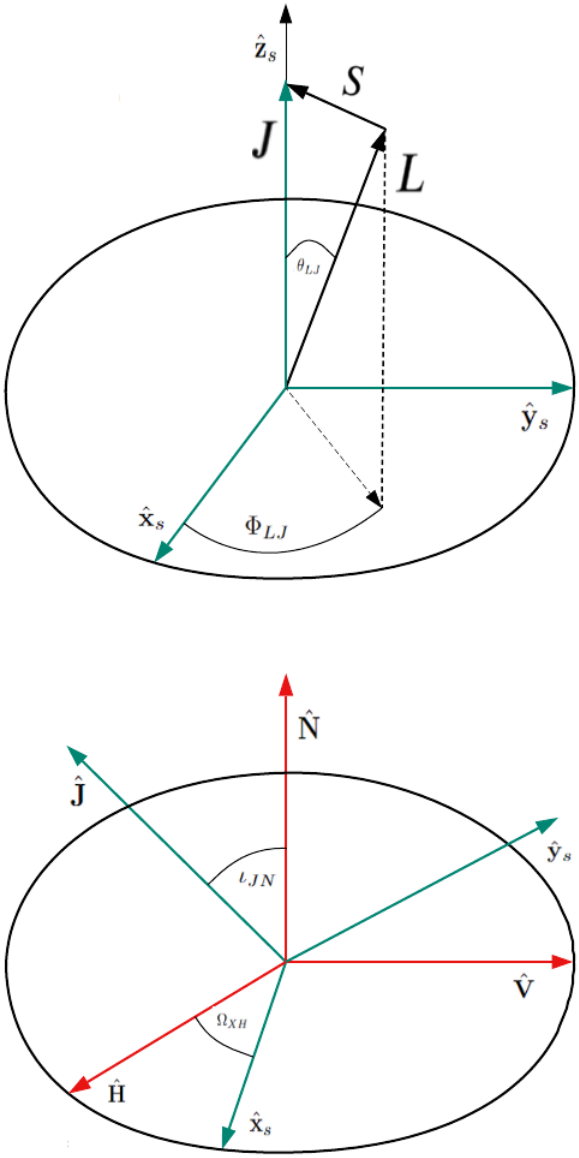


Figure 5.1. Schematic depictions of the sky and source coordinate frames. The sky frame, shown in red in the bottom panel, is defined such that the unit vector $\hat{\boldsymbol{N}}$ pointing from the detector to the sky location of the GW source is along the z axis, the unit vector $\hat{\boldsymbol{V}}$ points towards the zenith at the detector location is along the y axis, and the unit vector $\hat{\boldsymbol{H}} = \hat{\boldsymbol{V}} \times \hat{\boldsymbol{N}}$ completes the orthonomal triad. The source frame is defined such that the total angular momentum $\hat{\boldsymbol{J}}$ is along the Z_S axis, $\hat{\boldsymbol{N}} \times \hat{\boldsymbol{J}}$ is along the X_S axis, and $\hat{\boldsymbol{Y}}_S = \hat{\boldsymbol{Z}}_S \times \hat{\boldsymbol{X}}_S$ completes the orthonormal triad.

5.2.2 Taxonomy of BBH precession and nutation

The gravitational waveform described in the previous subsection can be determined once the frequency-dependent direction of the orbital angular momentum $\hat{\boldsymbol{L}}(f)$ is specified. This problem was considered in Gangardt et al. (Gangardt et al., 2021) which created a taxonomy of spin precession. Following (Apostolatos et al., 1994), this work defined precession as *simple* when the direction of the total angular momentum $\hat{\boldsymbol{J}}$ is approximately constant during the inspiral. This is an excellent approximation except in the special cases of transitional precession (Apostolatos et al., 1994) or nutational resonances (Zhao et al., 2017). In the generic case of simple precession, misaligned BBH spins make the orbital angular momentum $\boldsymbol{L}$ to precess about the total angular momentum $\boldsymbol{J}$ with polar angle θ_{LJ} and azimuthal angle Φ_{LJ} that both vary on the precession timescale. The oscillation of θ_{LJ} on the precession timescale is called *nutation*. In the special case of regular precession, nutation vanishes and both θ_{LJ} and the precession frequency $\Omega_{LJ} \equiv d\Phi_{LJ}/dt$ vary on the longer radiation-reaction timescale.

Nine general coordinates would be needed to specify $\boldsymbol{L}$ and the individual spins $\boldsymbol{S}_1$ and $\boldsymbol{S}_2$ of a precessing BBH system. However, for the case of simple precession, Kesden et al. (Kesden et al., 2015) recognized that, on the precession timescale, the constancy of $\boldsymbol{J}$, $\boldsymbol{L}$, $\boldsymbol{S_1}$, $\boldsymbol{S_2}$, and the projected effective spin $\chi_{\rm eff}$ (Damour, 2001; Racine, 2008) imply that such systems only possess two degrees of freedom. These degrees of freedom can be described by one intrinsic parameter, the total spin magnitude $S = |\boldsymbol{S}_1 + \boldsymbol{S}_2|$, and one extrinsic parameter, a global rotation of the BBH system by an angle Φ_{LJ} about $\boldsymbol{J}$. As indicated in the top panel of Fig. 5.1, oscillation of S between its extrema S_- and S_+ on the precession timescale corresponds to nutation, the oscillation of

$$\theta_{LJ}(S) = \cos^{-1}\left(\frac{J^2 + L^2 - S^2}{2JL}\right) \tag{5.14}$$

between its extrema $\theta_{LJ\pm} = \theta_{LJ}(S_\pm)$. The duration of this oscillation defines the nutation period

$$\tau = 2\int_{S_-}^{S_+} \frac{dS}{|dS/dt|}\,. \tag{5.15}$$

As S describes the only intrinsic degree of freedom of the system, one can define the *precession average* of any quantity X varying on the precession timescale as

$$\langle X\rangle \equiv \frac{2}{\tau}\int_{S_-}^{S_+} \frac{X(S)}{|dS/dt|}dS\,. \tag{5.16}$$

In addition to the polar angle $\theta_{LJ}(S)$, such quantities include the precession frequency $\Omega_{LJ}(S) = d\Phi_{LJ}/dt$. Precessional dynamics in this formalism are described in much greater detail in (Gerosa et al., 2015) which also provides expressions for $S_\pm$, dS/dt, and $\Omega_{LJ}(S)$.

This formalism was used in (Gangardt et al., 2021) to describe the evolution of $\hat{\boldsymbol{L}}$ in terms of five new precession parameters in the generic case of simple precession: (1) the precession amplitude $\langle\theta_{LJ}\rangle$, (2) the precession frequency $\langle\Omega_{LJ}\rangle$, (3) the nutation amplitude $\Delta\theta_{LJ} = (\theta_{LJ+} - \theta_{LJ-})/2$, (4) the nutation frequency $\omega = 2\pi/\tau$, and (5) the variation of the precession frequency $\Delta\Omega_{LJ} = [\Omega_{LJ}(S_+) - \theta_{LJ}(S_-)]/2$.

These parameters have several advantages over other approaches to characterizing generic simple precession. By describing the evolution of $\hat{\boldsymbol{L}}$, they have a more direct connection the the inspiral waveform (5.9) than the individual spins $\boldsymbol{S}_i$. They also vary on the longer radiation-reaction timescale, making parameter estimates less sensitive to the reference frequency at which they are specified. They also provide a richer and more accurate description of generic simple precession than any single parameter like the effective precession spin $\chi_{\rm p}$ (Hannam et al., 2014; Schmidt et al., 2015; Gerosa et al., 2021) ever could. Even the case of regular precession considered in this paper is two dimensional, being described by the precession amplitude $\langle\theta_{LJ}\rangle$ and frequency $\langle\Omega_{LJ}\rangle$. In the next section, we incorporate these parameters into Eq. (5.9) to create a toy model of a regularly precessing waveform.

5.3 A new model of regularly precessing waveforms

5.3.1 Defining the new model using precession parameters

We are now ready to construct our regularly precessing waveform. We need to specify the direction of the orbital angular momentum $\hat{\boldsymbol{L}}$ as a function of GW frequency f and our precession parameters $\langle\theta_{LJ}\rangle$ and $\langle\Omega_{LJ}\rangle$. Once this is determined, the expression $\hat{\boldsymbol{L}}(f)$ can be inserted into the polarization angle ψ of Eq. (5.3), the GW amplitude B of Eq. (5.10), and the GW phase $2\delta\Phi - \phi_p$ of Eqs. (5.11) and (5.12) to obtain the waveform $\tilde{h}(f)$. The direction of the orbital angular momentum is given by

$$\hat{\boldsymbol{L}} = \sin\theta_{LJ}[\cos\Phi_{LJ}\hat{\boldsymbol{X}}_S + \sin\Phi_{LJ}\hat{\boldsymbol{Y}}_S] + \cos\theta_{LJ}\hat{\boldsymbol{Z}}_S\,, \tag{5.17}$$

where the *source frame* $\{\hat{\boldsymbol{X}}_S, \hat{\boldsymbol{Y}}_S, \hat{\boldsymbol{Z}}_S\}$ is defined such that $\hat{\boldsymbol{Z}}_S = \hat{\boldsymbol{J}}$ and $\hat{\boldsymbol{X}}_S$ is along the line of ascending node of the plane perpendicular to $\hat{\boldsymbol{J}}$ with respect to the plane of the sky. This frame is depicted in Fig. 5.1 and described in greater detail in Appendix A.

The precession parameters $\langle\theta_{LJ}\rangle$ and $\langle\Omega_{LJ}\rangle$ defined in the previous section are only constant on the precession timescale, but the angles θ_{LJ} and Φ_{LJ} must be specified on the radiation-reaction timescale as the BBH inspirals through the sensitivity band of the GW detector. For simplicity, we assume that these parameter maintain their lowest PN order frequency dependence as given by Eqs. (5) and (9) in (Gangardt et al., 2021) all the way to merger,

$$\langle\theta_{LJ}\rangle = 0.1\tilde{\theta}\left(\frac{f}{f_{\rm cut}}\right)^{1/3}\,, \tag{5.18a}$$

$$\langle\Omega_{LJ}\rangle = 10^3\,{\rm Hz}\;\tilde{\Omega}\left(\frac{f}{f_{\rm cut}}\right)^{5/3}\left(\frac{M}{M_\odot}\right)^{-1}\,, \tag{5.18b}$$

where the normalizations were chosen such that the dimensionless precession amplitude $\tilde{\theta}$ and dimensionless precession frequency $\tilde{\Omega}$ are constants of order unity. Another way to derive

the above Equations can be found in Appendix. B. We directly equate θ_{LJ} in Eq. (5.17) with $\langle\theta_{LJ}\rangle$ in Eq. (5.18a), while Φ_{LJ} is obtained by integrating $\langle\Omega_{LJ}\rangle$ in Eq. (5.18b),

$$\Phi_{LJ} = \gamma_p + \int_{f_{\rm min}}^{f} \langle\Omega_{LJ}\rangle \left(\frac{df'}{dt}\right)^{-1} df' \,, \tag{5.19}$$

where $f_{\rm min} = 20\,{\rm Hz}$ is the floor of the LVK sensitivity band, the integration constant γ_p is the value of Φ_{LJ} when the binary enters this sensitivity band, and

$$\frac{df}{dt} = \frac{96}{5}\pi\eta f^2 u^5 \left[1 - \left(\frac{743}{336} + \frac{11}{4}\eta\right) u^2 + 4\pi u^3\right] \,, \tag{5.20}$$

is the rate at which the GW frequency increases during the inspiral at 1.5 PN order (Cutler and Flanagan, 1994).

Fig. 5.2 depicts the precession of $\hat{\boldsymbol{L}}$ according to Eqs. (5.17) - (5.19) in the detector frame for equal-mass BBHs with chirp mass $\mathcal{M} = 10M_\odot$. As the GW frequency f increases from $f_{\rm min} = 20\,{\rm Hz}$ to $f_{\rm cut} = 190\,{\rm Hz}$, $\hat{\boldsymbol{L}}$ spirals in a cone about the direction of the total angular momentum $\hat{\boldsymbol{J}}$ with increasing opening angle. The left, middle, and right panels in the top row of this figure show three different directions $\hat{\boldsymbol{J}}$ specified in the detector frame in Table 5.1. Each panels shows three choices of the dimensionless precession amplitude $\tilde{\theta}$ approximately equal to the $5^{\rm th}$, $50^{\rm th}$, and $95^{\rm th}$ percentiles of its distribution for equal-mass, maximally spinning BBHs which are respectively $\tilde{\theta} = 0.95$, 4.23, and 7.75. In the left, middle, and bottom panels of the bottom row of this figure, the direction $\hat{\boldsymbol{J}}$ remains fixed to that of System 2, but the dimensionless precession amplitude $\tilde{\theta}$, dimensionless precession frequency $\tilde{\Omega}$, and azimuthal integration constant γ_p are successively varied. The three choices of $\tilde{\Omega}$ are approximately equal to the $5^{\rm th}$, $50^{\rm th}$, and $95^{\rm th}$ percentiles of its distribution for equal-mass, maximally spinning BBHs which are respectively $\tilde{\Omega} = 1.58$, 2.12, and 2.40, while γ_p is expected to have a flat distribution. The effects of these changes on the precession of $\hat{\boldsymbol{L}}$ are readily apparent; $\tilde{\theta}$ is proportional to the opening angle of the cone, $\tilde{\Omega}$ is proportional to the total angle precessed about the cone during the inspiral, and γ_p corresponds to a global rotation of the system about $\hat{\boldsymbol{J}}$.

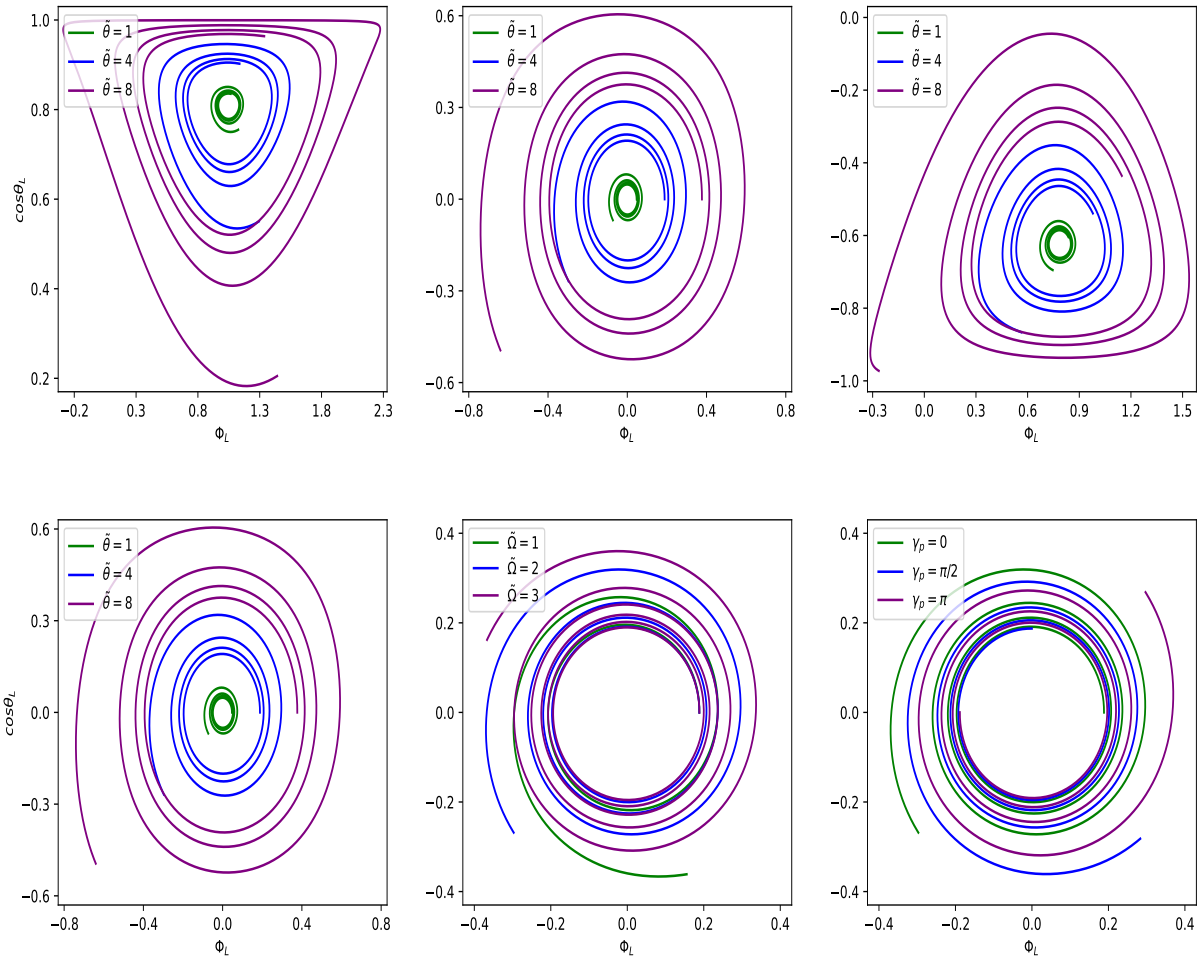


Figure 5.2. Precession of the spherical coordinates θ_L and Φ_L that specify the direction of orbital angular momentum $\boldsymbol{L}$ with respect to the detector frame. The BBHs have equal masses and chirp mass $\mathcal{M} = 20M_\odot$. The top row shows this precession for dimensionlesss precession frequency $\tilde{\Omega} = 2$, azimuthal integration constant $\gamma_p = 0$, and dimensionless precession amplitude $\tilde{\theta} = 1$ (green), 4 (blue), and 8 (purple). The left, middle, and right panels of the top row correspond to different directions of the total angular momentum $\boldsymbol{J}$ listed as Systems 1, 2, and 3 in Table 5.1. The bottom row show the precession of $\boldsymbol{L}$ for System 2. The left panel is a copy of the middle panel of the top row, the middle panel has $\tilde{\theta} = 4$, $\gamma_p = 0$, and $\tilde{\Omega} = 1$ (green), 2 (blue), and 3 (purple), and the bottom panel has $\tilde{\theta} = 4$, $\tilde{\Omega} = 2$, and $\gamma_p = 0$ (green), $\pi/2$ (blue), and π (purple).

We already mentioned the values of our dimensionless precession amplitude and frequency for the 5th, 50th, and 95th percentiles of its distribution for equal-mass, maximally spinning

Table 5.1. Spherical coordinates θ_J, Φ_J of the total angular momentum $\boldsymbol{J}$ in the detector frame, inclination ι_{JN} between $\hat{\boldsymbol{J}}$ and the sky location $\hat{\boldsymbol{N}}$, and polarization angle ψ_0 for zero precession amplitude. All three BBH systems have equal-mass BBHs with chirp mass $\mathcal{M} = 10M_\odot$ and sky location $\theta_S = \pi/4, \Phi_S = 0$.

Parameter	System 1	System 2	System 3
θ_J	$\pi/5$	$\pi/2$	$8\pi/9$
Φ_J	$\pi/3$	0	$\pi/4$
$\cos\iota_{JN}$	0.561	0.707	-0.493
ψ_0	0.621	$\pi/2$	-1.29

BBHs. We obtain these values at $r = 10M$ from Fig. 3 in (Gangardt et al., 2021). We employ a change of variables and set $(\pi M f)^{1/3} = (M/r)^{1/2}$, and by using Eqs. (5.18a), (5.18b) we can calculate the valid parameter space and the corresponding percentile levels.

Futhermore, a necessary but insufficient condition for precession to be detected with confidence is for the number of precession cycles

$$N_{\rm pre} = \frac{1}{2\pi}[\Phi_{LJ}(f_{\rm cut}) - \Phi_{LJ}(f_{\rm min})] \tag{5.21}$$

occurring within the sensitivity band of the detector to be greater than unity. In Fig. 5.3, the dot-dashed, dashed, and dotted horizontal lines show the $5^{\rm th}$, $50^{\rm th}$, and $95^{\rm th}$ percentiles of the distribution of the dimensionless precession frequency $\tilde{\Omega}$ for maximal spins (left panel) and equal-mass binaries (right panel). We see that the median values of $\tilde{\Omega}$ are weakly sensitive to both the mass ratio q and spin magnitudes χ_i, and that dispersion in $\tilde{\Omega}$ increases with χ_i for equal-mass BBHs. This is consistent with the behavior of $\langle\Omega_{LJ}\rangle$ at $r = 10M$ ($u = 10^{-1/2}$) shown in the middle right panel of Fig. 3 in (Gangardt et al., 2021). The solid curves show the values of $\tilde{\Omega}$ for which $N_{\rm pre} = 1$. The portions of parameter space above these curves are more likely to exhibit detectable precession. We see that, for a wide range of mass ratios and spins, a majority of binaries satisfy the criterion $N_{\rm pre} \geq 1$ for $M \lesssim 45M_\odot$, and that 5% of the distribution could have detectable precession for $q \geq 0.5$, $\chi_i \approx 1$, and $M \lesssim 60M_\odot$.

The dependence of the amplitude $B(f)$ on our three new parameters is shown in Fig. 5.4. We choose equal-mass BBHs with chirp mass $\mathcal{M} = 20M_\odot$ with a sky location $\theta_S = \pi/4$,

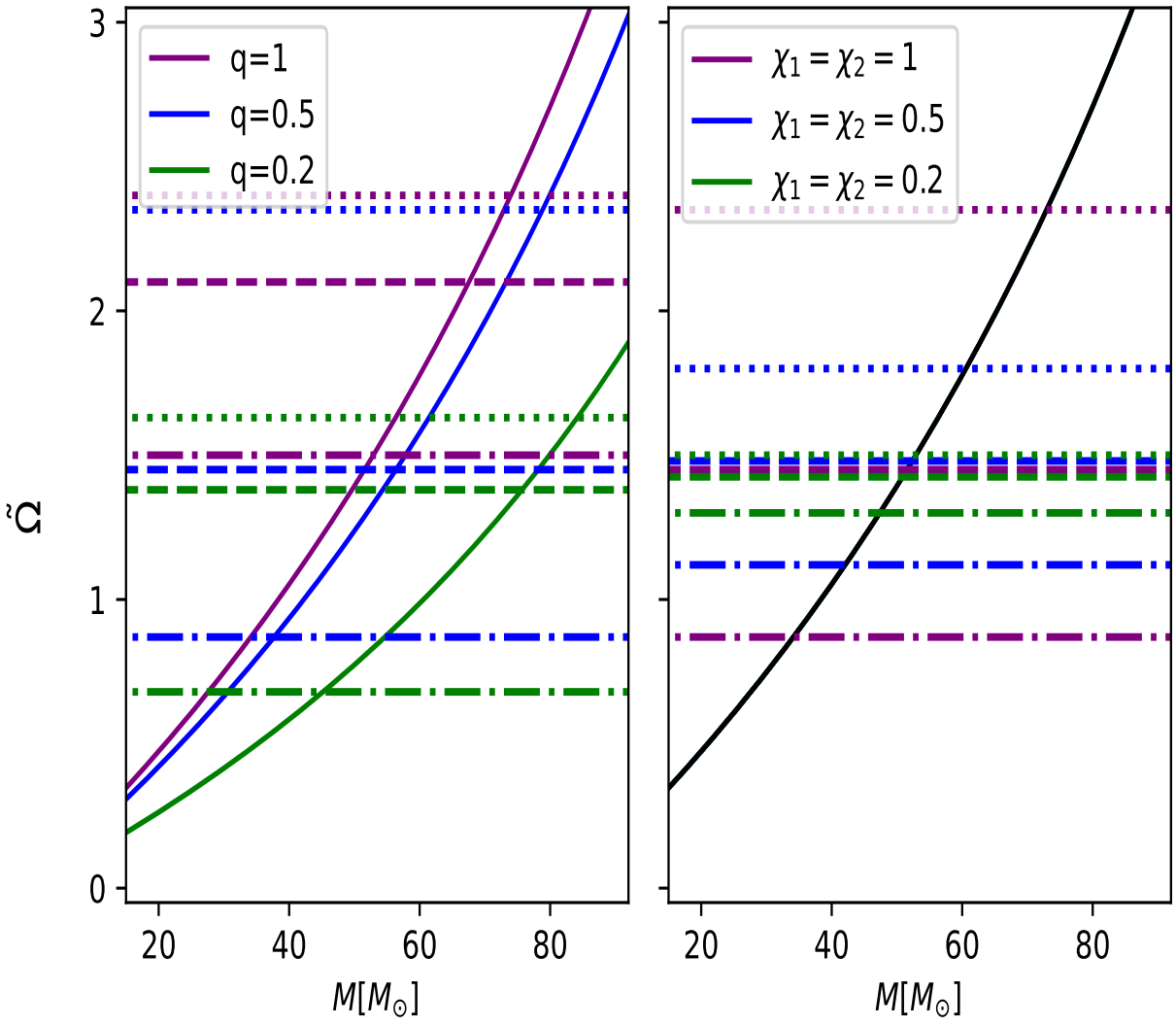


Figure 5.3. Dimensionless precession frequency $\tilde{\Omega}$ as a function of total binary mass M. In the left panel, the dot-dashed, dashed, and dotted horizontal lines show the 5th, 50th, and 95th percentiles of the $\tilde{\Omega}$ distribution for BBHs with isotropic maximal spins and mass ratios $q = 1$ (purple), $q = 0.5$ (blue), and $q = 0.2$ (green). In the right panel, the dot-dashed, dashed, and dotted horizontal lines show the 5th, 50th, and 95th percentiles of the $\tilde{\Omega}$ distribution for equal-mass BBHs with dimensionless spins magnitudes $\chi_i = 1$ (purple), $\chi_i = 0.5$ (blue), and $\chi_i = 0.2$ (green). The solid curves in both panels show the minimum value of $\tilde{\Omega}$ for which at least one precession cycle occurs after the BBHs enter the LVK sensitivity band at a GW frequency of 20 Hz.

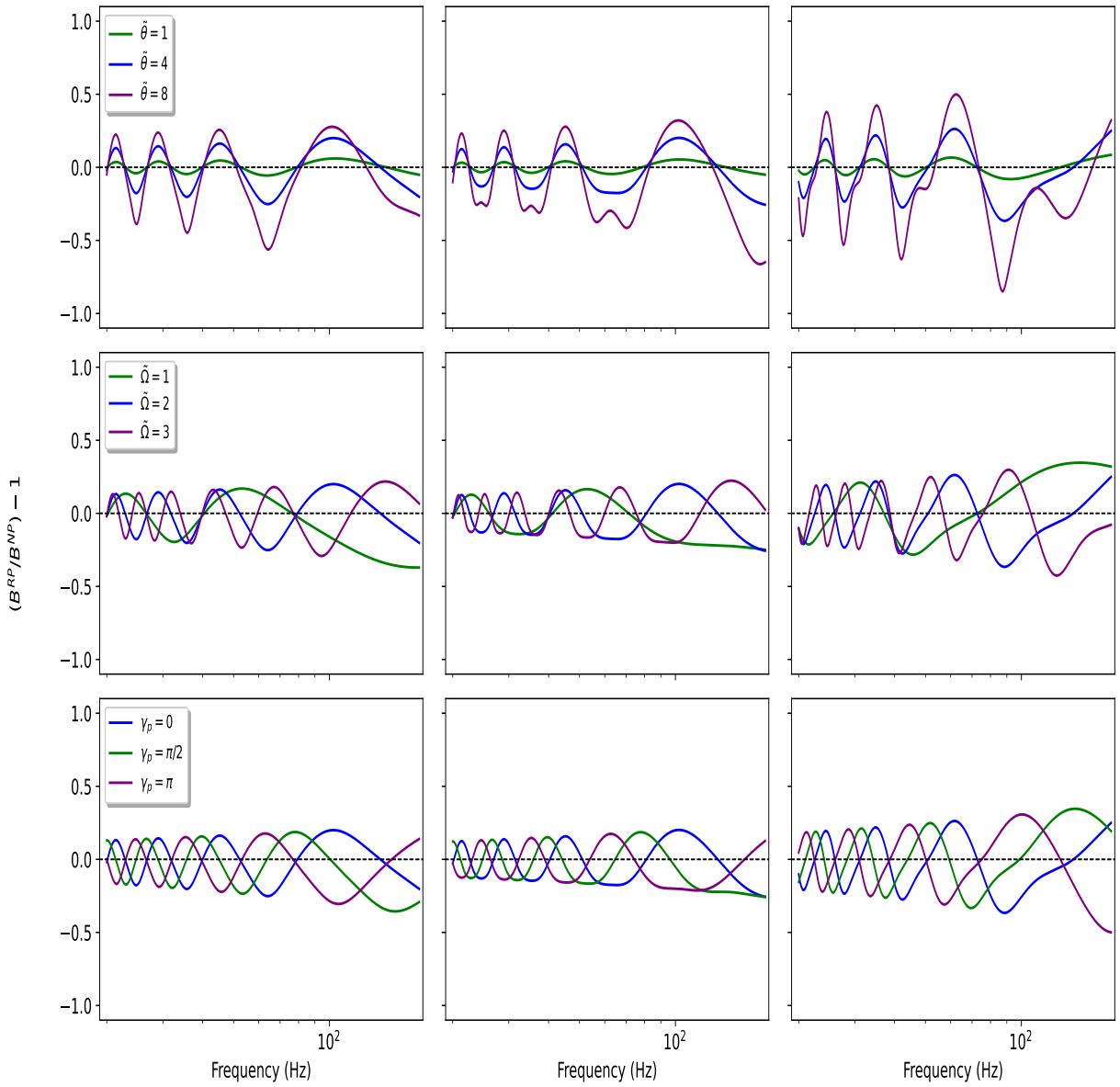


Figure 5.4. The ratio of the GW amplitude B given by Eq. (5.10) in our regularly precessing model (RP) to a non-precessing (NP) model with the same BBH masses. The left, middle, and right panels correspond to directions of the total angular momentum $\hat{\boldsymbol{J}}$ listed as Systems 1, 2, and 3 in Table 5.1. The top panels fix $\tilde{\Omega}=2$, $\gamma_p=0$ and vary $\tilde{\theta}$, the middle panels fix $\tilde{\theta}=4$, $\gamma_p=0$ and vary $\tilde{\Omega}$, and the bottom panels fix $\tilde{\theta}=4$, $\tilde{\Omega}=2$ and vary γ_p.

$\Phi_S=0$. This sky location implies a beam-pattern amplitude $C=3/4$ and phase $\alpha=0$ according to Eq. (5.4). The left, middle, and right columns correspond to three different choices of the direction of the total angular momentum $\boldsymbol{J}$ listed as Systems 1, 2, and 3 in

Table 5.1. Examination of Eq. (5.10) reveals that the GW amplitude B depends linearly on A, is explicitly proportional to $f^{7/6}$, and depends on f implicitly through the beam-pattern coefficients and the inner product $\hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{N}}$. We plot the ratio of the regular precessing (RP) over the non-precessing (NP) amplitude, in order to focus on the effect of our precession parameters on the amplitude and to demonstrate the aforementioned dependencies more clearly. Here, we have to note that since the three different Systems of Fig. 5.4 share the same source parameters but they have different choices for the direction of the total angular momentum $\boldsymbol{J}$ given in Table 5.1, we can safely assume that this different choice of parameters is the driving force behind the appearance of various peaks and troughs in the evolution of the amplitude ratios in each column. These behaviors are even more prevalent for increased values of $\tilde{\theta}$, as can be seen clearly on the top row. To explain this feature, we analyze Eq. (5.1) and conclude that there are various scenarios, in the NP case where some or all of the terms inside Eq. (5.10) vanish.

Two of these cases where the terms in our GW template can vanish are the *face-on* and *edge-on* cases. For example, we call a BBH system face-on when the angle between the total orbital angular momentum and the line of sight is $\iota_{JN} = 0$ and edge-on when $\iota_{JN} = \pi/2$. The expressions in our various frames for the angle ι_{JN} can be found in Appendix. A. For the face-on NP case we know that $\hat{\boldsymbol{J}} \parallel \hat{\boldsymbol{L}}$ and we get $\hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{N}} = 1$ inside Eq. (5.10). In addition, for a particular choice of the sky localization angles in the NP limit, we can make both $F_+ = 0$ and $F_\times = 0$ by making $C = 0$ from Eq. (5.4) or one of them by making the corresponding phase inside the sin and cos terms $= 0$ and $= \pi/2$. Having either of these terms equal to 0 affects the calculation of the amplitude both in the NP and the RP regime and that is one of the key factors that lead to these behaviors in Fig. 5.4 as we have already discussed. It also implies that the location of the BBH system in the sky, can in fact make either one of the terms that describe the GW polarizations vanish. This has already been well studied and stated in the literature, i.e. (Nissanke et al., 2010). It was shown that, we are observationally

biased towards detecting binaries, in which the orbital angular momentum is either aligned or anti-aligned with the line of sight.

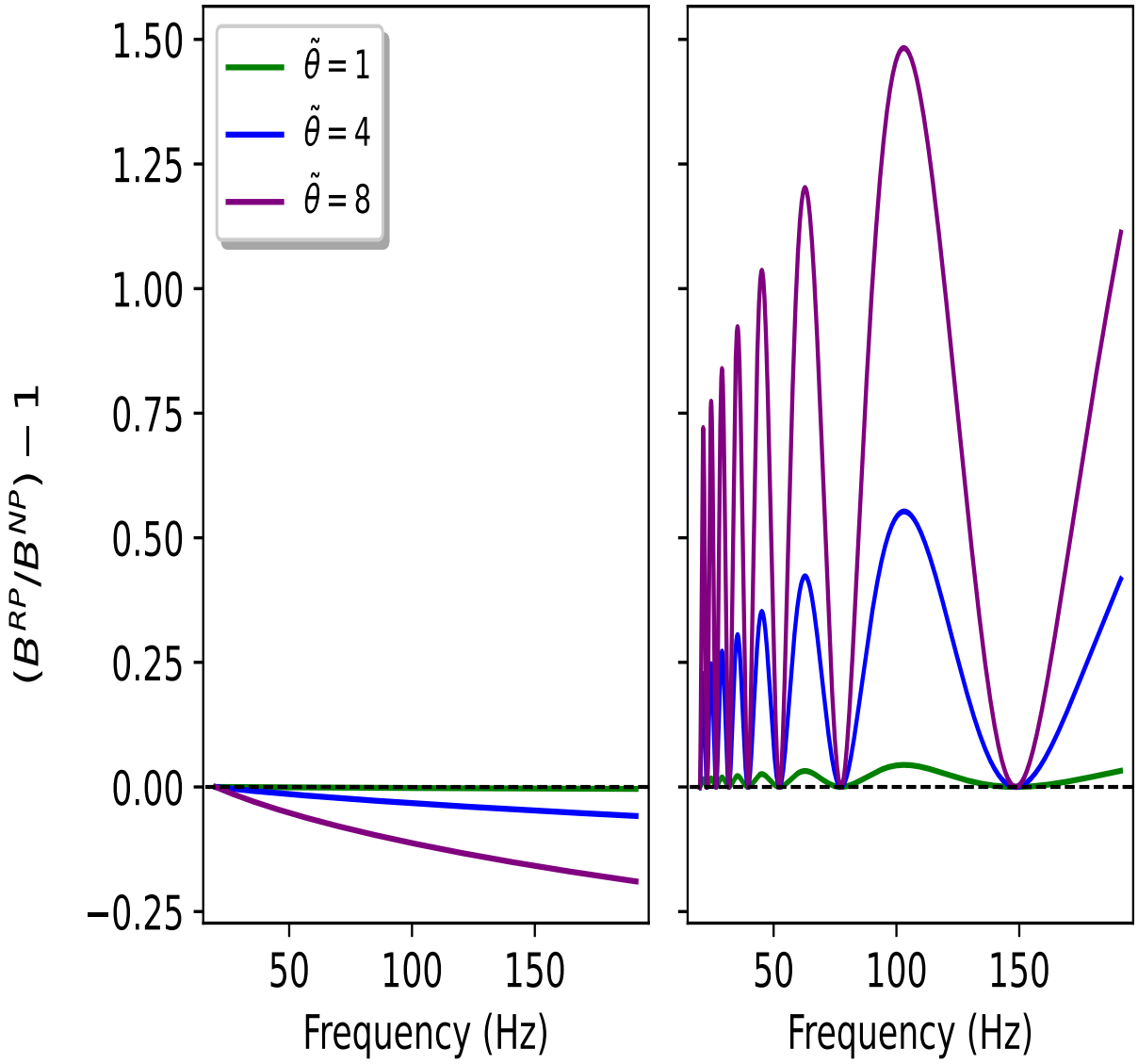


Figure 5.5. The ratio of the GW amplitude B given by Eq. (5.10) in our regularly precessing model (RP) to a non-precessing (NP) model with the same BBH masses of $\mathcal{M} = 10M_\odot$. The left[right] panel corresponds to directions of the total angular momentum $\hat{\boldsymbol{J}}$ and the line of sight $\hat{\boldsymbol{N}}$ that produce a face-on[edge-on] case, i.e. $\theta_S, \Phi_S, \theta_J, \Phi_J = (\pi/6, \pi/4, \pi/6, \pi/4)\big[\theta_S, \Phi_S, \theta_J, \Phi_J = (\pi/6, \pi/3, 2\pi/3, \pi/3)\big]$. Both panels fix $\tilde{\Omega} = 2$, $\gamma_p = 0$ and vary $\tilde{\theta}$.

Fig. (5.5) presents an example of a face-on and edge-on case. There exist infinite combinations of the four angles that specify the sky localization of the BBH, that can give us a face-on and an edge-on case respectively. In our example, we have picked the simplest cases for the angle choices, that is for the face-on case we have set $(\theta_S, \theta_J) = (\pi/6, \pi/6)$ and we have chosen the same random value for the azimuthal angles seeing as that is the easier way to make $\cos \iota_{JN} = 1$ Eq. (A.4). As for the edge-on case, we choose $(\theta_S, \theta_J) = (\pi/6, 2\pi/3)$ and we set the azimuthal angles equal to each other, $\Phi_S = \Phi_J = \pi/3$. Taking a closer look at the left panel of Fig. (5.5), we find out that the face-on case confirms our analysis. Even though, we have introduced RP through non-zero values of our precession parameters, the term $\sin \iota_{JN} = 0$ forces the term that gives us the periodic precessing behavior in Eq. (A.8a) to vanish. In turn, this transforms the ratios into lines with different slopes depending on the value of the precession amplitude. However, we mention that the GW phase will still be modulated on the precession timescale, even if the amplitude B is not.

Things get more interesting when we move over to the edge-on case and the right panel of Fig. (5.5). By closer examination, we understand that even though we have used the same value of $\tilde{\Omega}$, the number of precession cycles in band almost gets doubled. Taking a look at Eqs. (5.4a) and (5.3), we can check and see that $\alpha = 0$ and for the edge-on case and the frequency depdence inside ψ vanishes, so F_+ and $F_\times$ have constant values. Moving on to Eq. (A.8a), we also note that for the edge-on case the second term is equal to 0. That means, when we square the $\hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{N}}$ term in order to calculate Eq. (5.10), we only get a single $\sin^2 \Phi_{LJ}$ term inside B and that doubles the phase of our precessing system.

We have shown previously that by introducing RP and in turn frequency dependence to the inclination angle between the orbital angular momentum and the line of sight, the phase of the GW template gets modulated via two terms ϕ_p and $2\delta\Phi$ respectively. $\delta\Phi$ is given in Eq. (5.12) and ϕ_p was defined in Eq. (5.11). Figs. 5.6 and 5.7 show the regular precessing behavior of these terms for an equal-mass BBHs with chirp mass $\mathcal{M} = 10 M_\odot$.

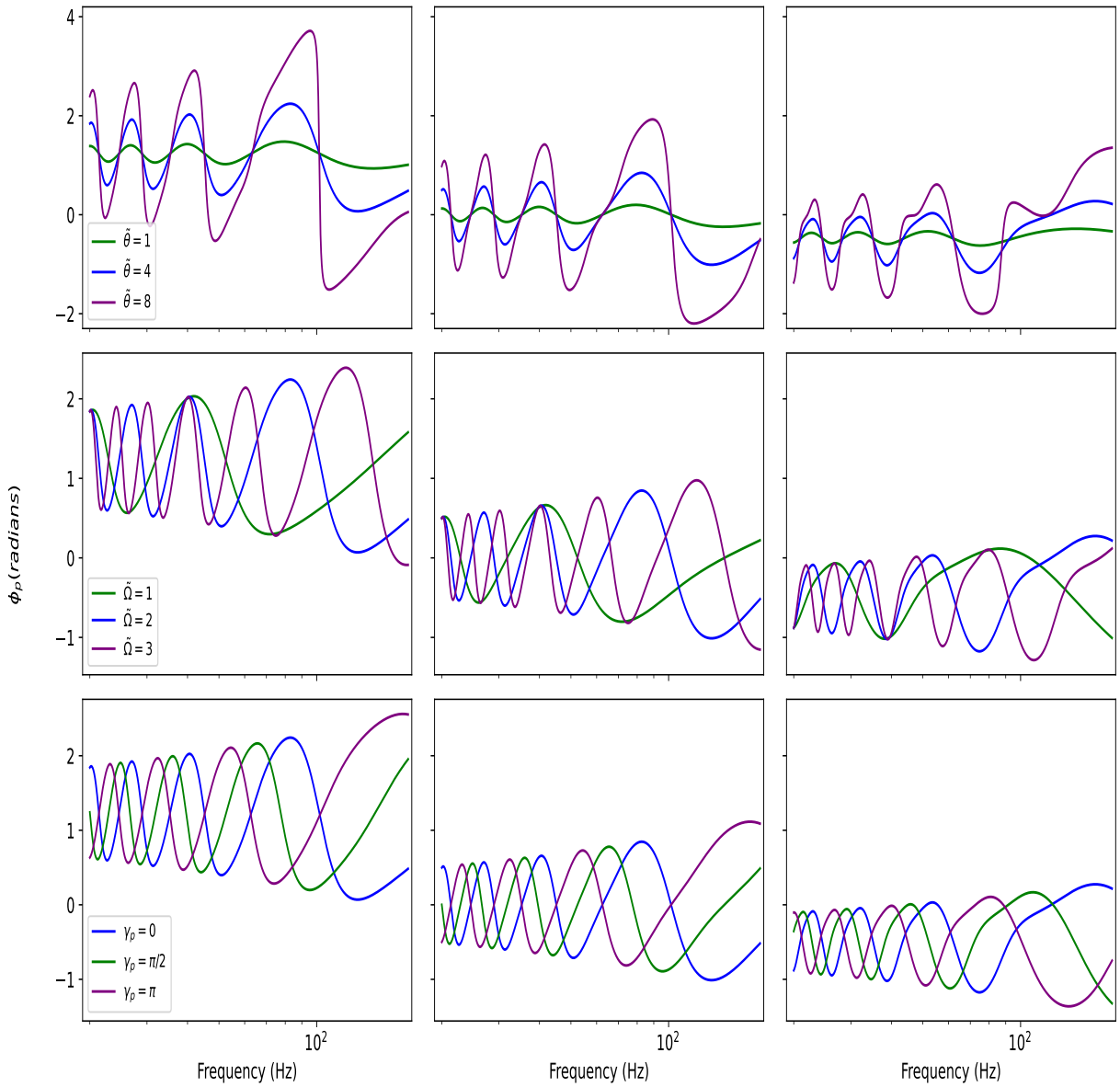


Figure 5.6. The evolution of the polarization phase ϕ_p given by Eq. (5.11) in our regularly precessing model (RP) with respect to the frequency. The left, middle, and right panels correspond to directions of the total angular momentum $\hat{\boldsymbol{J}}$ listed as Systems 1, 2, and 3 in Table 5.1. The top panels fix $\tilde{\Omega} = 2$, $\gamma_p = 0$ and vary $\tilde{\theta}$, the middle panels fix $\tilde{\theta} = 4$, $\gamma_p = 0$ and vary $\tilde{\Omega}$, and the bottom panels fix $\tilde{\theta} = 4$, $\tilde{\Omega} = 2$ and vary γ_p.

Again, the left, middle, and right columns of the two figures show three different directions of $\hat{\boldsymbol{J}}$ specified in the detector frame in Table 5.1. In both terms, but more noticeably in the polarization angle ϕ_p, we see that different choices of the angles produce different behaviors.

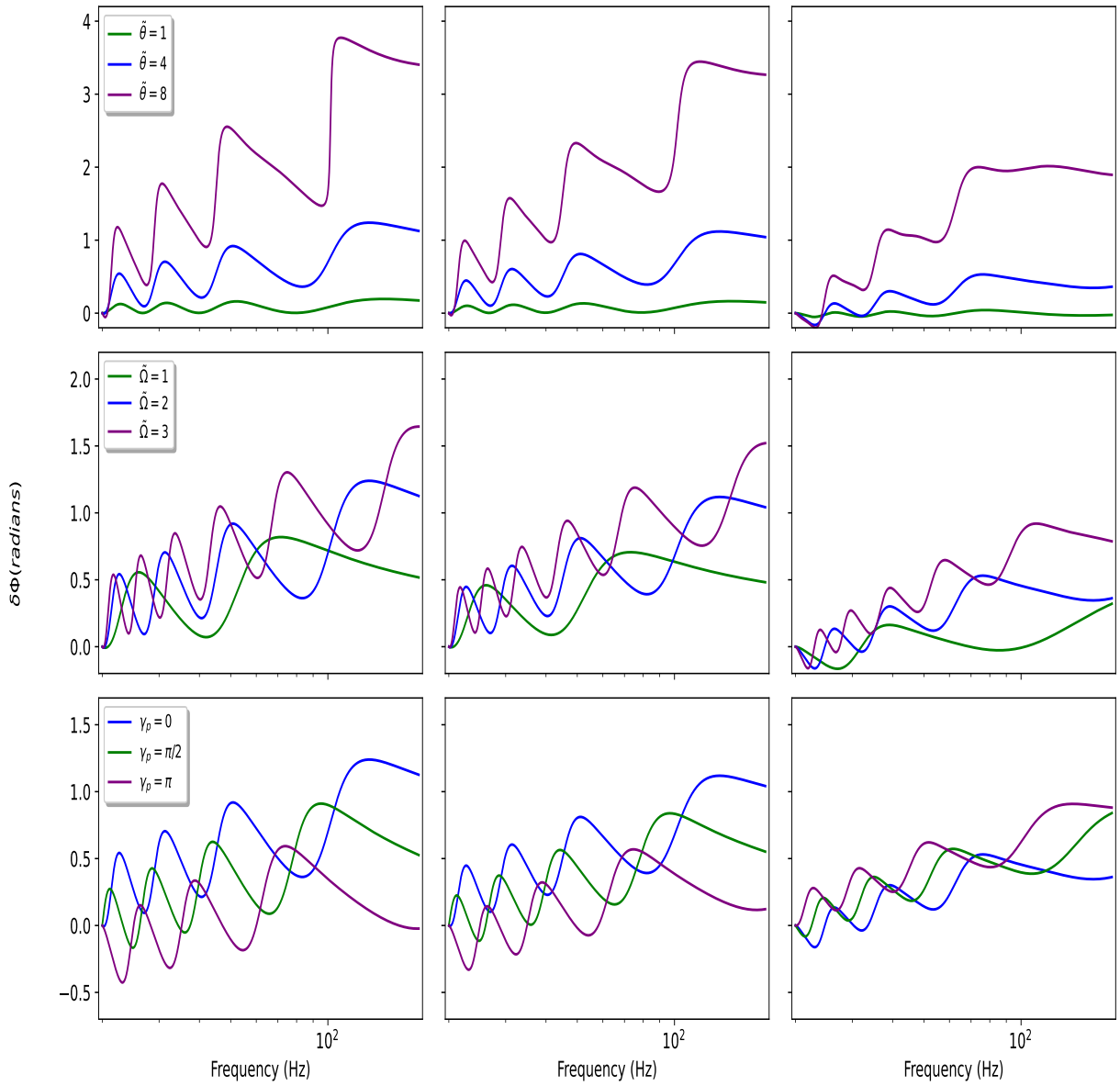


Figure 5.7. The evolution of the phase correction term $\delta\Phi$ given by Eq. (5.12) in our regularly precessing model (RP) with respect to the frequency. The left, middle, and right panels correspond to directions of the total angular momentum $\hat{\boldsymbol{J}}$ listed as Systems 1, 2, and 3 in Table 5.1. The top panels fix $\tilde{\Omega}=2$, $\gamma_p=0$ and vary $\tilde{\theta}$, the middle panels fix $\tilde{\theta}=4$, $\gamma_p=0$ and vary $\tilde{\Omega}$, and the bottom panels fix $\tilde{\theta}=4$, $\tilde{\Omega}=2$ and vary γ_p.

By looking at Eqs. (5.11) and (5.12), it is again evident that the values of the beam-pattern coefficients and more importantly $\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}}$ affect the phase-correction terms in various degrees.

However, we find it important to note the one big difference in these two phase terms. The ϕ_p term is calculated from Eq. (5.11) as the inverse tangent of our terms, so its value closely resembles a periodic behavior. Going back to the calculation of the $\delta\Phi$ term we had to solve the integral from (Apostolatos et al., 1994)

$$\delta\Phi(f) = \int_{f_{\min}}^{f} \left(\frac{\hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{N}}}{1 - (\hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{N}})^2} \right) (\hat{\boldsymbol{L}} \times \hat{\boldsymbol{N}}) \cdot \frac{d\hat{\boldsymbol{L}}}{df} df' \ . \tag{5.22}$$

By making a change of variables to the integrand, that the precession average of Eq. (5.12) is non-vanishing. That implies, that when we acquire the definition of $\delta\Phi$ from Eq. (5.12) and we plot it with respect to the frequency, we would expect to not only have a periodic contribution but also a secular one. Fig. 5.7 confirms our expectations and shows that not only do we have a periodic behavior for $\delta\Phi$ but its value is also increasing the more time we spend in the frequency band of the detector.

5.4 Mismatch analysis

In this section, we will employ the matched-filtering tools on our new RP 'toy' model and arrive at several useful results and conclusions. An inner product on the parameter vector space between two signals expressed in the frequency domain, $\tilde{h}(f)$, $\tilde{g}(f)$, is given by (Finn, 1992; Wiseman, 1992)

$$\langle h|g\rangle = 4\Re \int_{f_{\min}}^{f_{\rm cut}} \frac{h(f)g^*(f)}{S_n(f)} df \,, \tag{5.23}$$

where $S_n(f)$ is the power spectral density (PSD) of the aLIGO detector.

We can then use Eq. (5.23) to compute the signal-to-noise ratio (SNR). The SNR provides a measure of the strength of the GW signal arriving at the detector compared to the background noise. The SNR of a waveform signal h_s reconstructed with a template h_t is (Cutler and Flanagan, 1994)

$$\rho = \langle h_s|h_t\rangle^{1/2} \,. \tag{5.24}$$

This SNR is maximized when the template matches the source, i.e. $h_t = h_s$.

We can also define the overlap between the two signals such that (Cutler and Flanagan, 1994)

$$\mathcal{O}(h_s, h_t) = \frac{\langle h_s | h_t \rangle}{\sqrt{\langle h_s | h_s \rangle \langle h_t | h_t \rangle}}\,, \tag{5.25}$$

where the overlap is constrained to be $\mathcal{O} \leq 1$.

The match M is defined as the overlap maximized over the time and the phase of coalescence

$$M(h_s, h_t) = \max_{t_c, \phi_c} \mathcal{O}(h_s, h_t)\,, \tag{5.26}$$

while the mismatch ϵ is just

$$\epsilon(h_s, h_t) = 1 - M\,, \tag{5.27}$$

where we use the PYCBC.FILTER package (Nitz et al., 2022) to evaluate the mismatch between waveforms and to pick various PSD. The two waveforms h_1 and h_2 can be distinguished when their mismatch satisfies the Lindblom criterion (Lindblom et al., 2008)

$$\epsilon(h_1, h_2) \gtrsim \frac{1 - (\rho_1 - \rho_2)^2}{2\rho_1\rho_2} \approx \frac{1}{2\rho^2}\,, \tag{5.28}$$

where the latter equality holds for $\rho_1, \rho_2 \approx \rho$.

Fig. (5.8) shows plots of mismatch ϵ between a RP GW source and a NP template used by a ground based L-shaped detector as a function of the precession amplitude $\tilde{\theta}$ (left panels) and the precession frequency $\tilde{\Omega}$ (right panels). When plotting each column, we keep the source parameters the same although we change the chirp mass of the BBH system from $10M_\odot$ (top panels) to $20M_\odot$ (bottom panels). As for the choice of the angles that define the sky location of our BBH, we follow System II from Table 5.1. In addition, when we calculate ϵ for the left column, we pick $\tilde{\Omega} = 1, 2, 3$ corresponding to reasonable values for our

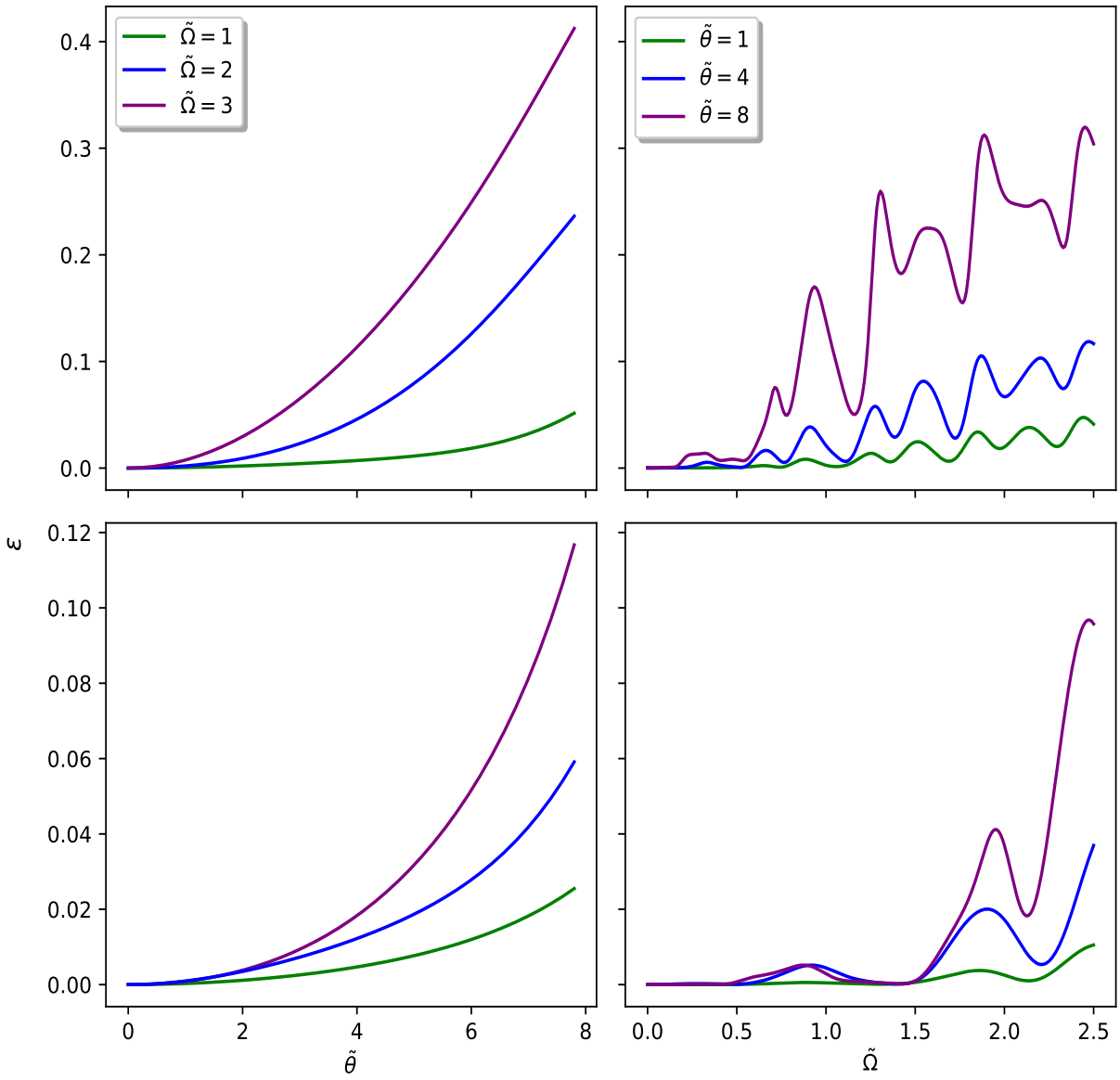


Figure 5.8. The mismatch ϵ given by Eq. (5.27) between a regularly precessing(RP) GW source and a non-precessing(NP) GW template with respect to the normalized precession amplitude $\tilde{\theta}$ (left panels) and the normalized precession frequency $\tilde{\Omega}$ (right panels). All four panels correspond to the direction of the total angular momentum $\hat{\boldsymbol{J}}$ listed as System 2 in Table 5.1. The sky location $\hat{\boldsymbol{N}}$ is specified by $(\theta_S, \Phi_S) = (\pi/4, 0)$, and the precession integration constant is $\gamma_p = 0$. The values of the chirp mass are $\mathcal{M} = 10M_\odot$ (top panels) and $\mathcal{M} = 20M_\odot$ (bottom panels). The green, blue, and purple curves show $\tilde{\Omega} = 1$, 2, and 3 (left panels) and $\tilde{\theta} = 1$, 4, and 8 (right panels).

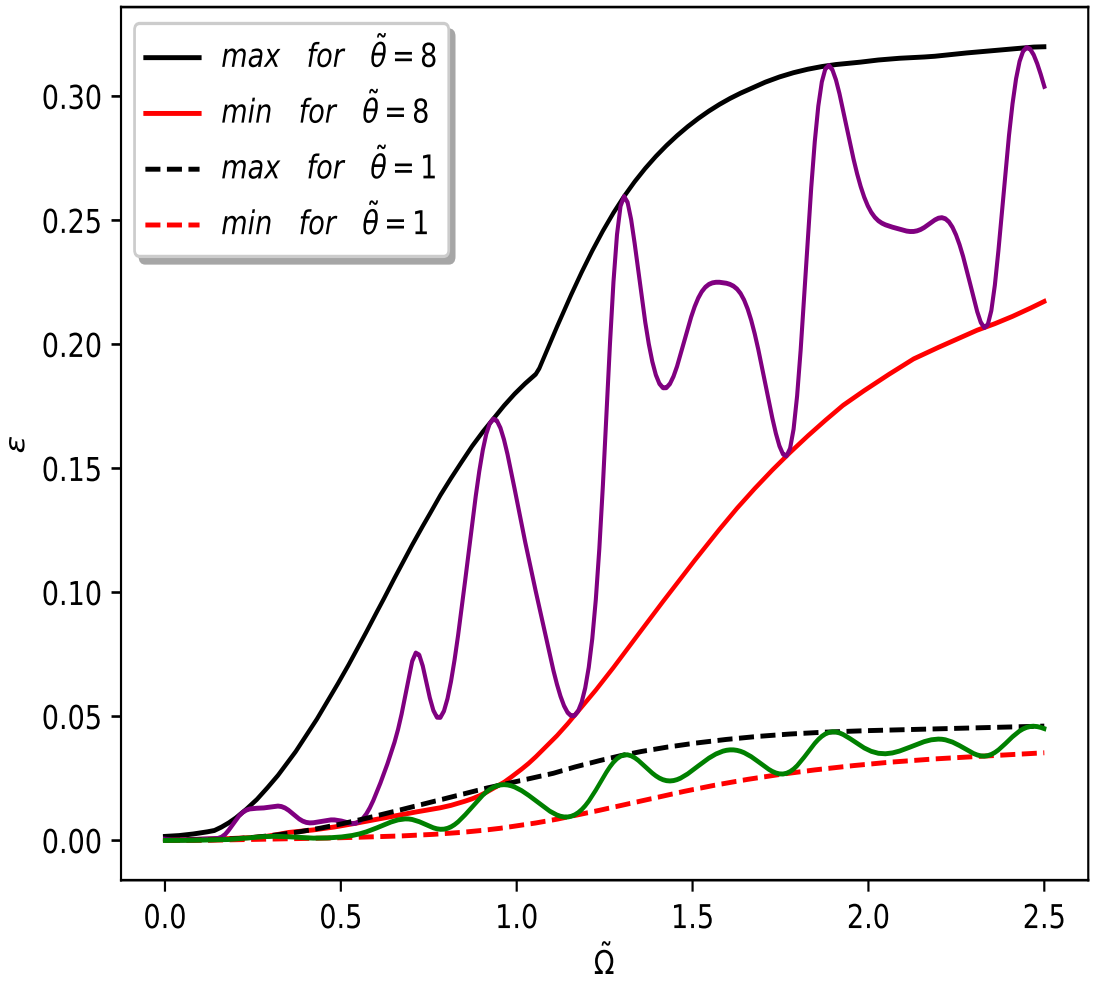


Figure 5.9. The evolution of the mismatch ϵ given by Eq. (5.27) between a regularly precessing(RP) GW source and a non-precessing(NP) GW template with respect to the normalized precession frequency $\tilde{\Omega}$. The direction of the total angular momentum $\hat{\boldsymbol{J}}$ is listed as System 2 in Table 5.1 and the direction of the line of sight $\hat{\boldsymbol{N}}$ is specified by $(\theta_S, \Phi_S) = (\pi/4, 0)$ and for the chirp mass we have chosen $\mathcal{M} = 10M_\odot$. The purple (green) solid line denotes a value of $\tilde{\theta} = 8$ ($\tilde{\theta} = 1$). γ_p is varied from 0 to 360 for each value of $\tilde{\Omega}$ and the maximum and minimum value of the mismatch is obtained independently. The black(red) solid line is showing the maximum(minimum) envelope for the mismatch values of each individual γ_p for $\tilde{\theta} = 8$, while the dashed lines are calculated for $\tilde{\theta} = 1$.

parameter, while we follow the same method for the right panels and we calculate ϵ for the different values of $\tilde{\theta}$. Fig. 5.8 confirms that $\epsilon \to 0$, when we have either $\tilde{\theta} \to 0$ or $\tilde{\Omega} \to 0$ and that brings us back to the non-precessing GW template.

Moreover, we understand that by increasing the chirp mass of the BBH system we are decreasing the upper limit $f_{\rm cut}$ of the integral calculations in Eq. (5.23), while the $f_{\rm min}$ is again set to 20 Hz corresponding to the bottom of LVK sensitivity. That implies that the BBH system is spending less time in the observable frequency band of the detector and that in turn is decreasing the number of precession cycles we are going to detect for the same value of $\tilde{\Omega}$. By doubling the chirp mass between calculations we are cutting the $f_{\rm cut}$ in half. This assumption is confirmed when we take a look at the maximum values of the mismatches in each case. The other main difference between the two columns of Fig. 5.8 is the dependence of ϵ on the two of our three RP parameters. For the left panels of Fig. 5.8, the dependence of ϵ on $\tilde{\theta}$ for constant $\tilde{\Omega}$ is fairly easy to interpret. However, the behavior of ϵ in the right panels presents features that are difficult to interpret. The main reason for the oscillations is that the modulations in the waveform move in and out of the bucket of maximum LVK sensitivity as $\tilde{\Omega}$ increases, while another significant one reason is that the number of precession cycles is monotonically increasing when we increase $\tilde{\Omega}$.

Fig. 5.9 attempts to clarify this complicated behavior by showing the evolution of the mismatch ϵ given by Eq. (5.27) in our regularly precessing model (RP) with respect to the normalized precession frequency $\tilde{\Omega}$. The purple (green) solid line denotes a value of $\tilde{\theta} = 8$ ($\tilde{\theta} = 1$). γ_p is varied from 0 to 2π for each value of $\tilde{\Omega}$ and the maximum and minimum value of the mismatch is obtained independently. Then for every mismatch calculation for the different angles of γ_p we pick the minimum and the maximum value of the mismatch. We then use these values to produce maximum-minimum envelopes around the values of the mismatch. We then see that these envelopes cover almost exactly the mismatches between a NP and our RP model, independently of the various peaks and troughs. This confirms our

hypothesis, that even though we refer to γ_p as a nuisance parameter, it also corresponds to a global rotation of the system about $\hat{\boldsymbol{J}}$. This rotation moves the GW template as seen in Fig. 5.4 to the left and that makes the mismatch find the *sweet spot* of the *bucket* of the PSD used and that is the driving force behind the maximum envelope. In the same manner, by varying γ_p, we also find that our global rotation brings the two GW templates almost on top of one another, minimizing the value for the mismatch.

Finally, Fig. 5.10 provides a look at the prospect of distinguishing a precessing GW signal in the future runs of LVK as well as with 3G detectors. Eq. (5.28) provides us with a criterion to distinguish two waveforms. In Fig. 5.10, we employ that exact criterion. The red solid lines correspond to the same calculation of the mismatch between a RP BBH source and a NP template, at the bottom right panel of Fig. 5.10 for values of $\tilde{\theta} = 1, 8$ for the right and left panel respectively, while we keep $\tilde{\Omega} = 2$ and $\gamma_p = 0$ constant. The blue lines show the calculation of the mismatch for the same values of our parameters but this time we use the noise spectrum of proposed future detector, Cosmic Explorer(CE) (Maggiore et al., 2020). Here, we have to note that we calculated the mismatches for both the CE and the Einstein telescope (Maggiore et al., 2020). However, we present the results concerning only CE. The reason for that being that our calculations bore similar results so it would be a redundancy to plot both of them at the same time.

Looking at Fig. 5.10, the dot-dashed, dashed, and dotted blue horizontal lines show the value of our criterion, $1/2\rho^2$, for a different choice of luminosity distance D_L, calculated for a flat universe. Using the criterion, we can see that if the mismatch between a precessing and a non-precessing GW signal lies above these three different lines, that implies that precession can be clearly distinguished. In addition, we can see that the difference in design sensitivity and in turn the difference in the mismatch calculations, is huge when we compare aLIGO to the future detectors. The two main reasons behind that is the difference spectral noise density they are using and that the lower frequency limit where a BBH can enter the band

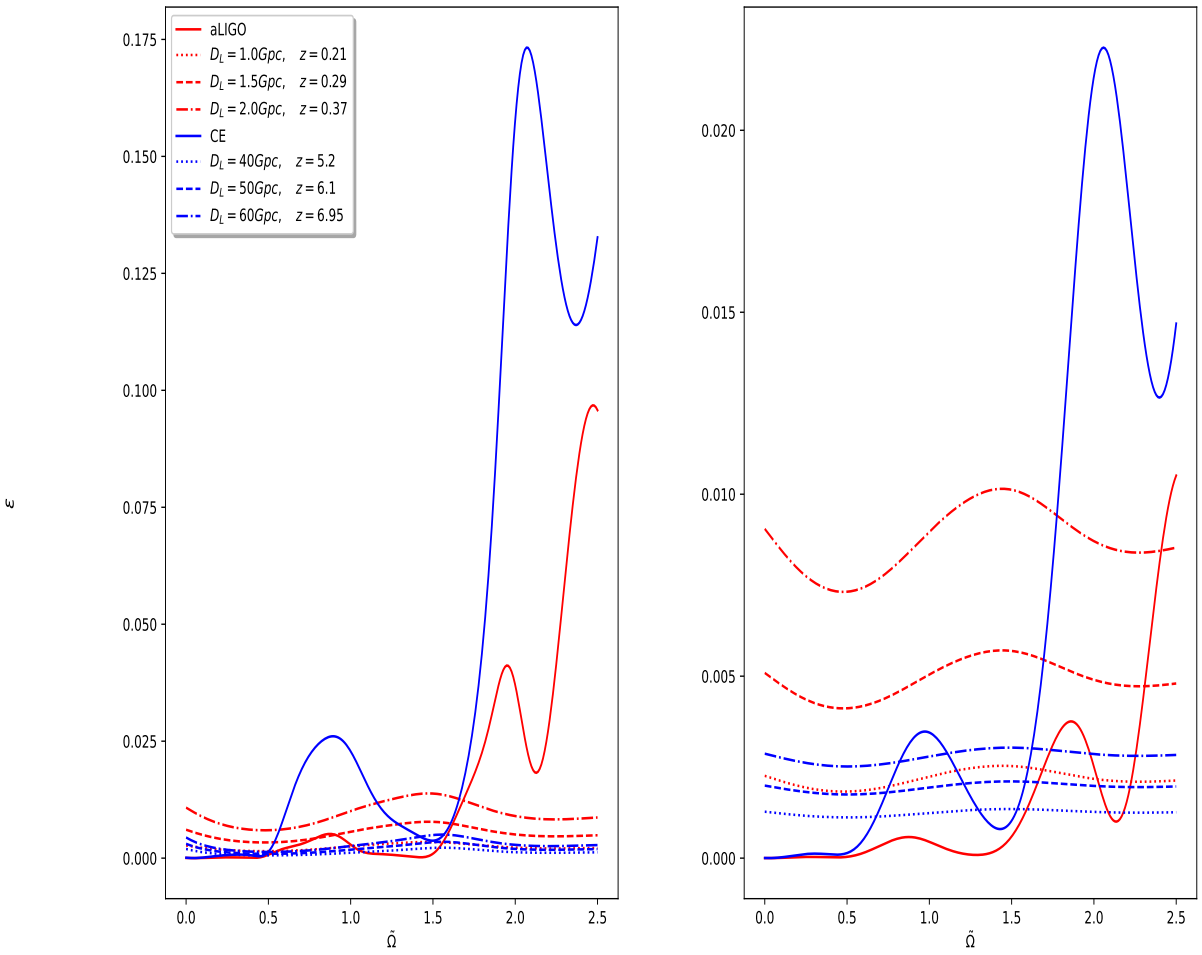


Figure 5.10. The evolution of the mismatch ϵ given by Eq. (5.27) between a regularly precessing(RP) GW source and a non-precessing(NP) GW template with respect to the normalized precession frequency $\tilde{\Omega}$. The direction of the total angular momentum $\hat{\boldsymbol{J}}$ is listed as System 2 in Table 5.1 and the direction of the line of sight $\hat{\boldsymbol{N}}$ is specified by $(\theta_S, \Phi_S) = (\pi/4, 0)$ and for the chirp mass we have chosen $\mathcal{M} = 20M_\odot$. The blue(red) solid line denote that we are using the PSD of the Cosmic Explorer (aLIGO) detector for the calculation of our mismatch. The BBHs enter the Cosmic Explorer (aLIGO) sensitivity band at a GW frequency of 10Hz (20Hz). The solid lines have a value of $\tilde{\theta} = 8$ ($\tilde{\theta} = 1$) for the left (right) panel and a fixed $\gamma_p = 0$. The dot-dashed, dashed, and dotted blue horizontal lines show the value of $1/2\rho^2$ for a different choice of luminosity distance D_L. The different choices are shown in the legend.

is 20Hz for aLIGO and 10Hz for CE. Finally, we can see that our new proposed regular precessing waveform can be used to distinguish precession not only for BBH systems lying in the aLIGO sensitivity band to close distances but also it is our hope that it can be employed for future detectors to distinguish precessing system up to very high redshift.

5.5 Discussion

During the inspiral part of a BBH coalescence, if the individual spins of the BHs that comprise a BBH system are non-zero and misaligned with one another and the angular momentum $\hat{\boldsymbol{L}}$, they cause the orbital plane to precess and nutate. These spin-orbit and spin-spin coupling effects are the reason behind the introduction of modulations to the GW template's amplitude and phase at various PN orders. These modulations have to be modelled as accurately as possible in order to extract the necessary information concerning the parameters that describe the BBH system in the current LVK runs as well as the next 3G detectors (Pürrer and Haster, 2020). In our previous study (Gangardt et al., 2021), we have introduced a taxonomy that incorporates all the different cases that can cause precession and nutation to happen, came up with 5 novel phenomenological parameters in the inspiral PN regime, and proposed that these parameters can model precession and nutation effects inside a GW template.

In this Chapter, we investigate this proposal by constructing a regular precessing GW template, described by nine parameters shared with the non-precessing case, seven extrinsic ones: the source's polar and azimuthal angles (θ_S, Φ_S) that fully describe the line of sight vector, the angles that define the total angular momentum in the detector frame (θ_J, Φ_J), the luminosity distance D_L, the time and phase of coalescence t_c, ϕ_c and two intrinsic ones: the chirp mass $\mathcal{M}$ and the symmetric mass ratio η, plus three of our novel parameters: two intrinsic ones; the precession amplitude $\langle\theta_{LJ}\rangle$ and the precession frequency $\langle\Omega_{LJ}\rangle$ given in Eqs. (5.18a) and (5.18b) respectively, and one extrinsic one; the precession integration

constant γ_p which corresponds to a global rotation of our BBH system about the total angular momentum $\hat{\boldsymbol{J}}$. The introduction of these three parameters inside the GW template also introduces a frequency dependence to the GW template's amplitude in Eq. (5.10) through its terms, $\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}}$, F_+, $F_\times$ and its phase via two terms, ϕ_p, $2\delta\Phi$ from Eqs. (5.11) and (5.12). Fig. 5.2 shows the imprint of precession on the spherical coordinates θ_L and Φ_L that specify the direction of orbital angular momentum $\boldsymbol{L}$ with respect to the detector frame from Eq. (A.7), while Fig. 5.4 shows the evolution of the GW amplitude B with respect to the GW frequency through the three new precession parameters. Furthermore, Figs. 5.6 and 5.7 show the same exact evolution but this time of the GW phase in the regular precessing regime through the inclusion of our three new parameters.

We have mentioned before, that in our previous study (Gangardt et al., 2021) we presented a new set of parameters that geometrically characterize the precession and nutation of the orbital angular momentum $\boldsymbol{L}$ with respect to the total angular momentum $\boldsymbol{J}$: 1) the precession amplitude $\langle\theta_{LJ}\rangle$, (2) the precession frequency $\langle\Omega_{LJ}\rangle$, (3) the nutation amplitude $\Delta\theta_{LJ} = (\theta_{LJ+} - \theta_{LJ-})/2$, (4) the nutation frequency $\omega = 2\pi/\tau$, and (5) the variation of the precession frequency $\Delta\Omega_{LJ} = [\Omega_{LJ}(S_+) - \theta_{LJ}(S_-)]/2$. Our numerical investigation into these five parameters demonstrates that the nutation amplitude is largest for BBHs with moderate mass ratios $q \approx 0.6$, high spin magnitudes $\chi_i \gtrsim 0.5$, and spin orientations for which $\boldsymbol{J} \parallel \boldsymbol{L}$ at some point in the inspiral. GW events from BBHs satisfying these conditions might provide an optimal perspective to distinguish the effects of precession and nutation, and to observationally constrain our five parameters. Foundational preliminary investigations into this was the subject of (Gangardt et al., 2022), where the five precesssion/nutation parameters in the 90 O3 events were constrained some constraints on precession were found, but none on nutation. As we have shown especially in Fig. 5.10 future GW detectors will uncover a much larger number of BBH systems at higher signal-to-noise ratios and redshift, and the effects of precession and nutation will be detected unambiguously. We hope that our new

regular precessing GW template will aid in the characterization of these systems, and help push the frontiers of GW astronomy.

Finally, it is well established theoretically that BBHs can form through two main formation channels (Mapelli, 2020), either the dynamical or isolated channel. Theoretical predictions from the dynamic channel indicate randomly distributed spins, while even though the isolated channel is expected to produce BBHs with aligned spins, studies have shown that there can exist special cases of spin misalignment (Kalogera, 2000; Gerosa et al., 2018). Detailed simulations of dense stellar clusters indicate that BBHs from the dynamical channel generally form with isotropically oriented spins (Benacquista and Downing, 2013). Meanwhile, binary population synthesis models of BBHs from the isolated channel show that their spins are likely to be mostly aligned with the binary orbital angular momentum as natal kicks at birth may provide misalignment (Postnov and Yungelson, 2014). Both formation channels are subject to various astrophysical uncertainties, implying that GW detections and especially the accurate modelling of the spin effects are important for probing the prevalence and details of these formation channels. M. Kesden and N. Steinle have already completed preliminary work on this subject by using our regular precessing parameters (Steinle and Kesden, 2022), which specifically showed how these parameters depend on binary stellar evolution in the isolated channel.

CHAPTER 6

CONCLUSIONS

Just like electromagnetic (EM) waves, gravitational waves (GWs) can be gravitationally lensed by an intervening massive object, such as a compact object, galaxy, or galaxy cluster, acting as a lens. While the theory behind the lensing of GWs is similar to that of EM lensing, the methods to detect lensing signatures are entirely different due to fundamentally different sources and detectors. The first difference is in the applicability of the geometrical-optics approximation, which is typically valid for the vast majority of EM lensing observations since the wavelengths are much smaller than the Schwarzschild radius of the lens. However, it is only sometimes valid for GWs since ground-based detectors make observations at much lower frequencies $(10 - 10^4)$ Hz, leading to non-negligible wave-optics effects. The second difference is that GWs emitted by compact binaries, unlike most EM sources, are coherent; this is predicted to cause distinctive interference between lensed images when the signals overlap at the observer (Schneider et al., 1992; Takahashi and Nakamura, 2003). In its current state, the Advanced LIGO-VIRGO-KAGRA (LVK) (Aasi et al., 2015; Acernese et al., 2015; Somiya, 2012; Aso et al., 2013; Akutsu et al., 2021) network cannot detect the gravitational lensing of GW. However, with the increased sensitivity of the ground-based detector network and future third-generation (3G) detectors such as the Cosmic Explorer (CE) (Evans et al., 2021), the Einstein Telescope (ET) (Maggiore et al., 2020), the Deci-Hertz Interferometer Gravitational Wave Observatory (DECIGO) (Kawamura et al., 2021), and the Laser Interferometer Space Antenna (LISA) (Barausse et al., 2020), the number of observed GW events will increase dramatically, as will the probability of observing lensing of GW. Detection of GW lensing could facilitate several scientific studies, such as probing the dark matter in galaxies acting as lenses (Tambalo et al., 2022), testing general relativity (GR) through constraints from GW polarization content (Goyal et al., 2021), and localizing the host galaxy of the GW event to sub-arcsecond precision (Hannuksela et al., 2020).

The lensed waveform is given by the product of the lensing amplification factor F and the unlensed waveform. For many axisymmetric lens models, such as the point mass and singular isothermal sphere that we consider, F can be calculated in terms of two lens parameters, the lens mass M_L and source position y. In the geometrical-optics approximation, lensing in these models produces at most two discrete images which can be parameterized by two image parameters, the flux ratio I and time delay Δt_d between images (Ali et al., 2022). In the macrolensing regime for which Δt_d is large compared to the time T they spend within the sensitivity band of GW detectors, it is natural to parameterize lensing searches in terms of these image parameters. The functional dependence of the lensed signal on these image parameters is far simpler, facilitating data analysis for events with modest signal-to-noise ratios, and constraints on I and Δt_d can be inverted to constrain M_L and y for any lens model. In the thesis we show that this use of image parameters can be extended to the microlensing regime ($\Delta t_d < T$) in which the two interfering images are observed as a single GW event. We use image parameters to determine the detectability of gravitational lensing in GW the microlensing regime and find that for GW events with signal-to-noise ratios ρ and total mass M, lensing should in principle be identifiable for flux ratios $I \gtrsim 2\rho^{-2}$ and time delays $\Delta t_d \gtrsim M^{-1}$ (Ali et al., 2022).

The analysis of GW lensing with the axisymmetric lens models is relatively straightforward compared to the non-axisymmetric lens. However, it is essential to employ non-axisymmetric lens models, such as the singular isothermal ellipsoid (SIE) lens, to understand the lensing of GWs in a more realistic setting. The use of elliptical mass distributions is motivated by observational constraints on the shapes of galaxies and their dark matter halos as well as the existence of EM strong lensing configurations with more than 2 images (Ohanian, 1983; Turner et al., 1984; Blandford and Narayan, 1986). An SIE can produce four images when the source and lens are well aligned in projection. The cross-section for producing at least two images depends on the velocity dispersion (or mass) of the lens, and the four-image

cross-section depends on the lens ellipticity (Schneider et al., 1992). From the perspective of GW lensing, one of the challenges with elliptical potentials is that, for a particular source position, the locations of the images on the image plane can lead to a non-negligible diffraction effect, and the geometrical-optics approximation becomes invalid. Therefore, following (Takahashi, 2004), the quasi-geometrical optics approximation (QGO) is employed in the two-image and four-image regime. Using QGO, we calculate the lower bound on the lens mass, $M_{Lz(min)}$, below which geometrical-optics approximation breaks down. We analyze $M_{Lz(min)}$ as a function of source position y for cusp and fold image configurations of the SIE lens. In the two-image regime, both images can satisfy the geometrical-optics approximation if the lens mass is above $M_{Lz(min)} \sim 10^5 M_\odot$ for $y \sim 0.2 - 0.9$. In the four-image regime, the $M_{Lz(min)} \sim 10^5 M_\odot$, similar to the two-image regime, however the range of y varies with the image configurations. We also perform match-filtering analysis between GW sources (lensed by an SIE, producing four images) and (i) unlensed templates and (ii) two types of two-image lensed templates, where the images have the same or different Morse index. We use image parameters to derive the analytical expressions to explain the mismatch behaviors for different choices of templates.

Binary black hole (BBH) systems comprise two black holes (BHs) closely orbiting each other. The spins of the individual BHs are expected to have non-zero values and to be misaligned with the orbital angular momentum leading to the precessional and nutational effects. These effects induce modulations in the amplitude and phase of the GWs. These effects are parameterized by two spin parameters, $\chi_{\rm eff}$, $\chi_{\rm p}$ (The LIGO Scientific Collaboration et al., 2021). Gravitational waveform templates constructed by employing these two spin parameters are used in parameter estimation analysis. In a recent study, claims have been made that precession has been detected in the GW200129 event observed during the O3 run of LIGO-Virgo collaboration (Hannam et al., 2022). However, the issue of data quality regarding this particular GW event makes it a 'curious' case (Payne et al., 2022) and a

consensus has not been reached regarding the detection of the precession effect. In this study, we quantify the precessional and nutational effects by introducing five new phenomenological parameters. These parameters are expected to provide a comprehensive understanding of the amplitude and frequency modulation, especially from the geometrical perspective. We also analyse the mismatch between the templates to determine the minimum signal-to-noise ratio needed to detect the precession for upcoming observing runs of LIGO and proposed third generational future detectors.

The work presented in this thesis is essential for several other investigations. One of them could be combining the lensing observations of GWs and EMs. In the era of multi-messenger astrophysics, GW and EM lensing observations will complement each other and allow us to learn more from joint observations than from either one individually. GW detectors, essentially working as antennas, have an excellent temporal resolution, but lack the spatial resolution required to distinguish the multiple images that form in the geometric-optics limit. On the other hand, EM telescopes have excellent spatial resolution allowing them to measure flux ratios, positions, and parity of images. Combining EM observations of lens systems with associated GW lensing could provide much new information. EM data could include, e.g., radio, infrared, optical or high-energy observations of lensed host galaxies, lensed transient EM counterparts and foreground lenses. Multi-messenger observations promise to: (1) localize binary mergers with sub-arcsecond precision (Yu et al., 2020), (2) test GR by measuring the speed and polarization content of GWs (Baker and Trodden, 2017; Collett and Bacon, 2017; Mukherjee et al., 2020), (3) infer lens and source properties (Lai et al., 2018; Diego, 2020; Oguri and Takahashi, 2020), and (4) constrain the Hubble constant by combining time delays of GW and EM lensing (Treu and Marshall, 2016).

The precession of a compact binary source is scientifically important; it modulates the amplitude of the GW polarizations, adding a periodic structure that encodes valuable information about the binary's members. In the geometrical-optics approximation, gravitational

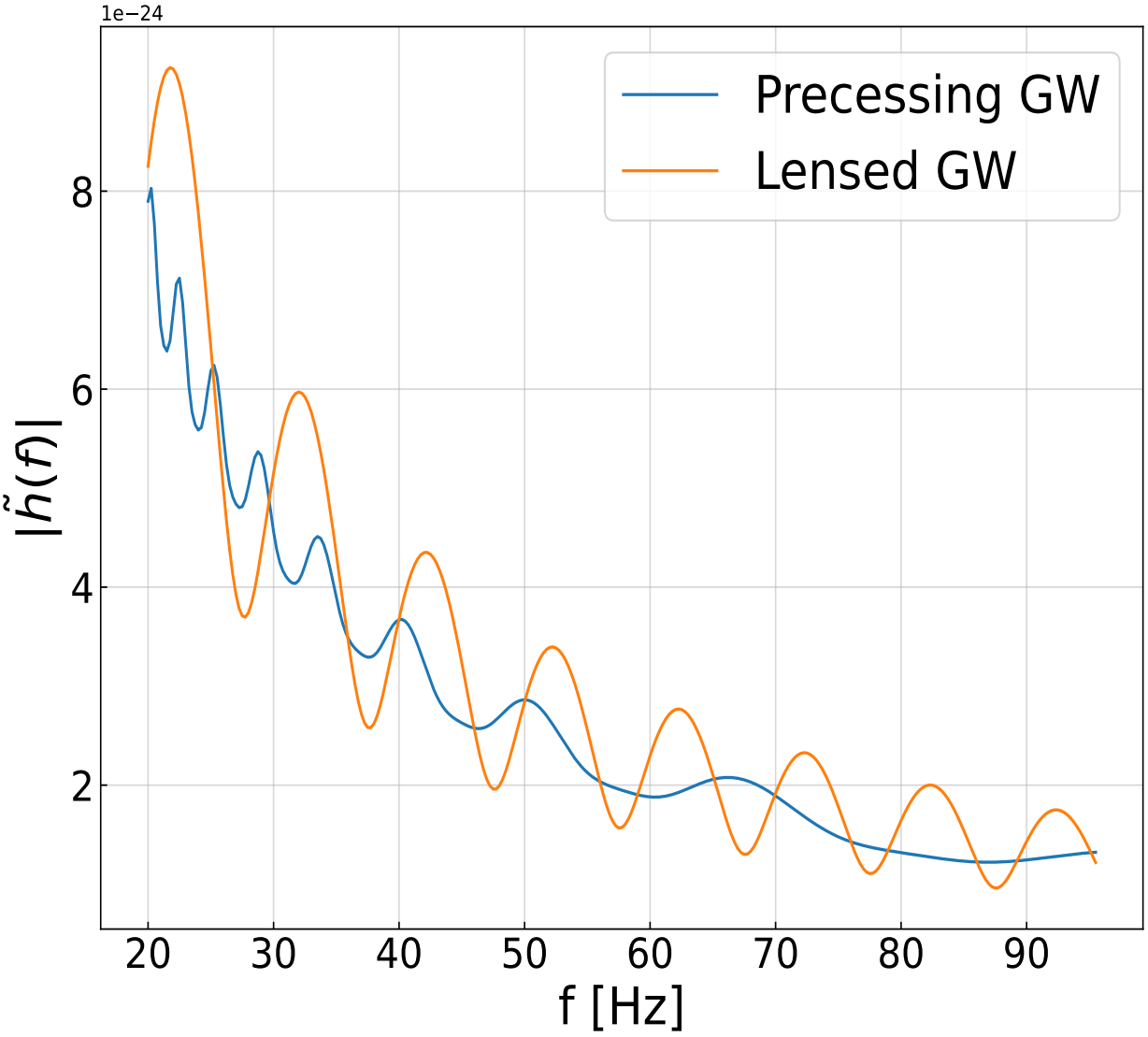


Figure 6.1. Lensed gravitational waveform (orange) constructed for Point-Mass lens with fiducial lens parameters $y = 1.2$ and $M_{Lz} = 2 \times 10^3 M_\odot$. Precessing gravitational waveform (blue) constructed with fiducial precessing parameters $\tilde{\theta} = 2.0$, $\tilde{\Omega} = 10.0$, and $\gamma_p = 0.8$.

lensing also introduces modulations in the waveform. However, the modulations imprinted on the GW by strong lensing and precession are quantitatively different as shown in Fig. 6.1. In the scenario where we simultaneously observe these effects, an intriguing question is: What is the SNR needed to distinguish between lensing, precessional, or combined modulation? It will be worthwhile to investigate the SNR as a function of source and lens parameters and for various detectors. For example, it will be crucial to distinguish lensed and precessing GWs in the low-frequency regime where the wave-optics effect is dominant. The understanding of this regime is essential for detectors like LISA, which is sensitive in the low-frequency band $(10^{-4} - 10^{-1})$ Hz.

The current estimates of the number of GW sources and the distribution of lenses in the universe predict a non-trivial chance of detecting gravitationally lensed GW in the O4 run of the LVK network. This number will be "astronomically" high when the 3G detectors come online, and with this GW astrophysics will enter the *big data* era. The current template-based search methods to identify lensed GWs is time-consuming and computationally intensive as it requires processing billions of pairs of GW events. Therefore it is important to explore alternatives to traditional template-based searches, and one of the candidates is Deep Learning or Neural Networks (Kim et al., 2022). Convolutional Neural Networks (CNN), a particular class of neural network, is proven to be a robust algorithm for image classification by recognizing patterns in images. One can employ several deep-learning techniques to explore the detection and parameter estimation of lensed GW events.

APPENDIX A

THREE REFERENCE FRAMES

We use three distinct reference frames to calculate our regularly precessing waveforms.

A.1 Detector frame

The *detector frame* is chosen such that its basis vectors $\hat{\boldsymbol{X}}_D$ and $\hat{\boldsymbol{Y}}_D$ point along the arms of the interferometer, and $\hat{\boldsymbol{Z}}_D = \hat{\boldsymbol{X}}_D \times \hat{\boldsymbol{Y}}_D$ points towards the zenith at the detector location. The sky location of the binary $\hat{\boldsymbol{N}}$ and the direction of the total angular momentum $\hat{\boldsymbol{J}}$ are specified respectively by the angles (θ_S, Φ_S) and (θ_J, Φ_J) with respect to this frame:

$$\hat{\boldsymbol{N}} = \sin\theta_S(\cos\Phi_S\hat{\boldsymbol{X}}_D + \sin\Phi_S\hat{\boldsymbol{Y}}_D) + \cos\theta_S\hat{\boldsymbol{Z}}_D\,, \tag{A.1a}$$

$$\hat{\boldsymbol{J}} = \sin\theta_J(\cos\Phi_J\hat{\boldsymbol{X}}_D + \sin\Phi_J\hat{\boldsymbol{Y}}_D) + \cos\theta_J\hat{\boldsymbol{Z}}_D\,. \tag{A.1b}$$

A.2 Sky frame

The *sky frame* is chosen such that $\hat{\boldsymbol{N}}$ points into the plane of the sky (opposite to the direction of GW propagation), $\hat{\boldsymbol{V}}$ points towards the zenith, and $\hat{\boldsymbol{H}} = \hat{\boldsymbol{V}} \times \hat{\boldsymbol{N}}$ is parallel to the horizon:

$$\hat{\boldsymbol{H}} = -\sin\Phi_S\hat{\boldsymbol{X}}_D + \cos\Phi_S\hat{\boldsymbol{Y}}_D\,, \tag{A.2a}$$

$$\hat{\boldsymbol{V}} = -\cos\theta_S(\cos\Phi_S\hat{\boldsymbol{X}}_D + \sin\Phi_S\hat{\boldsymbol{Y}}_D) + \sin\theta_S\hat{\boldsymbol{Z}}_D\,. \tag{A.2b}$$

A.3 Source frame

The *source frame* is chosen such that $\hat{\boldsymbol{Z}}_S$ points along the direction of the total angular momentum $\hat{\boldsymbol{J}}$,

$$\hat{\boldsymbol{X}}_S = \frac{\hat{\boldsymbol{N}} \times \hat{\boldsymbol{J}}}{|\hat{\boldsymbol{N}} \times \hat{\boldsymbol{J}}|} \tag{A.3}$$

points along the line of ascending node of the plane perpendicular to $\boldsymbol{J}$ with respect to the sky plane spanned by $\hat{\boldsymbol{H}}$ and $\hat{\boldsymbol{V}}$, and $\hat{\boldsymbol{Y}}_S = \hat{\boldsymbol{Z}}_S \times \hat{\boldsymbol{X}}_S$. We define the inclination ι_{JN} as the angle between $\hat{\boldsymbol{J}}$ and $\hat{\boldsymbol{N}}$,

$$\cos\iota_{JN} = \hat{\boldsymbol{J}}\cdot\hat{\boldsymbol{N}} = \sin\theta_J\sin\theta_S\cos(\Phi_J-\Phi_S) + \cos\theta_J\cos\theta_S\,, \tag{A.4}$$

and the longitude of ascending node Ω_{XH} such that the source and sky frames are related by

$$\hat{\boldsymbol{X}}_S = \cos\Omega_{XH}\hat{\boldsymbol{H}} + \sin\Omega_{XH}\hat{\boldsymbol{V}}\,, \tag{A.5a}$$

$$\hat{\mathbf{Y}}_S = \cos\iota_{JN}(-\sin\Omega_{XH}\hat{\mathbf{H}} + \cos\Omega_{XH}\hat{\mathbf{V}}) + \sin\iota_{JN}\hat{\mathbf{N}}\,, \tag{A.5b}$$

$$\hat{\mathbf{Z}}_S = \sin\iota_{JN}(\sin\Omega_{XH}\hat{\mathbf{H}} - \cos\Omega_{XH}\hat{\mathbf{V}}) + \cos\iota_{JN}\hat{\mathbf{N}}\,. \tag{A.5c}$$

In the source and sky frames, the direction of the orbital angular momentum $\hat{\boldsymbol{L}}$ is given by

$$\hat{\boldsymbol{L}} = \sin\theta_{LJ}[\cos\Phi_{LJ}\hat{\boldsymbol{X}}_S + \sin\Phi_{LJ}\hat{\boldsymbol{Y}}_S] + \cos\theta_{LJ}\hat{\boldsymbol{Z}}_S\,, \tag{A.6a}$$

$$\begin{aligned}
&= [\sin\theta_{LJ}(\cos\Phi_{LJ}\cos\Omega_{XH} - \sin\Phi_{LJ}\cos\iota_{JN}\sin\Omega_{XH}) + \sin\iota_{JN}\sin\Omega_{XH}\cos\theta_{LJ}]\hat{\mathbf{H}} \\
&\quad + [\sin\theta_{LJ}(\cos\Phi_{LJ}\sin\Omega_{XH} + \sin\Phi_{LJ}\cos\iota_{JN}\cos\Omega_{XH}) - \sin\iota_{JN}\cos\Omega_{XH}\cos\theta_{LJ}]\hat{\mathbf{V}} \\
&\quad + (\sin\theta_{LJ}\sin\Phi_{LJ}\sin\iota_{JN} + \cos\theta_{LJ}\cos\iota_{JN})\hat{\mathbf{N}}\,.
\end{aligned} \tag{A.6b}$$

The source and sky frames are depicted in green and red respectively in Fig. 5.1. In our regularly precessing waveform model, only the angles Φ_{LJ} and θ_{LJ} are time dependent, and they vary on the precession and radiation-reaction timescales respectively according to

Eqs. (5.19) and (5.18a). Finally, $\hat{\boldsymbol{L}}$ in the detector frame is expressed

$$\begin{aligned}\hat{\mathbf{L}} = [&- \sin\theta_{LJ}\cos\Phi_{LJ}\sin\Omega_E\cos\theta_s\cos\Phi_s - \sin\theta_{LJ}\cos\Phi_{LJ}\cos\Omega_E\sin\Phi_s \\ &+ \sin\theta_{LJ}\sin\Phi_{LJ}\cos\iota_E\sin\Omega_E\sin\Phi_s - \sin\theta_{LJ}\sin\Phi_{LJ}\cos\iota_E\cos\Omega_E\cos\theta_s\cos\Phi_s \\ &+ \sin\theta_{LJ}\sin\Phi_{LJ}\sin\iota_E\sin\theta_s\cos\Phi_s + \cos\theta_{LJ}\sin\theta_J\cos\Phi_J]\hat{\mathbf{X}}_D \\ &+ [\sin\theta_{LJ}\cos\Phi_{LJ}\cos\Omega_E\cos\Phi_s - \sin\theta_{LJ}\cos\Phi_{LJ}\sin\Omega_E\cos\theta_s\sin\Phi_s \\ &- \sin\theta_{LJ}\sin\Phi_{LJ}\cos\iota_E\sin\Omega_E\cos\Phi_s - \sin\theta_{LJ}\sin\Phi_{LJ}\cos\iota_E\cos\Omega_E\cos\theta_s\sin\Phi_s \\ &+ \sin\theta_{LJ}\sin\Phi_{LJ}\sin\iota_E\sin\theta_s\sin\Phi_s + \cos\theta_{LJ}\sin\theta_J\sin\Phi_J]\hat{\mathbf{Y}}_D \\ &+ [\sin\theta_{LJ}\cos\Phi_{LJ}\sin\Omega_E\sin\theta_s + \sin\theta_{LJ}\sin\Phi_{LJ}\cos\iota_E\cos\Omega_E\sin\theta_s \\ &+ \sin\theta_{LJ}\sin\Phi_{LJ}\sin\iota_E\cos\theta_s + \cos\theta_{LJ}\cos\theta_J]\hat{\mathbf{Z}}_D\end{aligned} \tag{A.7}$$

A.4 GW strain

The observed GW strain given by Eq. (5.1) depends on the relative orientation of these three frames through the GW amplitude B of Eq. (5.10) and the detector beam-pattern coefficients F_+ and $F_\times$ of Eqs. (5.2a) and (5.2b). The latter depend on the polarization angle ψ of Eq. (5.3). The scalar products in these expressions are given in terms of the above angles as

$$\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{N}} = \sin\theta_{LJ}\sin\Phi_{LJ}\sin\iota_{JN} + \cos\theta_{LJ}\cos\iota_{JN}\,, \tag{A.8a}$$

$$\begin{aligned}\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{Z}}_D = &\sin\theta_{LJ}[\cos\Phi_{LJ}\sin\Omega_{XH}\sin\theta_S + \sin\Phi_{LJ}(\cos\iota_{JN}\cos\Omega_{XH}\sin\theta_S) \\ &+ \sin\iota_{JN}\cos\theta_S)] \\ &+ \cos\theta_{LJ}(\cos\iota_{JN}\cos\theta_S - \sin\iota_{JN}\cos\Omega_{XH}\sin\theta_S)\,,\end{aligned} \tag{A.8b}$$

$$\hat{\boldsymbol{N}}\cdot\hat{\boldsymbol{Z}}_D = \cos\theta_S\,, \tag{A.8c}$$

and the triple product is

$$\begin{aligned}\hat{\boldsymbol{N}} \cdot (\hat{\boldsymbol{L}} \times \hat{\boldsymbol{Z}}_D) &= \hat{\boldsymbol{Z}}_D \cdot (\hat{\boldsymbol{N}} \times \hat{\boldsymbol{L}}) \\ &= [\sin\theta_{LJ}(\cos\Phi_{LJ}\cos\Omega_{XH} - \sin\Phi_{LJ}\cos\iota_{JN}\sin\Omega_{XH}) \\ &\quad + \sin\iota_{JN}\sin\Omega_{XH}\cos\theta_{LJ}]\sin\theta_S\,. \end{aligned} \tag{A.9a}$$

APPENDIX B

DERIVATION OF THE PRECESSION PARAMETERS

The precession parameters $\langle\theta_{LJ}\rangle$ and $\langle\Omega_{LJ}\rangle$ however are only constant on the precession timescale, but the angles θ_{LJ} and Φ_{LJ} must be specified on the radiation-reaction timescale as the BBH inspirals through the sensitivity band of the GW detector. For simplicity, we assume that these parameters maintain their lowest PN order frequency dependence as given by Eqs. (9) and (10) of (Gangardt et al., 2021) all the way to merger. First, we use the multi-timescale analysis to express Eq. (5) in (Gangardt et al., 2021) in terms of the precession timescale:

$$\langle\Omega_{LJ}\rangle \propto t_{pre}^{-1}\,, \tag{B.1}$$

Now Ω_{LJ} can be expressed as a function of the GW frequency

$$M\langle\Omega_{LJ}\rangle \propto \left(\frac{r}{M}\right)^{-5/2} \propto x^{5/2}\,, \tag{B.2}$$

$$\langle\Omega_{LJ}\rangle \propto \pi^{5/3} f^{5/3} M^{2/3}\,, \tag{B.3}$$

To make things even more compact, we can use the f_{cut} from Eq. (2.30) and the fact that $M_\odot \sim 5\mu s$:

$$\langle\Omega_{LJ}\rangle = 10^3\,\mathrm{Hz}\;\tilde{\Omega}\left(\frac{f}{f_{\rm cut}}\right)^{5/3}\left(\frac{M}{M_\odot}\right)^{-1}\,, \tag{B.4}$$

Now, for the precession amplitude, if we take a look at Fig 5.1, we get that

$$\langle\theta_{LJ}\rangle \sim \frac{S}{L}\,, \tag{B.5}$$

where

$$S \sim \chi M^2\,, \tag{B.6}$$

$$L \sim \eta(M^3 r)^{1/2} \sim \eta M^2 x^{-1/2}\,, \tag{B.7}$$

Combining Eqs. (B.6), (B.7), (B.5)

$$\langle \theta_{LJ} \rangle = 0.1\tilde{\theta} \left(\frac{f}{f_{\rm cut}} \right)^{1/3} , \tag{B.8}$$

where the normalizations were chosen such that the dimensionless precession amplitude $\tilde{\theta}$ and dimensionless precession frequency $\tilde{\Omega}$ are constants of order unity.

REFERENCES

Aasi, J. et al. (2014). The NINJA-2 project: Detecting and characterizing gravitational waveforms modelled using numerical binary black hole simulations. *Class. Quant. Grav. 31*, 115004.

Aasi, J. et al. (2015). Advanced LIGO. *Class. Quant. Grav. 32*, 074001.

Abbott, B., R. Abbott, T. Abbott, M. Abernathy, F. Acernese, K. Ackley, C. Adams, T. Adams, P. Addesso, R. Adhikari, and et al. (2016, Feb). Observation of gravitational waves from a binary black hole merger. *Physical Review Letters 116*(6).

Abbott, B., R. Abbott, T. Abbott, F. Acernese, K. Ackley, C. Adams, T. Adams, P. Addesso, R. Adhikari, V. Adya, and et al. (2017, Oct). Gw170817: Observation of gravitational waves from a binary neutron star inspiral. *Physical Review Letters 119*(16).

Abbott, R., T. Abbott, F. Acernese, K. Ackley, C. Adams, N. Adhikari, R. Adhikari, V. Adya, C. Affeldt, D. Agarwal, et al. (2021). Gwtc-3: compact binary coalescences observed by ligo and virgo during the second part of the third observing run. *arXiv preprint arXiv:2111.03606*.

Abbott, R., T. D. Abbott, S. Abraham, F. Acernese, K. Ackley, A. Adams, C. Adams, R. X. Adhikari, V. B. Adya, C. Affeldt, and et al. (2021, Jun). Observation of gravitational waves from two neutron star–black hole coalescences. *The Astrophysical Journal Letters 915*(1), L5.

Abbott, R., T. D. Abbott, S. Abraham, F. Acernese, K. Ackley, A. Adams, C. Adams, R. X. Adhikari, LIGO Scientific Collaboration, and Virgo Collaboration (2021, December). Search for Lensing Signatures in the Gravitational-Wave Observations from the First Half of LIGO-Virgo's Third Observing Run. *923*(1), 14.

Acernese, F. et al. (2015). Advanced Virgo: a second-generation interferometric gravitational wave detector. *Class. Quant. Grav. 32*(2), 024001.

Adams, T., D. Buskulic, V. Germain, G. M. Guidi, F. Marion, M. Montani, B. Mours, F. Piergiovanni, and G. Wang (2016, September). Low-latency analysis pipeline for compact binary coalescences in the advanced gravitational wave detector era. *Classical and Quantum Gravity 33*(17), 175012.

Akutsu, T. et al. (2021). Overview of KAGRA: Detector design and construction history. *PTEP 2021*(5), 05A101.

Ali, S., E. Stoikos, M. Kesden, and L. King (2023). Identifying multiple images of gravitational-wave sources lensed by elliptical lensing potentials.

Ali, S., E. Stoikos, E. Meade, M. Kesden, and L. King (2022, October). Detectability of strongly lensed gravitational waves using model-independent image parameters. *arXiv e-prints*, arXiv:2210.01873.

Apostolatos, T. A., C. Cutler, G. J. Sussman, and K. S. Thorne (1994, Jun). Spin-induced orbital precession and its modulation of the gravitational waveforms from merging binaries. *Phys. Rev. D 49*, 6274–6297.

Asada, H., T. Hamana, and M. Kasai (2003, January). Images for an isothermal ellipsoidal gravitational lens from a single real algebraic equation. *397*, 825–829.

Aso, Y., Y. Michimura, K. Somiya, M. Ando, O. Miyakawa, T. Sekiguchi, D. Tatsumi, and H. Yamamoto (2013, Aug). Interferometer design of the kagra gravitational wave detector. *Phys. Rev. D 88*, 043007.

Aubin, F., F. Brighenti, R. Chierici, D. Estevez, G. Greco, G. M. Guidi, V. Juste, F. Marion, B. Mours, E. Nitoglia, O. Sauter, and V. Sordini (2021, May). The MBTA pipeline for detecting compact binary coalescences in the third LIGO-Virgo observing run. *Classical and Quantum Gravity 38*(9), 095004.

Baker, T. and M. Trodden (2017). Multimessenger time delays from lensed gravitational waves. *Phys. Rev. D 95*(6), 063512.

Barausse, E. et al. (2020). Prospects for Fundamental Physics with LISA. *Gen. Rel. Grav. 52*(8), 81.

Bartelmann, M. (2010, December). TOPICAL REVIEW Gravitational lensing. *Classical and Quantum Gravity 27*(23), 233001.

Basak, S., A. Ganguly, K. Haris, S. Kapadia, A. K. Mehta, and P. Ajith (2022, February). Constraints on Compact Dark Matter from Gravitational Wave Microlensing. *926*(2), L28.

Benacquista, M. J. and J. M. B. Downing (2013, March). Relativistic Binaries in Globular Clusters. *Living Reviews in Relativity 16*(1), 4.

Bildsten, L. and C. Cutler (1992, November). Tidal Interactions of Inspiraling Compact Binaries. *APJ 400*, 175.

Blanchet, L., S. Detweiler, A. Le Tiec, and B. F. Whiting (2009). High-accuracy comparison between the post-newtonian and self-force dynamics of black-hole binaries. *Mass and Motion in General Relativity*, 415–442.

Blandford, R. and R. Narayan (1986, November). Fermat's Principle, Caustics, and the Classification of Gravitational Lens Images. *310*, 568.

Blandford, R. D. and C. S. Kochanek (1987, October). Gravitational Imaging by Isolated Elliptical Potential Wells. I. Cross Sections. *321*, 658.

Bondarescu, R., H. Ubach, O. Bulashenko, and A. P. Lundgren (2022, November). Compact Binaries through a Lens: Silent vs. Detectable Microlensing for the LIGO-Virgo-KAGRA Gravitational Wave Observatories. *arXiv e-prints*, arXiv:2211.13604.

Bontz, R. J. and M. Haugan (1981). A diffraction limit on the gravitational lens effect. *Astrophysics and Space Science 78*, 199–210.

Bourassa, R. R. and R. Kantowski (1975, January). The theory of transparent gravitational lenses. *195*, 13–21.

Bourassa, R. R., R. Kantowski, and T. D. Norton (1973, November). The Spheroidal Gravitational Lens. *185*, 747–756.

Boyle, M. et al. (2019). The SXS Collaboration catalog of binary black hole simulations. *Class. Quant. Grav. 36*(19), 195006.

Bulashenko, O. and H. Ubach (2021, December). Lensing of gravitational waves: universal signatures in the beating pattern. *arXiv e-prints*, arXiv:2112.10773.

Bullock, J. S. and M. Boylan-Kolchin (2017, August). Small-Scale Challenges to the ΛCDM Paradigm. *55*(1), 343–387.

Buonanno, A. and T. Damour (1999, April). Effective one-body approach to general relativistic two-body dynamics. *59*(8), 084006.

Campanelli, M., C. O. Lousto, P. Marronetti, and Y. Zlochower (2006, March). Accurate Evolutions of Orbiting Black-Hole Binaries without Excision. *96*(11), 111101.

Cannon, K., S. Caudill, C. Chan, B. Cousins, J. D. E. Creighton, B. Ewing, H. Fong, P. Godwin, C. Hanna, S. Hooper, R. Huxford, R. Magee, D. Meacher, C. Messick, S. Morisaki, D. Mukherjee, H. Ohta, A. Pace, S. Privitera, I. de Ruiter, S. Sachdev, L. Singer, D. Singh, R. Tapia, L. Tsukada, D. Tsuna, T. Tsutsui, K. Ueno, A. Viets, L. Wade, and M. Wade (2021, June). GstLAL: A software framework for gravitational wave discovery. *SoftwareX 14*, 100680.

Cao, S., J. Qi, Z. Cao, M. Biesiada, W. Cheng, and Z.-H. Zhu (2022, March). Direct measurement of the distribution of dark matter with strongly lensed gravitational waves. *659*, L5.

Cao, S., J. Qi, Z. Cao, M. Biesiada, J. Li, Y. Pan, and Z.-H. Zhu (2019). Direct test of the FLRW metric from strongly lensed gravitational wave observations. *Sci. Rep. 9*(1), 11608.

Chang, K. and S. Refsdal (1979, December). Flux variations of QSO 0957 + 561 A, B and image splitting by stars near the light path. *282*(5739), 561–564.

Chang, K. and S. Refsdal (1984, March). Star disturbances in gravitational lens galaxies. *132*, 168–178.

Claeskens, J.-F. and J. Surdej (2002, January). Gravitational lensing in quasar samples. *10*(4), 263–311.

Clowe, D., M. Bradač, A. H. Gonzalez, M. Markevitch, S. W. Randall, C. Jones, and D. Zaritsky (2006, September). A Direct Empirical Proof of the Existence of Dark Matter. *648*(2), L109–L113.

Collett, T. E. and D. Bacon (2017). Testing the speed of gravitational waves over cosmological distances with strong gravitational lensing. *Phys. Rev. Lett. 118*(9), 091101.

Courbin, F., P. Saha, and P. L. Schechter (2002). Quasar Lensing. In F. Courbin and D. Minniti (Eds.), *Gravitational Lensing: An Astrophysical Tool*, Volume 608, pp. 1.

Cutler, C. and E. E. Flanagan (1994, Mar). Gravitational waves from merging compact binaries: How accurately can one extract the binary's parameters from the inspiral waveform? *Phys. Rev. D 49*, 2658–2697.

Cutler, C. and K. S. Thorne (2002, April). An Overview of Gravitational-Wave Sources. *arXiv e-prints*, gr–qc/0204090.

Dai, L., S.-S. Li, B. Zackay, S. Mao, and Y. Lu (2018, November). Detecting lensing-induced diffraction in astrophysical gravitational waves. *98*(10), 104029.

Dai, L., T. Venumadhav, and K. Sigurdson (2017, February). Effect of lensing magnification on the apparent distribution of black hole mergers. *95*(4), 044011.

Dalal, N. and C. S. Kochanek (2002, June). Direct Detection of Cold Dark Matter Substructure. *572*(1), 25–33.

Damour, T. (2001, Nov). Coalescence of two spinning black holes: An effective one-body approach. *Phys. Rev. D 64*, 124013.

Diego, J. M. (2020). Constraining the abundance of primordial black holes with gravitational lensing of gravitational waves at LIGO frequencies. *Phys. Rev. D 101*(12), 123512.

Dyson, F. W., A. S. Eddington, and C. Davidson (1920, January). A Determination of the Deflection of Light by the Sun's Gravitational Field, from Observations Made at the Total Eclipse of May 29, 1919. *Philosophical Transactions of the Royal Society of London Series A 220*, 291–333.

Einstein, A. (1905, January). Zur Elektrodynamik bewegter Körper. *Annalen der Physik 322*(10), 891–921.

Einstein, A. (1915, January). Die Feldgleichungen der Gravitation. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften (Berlin*, 844–847.

Einstein, A. (1916, January). Die Grundlage der allgemeinen Relativitätstheorie. *Annalen der Physik 354*(7), 769–822.

Evans, M., R. X. Adhikari, C. Afle, S. W. Ballmer, S. Biscoveanu, S. Borhanian, D. A. Brown, Y. Chen, R. Eisenstein, A. Gruson, et al. (2021). A horizon study for cosmic explorer: science, observatories, and community. *arXiv preprint arXiv:2109.09882*.

Ezquiaga, J. M., D. E. Holz, W. Hu, M. Lagos, and R. M. Wald (2021, March). Phase effects from strong gravitational lensing of gravitational waves. *103*(6), 064047.

Finn, L. S. (1992, December). Detection, measurement, and gravitational radiation. *46*(12), 5236–5249.

Finn, L. S. and D. F. Chernoff (1993, March). Observing binary inspiral in gravitational radiation: One interferometer. *47*(6), 2198–2219.

Forward, R. L. (1978, Jan). Wideband laser-interferometer graviational-radiation experiment. *Phys. Rev. D 17*, 379–390.

Gangardt, D., D. Gerosa, M. Kesden, V. De Renzis, and N. Steinle (2022, July). Constraining black-hole binary spin precession and nutation with sequential prior conditioning. *Phys. Rev. D. 106*(2), 024019.

Gangardt, D., N. Steinle, M. Kesden, D. Gerosa, and E. Stoikos (2021, Jun). A taxonomy of black-hole binary spin precession and nutation. *Phys. Rev. D 103*, 124026.

Gao, Z., X. Chen, Y.-M. Hu, J.-D. Zhang, and S.-J. Huang (2022, May). A higher probability of detecting lensed supermassive black hole binaries by LISA. *512*(1), 1–10.

García-Quirós, C., M. Colleoni, S. Husa, H. Estellés, G. Pratten, A. Ramos-Buades, M. Mateu-Lucena, and R. Jaume (2020, September). Multimode frequency-domain model for the gravitational wave signal from nonprecessing black-hole binaries. *102*(6), 064002.

Gavazzi, R., T. Treu, J. D. Rhodes, L. V. E. Koopmans, A. S. Bolton, S. Burles, R. J. Massey, and L. A. Moustakas (2007, sep). The sloan lens ACS survey. IV. the mass density profile of early-type galaxies out to 100 effective radii. *The Astrophysical Journal 667*(1), 176–190.

Gerosa, D., E. Berti, R. O'Shaughnessy, K. Belczynski, M. Kesden, D. Wysocki, and W. Gladysz (2018, October). Spin orientations of merging black holes formed from the evolution of stellar binaries. *Phys. Rev. D. 98*(8), 084036.

Gerosa, D., M. Kesden, U. Sperhake, E. Berti, and R. O'Shaughnessy (2015, Sep). Multi-timescale analysis of phase transitions in precessing black-hole binaries. *Phys. Rev. D 92*, 064016.

Gerosa, D., M. Mould, D. Gangardt, P. Schmidt, G. Pratten, and L. M. Thomas (2021, March). A generalized precession parameter χ_p to interpret gravitational-wave data. *Phys. Rev. D. 103*(6), 064067.

Gilman, D., S. Birrer, A. Nierenberg, T. Treu, X. Du, and A. Benson (2020, February). Warm dark matter chills out: constraints on the halo mass function and the free-streaming length of dark matter with eight quadruple-image strong gravitational lenses. *491*(4), 6077–6101.

Goyal, S., K. Haris, A. K. Mehta, and P. Ajith (2021, Jan). Testing the nature of gravitational-wave polarizations using strongly lensed signals. *Phys. Rev. D 103*, 024038.

Grote, H. and LIGO Scientific Collaboration (2010, April). The GEO 600 status. *Classical and Quantum Gravity 27*(8), 084003.

Guo, X. and Y. Lu (2022, Jul). Probing the nature of dark matter via gravitational waves lensed by small dark matter halos. *Phys. Rev. D 106*, 023018.

Hanna, C., S. Caudill, C. Messick, A. Reza, S. Sachdev, L. Tsukada, K. Cannon, K. Blackburn, J. D. E. Creighton, H. Fong, P. Godwin, S. Kapadia, T. G. F. Li, R. Magee, D. Meacher, D. Mukherjee, A. Pace, S. Privitera, R. K. L. Lo, and L. Wade (2020, Jan). Fast evaluation of multidetector consistency for real-time gravitational wave searches. *Phys. Rev. D 101*, 022003.

Hannam, M., C. Hoy, J. E. Thompson, S. Fairhurst, V. Raymond, M. Colleoni, D. Davis, H. Estellés, C.-J. Haster, A. Helmling-Cornell, S. Husa, D. Keitel, T. J. Massinger, A. Menéndez-Vázquez, K. Mogushi, S. Ossokine, E. Payne, G. Pratten, I. Romero-Shaw, J. Sadiq, P. Schmidt, R. Tenorio, R. Udall, J. Veitch, D. Williams, A. B. Yelikar, and A. Zimmerman (2022, October). General-relativistic precession in a black-hole binary. *NAT 610*(7933), 652–655.

Hannam, M., P. Schmidt, A. Bohé, L. Haegel, S. Husa, F. Ohme, G. Pratten, and M. Pürrer (2014, Oct). Simple model of complete precessing black-hole-binary gravitational waveforms. *Phys. Rev. Lett. 113*, 151101.

Hannuksela, O. A., T. E. Collett, M. Çalışkan, and T. G. F. Li (2020). Localizing merging black holes with sub-arcsecond precision using gravitational-wave lensing. *Mon. Not. Roy. Astron. Soc. 498*(3), 3395–3402.

Hannuksela, O. A., K. Haris, K. K. Y. Ng, S. Kumar, A. K. Mehta, D. Keitel, T. G. F. Li, and P. Ajith (2019, March). Search for Gravitational Lensing Signatures in LIGO-Virgo Binary Black Hole Events. *874*(1), L2.

Hild, S., M. Abernathy, F. e. Acernese, P. Amaro-Seoane, N. Andersson, K. Arun, F. Barone, B. Barr, M. Barsuglia, M. Beker, et al. (2011). Sensitivity studies for third-generation gravitational wave observatories. *Classical and Quantum gravity 28*(9), 094013.

Hoy, C., S. Fairhurst, M. Hannam, and V. Tiwari (2022, March). Understanding How Fast Black Holes Spin by Analyzing Data from the Second Gravitational-wave Catalogue. *APJ 928*(1), 75.

Hulse, R. A. and J. H. Taylor (1975, January). Discovery of a pulsar in a binary system. *apjl 195*, L51–L53.

Jung, S. and C. S. Shin (2019, February). Gravitational-Wave Fringes at LIGO: Detecting Compact Dark Matter by Gravitational Lensing. *122*(4), 041103.

Kalogera, V. (2000, September). Spin-Orbit Misalignment in Close Binaries with Two Compact Objects. *APJ 541*(1), 319–328.

Kassiola, A. and I. Kovner (1993, November). Elliptic Mass Distributions versus Elliptic Potentials in Gravitational Lenses. *417*, 450.

Kawamura, S. et al. (2021). Current status of space gravitational wave antenna DECIGO and B-DECIGO. *PTEP 2021*(5), 05A105.

Keeton, C. R., S. Mao, and H. J. Witt (2000, July). Gravitational Lenses with more than Four Images. I. Classification of Caustics. *537*(2), 697–707.

Kesden, M., D. Gerosa, R. O'Shaughnessy, E. Berti, and U. Sperhake (2015, Feb). Effective potentials and morphological transitions for binary black hole spin precession. *Phys. Rev. Lett. 114*, 081103.

Kidder, L. E. (1995, July). Coalescing binary systems of compact objects to (post)$^{5/2}$-Newtonian order. V. Spin effects. *Phys. Rev. D. 52*(2), 821–847.

Kidder, L. E., C. M. Will, and A. G. Wiseman (1992, sep). Innermost stable orbits for coalescing binary systems of compact objects. *9*(9), L125–L131.

Kim, K., J. Lee, O. A. Hannuksela, and T. G. F. Li (2022, October). Deep Learning-based Search for Microlensing Signature from Binary Black Hole Events in GWTC-1 and -2. *938*(2), 157.

Klimenko, S. and G. Mitselmakher (2004, sep). A wavelet method for detection of gravitational wave bursts. *Classical and Quantum Gravity 21*(20), S1819.

Klimenko, S., G. Vedovato, M. Drago, F. Salemi, V. Tiwari, G. A. Prodi, C. Lazzaro, K. Ackley, S. Tiwari, C. F. Da Silva, and G. Mitselmakher (2016, Feb). Method for detection and reconstruction of gravitational wave transients with networks of advanced detectors. *Phys. Rev. D 93*, 042004.

Kochanek, C. S. (1991a, November). Systematic Effects in Lens Inversions: 1 Exact Models for 0957+561. *382*, 58.

Kochanek, C. S. (1991b, June). The Implications of Lenses for Galaxy Structure. *373*, 354.

Kochanek, C. S. and R. D. Blandford (1987, October). Gravitaional Imaging by Isolated Elliptical Potential Wells. II. Probability Distributions. *321*, 676.

Kochanek, C. S., R. D. Blandford, C. R. Lawrence, and R. Narayan (1989, May). The ring cycle : an iterative lens reconstruction technique applied toMG 1131+0456. *238*, 43–56.

Kormann, R., P. Schneider, and M. Bartelmann (1994, April). Isothermal elliptical gravitational lens models. *284*, 285–299.

Kovner, I. (1987a, February). Marginal gravitational lenses of large separation: probing superclusters. *325*(6104), 507–509.

Kovner, I. (1987b, January). The Quadrupole Gravitational Lens. *312*, 22.

Lai, K.-H., O. A. Hannuksela, A. Herrera-Martín, J. M. Diego, T. Broadhurst, and T. G. F. Li (2018). Discovering intermediate-mass black hole lenses through gravitational wave lensing. *Phys. Rev. D 98*(8), 083005.

Lawrence, J. K. (1971, January). Focusing of gravitational radiation by interior gravitational fields. *Nuovo Cimento B Serie 6B*, 225–235.

Lee, M., D. Gretz, S. Steppel, and J. Weber (1976). Gravitational Radiation Detector Observations in 1973 and 1974. *Phys. Rev. D 14*, 893–906.

Li, S.-S., S. Mao, Y. Zhao, and Y. Lu (2018). Gravitational lensing of gravitational waves: A statistical perspective. *Mon. Not. Roy. Astron. Soc. 476*(2), 2220–2229.

Li, Y., X. Fan, and L. Gou (2019). Constraining Cosmological Parameters in the FLRW Metric with Lensed GW+EM Signals. *Astrophys. J. 873*(1), 37.

Liao, K., M. Biesiada, and X.-L. Fan (2019, April). The Wave Nature of Continuous Gravitational Waves from Microlensing. *875*(2), 139.

Liao, K., X.-L. Fan, X.-H. Ding, M. Biesiada, and Z.-H. Zhu (2017). Precision cosmology from future lensed gravitational wave and electromagnetic signals. *Nature Commun. 8*(1), 1148. [Erratum: Nature Commun. 8, 2136 (2017)].

Lindblom, L., B. J. Owen, and D. A. Brown (2008, Dec). Model waveform accuracy standards for gravitational wave data analysis. *Phys. Rev. D 78*, 124020.

Maggiore, M. et al. (2020). Science Case for the Einstein Telescope. *JCAP 03*, 050.

Maggiore, M., C. V. D. Broeck, N. Bartolo, E. Belgacem, D. Bertacca, M. A. Bizouard, M. Branchesi, S. Clesse, S. Foffa, J. García-Bellido, S. Grimm, J. Harms, T. Hinderer, S. Matarrese, C. Palomba, M. Peloso, A. Ricciardone, and M. Sakellariadou (2020, mar). Science case for the einstein telescope. *Journal of Cosmology and Astroparticle Physics 2020*(03), 050.

Mao, S. and P. Schneider (1998, April). Evidence for substructure in lens galaxies? *295*(3), 587–594.

Mapelli, M. (2020, July). Binary black hole mergers: formation and populations. *Frontiers in Astronomy and Space Sciences 7*, 38.

Matsunaga, N. and K. Yamamoto (2006, jan). The finite source size effect and wave optics in gravitational lensing. *Journal of Cosmology and Astroparticle Physics 2006*(01), 023–023.

Meneghetti, M. (2021). *Introduction to Gravitational Lensing: With Python Examples*. Lecture Notes in Physics. Springer International Publishing.

Messick, C., K. Blackburn, P. Brady, P. Brockill, K. Cannon, R. Cariou, S. Caudill, S. J. Chamberlin, J. D. E. Creighton, R. Everett, C. Hanna, D. Keppel, R. N. Lang, T. G. F. Li, D. Meacher, A. Nielsen, C. Pankow, S. Privitera, H. Qi, S. Sachdev, L. Sadeghian, L. Singer, E. G. Thomas, L. Wade, M. Wade, A. Weinstein, and K. Wiesner (2017, Feb). Analysis framework for the prompt discovery of compact binary mergers in gravitational-wave data. *Phys. Rev. D 95*, 042001.

Mondal, S., S. Ali, S. Shahul, N. Banerjee, and G. M. Hossain (2021, July). Propagation of gravitational waves in various cosmological backgrounds. *General Relativity and Gravitation 53*(7), 64.

Mortlock, D. J., R. L. Webster, and P. J. Francis (1999, November). Binary quasars. *309*(4), 836–846.

Mroue, A. H. et al. (2013). Catalog of 174 Binary Black Hole Simulations for Gravitational Wave Astronomy. *Phys. Rev. Lett. 111*(24), 241104.

Mukherjee, S., B. D. Wandelt, and J. Silk (2020). Probing the theory of gravity with gravitational lensing of gravitational waves and galaxy surveys. *Mon. Not. Roy. Astron. Soc. 494*(2), 1956–1970.

Nakamura, T. T. (1998, Feb). Gravitational lensing of gravitational waves from inspiraling binaries by a point mass lens. *Phys. Rev. Lett. 80*, 1138–1141.

Nakamura, T. T. and S. Deguchi (1999, January). Wave Optics in Gravitational Lensing. *Progress of Theoretical Physics Supplement 133*, 137–153.

Narasimha, D., K. Subramanian, and S. M. Chitre (1982, September). Spheroidal gravitational lenses and the triple quasar. *200*, 941–950.

Narasimha, D., K. Subramanian, and S. M. Chitre (1984, August). Gravitational lens models for the triple radio source MG 2016+112. *283*, 512–514.

Nissanke, S., D. E. Holz, S. A. Hughes, N. Dalal, and J. L. Sievers (2010, December). Exploring Short Gamma-ray Bursts as Gravitational-wave Standard Sirens. *APJ 725*(1), 496–514.

Nitz, A., I. Harry, D. Brown, C. M. Biwer, J. Willis, T. D. Canton, C. Capano, T. Dent, L. Pekowsky, A. R. Williamson, S. De, M. Cabero, B. Machenschalk, D. Macleod, P. Kumar, S. Reyes, dfinstad, F. Pannarale, S. Kumar, T. Massinger, M. Tápai, L. Singer, G. S. C. Davies, S. Khan, S. Fairhurst, A. Nielsen, S. Singh, K. Chandra, shasvath, and veronica villa (2022, March). gwastro/pycbc: v2.0.2 release of pycbc.

Nitz, A. H., T. Dent, T. D. Canton, S. Fairhurst, and D. A. Brown (2017, nov). Detecting binary compact-object mergers with gravitational waves: Understanding and improving the sensitivity of the PyCBC search. *The Astrophysical Journal 849*(2), 118.

Oguri, M. (2010, August). The Mass Distribution of SDSS J1004+4112 Revisited. *62*, 1017.

Oguri, M. (2018a). Effect of gravitational lensing on the distribution of gravitational waves from distant binary black hole mergers. *Mon. Not. Roy. Astron. Soc. 480*(3), 3842–3855.

Oguri, M. (2018b, Aug). Effect of gravitational lensing on the distribution of gravitational waves from distant binary black hole mergers. *Monthly Notices of the Royal Astronomical Society 480*(3), 3842–3855.

Oguri, M. (2019, December). Strong gravitational lensing of explosive transients. *Reports on Progress in Physics 82*(12), 126901.

Oguri, M. and R. Takahashi (2020). Probing Dark Low-mass Halos and Primordial Black Holes with Frequency-dependent Gravitational Lensing Dispersions of Gravitational Waves. *Astrophys. J. 901*(1), 58.

Ohanian, H. C. (1974, June). On the Focusing of Gravitational Radiation. *International Journal of Theoretical Physics 9*(6), 425–437.

Ohanian, H. C. (1983, August). The caustics of gravitational 'lenses'. *271*, 551–555.

Ossokine, S., A. Buonanno, S. Marsat, R. Cotesta, S. Babak, T. Dietrich, R. Haas, I. Hinder, H. P. Pfeiffer, M. Pürrer, C. J. Woodford, M. Boyle, L. E. Kidder, M. A. Scheel, and B. Szilágyi (2020, Aug). Multipolar effective-one-body waveforms for precessing binary black holes: Construction and validation. *Phys. Rev. D 102*, 044055.

Owen, B. J. and B. S. Sathyaprakash (1999, Jun). Matched filtering of gravitational waves from inspiraling compact binaries: Computational cost and template placement. *Phys. Rev. D 60*, 022002.

Pagano, G., O. A. Hannuksela, and T. G. F. Li (2020, November). LENSINGGW: a PYTHON package for lensing of gravitational waves. *643*, A167.

Payne, E., S. Hourihane, J. Golomb, R. Udall, D. Davis, and K. Chatziioannou (2022, November). Curious case of GW200129: Interplay between spin-precession inference and data-quality issues. *106*(10), 104017.

Peters, P. C. (1974, Apr). Index of refraction for scalar, electromagnetic, and gravitational waves in weak gravitational fields. *Phys. Rev. D 9*, 2207–2218.

Poisson, E. and C. M. Will (1995, Jul). Gravitational waves from inspiraling compact binaries: Parameter estimation using second-post-newtonian waveforms. *Phys. Rev. D 52*, 848–855.

Postnov, K. A. and L. R. Yungelson (2014, May). The Evolution of Compact Binary Star Systems. *Living Reviews in Relativity 17*(1), 3.

Pratten, G., C. García-Quirós, M. Colleoni, A. Ramos-Buades, H. Estellés, M. Mateu-Lucena, R. Jaume, M. Haney, D. Keitel, J. E. Thompson, and S. Husa (2021, May). Computationally efficient models for the dominant and subdominant harmonic modes of precessing binary black holes. *Phys. Rev. D. 103*(10), 104056.

Pürrer, M. and C.-J. Haster (2020, May). Gravitational waveform accuracy requirements for future ground-based detectors. *Phys. Rev. Res. 2*, 023151.

Racine, E. (2008, Aug). Analysis of spin precession in binary black hole systems including quadrupole-monopole interaction. *Phys. Rev. D 78*, 044021.

Riles, K. (2013, January). Gravitational waves: Sources, detectors and searches. *Progress in Particle and Nuclear Physics 68*, 1–54.

Sabín, C., J. Kohlrus, D. E. Bruschi, and I. Fuentes (2015, May). Thermal noise in BEC-phononic gravitational wave detectors. *arXiv e-prints*, arXiv:1505.01302.

Sathyaprakash, B. S. and B. F. Schutz (2009, Mar). Physics, astrophysics and cosmology with gravitational waves. *Living Reviews in Relativity 12*(1).

Schmidt, P., F. Ohme, and M. Hannam (2015, Jan). Towards models of gravitational waveforms from generic binaries: Ii. modelling precession effects with a single effective precession parameter. *Phys. Rev. D 91*, 024043.

Schneider, P. (1985, February). A new formulation of gravitational lens theory, time-delay, and Fermat's principle. *143*(2), 413–420.

Schneider, P., J. Ehlers, and E. E. Falco (1992). *Gravitational Lenses*.

Schramm, T. (1990, May). Realistic elliptical potential wells for gravitational lens models. *231*(1), 19–24.

Sereno, M., P. Jetzer, A. Sesana, and M. Volonteri (2011). Cosmography with strong lensing of LISA gravitational wave sources. *Mon. Not. Roy. Astron. Soc. 415*, 2773.

Sereno, M., A. Sesana, A. Bleuler, P. Jetzer, M. Volonteri, and M. C. Begelman (2010, December). Strong Lensing of Gravitational Waves as Seen by LISA. *105*(25), 251101.

Somiya, K. (2012). Detector configuration of KAGRA: The Japanese cryogenic gravitational-wave detector. *Class. Quant. Grav. 29*, 124007.

Steinle, N. and M. Kesden (2022, September). Signatures of spin precession and nutation in isolated black-hole binaries. *Phys. Rev. D. 106*(6), 063028.

Stoikos, E., S. Ali, M. Kesden, and L. King (2023). Quantifying precessional effects on binary black-hole systems by introducing a generic spinning waveform template .

Takahashi, R. (2004, September). Quasi-geometrical optics approximation in gravitational lensing. *423*, 787–792.

Takahashi, R. (2017, January). Arrival Time Differences between Gravitational Waves and Electromagnetic Signals due to Gravitational Lensing. *835*(1), 103.

Takahashi, R. and T. Nakamura (2003, Oct). Wave effects in the gravitational lensing of gravitational waves from chirping binaries. *The Astrophysical Journal 595*(2), 1039–1051.

Tambalo, G., M. Zumalacárregui, L. Dai, and M. Ho-Yeuk Cheung (2022, December). Gravitational wave lensing as a probe of halo properties and dark matter. *arXiv e-prints*, arXiv:2212.11960.

The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, R. Abbott, T. D. Abbott, F. Acernese, et al. (2021, November). The population of merging compact binaries inferred using gravitational waves through GWTC-3. *arXiv e-prints*, arXiv:2111.03634.

Thorne, K. S. (1980, Apr). Multipole expansions of gravitational radiation. *Rev. Mod. Phys. 52*, 299–339.

Treu, T. and P. J. Marshall (2016, July). Time delay cosmography. *24*(1), 11.

Turner, E. L., J. P. Ostriker, and I. Gott, J. R. (1984, September). The statistics of gravitational lenses : the distributions of image angular separations and lens redshifts. *284*, 1–22.

Usman, S. A., A. H. Nitz, I. W. Harry, C. M. Biwer, D. A. Brown, M. Cabero, C. D. Capano, T. D. Canton, T. Dent, S. Fairhurst, M. S. Kehl, D. Keppel, B. Krishnan, A. Lenon, A. Lundgren, A. B. Nielsen, L. P. Pekowsky, H. P. Pfeiffer, P. R. Saulson, M. West, and J. L. Willis (2016, oct). The PyCBC search for gravitational waves from compact binary coalescence. *Classical and Quantum Gravity 33*(21), 215004.

Varma, V., S. E. Field, M. A. Scheel, J. Blackman, D. Gerosa, L. C. Stein, L. E. Kidder, and H. P. Pfeiffer (2019, Oct). Surrogate models for precessing binary black hole simulations with unequal masses. *Phys. Rev. Res. 1*, 033015.

Wainstein, L. A. and V. D. Zubakov (1970). *Extraction of Signals from Noise*. (Prentice-Hall, Englewood Cliffs, NJ.

Weber, J. (1969, June). Evidence for Discovery of Gravitational Radiation. *22*(24), 1320–1324.

Weber, J. (1970). Anisotropy and polarization in the gravitational-radiation experiments. *Phys. Rev. Lett. 25*, 180–184.

Wempe, E., L. V. Koopmans, A. Wierda, O. A. Hannuksela, and C. v. d. Broeck (2022). A lensing multi-messenger channel: Combining ligo-virgo-kagra lensed gravitational-wave measurements with euclid observations. *arXiv preprint arXiv:2204.08732*.

Will, C. M. and A. G. Wiseman (1996, October). Gravitational radiation from compact binary systems: Gravitational waveforms and energy loss to second post-Newtonian order. *Phys. Rev. D. 54*(8), 4813–4848.

Wiseman, A. G. (1992, Aug). Coalescing binary systems of compact objects to (post)$^{5/2}$-newtonian order. ii. higher-order wave forms and radiation recoil. *Phys. Rev. D 46*, 1517–1539.

Wright, M. and M. Hendry (2022, August). Gravelamps: Gravitational Wave Lensing Mass Profile Model Selection. *935*(2), 68.

Xu, F., J. M. Ezquiaga, and D. E. Holz (2022, April). Please Repeat: Strong Lensing of Gravitational Waves as a Probe of Compact Binary and Galaxy Populations. *929*(1), 9.

Yeung, S. M. C., M. H. Y. Cheung, J. A. J. Gais, O. A. Hannuksela, and T. G. F. Li (2021, December). Microlensing of type II gravitational-wave macroimages. *arXiv e-prints*, arXiv:2112.07635.

Young, P. (1981, March). Q0957+561 : effects of random stars on the gravitational lens. *244*, 756–767.

Young, P., R. S. Deverill, J. E. Gunn, J. A. Westphal, and J. Kristian (1981, March). The triple quasar Q1115+080A, B, C : a quintuple gravitational lens image. *244*, 723–735.

Yu, H., P. Zhang, and F.-Y. Wang (2020). Strong lensing as a giant telescope to localize the host galaxy of gravitational wave event. *Mon. Not. Roy. Astron. Soc. 497*(1), 204–209.

Zhao, X., M. Kesden, and D. Gerosa (2017, July). Nutational resonances, transitional precession, and precession-averaged evolution in binary black-hole systems. *Phys. Rev. D. 96*(2), 024007.

BIOGRAPHICAL SKETCH

Saif Ali was born in Kolkata, India. He was accepted to Indian Institute of Science Education and Research, Kolkata, in 2013 where he received his Bachelor of Science and Master of Science with a major in Physics in 2018. During this period he received the INSPIRE Fellowship from the Department of Science and Technology, Government of India. He did his master's thesis, "*Propagation of Gravitational Waves in an Anisotropic Background*", with Dr. Golam Mortuza Hossain and Dr. Narayan Banerjee. He then entered the physics graduate program at The University of Texas at Dallas in 2018. There he obtained his Master of Science in 2020 and then his PhD in Physics in 2023. He wrote his PhD dissertation, "*Gravitational Lensing of Gravitational Waves: Effects of Different Lens Models*", under the supervision of Dr. Lindsay King and Dr. Michael Kesden.

CURRICULUM VITAE

Saif Ali

September 8, 2023

Contact Information:

Department of Physics
The University of Texas at Dallas
800 W. Campbell Rd.
Richardson, TX 75080-3021, U.S.A.

Voice: (682) 252-2645
Email: sxa180025@utdallas.edu

Educational History:

PhD Physics June 2020 – Present
The University of Texas at Dallas
Gravitational Lensing of Gravitational Waves: Effects of Different Lens Models

MS Physics Aug 2018 – May 2020
The University of Texas at Dallas

BS-MS Physics Aug 2013 – May 2018
Indian Institute of Science Education and Research, Kolkata
Propagation of Gravitational Waves in an Anistropic Background

Employment History:

Graduate Research Assistant, The University of Texas at Dallas June 2021 - Present
Graduate Teaching Assistant, The University of Texas at Dallas Aug 2018 - May 2021

External Talks:

Contributed talk, American Physical Society, New York April 2022
Research project talk, Department of Physics, Osaka University July 2017
Research project talk, Department of Physics, Indian Institute of Science June 2016

Publications:

Saif Ali, Evangelos Stoikos, Michael Kesden, and Lindsay King, to be submitted April 2023 (Ali et al., 2023)
Evangelos Stoikos, **Saif Ali**, Nathan Steinle, Michael Kesden, and Lindsay King, to be submitted April 2023 (Stoikos et al., 2023)
Saif Ali, Evangelos Stoikos, Evan Meade, Michael Kesden, and Lindsay King (Ali et al., 2022)
Sushovan Mondal, **Saif Ali**, Narayan Banerjee, Golam Mortuza Hossain (Mondal et al., 2021)